Signal Analysis
&
Signal Manipulation

Lecture Notes

by:

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This document has a long tradition and is evolved through the years. Major modifications are made in the previous years which made us decide to put only the main authors on the title page. Since in the past also other authors contributed to the document in various stages we want to mention them here: dr. ir. H. Wormeester, dr. ir. B.H.L Betlem, dr. H.K. Hemmes, prof. dr. S.G. Lemay and prof. dr. H.J. Zwart and Ir. I.C.W.T.A. van Veldhoven. In part III C.G. Zeinstra added about six additional problems to each chapter.



Version history of this document:

v1.0 First version in the new form concentrated on *Signals*.

Preface

Preface to edition v1.0
This reader has a long tradition, with many modifications.
This reader is a subset of the reader *Modelling, Analysis and Control of Dynamic Systems*, that was used in module 4 and module 5 for the courses: *Engineering Systems*, *Dynamical Systems*, *Models* and *Signals*. For last year's module 4, a significant modificaion of the reader is initiated with more emphasis on the *Modeling*-part of Advanced Engineering. Former Part-I of the reader used to be an introduction for *Models* as well as *Signals*. So for this reader, that concentrates on the *Signals* part, a subset of chapters from Part-I and Part-III is "glued" together to present a reader with all the material relevant for *Signals*.
Since this is the first step in the modification, some of the chapters might rely on the material of the *Models*-related material. Some effort is put in the replacement of hard links to the *Models* sections to avoid a direct relation to that material. However some of the examples or problems might require some knowledge of that material. In later iterations this will be improved to work to a consistent self-containing document on Signal Analysis and Manipulation.
In *Models* the emphasis is on describing physical systems by means of a mathematical model or system. Signals are used to test the model or system. In *Signals* the emphasis is on signals, their analysis and the design of systems to manipulate the signals: filters. Needless to say that there is a strong parallel between the two fields. It is more the approach that makes them different, which means that both fields share a lot of tools.
Due to the origin of this reader, you might find many problems related to the modeling part. In many cases you can skip these problems. In later reversions of this reader this will be corrected.
An other effect that the student will encounter is the accompanying code files for the various examples and problems. Since the time is too short to produce a consistent set of material, we will provide a translation table for the code files of the examples and problems. The answers and solutions for the tutorials are consistent with the reader since they are in the same source files with the problems.

jeroen

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Part I

Analysis of Systems and Signals

Chapter 1

Linear differential equations

Many phenomena in physics and chemistry are conveniently modeled by *ordinary differential equations* (ODEs). ODEs contain one or more functions of one independent variable and its derivatives. (The term *ordinary* is used instead of the term *partial* differential equation to differentiate ODEs from PDEs, which have terms that depend on more than one independent variable.) The ubiquitous example of an ODE is Newton's second law:

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = F(t),$$

where $y(t)$ is the position of the mass and $F(t)$ the force exerted on the mass.
Another example is that of heat flow and accumulation, leading to, for instance, the change in the temperature reading on a thermometer:

$$T^{(1)}_{\mathrm{merc.}}(t) = \frac{T_{\mathrm{env}}(t) - T_{\mathrm{merc.}}(t)}{\tau}, \tag{1.1}$$

where $T_{\mathrm{merc.}}(t)$ and $T_{\mathrm{env}}(t)$ denote the temperature of the mercury and the temperature of the environment, respectively. Here the number τ has dimension time, and is a combination of several physical parameters.
In these two examples, respectively, we assume that the force $F(t)$ and the temperature $T_{\mathsf{env}}(t)$ are known and that we are interested in the resulting $y(t)$ and $T_{\mathsf{merc.}}(t)$. In general, explicit solutions to ODE's are hard to find and often they are even unknown. In this chapter, however, we consider a class of ODE's that have a more or less explicit solution. This class of ODE's is useful to analyse so that we can develop an understanding of the dynamics of systems and their time constant. In this chapter, we consider linear ODEs with constant coefficients:

$$\begin{aligned} a_n y^{(n)}(t) + a_{n-1}y^{(n-1)}(t) + \cdots + a_1 y^{(1)}(t) + a_0 y(t) \\ = b_m x^{(m)}(t) + b_{m-1}x^{(m-1)}(t) + \cdots + b_1 x^{(1)}(t) + b_0 x(t). \end{aligned} \tag{1.2}$$

Here $a_0, a_1, \ldots, a_n, b_0, \ldots,$ and b_m are real numbers and $x(t)$ is a function that we assume we know. This function $x(t)$ is referred to as the *input* and the resulting function $y(t)$ that we are after is the *output* of the system, i.e. the solution of the ODE.
In the formulation of the ODE, the nth differentiation with respect to time is written as $y^{(n)}(t)$. Four different notations for differentiation are commonly employed, so get acquainted with all of these:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= y^{(1)}(t) = y^{'} = \dot{y} \\ \frac{\mathrm{d}^2y}{\mathrm{d}t^2} &= y^{(2)}(t) = y^{''} = \ddot{y} \\ \frac{\mathrm{d}^ny}{\mathrm{d}t^n} &= y^{(n)}(t) \end{aligned}$$

An ODE with constant coefficients is *time invariant*. This means that if the input is shifted by an amount of time the only thing that changes is that the output is also shifted by the same amount of time, but the shape of the output stays the same. An ODE is only time invariant if the coefficients of the ODE are constant, so that only the input and output depend on time. Obviously this class of ODE is also *linear*. That is, if $y_1(t)$ and $y_2(t)$ are the outputs resulting from inputs $x_1(t)$ and $x_2(t)$, then the output to input $x_3(t) = x_1(t) + x_2(t)$ is simply $y_3(t) = y_1(t) + y_2(t)$. Finally, we will also insist that physical systems should be *causal* , the technical term for saying that the output in the present should not depend on the input in the future. These three characteristics embody the exact mathematical expression that describes so-called *LTI* systems.

1.0.1 Coupled first-order ODE's

The presentation of equation 1.2 of an nth-order differential equation is in the representation of a so called input-output equation, where $y(t)$ is the output of the system and $x(t)$ is the input of the system. A single higher-order differential equation can only have one input and one output – also called a SISO system.
The other extreme is a set of n first-order differential equations that are coupled. In most general sense this is represented by:

$$y_i^{(1)}(t) = \sum_{j \neq i}^{n} a_{i,j} y_i(t) + \sum_{j=0}^{m} b_{i,j} x_j(t) \tag{1.3}$$

This set of equations can also be written in matrix form:

$$\vec{y}^{(1)}(t) = \mathbb{A}\vec{y}(t) + \mathbb{B}\vec{x}(t) \tag{1.4}$$

Where $\vec{y}(t)$ and $\vec{x}(t)$ are the output and input vector of dimensions n and m respectively and $\mathbb{A}$ the $n \times n$ matrix and $\mathbb{B}$ the $n \times m$ matrix.
The advantage of this representation is that is is close to the construction of physics models from element equations and conservation and continuity equations. This will keep you away from tedious algebraic manipulations the set to a single differential equation where all variables are rewritten in a single output variable and its derivatives and an input variable with its derivatives.

Example 1.0.1. *equivalence of a second order differential equation and set of 2 first-order differential equations*
Let us start with the set of two coupled first-order differential equations:

$$\begin{aligned} y_1^{(1)}(t) &= -2y_1(t) + y_2(t) \\ y_2^{(1)}(t) &= -y_2(t) + y_1(t) + x(t) \end{aligned} \tag{1.5}$$

This is equivalent with (by taking the derivative of the first equation, substitute the second one to eliminate $y_2^{(1)}(t)$, and substitute the first one in the result to eliminate $y_2(t)$):

$$y_1^{(2)}(t) + 3y_1^{(1)}(t) + y_1(t) = x(t) \tag{1.6}$$

For completeness we can also take the reverse route of splitting the input-output differential equation in two first-order differential equations; define $y_3(t) = y_1^{(1)}(t)$ to pull the equation apart:

$$\begin{aligned} y_1^{(1)}(t) &= y_3(t) \\ y_3^{(1)}(t) &= -3y_3(t) - y_1(t) + x(t) \end{aligned} \tag{1.7}$$

Note that the reverse route does not give the original set of coupled first-order differential equations back, which illustrates the fact that there is no one unique set of first order equations.
In general is the result of a transformation from a high-order differential equation to a set of coupled first-order equations a set of variables that are the derivative of on single variable, except for the lowest one, that is a linear combination of all the other variables.

□

1.1 Solving sets of linear time invariant differential equations

Before we solve a set of linear time invariant differential equations, we first point out some of the strategies to follow.
In this chapter we will use traditional methods to solve the differential equations. This involves a few sequential steps: first a homogeneous solutions is found in the form of a linear combination of a set of exponentials. In case of an input signal, a particular solution has to be found too, and added to the homogeneous solution. Initial conditions will be used to determine the coefficients of the linear combination. The total sum, with the appropriate coefficients is the solution.
Since traditional method is not very efficient, we will introduce the Laplace Transformation in chapter 3 to transform the set of differential equations in a set of linear equations. This set is easier to solve using techniques of Linear Algebra. The found solution can subsequently be transformed back to the time domain to obtain the overall solution. Since also the input and output signals can be described in the Laplace domain, an integrated approach is presented.
Both approaches will reveal characteristics that are included in the set of differential equations. In the chapter 2 on system response in the time domain and chapter 4 on transfer functions, we will highlight the characteristics from different viewpoints. These chapters will also provide you with tools to analyse and characterise systems. The techniques used will be applied on mechanical and electrical systems in this first part of the reader. Cross-references will be made between characteristics of a set of differential equations, transfer functions and time responses of systems.
For illustrative purposes we start with the direct solving method of differential equations.

1.2 Homogenous solution — natural response

The LTI ODE's describe the dynamics of physical systems. A system in equilibrium will not show any dynamic behaviour, which i s equivalent with all time derivatives set to zero. The set op differential equations will reduce to a set of linear algebraic equations, describing the equilibrium. There a two ways to get the system the system out of equilibrium: start with initial conditions that are not an equilibrium and/or a non-zero input signal. In this section we treat the first situation. Since there is no input to the system, all characteristics are related to the system itself and gives a good picture of the system.
Mathematically this is described by setting the input $x(t)$ and all its derivatives to zero. The general ODE simplifies to the *homogenous equation*:

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y^{(1)}(t) + a_0 y(t) = 0. \tag{1.8}$$

A solution $y_{\text{hom}}(t)$ of the homogeneous equation is sometimes referred to as a *natural response* as it is free of influence by an input signal. The zero function $y_{\text{hom}}(t) = 0$ clearly is a possible solution, but generally there are more.
The homogeneous ODE contains the dynamic characteristics of the system. In this section we will look at a first-order system and a second-order system. In fact this is all we need to derive the characteristic parameters from higher order systems.
The solutions we will derive below are only relate the characteristics to the differential equations. The method of setting up differential equations and finding solutions will be replaced by more efficient solutions later in this reader. Therefore we will only treat a few specific situations below.

Example 1.2.1. *First-order system*
The homogeneous equation of

$$y^{(1)}(t) + a_0 y(t) = b_0 x(t)$$

is

$$y^{(1)}(t) + a_0 y(t) = 0.$$

The general solution of this homogeneous equation for a real-valued solution is

$$y_{\text{hom}}(t) = c\,\mathrm{e}^{-a_0 t}, \qquad c \in \mathbb{R}.$$

It is easy to verify by direct subsitution into the homogeneous equation that for any $c \in \mathbb{R}$ the exponential function $y_{\text{hom}}(t) = c\,\mathrm{e}^{-a_0 t}$ is indeed a solution of the homogeneous equation and one can prove that no other solution exists. While the value of c appears arbitrary, in practice we will usually be constrained by so-called *initial conditions*: if the value of $y_{\text{hom}}(t)$ is known at $t = 0$, then there is only one possible choice for c since $y_{\text{hom}}(0) = c$.
From the differential equation as well as from the solution we conclude the the unit of a_0 must be [1/s]. Another way of saying is that $\tau \equiv 1/a_0$ is a characteristic time of the system. This parameter τ describes the dynamics of the system. Note that the other parameter c is only a scaling factor, determined by the initial condition of the system.
If we go back to the homogeneous part of equation of (1.1) by setting the input $T_{\text{env}}(t)$ to be zero we obtain:

$$T^{(1)}_{\text{merc.}}(t) + \frac{1}{\tau} T_{\text{merc.}}(t) = 0.$$

Its general solution is:

$$T_{\text{hom,merc.}}(t) = c\,\mathrm{e}^{-t/\tau}, \qquad c \in \mathbb{R}.$$

Since $\tau > 0$ these solutions all converge exponentially fast to zero as $t \to \infty$. This model thus says that setting the temperature of the environment $T_{\text{env}}(t)$ in (1.1) to zero causes the temperature of the mercury $T_{\text{merc.}}(t)$ to converge to zero exponentially fast with a characteristic time τ. □

Motivated by this example it is tempting to hypothesize that higher-order homogeneous equations (1.8) also allow exponential solutions $y(t) = \mathrm{e}^{st}$ as well[1]. Indeed they do: substitution of $y(t) = \mathrm{e}^{st}$ into the left-hand side of (1.8) gives

$$\begin{aligned}
&a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y^{(1)}(t) + a_0 y(t) = 0 \\
&a_n s^n\,\mathrm{e}^{st} + a_{n-1} s^{n-1}\,\mathrm{e}^{st} + \cdots + a_1 s\,\mathrm{e}^{st} + a_0\,\mathrm{e}^{st} = 0 \\
&\left(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0\right)\mathrm{e}^{st} = 0.
\end{aligned}$$

This is function is equal to zero if and only if s has such a value that

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0. \tag{1.9}$$

since e^{st} is not zero in general. Equation 1.9 is known as the *characteristic equation*. It is a polynomial equation in s. The fundamental theorem of algebra states that a polynomial equation of degree n has exactly n roots $s_1, \ldots, s_n \in \mathbb{C}$. Although the roots are in general not necessarily all different, we first concentrate on cases where this condition is satisfied. Since the values of s determine the dynamics of the system we will explore a few situations.
Every root $s_i \in \mathbb{C}$ gives rise a homogeneous solution

$$y_{\text{hom},i}(t) = c_i\,\mathrm{e}^{s_i t}\,.$$

[1] In order to make this chapter consistent with the rest of the reader, beginning with chapter3 we use here the variable s in the exponential solution instead of λ, as is often seen in mathematical text books (and also used before (<v6.0) in this reader).

Any superposition of solutions is a solution as well. So in fact

$$y_{\text{hom}}(t) = c_1\,\mathrm{e}^{s_1 t} + c_2\,\mathrm{e}^{s_2 t} + \cdots + c_n\,\mathrm{e}^{s_n t} = \sum_{i=1}^{n} c_n \mathrm{e}^{s_i t}$$

is a solution of the homogeneous equation for any $c_1, c_2, \ldots, c_n \in \mathbb{C}$. The actual values of c_i are determined by the initial conditions of the variables. Note that you need as many initial conditions as the order of the system in order to get a unique solution.

Example 1.2.2. *First-order system*
The characteristic equation of

$$y^{(1)}(t) + a_0 y(t) = 0$$

is $s + a_0 = 0$. It has one root, $s_1 = -a_0$ hence by the above

$$y_{\text{hom}}(t) = c_1\,\mathrm{e}^{-a_0 t}, \qquad c_1 \in \mathbb{C}.$$

For physical reasons the constant c_1 usually turns out to be real, but mathematically any complex number satisfies the equation. □

Example 1.2.3. *Second-order system*
Consider

$$y^{(2)}(t) + 3y^{(1)}(t) + 2y(t) = 0. \tag{1.10}$$

Its characteristic polynomial is

$$s^2 + 3s + 2.$$

Now we have two characteristic roots, $s_1 = -1$ en $s_2 = -2$ and hence

$$y_{\text{hom}}(t) = c_1\,\mathrm{e}^{-t} + c_2\,\mathrm{e}^{-2t}, \qquad c_1, c_2 \in \mathbb{C}$$

are solutions of the homogeneous equation. Note that both $c_1\,\mathrm{e}^{-t}$ and $c_2\,\mathrm{e}^{-2t}$ converge to zero exponentially fast but e^{-2t} converges *much* faster than e^{-t} so as time progresses $c_1\,\mathrm{e}^{-t}$ dominates over $c_2\,\mathrm{e}^{-2t}$ (provided that c_1 is non zero). That is, the duration of the response is set by the slowest mode. We say that the slowest *mode* $c_1\,\mathrm{e}^{-t}$ dominates the behaviour of the natural response. □

Example 1.2.4. *Third-order system*
Consider the ODE

$$y^{(3)}(t) + 3y^{(2)}(t) + 2y^{(1)}(t) = x(t).$$

For the homogeneous part the term due to the input $x(t)$ is irrelevant and we have similarly as in the previous example a characteristic polynomial whose degree equals the highest order of differentiation,

$$s^3 + 3s^2 + 2s.$$

This we can factor as $s(s^2 + 3s + 2)$ and the polynomial of degree two that remains in turn factors as $s^2 + 3s + 2 = (s + 1)(s + 2)$. We thus found three characteristic roots

$$s_1 = 0, \quad s_2 = -1, \quad s_3 = -2$$

and so the general solution of the homogeneous equation is

$$y_{\text{hom}}(t) = c_1\,\mathrm{e}^{0t} + c_2\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{-2t} = c_1 + c_2\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{-2t}$$

with c_1, c_2, c_3 arbitrary constants. □

Example 1.2.5. *Complex roots – only imaginary*
The ODE

$$y^{(2)}(t) + y(t) = 0 \tag{1.11}$$

has characteristic polynomial $A(s) := s^2 + 1$. Its roots are complex, $s_1 = -\mathrm{j}$ and $s_2 = +\mathrm{j}$ and hence the complete solution of (1.11) is given by

$$y(t) = c_1\,\mathrm{e}^{-\mathrm{j}t} + c_2\,\mathrm{e}^{\mathrm{j}t}, \qquad c_1, c_2 \in \mathbb{C}.$$

We are typically only interested in real solutions. These can identified using Euler's formula $\mathrm{e}^{\mathrm{j}\omega} = \cos(\omega) + \mathrm{j}\sin(\omega)$,

$$\begin{aligned} y(t) &= c_1\,\mathrm{e}^{-\mathrm{j}t} + c_2\,\mathrm{e}^{\mathrm{j}t} \\ &= c_1\left[\cos(-t) + \mathrm{j}\sin(-t)\right] + c_2\left[\cos(t) + \mathrm{j}\sin(t)\right] \\ &= [c_1 + c_2]\cos(t) + [-\mathrm{j}c_1 + \mathrm{j}c_2]\sin(t). \end{aligned} \tag{1.12}$$

Defining $\tilde{c}_1 = c_1 + c_2$ and $\tilde{c}_2 = -\mathrm{j}c_1 + jc_2$, allows us to rewrite equation 1.12 as

$$y(t) = \tilde{c}_1\cos(t) + \tilde{c}_2\sin(t). \tag{1.13}$$

These are real valued functions if and only if $\tilde{c}_1$ and $\tilde{c}_2$ are real numbers. □

Two more examples complete the picture.

Example 1.2.6. *Complex roots*
The homogeneous equation

$$y^{(2)}(t) + 4y^{(1)}(t) + 13y(t) = 0$$

has characteristic polynomial $s^2 + 4s + 13$ with roots at

$$s_1 = -2 + 3\mathrm{j}, \quad s_2 = -2 - 3\mathrm{j}.$$

The real part of the root is -2 (for both roots) and their imaginary parts are ± 3. Initially we express the complete set of homogeneous solutions as before as a sum of exponential functions,

$$y(t) = c_1\,\mathrm{e}^{(-2+3\mathrm{j})t} + c_2\,\mathrm{e}^{(-2-3\mathrm{j})t}.$$

Once again, using Euler's formula, this can be rewritten in real form

$$\begin{aligned} y(t) &= c_1\,\mathrm{e}^{(-2+3\mathrm{j})t} + c_2\,\mathrm{e}^{(-2-3\mathrm{j})t} \\ &= c_1\,\mathrm{e}^{-2t}\left[\cos(3t) + \mathrm{j}\sin(3t)\right] + c_2\,\mathrm{e}^{-2t}\left[\cos(-3t) + \mathrm{j}\sin(-3t)\right] \\ &= [c_1 + c_2]\,\mathrm{e}^{-2t}\cos(3t) + [\mathrm{j}c_1 - \mathrm{j}c_2]\,\mathrm{e}^{-2t}\sin(3t). \end{aligned}$$

Defining $\tilde{c}_1 = c_1 + c_2$ and $\tilde{c}_2 = \mathrm{j}c_1 - \mathrm{j}c_2$, we have

$$y(t) = \tilde{c}_1\,\mathrm{e}^{-2t}\cos(3t) + \tilde{c}_2\,\mathrm{e}^{-2t}\sin(3t).$$

It is the real part $\mathrm{Re}(s_1) = \mathrm{Re}(s_2) = -2$ that reappears as the exponent in the exponential function, whereas the imaginary part $\mathrm{Im}(s_1) = -\mathrm{Im}(s_2) = 3$ reappears as the angular frequency of the sine and cosine. □

Example 1.2.7. *Coinciding roots*
The situation is slightly different if characteristic roots coincide. For instance the ODE

$$y^{(3)}(t) + 4y^{(2)}(t) + 5y^{(1)}(t) + 2y(t) = 0 \tag{1.14}$$

has characteristic polynomial $A(s) := s^3 + 4s^2 + 5s + 2 = (s+1)^2(s+2)$. It has three roots

$$s_1 = -1, \quad s_2 = -1, \quad s_3 = -2$$

of which the first two coincide. We know that

$$y(t) = \underbrace{c_1\,\mathrm{e}^{-t} + c_2\,\mathrm{e}^{-t}}_{(c_1+c_2)\,\mathrm{e}^{-t}} + c_3\,\mathrm{e}^{-2t}$$

are solutions to the homogeneous equation but there are more. Since -1 appears *twice* the general form of the homogeneous solution takes the form (no derivation, but you can check that this form satisfies the differential equation):

$$[\tilde{c}_1 + \tilde{c}_2 t]\,\mathrm{e}^{-t} + \tilde{c}_2\,\mathrm{e}^{-2t}\,.$$

The pole plot is shown in Fig. 1.1(c). □

1.2.1 Characteristics and representation

In general, if the multiplicity of s_1 is m then a term $[c_1 + c_2 t + \cdots c_m t^{m-1}]\,\mathrm{e}^{s_1 t}$ with m degrees of freedom $c_1, c_2, \ldots, c_m$ shows up in the homogeneous solution. The complete result is:

Theorem 1.2.1. *General solution of homogeneous equation*
Factor the $A(s) := a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$ as

$$A(s) = a_n(s - s_1)^{n_1}(s - s_2)^{n_2}\cdots(s - s_m)^{n_m}$$

with all s_k mutually distinct and a multiplicity of n_k. Then $y(t)$ is a solution of the homogeneous equation (1.8) if and only if it is of the form

$$y(t) = \sum_{k=1}^{m} \underbrace{(c_{k,0} + c_{k,1}t + \cdots + c_{k,n_k-1}t^{n_k-1})}_{\text{polynomial of degree } n_k - 1}\,\mathrm{e}^{s_k t} \tag{1.15}$$

$$= \sum_{k=1}^{m} \left(\sum_{j=0}^{n_k-1} c_{k,j} t^j \right) \mathrm{e}^{s_k t} \tag{1.16}$$

with $c_{k,j} \in \mathbb{C}$. □

The total degree of freedom (*i.e.* the total number of coefficients $c_{k,j}$) equals the order n of the equation.
Often characteristic roots s_i are referred to as *poles* or *eigenvalues* for reasons that will become clear later when we treat transfer functions.
Due to the fact that we are only treating real physical problems, the coefficients of the characteristic equation, of which the roots are the poles of the system, we only have to deal with real coefficients. This has a consequence on the kind of poles we can derive from the characteristic equation.

Theorem 1.2.2. The roots of the characteristic equation $a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$ are only real valued or pairs of complex valued with their complex conjugate.

□

A consequence of this theorem is that the complex conjugated pairs can be interpreted as a characteristic frequency and a damping instead of two (complex) characteristic times. At this point in the reader we only use this representation to make it consistent with later parts in the reader and to give complex characteristic times a more meaningful description.
A complex conjugated pair of poles can be written in terms of real coefficients a and b: $s_{1,2} = a \pm \mathrm{j}b$ with a and $b \in \mathbb{R}$, than the characteristic equation becomes:

$$s^2 - 2as + a^2 + b^2 = 0 \tag{1.17}$$

with only real coefficients. If we now define: $a^2 + b^2 \equiv \omega_n^2$ with ω_n the natural frequency and $2a \equiv -2\zeta\omega_n$ with ζ the damping coefficient, we get:

$$a = -\omega_n\zeta \tag{1.18}$$

$$b = \pm\omega_n\sqrt{1-\zeta^2} \tag{1.19}$$

From this definition it is obvious that if ζ is between zero and one, the coefficient b is real and we have an oscillation with a frequency ω_d:

$$\omega_d \equiv \omega_n\sqrt{1-\zeta^2} \tag{1.20}$$

and a damping of $\omega_n\zeta$. Note that the sign of the value of a is negative resulting in a real negative argument in the exponential of the homogeneous solution of the differential equation.
In case $\zeta > 1$ the coefficient b is not real any more; b is pure imaginary and therefore we get two real arguments in the exponential; remember that $s_{1,2} = a \pm \mathrm{j}b$. The result is that we can factor the characteristic equation to $(s - s_1)(s - s_2)$, *i.e.* two real solutions, corresponding to two characteristic times. Note that this is consistent with equation 1.17, although we assumed b to be real.
It might be convenient to make a graphical representation of the poles of a system, especially in more complex higher-order systems. This is done in a so called *pole plot*. A pole plot is a plot in the complex plane with $\times$ indicating the location of the poles (i.e. characteristic roots) of the ODE. For instance the pole plot of the ODE of three examples is shown in figure 1.1; see caption for the references to the examples.

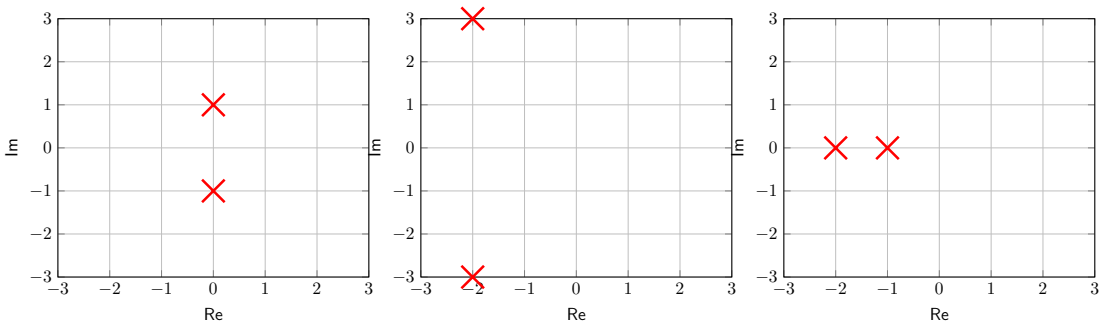


Figure 1.1: Three pole plots: Examples 1.2.5, 1.2.6 and 1.2.7.

1.3 General solution for non-zero input

This section is for completeness to obtain a full solution to the input-output differential equation. In fact this method is not used in this course, since other methods are more efficient. Only the basis is presented with some examples to get a general idea of this method.
For arbitrary inputs $x(t)$ it is hard to obtain closed form expressions of the output $y(t)$ but for some specific inputs $x(t)$ it is possible and this allows us to gain some further insight into the dynamical properties of the system. We restrict ourselves to inputs that are polynomial or exponential. In

such cases one can try a particular output $y_{\text{part}}(t)$ that respectively is polynomial and exponential as well. Once we found one particular solution, the general solution set is obtained by adding the set of homogeneous solutions,

$$y(t) = y_{\text{hom}}(t) + y_{\text{part}}(t).$$

Example 1.3.1. *Polynomial input*
Consider

$$y^{(3)}(t) + 4y^{(2)}(t) + 5y^{(1)}(t) + 2y(t) = x(t), \qquad x(t) = 4t.$$

The right-hand side of the ODE has degree 1, so we try a particular solution of degree 1 as well,

$$y_{\text{part}}(t) = \beta_0 + \beta_1 t.$$

For this choice, the left-hand side of the ODE becomes

$$y^{(3)}_{\text{part}}(t) + 4y^{(2)}_{\text{part}}(t) + 5y^{(1)}_{\text{part}}(t) + 2y_{\text{part}}(t) = 0 + 0 + 5\beta_1 + 2\beta_0 + 2\beta_1 t.$$

It equals $4t$ if $\beta_1 = 2$ and $\beta_0 = -5$, hence

$$y_{\text{part}}(t) = -5 + 2t$$

is one particular solution. The homogeneous solution was determined earlier $y_{\text{hom}}(t) = [c_1 + c_2 t]\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{-2t}$ so the general solution is

$$y(t) = y_{\text{hom}}(t) + y_{\text{part}}(t) = c_1\,\mathrm{e}^{-t} + c_2 t\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{-2t} - 5 + 2t.$$

The general solution has three degrees of freedom, c_1, c_2 and c_3. This freedom can be exploited to match, three initial conditions. For instance, suppose that the initial conditions are

$$y(0) = 0, \quad \dot{y}(0) = -3, \quad \ddot{y}(0) = 13.$$

This determines the coefficients c_1, c_2, c_3:

$$\begin{array}{rcrcl} y(0) = 0 & : & c_1 + c_3 - 5 & = & 0, \\ \dot{y}(0) = -3 & : & -c_1 + c_2 - 2c_3 + 2 & = & -3, \\ \ddot{y}(0) = -13 & : & c_1 - 2c_2 + 4c_3 & = & 13. \end{array}$$

The solution is $c_1 = -3, c_2 = 8$ and $c_3 = 8$. Given the initial conditions the solution $y(t)$ of the ODE is therefore unique, $y(t) = -3\,\mathrm{e}^{-t} + 8t\,\mathrm{e}^{-t} + 8\,\mathrm{e}^{-2t} - 5 + 2t$. □

Example 1.3.2. *Exponential input*
Consider the same ODE but now with an exponential input,

$$y^{(3)}(t) + 4y^{(2)}(t) + 5y^{(1)}(t) + 2y(t) = x(t), \qquad x(t) = \mathrm{e}^{st}.$$

We try $y_{\text{part}}(t) = A\,\mathrm{e}^{st}$, with A yet to be determined. The ODE then becomes

$$As^3\,\mathrm{e}^{st} + 4As^2\,\mathrm{e}^{st} + 5As\,\mathrm{e}^{st} + 2A\,\mathrm{e}^{st} = \mathrm{e}^{st}.$$

Division by e^{st} of both sides of the equation yields

$$A(s^3 + 4s^2 + 5s + 2) = 1,$$

hence $A = 1/(s^3 + 4s^2 + 5s + 2)$ and we get

$$y_{\text{part}}(t) = \frac{1}{s^3 + 4s^2 + 5s + 2}\,\mathrm{e}^{st}.$$

This works if and only when the characteristic equation $s^3 + 4s^2 + 5s + 2 \neq 0$ i.e. if and only if s is not a pole of the ODE. □

An important case is when the input is constant, *i.e.* $s = 0$,

$$x(t) = x_0.$$

Constant functions are special cases of polynomials and are also special cases of exponentials, so we contemplate a constant particular solution $y_{\text{part}}(t)$. Since every derivative of a constant signal is zero, the general ODE (1.2) reduces to the algebraic equation

$$a_0 y_{\text{part}}(t) = b_0 x(t).$$

That is,

$$y_{\text{part}}(t) = \frac{b_0}{a_0} x_0.$$

Example 1.3.3. *Mercury in constant environment*
Suppose the temperature T_{env} of the environment is constant. The constant particular solution of the mercury ODE

$$T^{(1)}_{\text{merc.}}(t) + \frac{1}{\tau} T_{\text{merc.}}(t) = \frac{1}{\tau} T_{\text{env}}$$

is then

$$T_{\text{merc.}}(t) = T_{\text{env}}$$

and the general solution therefore is

$$T_{\text{merc.}}(t) = T_{\text{env}} + c\,\mathrm{e}^{-t/\tau}, \qquad c \in \mathbb{R}.$$

It shows that a constant T_{env} makes the mercury temperature converge exponentially fast to the temperature of the environment. □

Next to the influence of the input on the output, the initial conditions are also important. This can of course be solved in the time domain as done above, but a more straightforward method for handling the specific input and the initial condition(s) is provided by the method of Laplace transforms. The further in depth treatment of initial values is therefore delayed to that chapter.

1.3.1 Fast and slow system

Figure 1.2 depicts three pole plots with the corresponding step responses. The first pole plot indicates a single pole at $s_1 = -1$. The time constant hence is $-1/s_1 = 1$ and indeed after 1 time unit the response approaches the final value to about 37%. The second pole plot is quite similar but now the pole as at $s_1 = -2$. The time constant is half that of the first case, and the step response is now twice as fast. The third pole plot indicates that there are *two* poles. Strictly speaking we do not yet know what the time constant is. However, based on the two poles $s_{1,2} = -1 \pm 2\mathrm{j}$, the homogeneous solutions take the form

$$y_{\text{hom}}(t) = \tilde{c}_1\,\mathrm{e}^{-t}\cos(2t) + \tilde{c}_2\,\mathrm{e}^{-t}\sin(2t).$$

It is the exponential term (and not the cosines and sines) that dictate how fast the signals converge to zero. We say therefore that its time constant is that of $c\,\mathrm{e}^{-t}$. In other words, we think of the time constant as being $\tau = 1$. In general, it is the real part of the poles that define the time constant; the imaginary parts play no role. The so-defined time constant does not correspond to an exact half-life of $\ln(2)\tau$ but it does give an indication of the speed of convergence.
In general, the *time constant* for high-order systems is defined as that of its *slowest* mode $c\,\mathrm{e}^{s_i t}$ in the homogeneous solution. Concretely, pick the pole s_* that is closest to the imaginary axis, then the time constant equals $-1/\operatorname{Re}(s_*)$ by definition. The notion of time constant only makes sense for asymptotically stable systems, i.e., systems whose homogeneous solution converges to zero at long times.

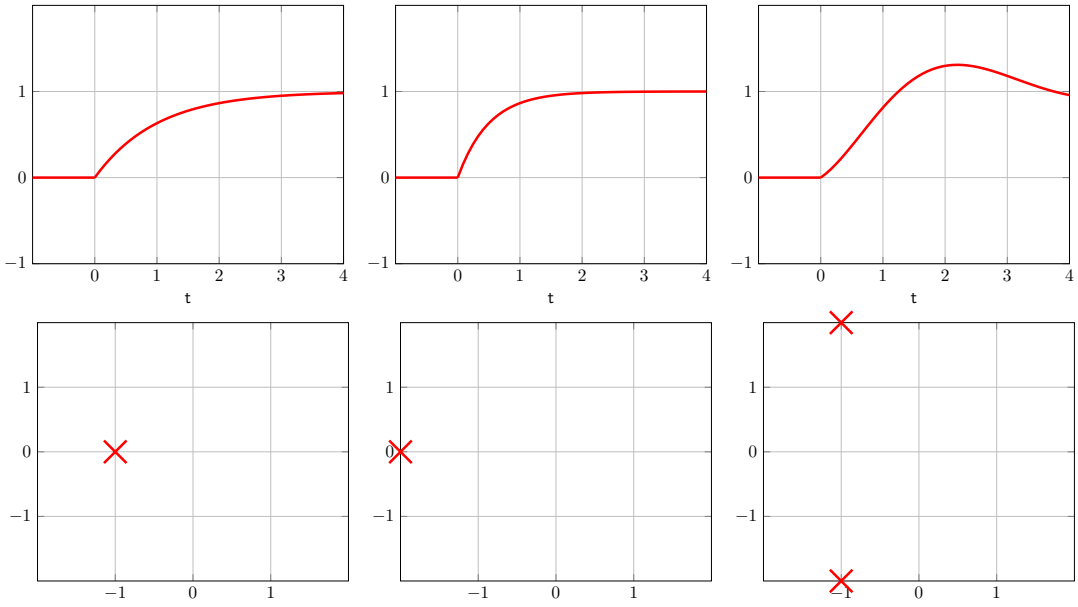


Figure 1.2: Three step responses and the corresponding pole plots.

1.4 Non-linear, time-varying ODEs

In this short section we want to stress that not all ODE's can be separated into homogeneous and particular solutions and that the technique of characteristic equations is not universally applicable . In fact this is not part of the course, however its intention is to explore the boundaries of the field. In relation to physical models, we will treat non-linear systems only in an approximated manner by means of linearisation of the equations.

Example 1.4.1. *Non-linear ODE*
Consider the *non*-linear ODE

$$y^{(1)}(t) + y(t)^2 = 0. \tag{1.21}$$

Attempting an exponential solution $y(t) = \mathrm{e}^{st}$ is futile,

$$s\,\mathrm{e}^{st} + \mathrm{e}^{2st} = 0$$

because for no s does this hold for all t.
The solution (obtained by different means) is $y(t) = (t+c)^{-1}$ with c an arbitrary constant (provided that $y(0) \neq 0$). □

The above is an example of a non-linear ODE. Below is an example of a time-varying ODE.

Example 1.4.2. *Time-varying ODE*
The time-varying ODE

$$y^{(1)}(t) + ty(t) = 0. \tag{1.22}$$

admits no exponential function $y(t) = \mathrm{e}^{st}$ as that would have resulted in

$$s\,\mathrm{e}^{st} + t\,\mathrm{e}^{st} = 0,$$

that is, $s+t = 0$, which clearly does not hold for some fixed s and all t. The solution is $y(t) = c\,\mathrm{e}^{-t^2/2}$, with c an arbitrary constant. □

1.5 Asymptotic stability

If all poles (i.e. characteristic roots) have strictly negative real parts then the contribution of the homogeneous part in the general solution $y(t) = y_{\text{part}}(t) + y_{\text{hom}}(t)$ converges to zero. In such situations, therefore, all possible solutions $y(t)$ converge towards any particular solution $y_{\text{part}}(t)$. This property is known as asymptotic stability.

Definition 1.5.1. *Asymptotic stability*
An ODE (1.2) is *asymptotically stable* if every solution $y_{\text{hom}}(t)$ of the homogeneous equation converges to zero. □

Not surprisingly there holds the following.

Theorem 1.5.2. *Asymptotic stability*
ODE (1.2) is asymptotically stable if all its poles have strictly negative real parts. □

If in ODE (1.2) we have $a_0 \neq 0$ and that the input is constant $x(t) = x_0$, then every solution $y(t)$ of the ODE converges towards the constant solution $y(t) = b_0/a_0 x_0$ if and only if the ODE is asymptotically stable. If the ODE is not asymptotically stable then $\mathrm{Re}(s_k) \geq 0$ for at least one k. In that case homogeneous solutions $y_{\text{hom}}(t) = c\,\mathrm{e}^{s_k t}$ exist that do not converge to zero as $t \to \infty$ and therefore not all solutions $y(t)$ converge to one single solution.

Example 1.5.1. *Stability of ODE*
If $x(t) = 1$ and $y^{(1)}(t) + 2y(t) = 2x(t)$ then by asymptotic stability of the ODE all solutions $y(t)$ converge towards the constant solution $y(t) = 2/2 = 1$. □

Summaries

Summary Chapter 1

- Linear Time-Invariant (LTI) differential equation:

$$a_n y^{(n)}(t)+a_{n-1}y^{(n-1)}(t) + \cdots + a_1 y^{(1)}(t) + a_0 y(t) = b_m x^{(m)}(t) + b_{m-1}x^{(m-1)}(t) + \cdots + b_1 x^{(1)}(t) + b_0 x(t). \quad (1.23)$$

- Solution of the homogeneous equation is a set of$\{\mathrm{e}^{s_i t}\}$, leading to the characteristic equation:

$$a_n s^n + a_{n-1}s^{n-1} + \ldots + a_1 s + a_0 = 0 \quad (1.24)$$

- The solutions of this characteristic equations are called poles and are in general complex valued. For every complex pole, a complex conjugate pole exists.
- Solution for a set of non-equal (complex) poles $\{s_i\}$

$$y(t) = \sum_{i=1}^{n} c_i \mathrm{e}^{s_i t} \quad (1.25)$$

- For a subspace of k equal solutions s_0:

$$y(t) = \sum_{i=0}^{k-1} c_i t^i \mathrm{e}^{s_0 t} \quad (1.26)$$

- Giving a total solution of:

$$y(t) = \sum_{i=0}^{k-1} c_i t^i \mathrm{e}^{s_0 t} + \sum_{i=k}^{n-1} c_i \mathrm{e}^{s_i t} \quad (1.27)$$

- Repeat the subspace solution for every set of equal solutions.
- The coefficients c_i are determined by the $(n-1)$ initial value conditions.

$$\left\{ y^{(i)}(t)\Big|_{t=0} \right\},\ i = 0, 1, \ldots, n-1 \quad (1.28)$$

- Poles – solutions of the characteristic equation – are graphically presented as $\times$-es in the complex plane.
- Real poles are associated with characteristic times: $\tau_i = -\frac{1}{s_i}$
- Complex poles (conjugate pairs) are associated with natural frequencies ω_n and damping factors ζ; for poles given by $s_{1,2} = a \pm \mathrm{j}b$ with a and $b \in \mathbb{R}$:

$$a = -\omega_n \zeta \quad (1.29)$$

$$b = \pm\omega_n \sqrt{1-\zeta^2} \quad (1.30)$$

Problems

A-type problems

Problem 1.1. Quarrel *(0:15h)*
Two students are quarrelling after their test for Calculus 1. They didn't agree on the answer to a problem involving the following homogeneous second-order ODE:

$$y^{(2)}(t) - 9y(t) = 0.$$

One of the students - Neil - asserts that all solutions of this equation diverge. The other one - Buzz - says that he found a solution that does converge to zero.

a. Now it's up to you to tell us who is right. Don't forget to motivative your answer, otherwise people won't believe you.

b. If Buzz is right, can you give the restrictions on the initial conditions we need in order to get a converging solution?

Problem 1.2. Root of characteristic equation
Consider the second-order homogeneous differential equation:

$$a_2 y^{(2)}(t) + a_1 y^{(1)}(t) + a_0 y(t) = 0$$

with a_0, a_1 and a_2 real coefficients. The general solution of this differential equation is:

$$y(t) = c_1 \mathrm{e}^{s_1 t} + c_2 \mathrm{e}^{s_2 t}$$

with c_1 and c_2 real constants and s_1 and s_2 the roots of the characteristic equation of the differential equation.
Express the solution:

- in terms of the real and imaginary part of the roots, *i.e.* combine the real and imaginary parts as good as possible. If needed distinguish between different situations. Explain whether there should be extra constraints for c_1 and c_2.
- in terms of the defined parameters for the natural frequency ω_n and damping coefficient ζ. If needed distinguish between different situations.

Problem 1.3. Poles *(0:15h)*
In figure 1.3 distinct positions of the two poles of the characteristic polynomial $A(s) = s^2 + a_1 s + a_0$ are shown. In figure 1.4 the corresponding solutions are shown. Which configuration of poles belongs to which solution?

Problem 1.4. Stability
Sketch the pole plot of the following differential equations, also indicate whether the solution is asymptotic stable or not:

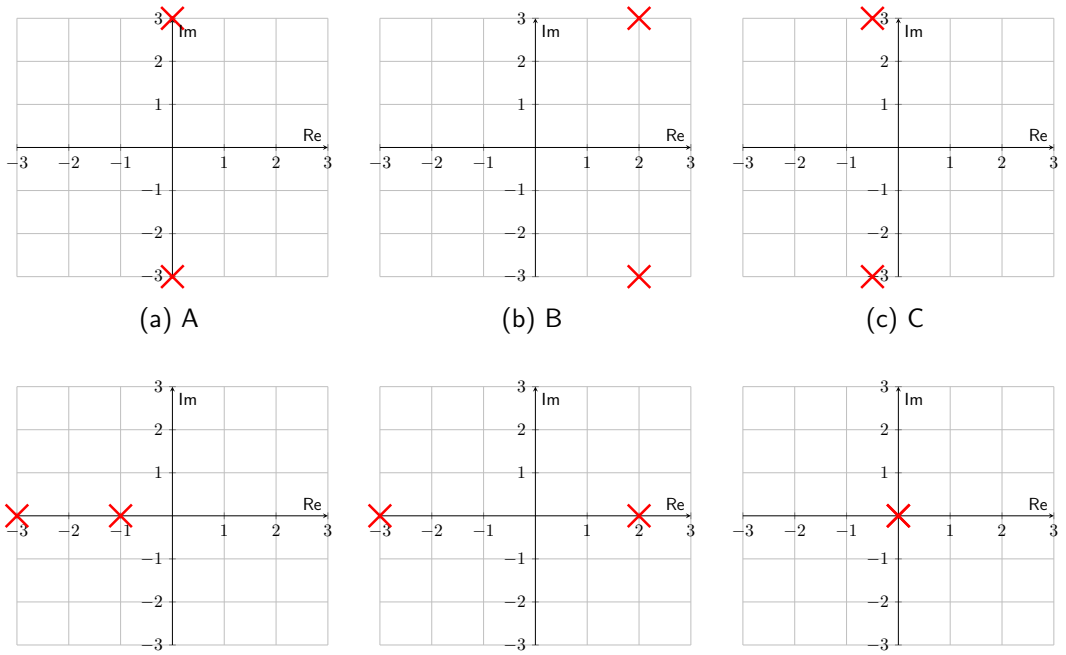
a. $2\frac{\mathrm{d}y}{\mathrm{d}t} + 8y = 10x.$

b. $4\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 8y = 10x.$

c. $4\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} - 8y = 10x.$

d. $4\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\frac{\mathrm{d}y}{\mathrm{d}t} + 8y = 10x.$

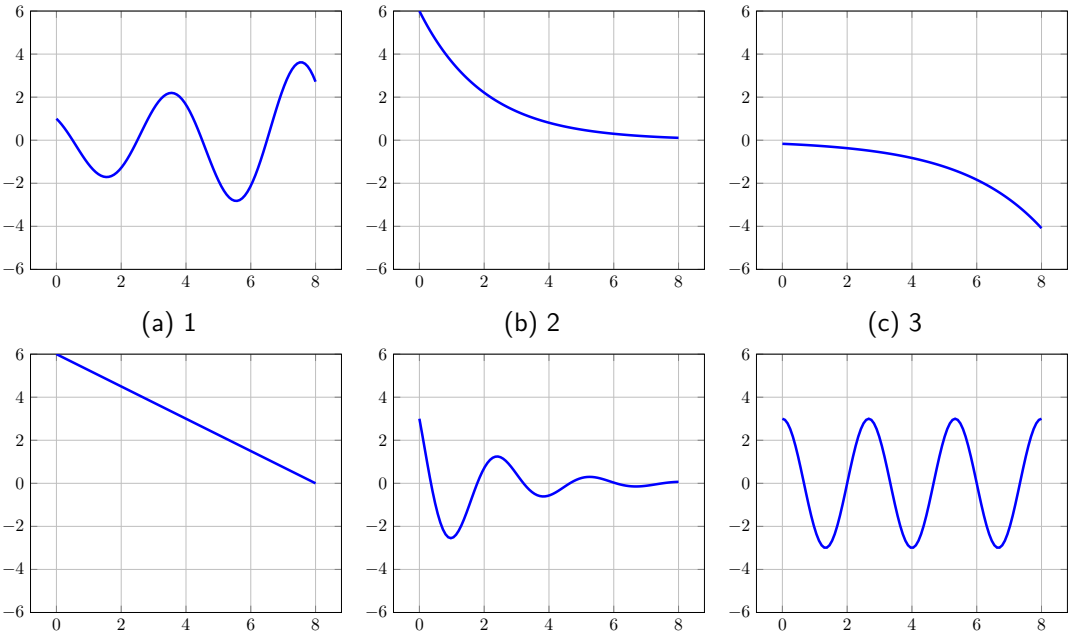
e. $4\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 8y = -10x.$



(a) A (b) B (c) C

(d) D (e) E (f) F

Figure 1.3: Six possible situations for the poles of $s^2 + a_1 s + a_0$



(a) 1 (b) 2 (c) 3

(d) 4 (e) 5 (f) 6

Figure 1.4: Six possible solutions of $\ddot{y}(t) + a_1 \dot{y}(t) + a_0 y(t) = 0$

B-type problems

Problem 1.5. Determining some constants *(0:20h)*
The solution of the differential equation

$$y^{(1)}(t) + a_0 y(t) = b_0 \qquad y(0) = 0$$

is shown in figure 1.5. Estimate the constants a_0 and b_0 from the graph without solving the DE.

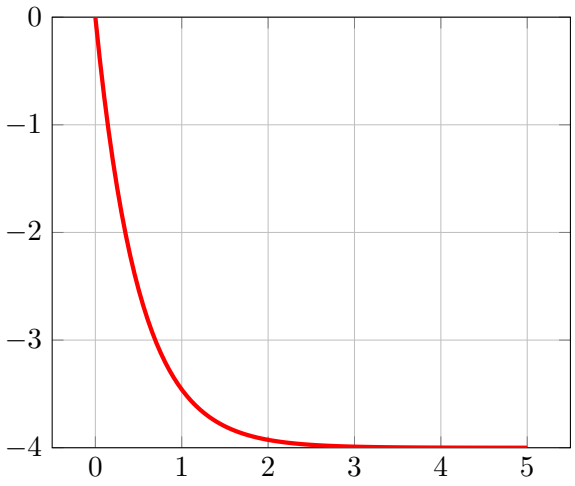


Figure 1.5: The solution $y(t)$

Problem 1.6. Relative importance
A system consists of a cascade of two first order systems, the first one with a characteristic root $s_1 = -1$ and the second one with $s_1 = -100$.

a. Give the homogeneous differential equation of this system.

b. Discuss under what conditions this system can be viewed as an effective first order system.

Chapter 2

System response in the time domain

2.1 Stationary and Dynamic response

Observation of the response of a system to an input provides direct information about the properties and values of the components of a system. If a system is taken from its equilibrium value and slides back to this equilibrium position by a single exponential time behaviour, the model that describes this system might well be a first order differential equation. The range of validity of this model is determined by the range (of various equilibria levels and excitations from this equilibrium) in which this single exponential behaviour is observed. The system response is thus an integral part of a modelling cycle that consists of identifying individual components, describing their interaction, find the overall behaviour, validate this behaviour from the system response and adapt the model. The reason for this cycle is that the influence of all parts of a system are not equally important. Also the specific values of components are not always known beforehand. Of course masses can be weighted and lengths can be measured, but an element with capacitive properties is not always easily taken out of an electrical network to measure its capacitance. Quantification of the system response can provide such values. Here an important distinction between stationary and dynamic behaviour has to be made. The stationary behaviour occurs when the energy distribution does not change with time. At first glance a very dull situation, yet very rich in information.
In the stationary situation all energy put in a system has to be dissipated instantly. Energy storage in for example a capacitor would immediately lead to a change in the voltage across this capacitor, i.e. a change of a quantity in time. The stationary behaviour in electrical system can tell us only something about the resistors, which also means that all other components have no influence and can thus be neglected. The stationary behaviour provides direct information on a limited part of the input-output equation, which in general form is, eq. 2.1:

$$\begin{aligned} a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y^{(1)}(t) + a_0 y(t) \\ = b_m x^{(m)}(t) + b_{m-1} x^{(m-1)}(t) + \cdots + b_1 x^{(1)}(t) + b_0 x(t). \end{aligned} \tag{2.1}$$

In a stationary situation nothing changes with time, so all time derivatives are zero. The stationary input-output equations thus simplifies to:

$$a_0 y(t) = b_0 x(t). \tag{2.2}$$

and the ratio b_0/a_0 follows by providing a stationary input $x(t)$. The (t) is left in this on purpose. A good experimentalist is not satisfied with only one value for $x(t)$ and will try several values to increase the accuracy of this ratio. After each new value sufficient time is taken to achieve the new stationary situation. However, there is a more important reason to vary the input, it helps to establish the range in which linear behaviour is observed. Let's take the example of a mass suspended by a spring, as shown in figure 2.1. In the stationary situation the force of the extended spring balances the gravitational force: $mg = k(z - z_0)$ with m the mass, g the gravitational constant, k the spring

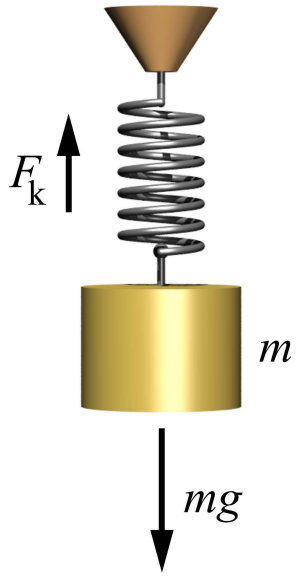


Figure 2.1: A mass suspended from a spring.

constant, z the length of the spring and z_0 its length without a mass attached. Figure 2.2 shows the extension of this spring as a function of the mass attached. The solid line is a linear fit and provides directly a good value of the spring constant ($k = 100$ N/m) and the observation that the spring behaves linearly up to an extension of 10 cm.

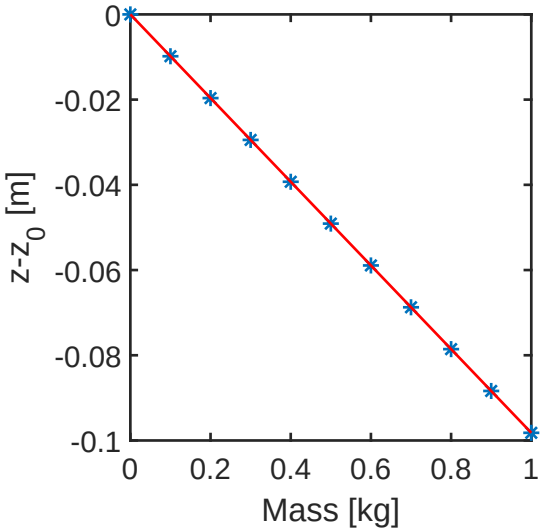


Figure 2.2: Extension of a spring as a function of the attached mass. The solid line is a linear fit through the data points. Here the extension z is measured with respect to the length without a mass attached, z_0

A second example is the response of a car. In a stationary situation the power supplied by the engine will lead to a stationary velocity which is determined by several factors, such as the drag by the air, rolling resistance, tire and internal resistance. An investigation of the relation between the required power for specific stationary velocities is given in figure 2.3. From this graph it is clear that a linear model will not suffice to describe the behaviour of such a car in this velocity range. Drag by the air is the main culprit and leads to this nonlinear behaviour.
The dynamic response is far more complicated as the complete interaction of a system comes into

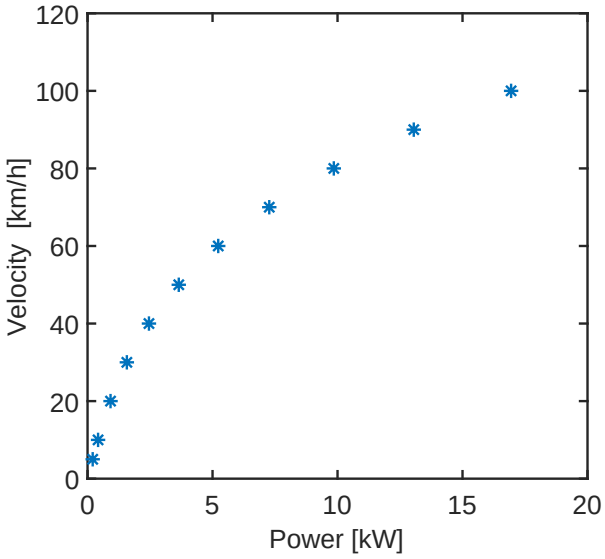


Figure 2.3: The velocity of a car as a function of the power supplied by the engine.

play. It can only provide qualitative and quantitative insight if it is exactly known how the system is excited. Typical excitations used are a step, which allows the observation of the response of a system when it transfers from one stationary state to another, and a pulse, which supplies a limited amount of energy in a limited amount of time. Both these excitations, and the corresponding step and impulse responses, are mathematically well defined.

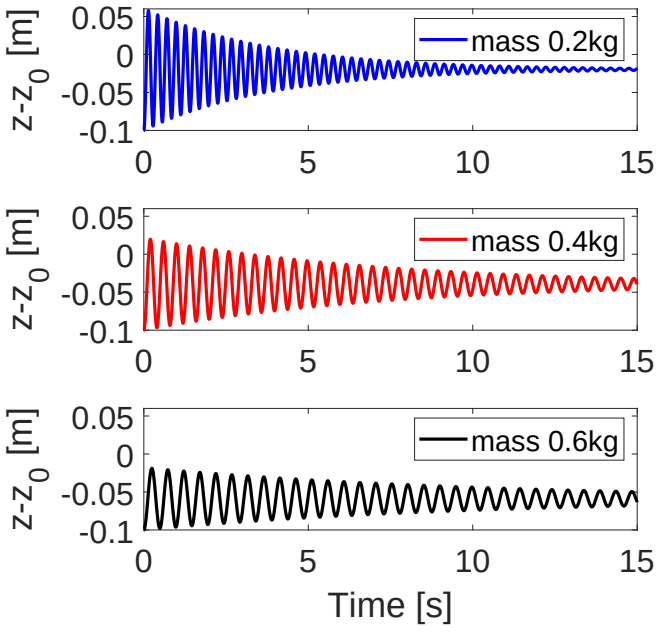


Figure 2.4: Step response of a mass spring system.

An example of a dynamic response is shown in figure 2.4, in which different masses are attached to a spring and released at an extension of the spring of $10\,\mathrm{cm}$. The response of this second-order mechanical system is a damped cosine function, as might be expected from the previous chapter. The oscillation period and the damping however clearly depend on the attached mass. Also note the difference in the extension at long times, the static extension already discussed in 2.2. The

behaviour observed in figure 2.4 is called a step response. The system is in a situation which is no longer stationary as it moves toward its new stationary value. During this movement no additional excitation takes place, i.e., the system moves directly around its new stationary value. In this chapter we try to relate the response of a system to standard excitations in terms of standard characteristics of that system; for an oscillating second order system these will turn out to be the natural frequency and the damping ratio. Typical excitations used to characterize this are the step and impulse responses. Other possibilities are the ramp response (a linear change in time of the input signal) or a sinusoidal input signal (usually measured for a broad range of frequencies).

2.2 Probe functions for dynamic systems

In order to determine the characteristics of systems and differential equations that describe them, a few standard functions are used:

- the unit-step function

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \tag{2.3}$$

- the Dirac-delta function, an infinitely sharp spike.

$$\delta(t) = \begin{cases} 0 & t < 0 \\ \infty & t = 0 \\ 0 & t > 0 \end{cases} \tag{2.4}$$

- the harmonic function

$$x(t) = \sin \omega t \tag{2.5}$$

The Dirac-delta function is a unit-valued function due to the fact that the integral over time is equal to one: see also section 2.2.2. The running integral of the Dirac-delta function results in the unit-step function. Reversely the derivative with respect to time of the unit-step function gives the Dirac-delta function. This relation can be convenient in the analysis of linear systems.
In the subsections below the characteristics and use of the probe functions is presented in more detail.

2.2.1 Step function

The function definition is presented in equation 2.3. This function is convenient since it contains a broad range of frequencies, and it starts and ends in a stationary state, not necessarily zero. This makes it suitable to explore the DC behaviour of the system. The function is graphically represented in figure 2.5.

2.2.2 Impulse function

The function definition is presented in equation 2.4. This function is convenient since it contains a broad range of frequencies, and has a limited energy content. The function starts and ends at zero, *i.e.* returning to its original value. The function is graphically represented in figure 2.6 as a sequence of pulse functions that only in the limit for the width $\Delta \to 0$ approaches the delta function $\delta(t)$.
To illustrate this, let us define the pulse function:

$$p(t) = \begin{cases} 0 & t < -\frac{1}{2}\Delta \\ \frac{1}{\Delta} & -\frac{1}{2}\Delta \leq t \leq \frac{1}{2}\Delta \\ 0 & t > \frac{1}{2}\Delta \end{cases} \tag{2.6}$$

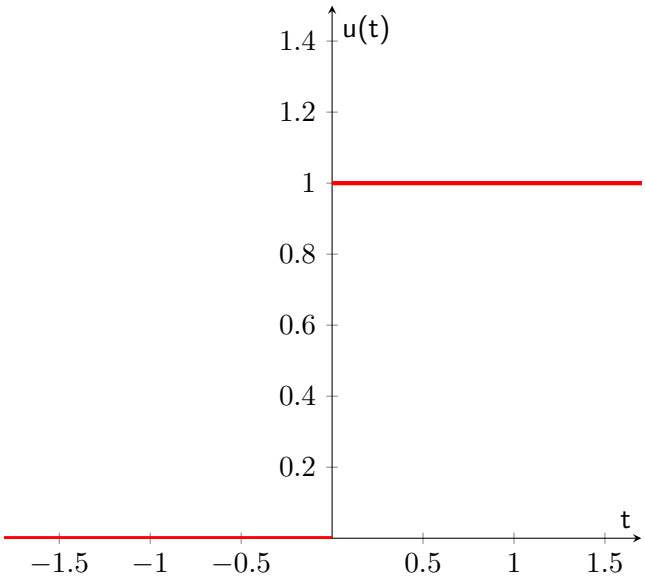


Figure 2.5: Unit-step function.

then:

$$\delta(t) = \lim_{\Delta \to 0} p(t) \tag{2.7}$$

This function can also be defined as the time derivative of the unit-step function $u(t)$.

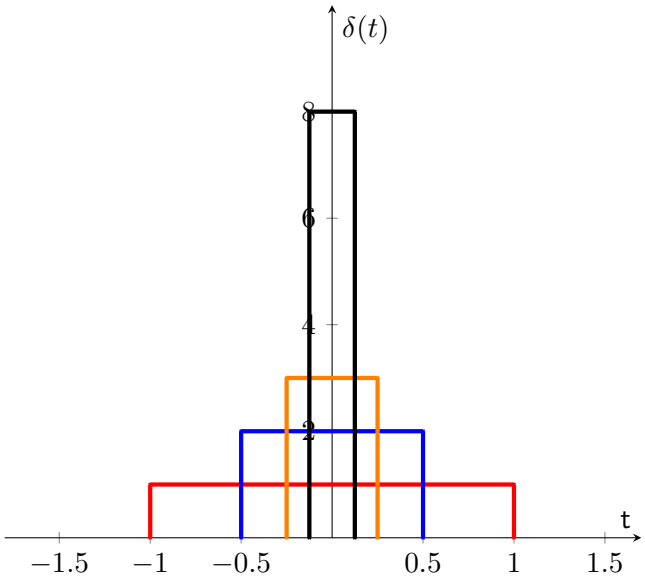


Figure 2.6: Impulse function of area 1 with different widths Δ

For this function the following relation holds:

$$\int_{-\infty}^{\infty} \delta(t) \mathrm{d}t = 1 \tag{2.8}$$

This can also be deduced from the definition in equations 2.7 and 2.6.
A pulse function having a width Δ might be useful in probing systems. In the case where Δ is larger than the largest characteristic time of the system, the pulse behaves like a double step function: a

step up followed by a step down. In the case where Δ is much smaller than the smallest characteristic time of the system, the pulse behaves as an impulse function, even if Δ does not strictly approach zero. This is what makes the δ-function so useful: while we cannot generate an infinitely sharp spike, $\delta(t)$ is a good approximation for any input that occurs much faster than the time constant of the system. Think of hitting a mass with a hammer to provide a sudden impulse, or sending a single 'ping' from a submarine's sonar and listening to the signals that come back.

2.2.3 Harmonic function

The function definition is presented in equation 2.5. This function is convenient since it contains only a single frequency. The function is graphically represented in figure 2.7.

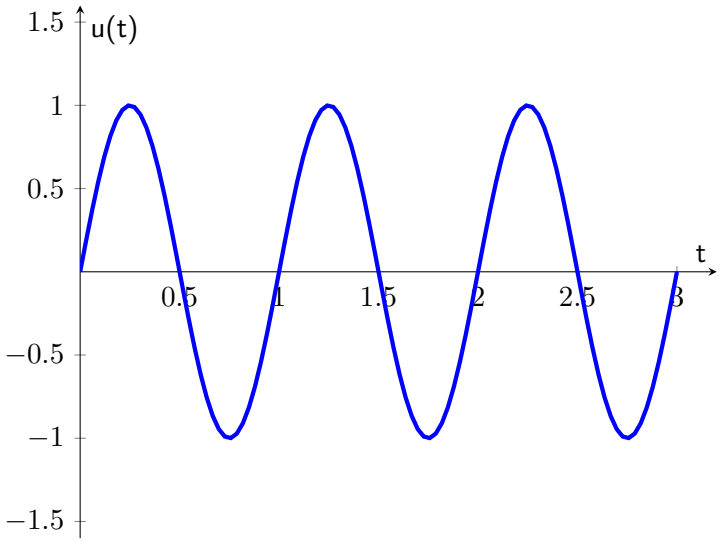


Figure 2.7: Harmonic function.

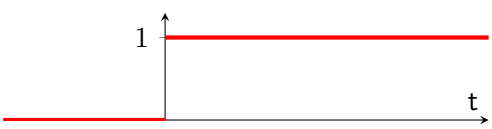
The advantage of this function and its use will become clear later when you learn about transfer functions.

2.2.4 Linear combination of probe functions

The advantage of linear systems is that the response to a linear combination of input functions is equal to the linear combination of the responses to the individual input functions. This is a very powerful tool in exploring the characteristics of a given LTI-system.
A convenient consequence of this is that in case $y(t)$ is the response to an input $x(t)$, then $y'(t)$ is the response to $x'(t)$. This also holds for the inverse operation, integration. This makes it easy to relate impulse and step responses of systems, since the functions are related by the integral and differential operator.

2.3 Step response of a first order system

The unit step input

$x(t) = u(t) :=$ 

models change in the environment from one constant value to another constant value. The resulting output $y(t)$ is known as the (unit) *step response* and therefore describes how the output reacts to the change from one constant environment to the other.

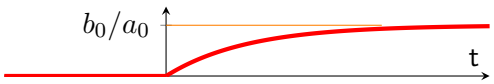
Example 2.3.1. Suppose the system was at rest for $t < 0$, that is, $y(t) = x(t) = 0$ for all $t < 0$. What happens with the solution $y(t)$ of

$$y^{(1)}(t) + a_0 y(t) = b_0 x(t)$$

if we suddenly increase the input to the new but constant value of $x(t) = 1$ for $t > 0$ (in other words, $x(t) = u(t)$)?
The general solution to this equation is:

$$y(t) = \frac{b_0}{a_0}\left(1 - \mathrm{e}^{-a_0 t}\right)$$

A graphical representation of the solution, where $a_0 > 0$ and $b_0 > 0$:



Solution via Laplace method.
We solve this problem by first transform this differential equation to an algebraic one via the Laplace Transform:

$$sY(s) - y(0) + a_0 Y(s) = \frac{b_0}{s}$$

This can be solved for $Y(s)$ in the complex frequency domain:

$$Y(s) = \frac{b_0}{s(s + a_0)} = \frac{k_1}{s} + \frac{k_2}{s + a_0}$$

where we already did the partial fraction decomposition. Solving for k_1 and k_2 we obtain:

$$k_1 = \left.\frac{b_0}{s + a_0}\right|_{s=0} = \frac{b_0}{a_0}$$
$$k_2 = \left.\frac{b_0}{s}\right|_{s=-a_0} = -\frac{b_0}{a_0}$$

Combining these results gives us the solution:

$$y(t) = \frac{b_0}{a_0}\left(1 - \mathrm{e}^{-a_0 t}\right)$$

This concludes this example. □

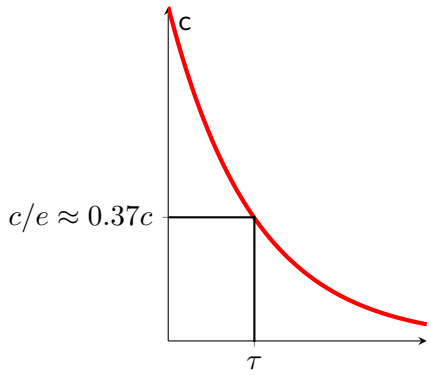
The example demonstrates once more that ODEs describe dynamical phenomena: the output $y(t)$ cannot follow the change in input immediately. It needs time to settle to a new steady-state value due to the change in input. A natural question now is: how fast does the output react? in other words, how fast is the system? Continuing with the last example, the difference between $y(t)$ and its steady-state value b_0/a_0 is the homogeneous solution,

$$c\,\mathrm{e}^{-t/\tau}, \qquad \tau = \frac{1}{a_0}.$$

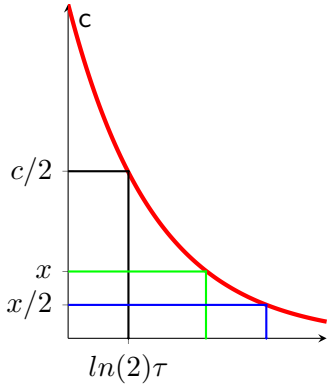
The number τ has dimension time and we define

Definition 2.3.1. *Time constant*
Signal $c\,\mathrm{e}^{-t/\tau}, \tau > 0$ is said to have *time constant* τ. Similarly, ODE $y^{(1)}(t) + a_0 y(t) = b_0 x(t)$ is said to have *time constant* $\tau = 1/a_0$, assuming it is positive. □

Exponential signals $c\,\mathrm{e}^{-t/\tau}$ have a well-defined half life. The *half-life* is defined as the time needed to reach a 50% reduction in magnitude. It is well defined for any $c\,\mathrm{e}^{-t/\tau}$ in the sense that the half-life does not depend on c and not on the initial time t_0, see figure 2.8(b). The half-life equals



(a) Attenaution of 37% after time τ



(b) Half life of $c\,\mathrm{e}^{-t/\tau}$ equals $\ln(2)\tau \approx 0.7\tau$. This holds for any two points that differ by a factor 2 in the response curve.

Figure 2.8: Two ways for determination of the timeconstant τ from a response curve.

$\ln(2)\tau \approx 0.7\tau$ and so is proportional to the time constant τ. Similarly, the time needed to reach a 90% reduction (to 10%, a common figure of merit in electronics) is proportional to τ as well, being given by $\ln(10)\tau$. Here are some more comparisons:

reduction to 50%	$\approx$	0.7τ	half life
reduction to 37%	$\approx$	τ	time constant
reduction to 10%	$\approx$	2.3τ	$(\ln(10) \approx 2.3)$
reduction to 1%	$\approx$	4.6τ	$(\ln(100) \approx 4.6)$.

The time constant therefore indicates how fast the system is, assuming it it is described by an asymptotically stable first order ODE $y^{(1)}(t)+a_0 y(t) = b_0 x(t)$. For higher order ODE's the definition is a bit ambiguous, as discussed earlier.

2.3.1 Analysis of a first order system

The value of the capacitance C of a capacitor can be evaluated with a proper time response. The capacitor is than set in series with a known resistor R_1. However consider the situation in which it is expected that the capacitor used is not ideal, it also shows resistor character, giving a small leakage current represented by a resistor R_2 in figure 2.9.
The model for this network is described by

$$C\frac{\mathrm{d}u_C}{\mathrm{d}t} = i_{R_1} - i_{R_2} = \frac{u_S - u_C}{R_1} - \frac{u_C}{R_2},$$

yielding the ODE

$$\frac{\mathrm{d}u_C}{\mathrm{d}t} + \left(\frac{1}{R_1 C} + \frac{1}{R_2 C}\right) u_C = \frac{\mathrm{d}u_C}{\mathrm{d}t} + \frac{R_1 + R_2}{R_1 R_2 C} u_C = \frac{1}{R_1 C} u_s.$$

The step response of a first order system has two characteristics: the rate of increase (or decrease) of the signal and the stationary value. First consider the stationary value, a situation in which changes

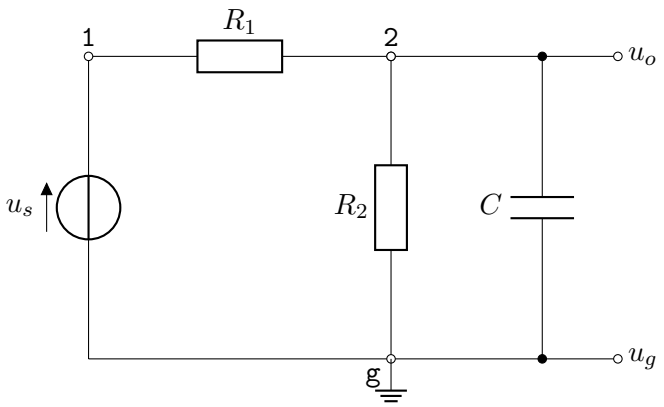


Figure 2.9: RC network

with time do not occur. This implies that the derivatives are zero and the ODE is transferred to the analytical relation:

$$\frac{R_1 + R_2}{R_1 R_2 C} u_C = \frac{1}{R_1 C} u_s,$$

Which can be simplified to

$$\frac{R_1 + R_2}{R_2} u_C = u_s$$

Measuring the ratio $u_C/u_s = R_2/(R_1 + R_2)$ thus provides information on the resistor values. Measuring this ratio for a range of input voltages allows to verify that the system behaves linearly, i.e. can be treated as an LTI system. However, if R_2 is much larger than R_1, this ratio reduces to 1, limiting the ability to discern the existence of R_2. This puts a constrain on the values of R_1 that can be used.
A step response for this system in which the voltage in changed from 0 to $1\,\mathrm{V}$ is given in figure 2.10. The characteristic response time $\tau = \frac{R_1 R_2 C}{R_1 + R_2}$ can be used to infer the value of the capacitance C. Various methods can be used to determine this value with different accuracy:

a. determine the slope of the response at the time the step is made

b. find the time at which 63% of the response is done, this time is approximately τ

c. find the time at which the value is within 1% of the final value, this time is approximately 4.6τ

d. fit a single exponential curve

e. plot the rate at which u_C changes ($\mathrm{d}u_C/\mathrm{d}t$) versus u_C. This plot should give a linear relation with slope $1/\tau$.

The virtue of the last method is that it uses the whole time response and that it provides a direct visual indication whether a first order model is appropriate. A deviation from the expected linear relation (that is more pronounced than the accuracy with which the derivative can be evaluated from noisy data) provides direct proof when this is not the case (and, if so, in which range). In principle a fit of an exponential provides the same information, but this requires attention to the evaluation of the accuracy of a fit.
In the present example a resistor of $1\,\mathrm{M\Omega}$ is used leading to a final value of $0.98\,\mathrm{V}$ which indicates a finite value for the internal resistance R_2 of about $50\,\mathrm{M\Omega}$. The time response gives $\tau = 0.1\,\mathrm{s}$. For $R_1 \ll R_2$ as in this case, $\tau \approx \frac{R_1 R_2 C}{R_2} = R_1 C$. This gives a value of $C = 0.1/10^6 = 10^{-7}\,\mathrm{F}$.

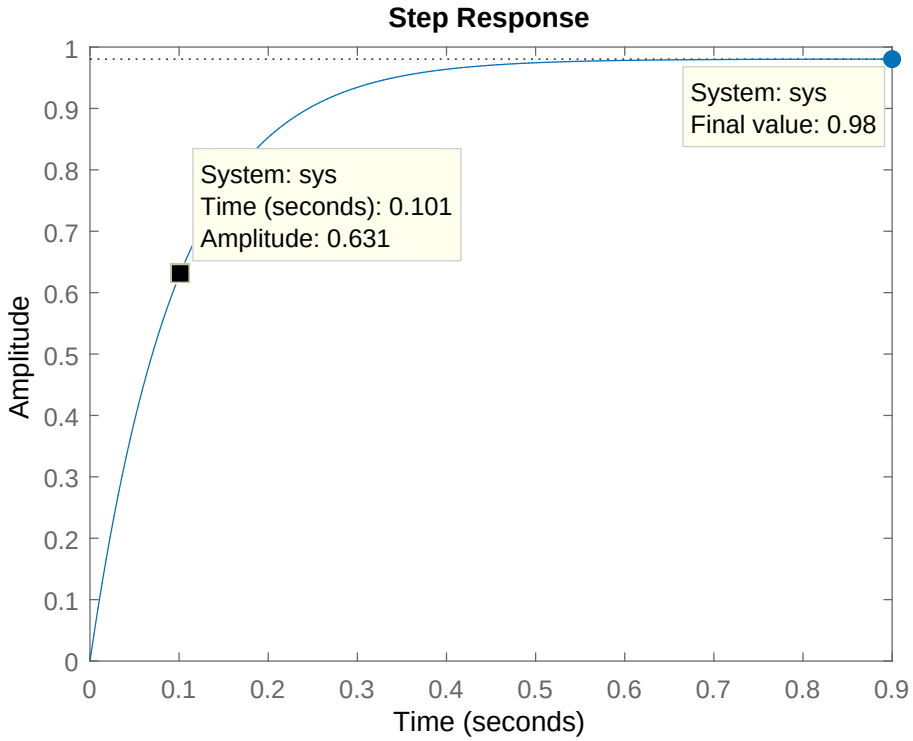


Figure 2.10: Time response of the RC network.

2.4 Step response of a second order systems

The input-output equation of a second order system with a step input is given by

$$a\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + b\frac{\mathrm{d}y}{\mathrm{d}t} + cy = u(t) \tag{2.9}$$

This differential equation is often written in a different form:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\zeta\omega_n\frac{\mathrm{d}y}{\mathrm{d}t} + {\omega_n}^2y = \frac{1}{a}u(t) \tag{2.10}$$

The reason for this different notation lies in the interpretation of the quantities ω_n as the natural resonance frequency and ζ as a damping factor, as shown further below. Both ω_n and $zeta$ are non-negative real numbers. For $\zeta = 0$ this differential equation describes an undamped second order system with an oscillation frequency ω_n (hence the name natural frequency) such as a frictionless mechanical pendulum or its electrical analogue, a passive network with a capacitor and an inductor. The characteristic equation is derived from this input-output equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The poles of this expression are:

$$p_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

From these poles we construct the solution of the homogeneous differential equation. Adding the particular solution, taking into account the input function (and initial conditions) we get the solution:

$$y(t) = \frac{1}{a\omega_n^2} + k_2\mathrm{e}^{p_2t} + k_3\mathrm{e}^{p_3t} \tag{2.11}$$

Solution via Laplace method.
In order to calculate the solution to this differential equation we first apply the Laplace Transform (note that the Laplace transform of the unit-step function is $1/s$), to obtain:

$$s^2Y(s) - sy(0) - y'(0) + 2\zeta\omega_n sY(s) - 2\zeta\omega_n y(0) + \omega_n^2 Y(s) = \frac{1}{as}$$

Re-shuffle the terms gives:

$$\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right) Y(s) = \frac{1}{as} + sy(0) + y'(0) + 2\zeta\omega_n y(0)$$

Solving for $Y(s)$ and setting all initial conditions to zero gives:

$$Y(s) = \frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2}\frac{1}{as}$$

The poles of this expression are:

$$\begin{aligned} p_{1,2} &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \\ p_3 &= 0 \end{aligned}$$

Where the third pole ($p_3 = 0$) is the result from the input function and not from the characteristic equation. Applying partial fraction decompositions gives:

$$Y(s) = \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2} + \frac{k_3}{s - p_3} \tag{2.12}$$

where:

$$\begin{aligned} k_1 &= (s - p_1)Y(s)|_{s=p_1} = \dots \\ k_2 &= (s - p_2)Y(s)|_{s=p_2} = \dots \\ k_3 &= (s - p_3)Y(s)|_{s=0} = \frac{1}{a\omega_n^2} \end{aligned} \tag{2.13}$$

At this point we did not write the full solutions, since the general solution does not give insight in the results. The General solution expressed in p_2, p_3, k_2 and k_3 will be:

$$y(t) = k_1 \mathrm{e}^{p_1 t} + k_2 \mathrm{e}^{p_2 t} + \frac{1}{a\omega_n^2}$$

The variation in solutions to this problem can be divided in three regions depending on a single parameter ζ in the expression of the poles. Poles p_1 and p_2 have a common term that is real: $-\zeta\omega_n$; and since this factor appears in the exponent of the solution, this, in general, will cause a damping of the system. The second part of the expression of the poles is more distinct. This results from the fact that the character of the poles change with this parameter ζ, the damping parameter:

- $0 \leq \zeta < 1 \;\Rightarrow\; \zeta^2 - 1 < 0 \;\Rightarrow\; \sqrt{\zeta^2 - 1} = \mathrm{j}\sqrt{1 - \zeta^2} \in \mathbb{C}$
- $\zeta = 1 \;\Rightarrow\; \zeta^2 - 1 = 0 \;\Rightarrow\; \sqrt{\zeta^2 - 1} \in \mathbb{R}$
- $\zeta > 1 \;\Rightarrow\; \zeta^2 - 1 > 0 \;\Rightarrow\; \sqrt{\zeta^2 - 1} \in \mathbb{R}$

In the first case we have two pure imaginary parts that combine with the other part to two complex poles, that are each others conjugate. In the second case we have to equal poles: $p_{1,2} = -\omega_n$. In the third case we obtain two real poles, that combine with the other part to two real poles.

The different character of the poles has an effect on the type of solutions. In order to give you an idea, we explore the distinction between two complex poles (case one) and two real poles (case three). For two complex poles we have the type of solution:

$$y(t) \propto \mathrm{e}^{(a+\mathrm{j}b)t} + \mathrm{e}^{(a-\mathrm{j}b)t} = \mathrm{e}^{at}\left(\mathrm{e}^{\mathrm{j}bt} + \mathrm{e}^{-\mathrm{j}bt}\right) = 2\mathrm{e}^{at}\cos(bt)$$

For two real poles we have the type of solution:

$$y(t) \propto \mathrm{e}^{at} + \mathrm{e}^{bt}$$

From this we observe that in the for two complex poles we have the combination of a real exponent and an oscillating function. For two real poles we have two exponential functions (no oscillations). In general this is a bit more complicated, since both initial exponentials have real pre-factors that need not be equal. However the principle holds. In case of two equal real poles, the solutions will be different; this is a spacial case. Below we will treat the three different cases separately.
In figure 2.11 the three different regions for $p_{1,2}$ are indicated for $\omega_n = 1$. From the solution we see that this picture scales with the value of ω_n. For $\zeta < 1$ (red line) the values of $p_{1,2}$ lie on a half circle with radius ω_n in the left half of the complex plane. For $\zeta > 1$ (blue line) the poles lie on the negative real axis. For the special case $\zeta = 1$ (green dot), both values of $p_{1,2}$ are -1.
It is essential that the real part of these solutions is negative as this ensures decay of the output with time. Positive real values will lead to growth of the output with time and are therefore unstable systems.

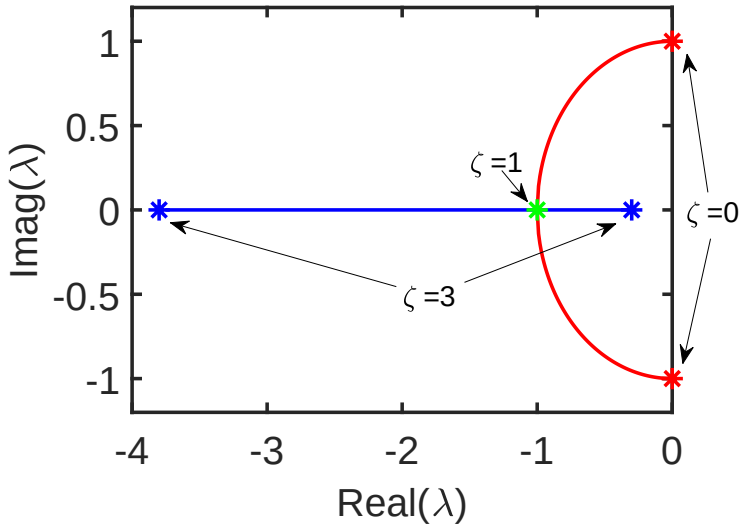


Figure 2.11: Solutions of the characteristic polynomial for the second order differential equations for ζ between 0 and 4. The value of $\omega_n$is set to 1. The red part is the underdamped case, $\zeta < 1$, The blue part is the overdamped case, $\zeta > 1$. The green point is the critically damped case $\zeta = 1$.

To display the characteristics of the solution we depict normalized situations. For eqn. 2.10 this corresponds to $1/(a\omega_n^2)$ being set to 1. At infinite time, $y(t)$ will then attain the value of the input step function, $y(t \to \infty) = 1$. As initial conditions $y(0) = 0$ and $y'(0) = 0$ are used.

2.4.1 Under-damped second-order system

If $\zeta < 1$, the two poles (roots of the characteristic equation) p_1 and p_2 are complex and are each other's conjugate, *i.e.*,

$$\begin{aligned} \mathrm{Re}(p_1) &= \mathrm{Re}(p_2) = -\zeta\omega_n \\ \mathrm{Im}(p_1) &= -\mathrm{Im}(p_2) = -\omega_n\sqrt{1-\zeta^2} \equiv -\omega_d \end{aligned} \tag{2.14}$$

giving:

$$p_{1,2} = -\omega_n\zeta \mp \mathrm{j}\omega_d$$

The real part of these solutions is associated with the attenuation of the oscillations in the solution. The magnitude of the imaginary part is the actual oscillation frequency ω_d that is seen in the response. Note that this frequency is always smaller than or equal to the natural frequency $\omega_d \leq \omega_n$.

$$\omega_n = \sqrt{\omega_n^2\zeta^2 + \omega_d^2} = \sqrt{\omega_n^2\zeta^2 + (1-\zeta^2)\omega_n^2} = \omega_n$$

For a value of the damping parameter $\zeta = 0$, the observed frequency ω_d becomes equal to the natural frequency ω_n, which corresponds to an undamped system; the poles are pure imaginary.

The solution of a unit-step response in which the input is raised from $u(t) = 0$ to 1 is then a damped oscillation, as already seen in Example 1.2.6.

$$y(t) = k_1' \,\mathrm{e}^{-\zeta\omega_n t}\cos(\omega_d t) + k_2' \,\mathrm{e}^{-\zeta\omega_n t}\sin(\omega_d t) + \frac{1}{a\omega_n^2}. \tag{2.15}$$

The term $1/(a\omega_n^2)$ is the result of the particular solution that results from the step function input. This is the asymptotic value at long times, since the other two terms vanish at large times.

The values of the other two terms k_1' and k_2' result from the partial fraction decomposition of the equation 2.12. In practice it is easier to calculate these values for specific cases than to give a general solution here.

2.4.2 Critically damped second order system

If $\zeta = 1$ the two poles (roots of the characteristic equation) are real and identical, *i.e.*:

$$p_1 = p_2 = -\omega_n \equiv -\frac{1}{\tau} \tag{2.16}$$

where we defined a characteristic time τ. The general solution is:

$$y(t) = \frac{\tau^2}{a} - \frac{\tau^2}{a}\mathrm{e}^{-t/\tau} - \frac{\tau^2}{a}t\mathrm{e}^{-t/\tau} \tag{2.17}$$

For positive values of τ this time response converges to τ^2/a for $t \to \infty$.

Solution via Laplace method.
In Laplace space, this leads to a special solution, since the poles are equal. The solution becomes:

$$Y(s) = \frac{1}{s^2 + 2\omega_n s + \omega_n^2}\frac{1}{as} = \frac{k_1}{s+\omega_n} + \frac{k_2}{(s+\omega_n)^2} + \frac{k_3}{s}$$

with the values for k_1, k_2 and k_3 calculated via:

$$\begin{aligned} k_1 &= \frac{\mathrm{d}}{\mathrm{d}s}\left((s+\omega_n)^2 Y(s)\right)\Big|_{s=-\omega_n} = -\frac{1}{a\omega_n^2} \\ k_2 &= (s+\omega_n)^2 Y(s)\Big|_{s=-\omega_n} = -\frac{1}{a\omega_n} \\ k_3 &= sY(s)|_{s=0} = \frac{1}{a\omega_n^2} \end{aligned}$$

Combining the above results gives:

$$y(t) = -\frac{1}{a\omega_n^2}\mathrm{e}^{-\omega_n t} - \frac{1}{a\omega_n}t\mathrm{e}^{-\omega_n t} + \frac{1}{a\omega_n^2} = \frac{1}{a\omega_n^2}\left(-\left(1+\omega_n t\right)\mathrm{e}^{-\omega_n t} + 1\right) \tag{2.18}$$

For positive values of ω_n this time response converges to $1/(a\omega_n^2)$ for $t \to \infty$. Since there is no oscillation involved in the solution often ω_n is replaced by $1/\tau$ as defined in equation 2.16, with τ the characteristic time of the system.

2.4.3 Over-damped second-order system

If $\zeta > 1$ the two poles (roots of the characteristic equation) are both real, and can be defined as two characteristic times[1]:

$$\begin{aligned} p_1 &= -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} = -\zeta\omega_n(1 - \sqrt{1 - 1/\zeta^2}) \equiv -\frac{1}{\tau_1} \\ p_2 &= -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} = -\zeta\omega_n(1 + \sqrt{1 - 1/\zeta^2}) \equiv -\frac{1}{\tau_2} \end{aligned} \tag{2.19}$$

with this definition of the characteristics times τ_1 and τ_2 we can relate the two time parameters to the natural frequency and damping:

$$\omega_n^2 = \frac{1}{\tau_1\tau_2} \qquad \zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1\tau_2}}$$

The solution is then given by:

$$y(t) = -\frac{\tau_1^2\tau_2}{a(\tau_1 - \tau_2)}\mathrm{e}^{-t/\tau_1} + \frac{\tau_1\tau_2^2}{a(\tau_1 - \tau_2)}\mathrm{e}^{-t/\tau_2} + \frac{\tau_1\tau_2}{a} \tag{2.20}$$

[1]For the definitions of the characteristic times, we used the numbering of the poles; due to the fact that $p_1 = 0$ in this case, there is no time associated with this pole and only two times can be identified in the system.

Solution via Laplace method
The solution in Laplace form becomes:

$$Y(s) = \frac{k_1\tau_1}{\tau_1 s + 1} + \frac{k_2\tau_2}{\tau_2 s + 1} + \frac{k_3}{as}$$

with expressions for:

$$k_1 = \left.\frac{\tau_1 s + 1}{\tau_1} Y(s)\right|_{s=-1/\tau_1} = -\frac{\tau_1^2\tau_2}{a(\tau_1 - \tau_2)}$$
$$k_2 = \left.\frac{\tau_2 s + 1}{\tau_2} Y(s)\right|_{s=-1/\tau_2} = \frac{\tau_1\tau_2^2}{a(\tau_1 - \tau_2)}$$
$$k_3 = sY(s)|_{s=0} = \frac{1}{a\omega_n^2} = \frac{\tau_1\tau_2}{a}$$

Combining this all, we get the solution:

$$\begin{aligned} y(t) &= -\frac{\tau_1^2\tau_2}{a(\tau_1 - \tau_2)}\mathrm{e}^{-t/\tau_1} + \frac{\tau_1\tau_2^2}{a(\tau_1 - \tau_2)}\mathrm{e}^{-t/\tau_2} + \frac{\tau_1\tau_2}{a} \\ &= \frac{\tau_1\tau_2}{a}\left(-\frac{\tau_1}{\tau_1 - \tau_2}\mathrm{e}^{-t/\tau_1} + \frac{\tau_2}{\tau_1 - \tau_2}\mathrm{e}^{-t/\tau_2} + 1\right) \end{aligned} \tag{2.21}$$

This is typical double exponential decay. The solution converges to $\tau_1\tau_2/a$ for $t \to \infty$.

A special situation occurs if the difference in the magnitude of the two poles is larger than an order of magnitude, *i.e.* $\zeta \gg 1$. In that case $\sqrt{\zeta^2 - 1} \approx \zeta - 1/(2\zeta)$ and the values of the poles are $p_1 = -\omega_n/2\zeta$ and $p_2 = -2\zeta\omega_n$. In this case the unit-step response effectively reduces to a first order response, since the response is dominated by the larger decay time. Also the pre-factor of the faster decay term in the solution is smaller with respect the the slower term. The ratio of the pre-factors is τ_1/τ_2 in favour of the slow term.
This can also be derived from the results above (here $\tau_1 \gg \tau_2$ is used), leading to:

$$y(t) \approx \frac{\tau_1\tau_2}{a}\left(1 - \frac{1}{1 - \tau_2/\tau_1}\mathrm{e}^{-t/\tau_1} + \frac{\tau_2}{\tau_1}\mathrm{e}^{-t/\tau_2}\right) \approx \frac{\tau_1\tau_2}{a}\left(1 - \mathrm{e}^{-t/\tau_1}\right) \tag{2.22}$$

It is easily seen that this is a first-order response with a characteristic time τ_1. The effect of τ_2 is too fast to be noticed in the step response of the system in this situation.

2.4.4 Normalized step response of a second order system

At this point we take an additional step in the analysis second-order LTI ODE's. The fact that we consider only LTI ODE's with real coefficients has as a consequence that the type of solutions is fully covered with the analysis of first-order and second-order differential equations. The distinct solutions obtained for a second-order system are in fact a model for the analysis. This motivates to explore a few characteristics of these systems even further. In order to make the analysis applicable to in specific situations, we first normalize the step-response of a second-order system.
Figure 2.12 shows the unit step response as a function of the damping factor ζ as derived in eqs. 2.15, 2.17 and 2.20. Note that for simplicity the stationary value is set to 1, equivalent with $a = 1/\omega_n^2 = \tau_1\tau_2$. To compare the various results, the time scale is normalized with the natural frequency ω_n. This figure is one of the key figures in the field of system theory as it highlights many features of second order systems. The transition from an oscillating system to a cascade of two first order responses is clearly seen. A closer look also shows that the maximum output occurs at a slightly earlier moment for smaller ζ, which is a direct consequence of the dependence of the imaginary part

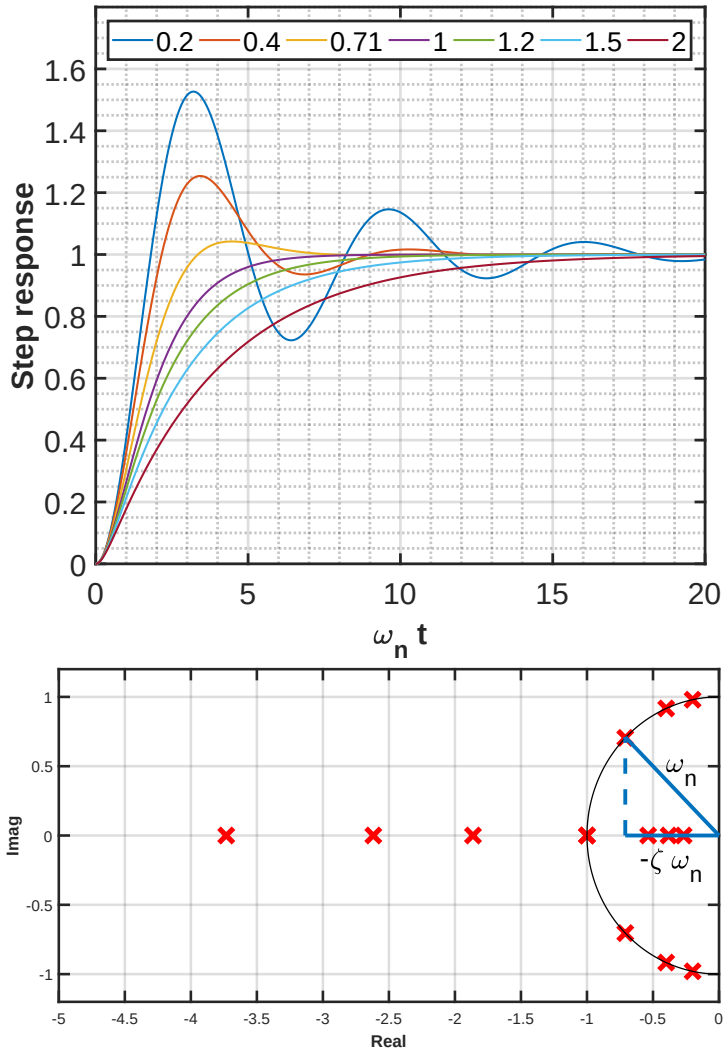


Figure 2.12: top: Step response of a second order system as a function of ζ with the time normalized by the resonance frequency ω_n. bottom: the poles of the systems with the step response shown in the top figure.

of the poles on the damping ζ. Also, the magnitude of the overshoot for the under-damped system decreases with an increasing value of ζ.
The pole plots associated to these response curves were made for $\omega_n = 1$. Because we are dealing with LTI systems, the solutions scale, *i.e.* if a two times larger step function input is used, the output response will be twice as large.

2.4.4.1 Time domain specifications

Figure 2.12 is a key figure because it can be used to determine the characteristic parameters of a system from a step response or in a design process to determine boundaries on system parameters. This is especially the case for underdamped systems. Overdamped systems can be regarded as a cascade of two first order systems and the one with the largest time constant will determine the time response, and as such are less complex. In practice the step response of underdamped systems can be characterized by 4 easily accessible experimental parameters that are illustrated in figure 2.13. These 4 parameters can also be formulated as requirements in a system design and are expressions of the effect of the system characteristics ω_n and ζ. These parameters are:

a. The **rise time** t_r is the time it takes the system to reach the vicinity of its new set point.

b. The **settling time** t_s is the time it takes the system transients to decay.

c. The **overshoot** M_p is the maximum amount the system overshoots its final value divided by its final value.

d. The **peak time** t_p is the time it takes the system to reach the maximum overshoot point.

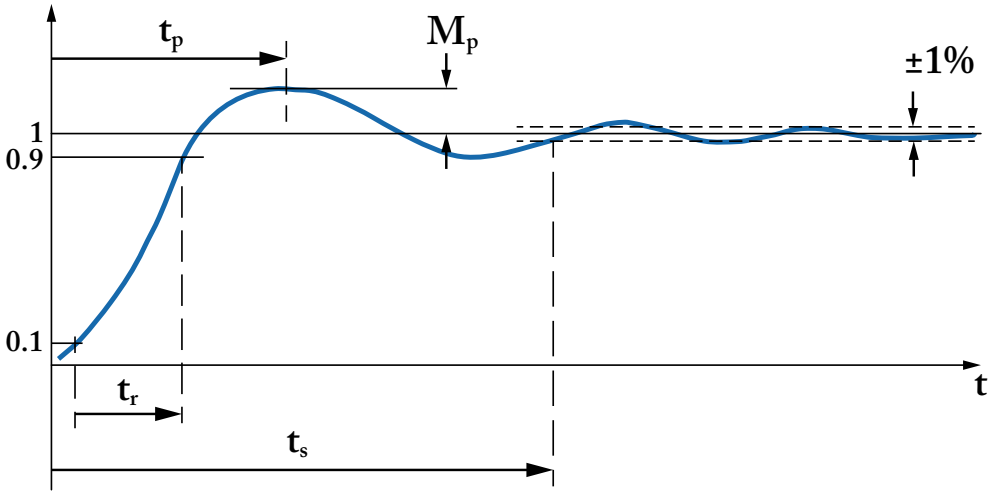


Figure 2.13: Characteristic parameters in second order step response, rise time, settling time, peak time and overshoot.

The relation of these 4 parameters to ω_n and ζ can be found from the analysis of the underdamped time evolution:

$$y(t) = 1 - \mathrm{e}^{-\zeta\omega_n t}\cos(\omega_d t) - \frac{\zeta\omega_n}{\omega_d}\,\mathrm{e}^{-\zeta\omega_n t}\sin(\omega_d t). \tag{2.23}$$

2.4.4.2 Rise time

An exact relation of the rise time is rather complex for second order systems as it depends on both ω_n and ζ. However a crude approximation can be made by inspection of figure 2.12. For underdamped system the rise time is only slightly depends on the damping. This is also obvious from eq. 2.23 which indicates that the initial phase has strong resemblance with the first part of a cosine function. If $\zeta = 0.5$ is taken as the typical situation, one can calculate that time from 0.1 to 0.9 takes approximately $\omega_n t_r = 1.8$. A rough relation between rise time and the system characteristics is thus

$$t_r \cong \frac{1.8}{\omega_n}. \tag{2.24}$$

2.4.4.3 Overshoot and peak time

The peak time is located at the maximum of the overshoot, when the derivative of $y(t)$ is zero.

$$\begin{aligned}\frac{\mathrm{d}y(t)}{\mathrm{d}t} &= \zeta\omega_n \,\mathrm{e}^{-\zeta\omega_n t}\left(\cos(\omega_d t) + \frac{\zeta\omega_n}{\omega_d}\sin(\omega_d t)\right) \\ &\quad - \mathrm{e}^{-\zeta\omega_n t}\left(-\omega_d \sin(\omega_d t) + \omega_d \frac{\zeta\omega_n}{\omega_d}\cos(\omega_d t)\right) = 0. \\ \frac{\mathrm{d}y(t)}{\mathrm{d}t} &= \mathrm{e}^{-\zeta\omega_n t}\left(\frac{\zeta^2\omega_n^2}{\omega_d}\sin(\omega_d t) + \omega_d \sin(\omega_d t)\right) = 0\end{aligned} \tag{2.25}$$

A zero value of eq 2.25 occurs when the argument of the $\sin$ function, $\omega_d t = n\pi$ a multiple of π. The first maximum and largest maximum is thus at

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \tag{2.26}$$

The value ω_d is the actual observed frequency, which is slight smaller than the natural frequency ω_n. Substitution of the peak time in the step response formula gives:

$$y(t_p) = 1 - \mathrm{e}^{-\pi\frac{\zeta\omega_n}{\omega_d}}\cos(\pi) - \frac{\zeta\omega_n}{\omega_d}\,\mathrm{e}^{-\zeta\omega_n t}\sin(\pi) = 1 + \mathrm{e}^{-\pi\frac{\zeta\omega_n}{\omega_d}}\,. \tag{2.27}$$

The overshoot M_p is defined as the amount the deflection from the final steady state value, normalized by the steady state value, which of course is only valid for $0 \le \zeta < 1$:

$$M_p = \frac{y(t_{max}) - y_{ss}}{y_{ss} - y(0)} = \mathrm{e}^{-\pi\frac{\zeta\omega_n}{\omega_d}}\,. \tag{2.28}$$

The measurement of $y(t_{peak})$ can than be used to evaluate the value of ζ, and with the value t_{peak} the natural frequency can also be evaluated. For many practical purposes this provides a fast way to determine the amount of damping for an underdamped system by making use of either this equation or its graphical display in figure 2.14.

2.4.4.4 Settling time

The setting time is usually defined as the time after which the variation of the transient is less than 1% around the steady state value, normalized to the difference between the initial value and the steady state value. For first order systems this is directly related to the decay time. In a similar fashion for underdamped second order systems t_s is defined as

$$\frac{y(t_s) - y_{ss}}{y_{ss} - y(0)} = \mathrm{e}^{-\zeta\omega_n t_s} = 0.01.$$

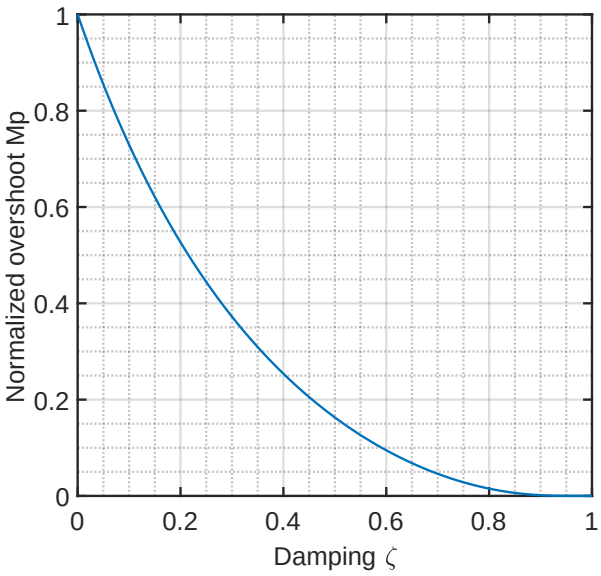


Figure 2.14: Normalized overshoot as a function of the damping ζ.

Therefore

$$\zeta\omega_n t_s = \ln\left(\frac{1}{0.01}\right) = 4.60517018598$$

or

$$t_s = \frac{4.60517018598}{\zeta\omega_n} \approx \frac{5}{\zeta\omega_n} \tag{2.29}$$

The value $\zeta\omega_n$ is exactly the negative real part of the pole of an underdamped second order system, see figure2.12. This is the value associated with the decay of the transient. The factor 5 is often used as an approximation, within 5 times the decay time, $1/(\zeta\omega_n)$, the system is within 1% of its final value.

2.4.5 Design of a second order system

The above parameters are very valuable in the design of a second order system. One usually has requirements that are in terms of the reaction time, *i.e.* t_r, the time in which a transient completed, and the maximum overshoot that is allowed. However these parameters are interconnected. By choosing two of them, the others are fixed. For instance, a fast rise time and a low overshoot determines immediately the settling time. The area in which poles of the system might be positioned in the pole plot is thus directly limited by design requirements.

2.5 Impulse response

The step response displays the reaction of a system when it changes from an initial to a final stationary situation and it allows to view and analyse the most prominent features of a dynamic system. The impulse response is of similar importance to study the dynamic response by observing of the reaction of a system to an impulse function.
Since the impulse also contains a broad spectrum of frequencies, like the step function, this signals is also a good one to probe the system response. The advantage over a step response is the limited amount of energy put into the system. In general causes a step function an amount of energy that goes to infinity with time, which is not the case with an impulse. Consequence is that stability requirements for an impulse input are less severe.

The impulse response is strongly related to the step-response. Due to the fact that we are dealing with linear systems, means that if we perform an operation on the input signal, the new output can be determined by applying the same operation on the original output. Since the impulse function is the derivative of the step function, the impulse response is the derivative of the step response. This also works the other way around: since the step function is the running integral of the impulse function, the step response is the running integral of the impulse response. Now you can construct the trend response yourself, knowing that the trend function $x(t) = \alpha t$ is the integral of the step function.

2.5.1 Impulse response of a physical system

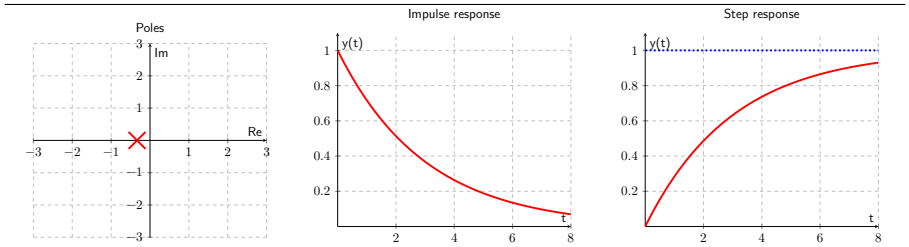
An impulse input in essence represents the supply of a well defined amount of energy to a system on a time scale that is much faster than the reaction time of that system. A stable system should eventually return to its initial state after application of the impulse. In table 2.1 the most important elementary pulse responses are displayed as well as the position of their poles in a pole plot.
A comparison of a) and b) shows the difference between a first order system and a cascade of two first order systems (overdamped second order). The second order system has a fast rise of the signal after the impulse is applied.
Figures b), c) and d) display various stable second-order systems. The under-damped case is easily recognized by its damped oscillation and the system properties can be inferred from the period and the attenuation. As can be inferred from the step response, the observation of an oscillation requires $\zeta < 1$. In e) and f), unstable second order systems are shown. An overview of the poles and the mathematical expressions of the responses in table 2.1 is given in table 2.2.

2.6 Beyond second-order systems.

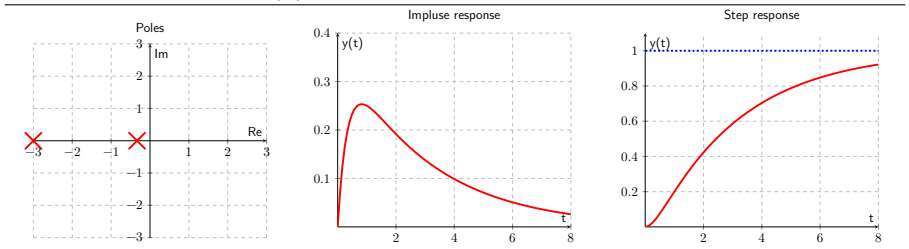
In principle the analysis of a system of third-order and higher is just an extension of the methods developed for first-order and second-order systems. As will be demonstrated in the next chapters, however, these systems can be treated as a combination of first-order and second-order systems. That is, the system can be broken up into subsystems. Second-order subsystems are only required if oscillations occur, *i.e.* a damping constant $\zeta < 1$, otherwise a combination of first-order subsystems suffices. They share that the real part of all solutions of the characteristic polynomial is negative. This assures that the solution decays with time. However, not all of these subsystems are equally relevant. Those with a large negative magnitude of the real part of the pole will decay much faster than those with a small value. The poles with a small real part, correspond to a long time and are therefore dominant in the time evolution as their effect is observed for the largest amount of time. The effect of poles with a large real part, correspond to short times and are too fast to be observed in the response signal.
An example of this is shown in the approximation of equation 2.20 the response of an over-damped second-order system to obtain equation 2.22. Here we could reduce a second order system to a first order system.
The initial conditions determine the actual strength of the various exponential decay functions. However, an order of magnitude difference in time constant leads to a very strong attenuation of exponential terms with small time constants. The consequence of this is that in many circumstances the dominant poles (roots of the characteristic polynomial) of a system will suffice to describe the system response. The representation of poles in a pole plot helps to identify the significant contributions.
The concept of dominant poles will be illustrated with the response of a third-order system with input-output model to make the connection to the coefficients in the differential equation:

$$a_3 \frac{\mathrm{d}^3 y}{\mathrm{d}t^3} + a_2 \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + a_1 \frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = x(t).$$

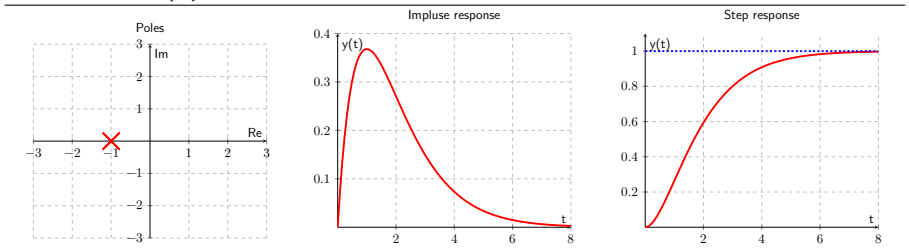
Table 2.1: Pole plot, impulse response and step response of second-order systems characterized by their poles



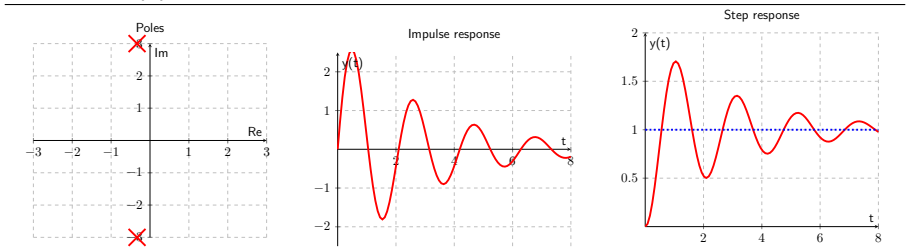
(a) First-order system – stable



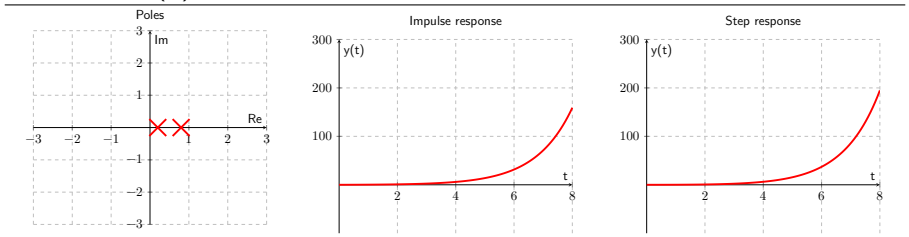
(b) Second-order system – overdamped – stable



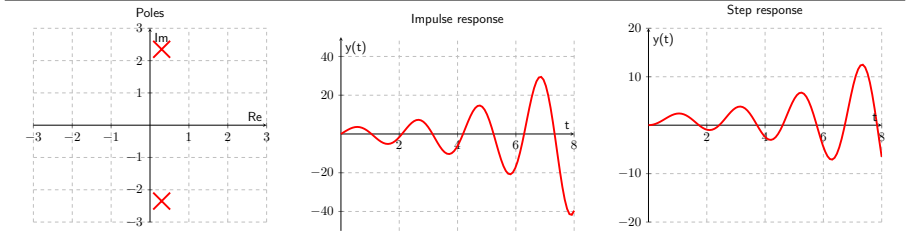
(c) Second-order system – critically damped – stable



(d) Second-order system – underdamped – stable



(e) Second-order system – overdamped – unstable



(f) Second-order system – underdamped – unstable

Table 2.2: Value of the poles and mathematical expressions of the responses for the graphs in table 2.1 with reference to the items.

ref.	Poles	Impulse response	Step response
(a)	$-\frac{1}{3}$	$y(t) = \mathrm{e}^{-x/3}$	$y(t) = 1 - \mathrm{e}^{-x/3}$
(b)	$\{-\frac{1}{3}, -3\}$	$y(t) = -\frac{3}{8}\left(\mathrm{e}^{-3t} + \mathrm{e}^{-t/3}\right)$	$y(t) = 1 - \frac{1}{8}\mathrm{e}^{-3t} - \frac{9}{8}\mathrm{e}^{-t/3}$
(c)	$\{-1, -1\}$	$y(t) = t\mathrm{e}^{-t}$	$y(t) = 1 - (1+t)\mathrm{e}^{-t}$
(d)	$\{-\frac{1}{3} + 3\mathrm{j}, -\frac{1}{3} - 3\mathrm{j}\}$	$y(t) = \frac{82}{27}\mathrm{e}^{-t/3}\sin(3t)$	$y(t) = 1 - \mathrm{e}^{-t/3}\left(\cos(3t) + \frac{1}{9}\sin(3t)\right)$
(e)	$\{\frac{1}{5}, \frac{4}{5}\}$	$y(t) = \frac{4}{15}\left(-\mathrm{e}^{t/5} + \mathrm{e}^{4t/5}\right)$	$y(t) = 1 - \frac{4}{3}\mathrm{e}^{t/5} + \frac{1}{3}\mathrm{e}^{4t/5}$
(f)	$\{\frac{1}{3} + 3\mathrm{j}, \frac{1}{3} - 3\mathrm{j}\}$	$y(t) = \frac{82}{27}\mathrm{e}^{t/3}\sin(3t)$	$y(t) = 1 + \mathrm{e}^{t/3}\left(-\cos(3t) + \frac{1}{9}\sin(3t)\right)$

Let us assume that the values of the coefficients only provide real negative roots, p_1, p_2, p_3, with the last root much further from the imaginary axis:

$$|p_3| \gg |p_1|\,, |p_3| \gg |p_2|$$

The characteristic polynomial is thus

$$(s - p_1)(s - p_2)(s - p_3) = 0 \tag{2.30}$$

$$s^3 - (p_1 + p_2 + p_3)s^2 + (p_1p_2 + p_1p_3 + p_2p_3)s - p_1p_2p_3 = 0 \tag{2.31}$$

and the differential equation can be written as

$$\frac{\mathrm{d}^3y}{\mathrm{d}t^3} - (p_1 + p_2 + p_3)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (p_1p_2 + p_1p_3 + p_2p_3)\frac{\mathrm{d}y}{\mathrm{d}t} - p_1p_2p_3y = \frac{x(t)}{a_3}$$

Dividing both sides of this DE by p_3 gives:

$$\frac{1}{p_3}\frac{\mathrm{d}^3y}{\mathrm{d}t^3} - (\frac{p_1 + p_2}{p_3} + 1)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (\frac{p_1p_2}{p_3} + p_1 + p_2)\frac{\mathrm{d}y}{\mathrm{d}t} - p_1p_2y = \frac{x(t)}{p_3a_3}.$$

The assumption regarding the magnitude of p_3 allows us to neglect the terms with p_3 in the denominator on the left side of the DE. This is a natural approximation for the term in front of the second and first derivative. The terms divided by p_3 are just much smaller. The third order differential is also set to zero as the value of p_3 makes this term much smaller than the other terms in the homogeneous equation. In contrast, the particular solution must be retained. The reason for this is that the solution of the homogeneous equation itself is already an important part of the response. The approximation thus leads to a second order differential equation,

$$-\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (p_1 + p_2)\frac{\mathrm{d}y}{\mathrm{d}t} - p_1p_2y = \frac{x(t)}{p_3a_3}.$$

The steady state response of this approximated ODE is similar to that of the original third order equation,

$$y_{ss} = -\frac{u_{ss}}{p_1p_2p_3}$$

2.7 Matching a model to measured data

In this section we focus on the matching between a mathematical model and measured data to determine the values of the parameters in the model. There are a few approaches to this matching of which we will discuss two: the first one, where the data is fit to the differential equation, and the second one where the data is fit to the general solution of the differential equation. Both methods have their advantages and drawbacks. After illustration of a set of data to a model we will compare the results of both methods.

Matching to the differential equation

Most simple and used in the past is to prepare the data in a way that these data can be matched with the differential equation directly.
For case of simplicity we illustrate the method on a first-order system:

$$y'(t) + ay(t) = x(t) \tag{2.32}$$

The data from a measurement on this system is presented in columns 1 and 2 of table 2.3. No input was supplied to the system ($x(t) = 0$), the system started from an initial value for $y(0)$ at $t = 0$.

Table 2.3: Measured data (column 1 and 2); and prepared data (column 3 and 4) for matching to the differential equation.

t [s]	y [...]	$\Delta y/\Delta t$	$(y_{i+1} + y_i)/2$
0	3.16	-0.55	2.61
2	2.06	-0.29	1.76
4	1.46	-0.21	1.26
6	1.06	-0.20	0.85
8	0.65	-0.10	0.55
10	0.44	-0.07	0.38
12	0.31		

In order to be able to process the data, we first rewrite the differential equation into:

$$\frac{\Delta y}{\Delta t} + ay = 0$$

or in terms of the discrete data points:

$$\frac{\Delta y}{\Delta t} = \frac{y_{i+1} - y_i}{t_{i+1} - t_i} = -ay_i$$

or to be more precise:

$$\frac{y_{i+1} - y_i}{t_{i+1} - t_i} = -a\frac{y_{i+1} + y_i}{2}$$

By taking the derivative at the center between two data points a systematic error is introduced due to the fact that the theoretical evolution of the derivative over the interval is not linear.
The results of these calculations on the data points is presented in the same table in column 3 and 4 respectively. A fit to a line $y = ax + b$ will result in the slope and the interception. The first being the slope equivalent with parameter a in the model and the second should be zero, and is a measure of the error. The result of the fit is:

$$\frac{\Delta y}{\Delta t} = -ay + b = -0.202y + 0.012$$

This means that the parameter of the differential equation is $a = 0.202$ for this model, corresponding to a characteristic time $\tau = 4.94$ [s]. In case we would have forced the line through $\{0, 0\}$ we would have obtained $a = 0.195$. This initial condition for the experimental data was: $y(0) = 3.16$. The result is presented in figure 2.15.

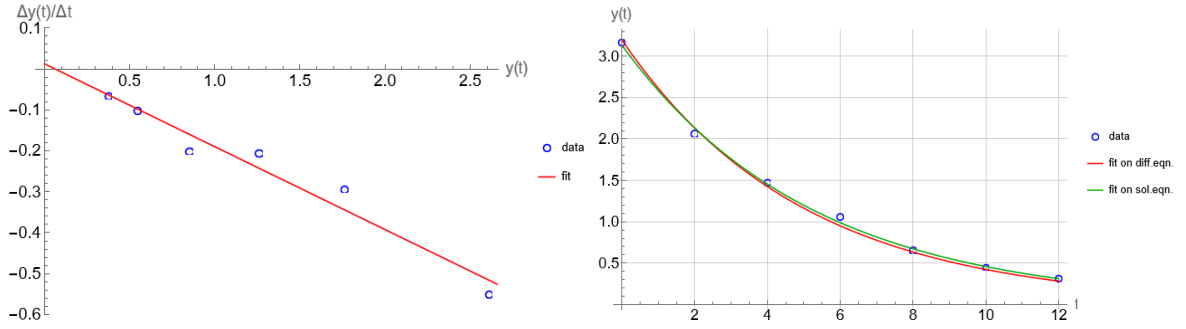


Figure 2.15: Result of the Model fitted to the data given in table 2.3. The left plot is the fit as being performed on the differential equation, the right plot is the result of the corresponding output function (red) as well as of the result of a fit to the general solution (green).

The advantage of using this method of finding parameters is that the deviation from the model is easier to detect. The relation between de derivative of $y(t)$ and the output itself must be linear as is dictated by the first-order differential equation 2.32. An additional advantage is that you directly derive the (inverse) of the time constant in the fit: it is the slope of the line in the left plot in figure 2.15.

Matching to the solution of the differential equation

The other method is based on the solution of the differential equation in general form, followed by fitting procedure to determine the values for the parameters. For this method we have to find a general solution to the differential equation first.
For the given example, data in table 2.3 and model in the differential equation 2.32, the general solution is:

$$y(t) = c_0 + c_1 \mathrm{e}^{-at} \tag{2.33}$$

with c_0, c_1 and a free parameters. Note that we allow for a non-zero offset of the solution. In case we would know that the solution has to go to zero for the time to infinity, we could leave out c_0 from the equation, leaving only two parameters to fit. Note that in the previous method, the offset is filtered out by means of only using the difference of the values.
The next step is fitting the data to the general solutions. Using a non-linear fit procedure on the given equation results is the values: $a = 0.1924$ and $c = 3.135$. The resulting response in presented in figure 2.15; green curve together with the fit of the other method (red curve). The corresponding time constant of parameter a is $\tau = 5.198$

Comparison of the two methods

In the right-hand plot in figure 2.15, a graphical representation of the fit results is given for both methods.
Note that the characteristic time values can be identified in the graph on the right side of the figure as the slope of $y(t)$ at $t = 0$: Draw a line from $y(0)$ to the intersection with the asymptote $y(t \to \infty) = 0$. The intersection is roughly at a value of $t = 5$.
The best way to compare the results of both methods is to calculate the chi-squared values (χ^2-value) of both methods. For the first method we obtain: $\chi^2 = 5.2 \cdot 10^{-4}$ and for the second method we

obtained: $\chi^2 = 2.8 \cdot 10^{-4}$. From this we conclude that the second method give better results as was expected.
Main advantage of the first method is a visual check whether the data fits the linear differential equation. However this method gives less accurate results with a systematic error[2].
Main advantage of the second method is that it produces better results, although you have to be careful in choosing the right general solution. An other advantage is that it is easier to apply on higher-order models. A drawback of the second method is a less clear visual relation between the data and the fit results, however this can be overcome by plotting the results on a log-scale: a multi-exponential would be easier to identify in this way.

[2]The fact that this method gives a systematic error is not proven here. The fact that the derivative obtained from the data is related to the center of the interval causes the systematic error due to the fact that the derivative does not vary linear over the interval.

Summaries

Summary Chapter 2

Characteristics in the time domain of systems of first and second order:

- First order
  - Characteristic equation:

$$\tau s + 1 = 0 \tag{2.34}$$

  - General Solution:

$$y(t) = C_1 \mathrm{e}^{t/\tau} + C_0 \tag{2.35}$$

- Second order:
  - Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \tag{2.36}$$

  - with characteristics:
    - Natural frequency: ω_n
    - Damping coefficient: ζ
    - Measured frequency: $\omega_d = \sqrt{1-\zeta^2}\omega_n$
    - Solution: $p_{1,2} = -\zeta\omega_n \pm \sqrt{1-\zeta^2}\omega_n \mathrm{j}$ with $|p_1| = |p_2| = \omega_n$.
  - General solution:

$$y(t) = \mathrm{e}^{-\zeta\omega_n t}\left(C_1 \mathrm{e}^{\mathrm{j}\omega_d t} + C_2 \mathrm{e}^{-\mathrm{j}\omega_d t}\right) + C_0 \tag{2.37}$$

  - Normalized step response:

$$t_r \cong \frac{1.8}{\omega_n} \tag{2.38}$$

$$t_s \approx \frac{5}{\zeta\omega_n} \tag{2.39}$$

$$M_p = e^{-\pi\frac{\zeta}{\sqrt{1-\zeta^2}}} \tag{2.40}$$

$$t_p = \frac{\pi}{\omega_d} \tag{2.41}$$

  - Figures 2.12 and 2.14 are part of this summary.

Problems

B-type problems

Problem 2.1. Ultimate simple model of an e-bike
The power in modelling is in an effective simplicity, *i.e.* reduce a complex system to a minimum of elements, that describe the system. In this problem the coasting of an e-bike is simplified to a mass m that slides over a ground with a friction coefficient b (linear friction $F_b = bv$). On this system a few measurements are done, of which the results are presented in table 2.4. The first two columns, relate the velocity to the applied power for a stationary (constant velocity) situation. The two other columns, relate the velocity to the elapsed time, from the moment of stopping the driving force.

a. Give a first order differential equation that describes this system for the situations described.

b. Derive the value of the damper coefficient b and the mass m from the values given in the table. Do this by plotting the measured values and using the script curvefitgui/fmlsq,

c. Show that the first-order linear model is sufficient to describe this system.

Table 2.4: Power-velocity reading on home trainer and the velocity attenuation during coasting.

velocity [km/h]	power [W]	time [s]	velocity [km/h]
10.2	24	0	28.8
18.0	75	10	19.3
23.5	128	20	12.9
27.8	179	30	8.7
32.3	241	40	5.8

Problem 2.2. Determine the capacitance of a capacitor.
You can determine the characteristics of electrical components with so called component testers like an Ohm meter for measuring the resistance. With a voltmeter, a known resistor and a battery at hand the capacitance of a capacitor can be determined. First the capacitor is charged with the battery. After disconnecting from the battery a resistor with a high value of resistance ($100\,\mathrm{k\Omega}$) is connected to the terminals of the capacitor resulting in a slow discharge of the capacitor. The voltmeter is connected to record the voltage drop with time. The resulting values are:

Table 2.5: Voltage drop over a capacitor

Time [s]	Voltage [V]
0	6.0
10	4.5
20	3.5
30	2.6
40	1.9
50	1.4
60	1.0

a. Sketch the electrical circuit used during discharge and derive the input-output equation.

b. Compare two methods to estimate a value of the capacitance and give a value of the capacitance.

c. Make a graph where you plot the measured data together with the result of the derived model.

A-type problem

Problem 2.3. Overshoot
Figure 2.16 shows the response of a second order electrical network with an unit step input.

a. Determine the damping factor and the natural frequency of this network.

b. Sketch the poles of this network.

c. Determine the differential equation that describes this network.

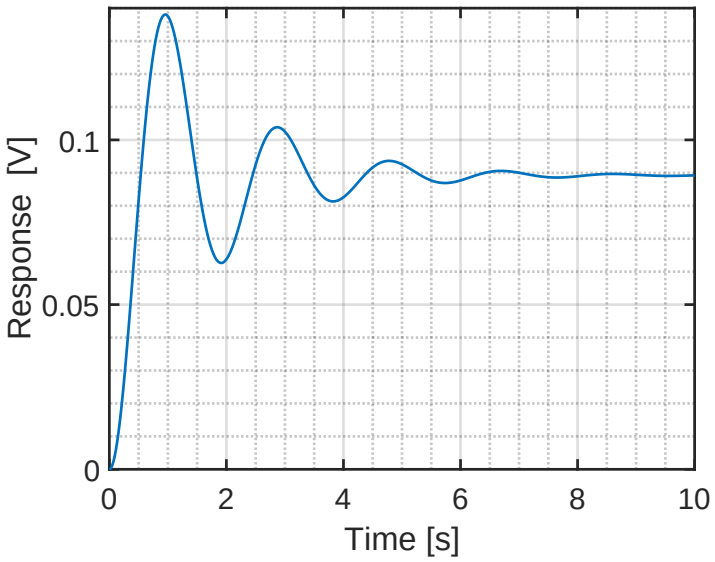


Figure 2.16: Response of an electrical network.

B-type problems

Problem 2.4. LCR-permutation *(0:45h)*
Figure 2.17 shows three electrical networks that can be obtained by a permutation of a capacitor, a resistor and a coil. These three networks show a very different response to an input signal and can be regarded as prototypical analogue filters.

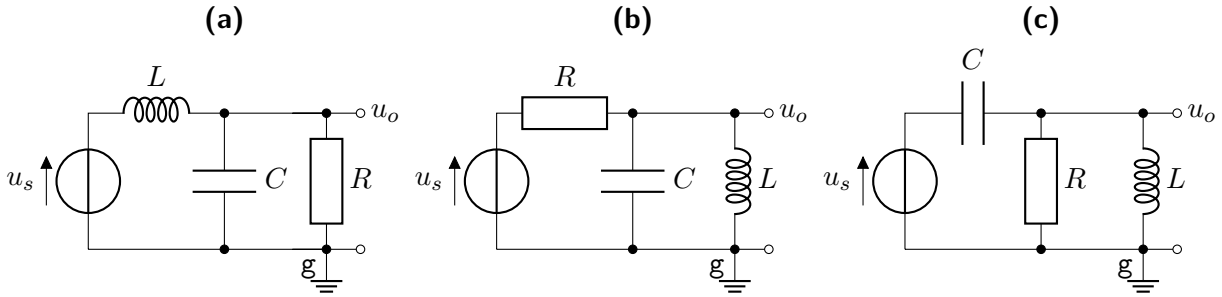


Figure 2.17: Three different configurations of a LRC network.

a. Determine for each of these systems the set of first order differential equations that describe the network.

b. Derive the model for these 3 systems, the input signal is u_s and the output signal is u_0. What is the difference between these three models?

c. Determine the poles of each system. How do these compare?

d. Derive a relation between the components to obtain critically damped systems.

e. Draw the poles for values of $C = 1\,\mu\mathrm{F}$, $L = 1\,\mathrm{mH}$ and $R = 1\,\mathrm{M}\Omega$.

f. Give the general solution of the system based on the calculated poles of the systems.

Problem 2.5. Dominant subsystem.
The step response of a third order system is measured, and found to be characterised by three roots, $p_1 = -10$, $p_2 = -1 - \mathrm{j}$, and $p_3 = -1 + \mathrm{j}$, with a steady state gain of 20.

a. Give the characteristic equation, and specify the differential equation of the input- output model of this third order system.

   Not all roots are equally relevant and the response can be represented by a sub-system which gives a very similar response.

b. Specify the characteristic equation, and specify the differential equation of the input-output model for the dominant subsystem.

c. Plot the evolution of the step response of the third order system as function of time. Compare this with the following characteristic values in the dominant subsystem: The maximum overshoot value, the time at which the signal first peaks, the period of the damped oscillation, and the steady state value.

d. Assume that all exponential functions have the same strength. Calculate and plot the response of the sub-system and that of the original system for $p1$ = -5, -10, -20. Investigate for what values of p_1 the approximation is valid, provide your criteria for your conclusion.

C-type problems

Problem 2.6. Input-Output model
Consider a linear time-invariant system that is characterised by an input-output equation for input $x(t)$ and output $y(t)$,

$$\frac{1}{\omega^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega}\frac{dy}{dt} + y = K\Big(\tau\frac{dx}{dt} + x\Big), \tag{2.42}$$

with fixed system parameters ω, ζ, and K, and an adjustable parameter τ.

a. Specify the characteristic equation for this system, and determine the values of ζ for which the system is (i) under-damped, (ii) critically damped, and (iii) over-damped. For each of these three cases make a sketch in which you indicate the location of the poles in the complex plane
   .

b. Assume that $\zeta = \sqrt{2}$. Derive the time-dependent output $y(t)$ of this system for $t > 0$, when the input is $x(t) = 1 - \exp(-t/\tau)$ for all $t \geq 0$ and the initial state is defined at $t = 0$ by the conditions $y = \frac{dy}{dt} = 0$.

In order to determine the system parameters ω, ζ, and K, we set $\tau = 0$ so the inhomogeneous part of Eq. (2.42) reduces to Kx. With this setting we perform a number of measurements using $x(t) = u(t)$. From the step response we get a steady-state value $\lim_{t\to\infty} y(t) = y_\infty$, an observed oscillation period T, and a normalised overshoot M.

c. Determine the three system parameters K, ω, and ζ in terms of these observed values for y_∞, T, and M.

d. For constant system parameters K, ω, and ζ, we increase τ to a finite value $\tau > 0$, and repeat the step-response experiment. Explain which of the observed values y_∞, T, and M would be different and which would be the same.

Chapter 3

Laplace Transform

In this chapter we introduce a sophisticated piece of mathematics called the Laplace Transform. For our present purposes, this is primarily a powerful tool for solving initial value problems by turning differential equations into algebraic equations that are easier to solve. The initial value conditions are also automatically folded into the calculation, which saves time and effort. All of this is achieved by re-expressing a function $y(t)$ into a different function of a different variable, $Y(s)$, solving the problem in this new language, then converting the answer back into the real $y(t)$ world.

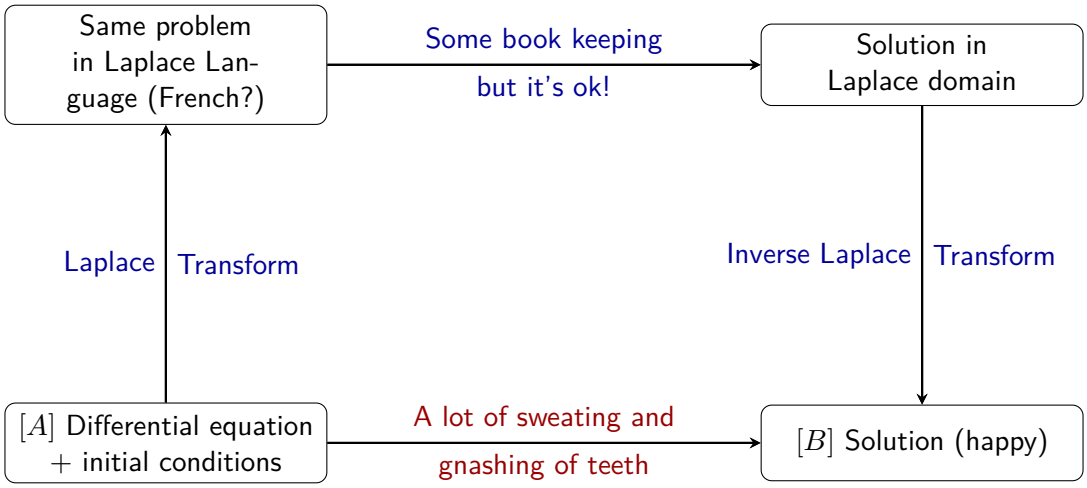


Figure 3.1: Two paths to solve differential equations from $[A]$ to $[B]$: the direct route and via the Laplace transform.

An additional use of the Laplace transform is the transformation of homogeneous ODE's to transfer functions. This part will be treated in chapter 4 for input-output equations and in chapter 6 for state-space models.

In situations where we are only interested in the modification of an input signal $x(t)$ by a system represented by a function $f(t)$ generating an output $y(t)$ we can set up a similar diagram to get from the input to the output of a system: see figure 3.2. Here $f(t)$ and $F(s)$ are the description of the system in the time and the frequency domain respectively. In the frequency domain the output is derived from the input by means of a multiplication with the system function $F(s)$, where as in the time domain the output is calculated from the input by means of a convolution of the input with the system function $f(t)$.

It is obvious that the functions $f(t)$ and $F(s)$ used in figure 3.2 are related to the set of differential equations describing the system.

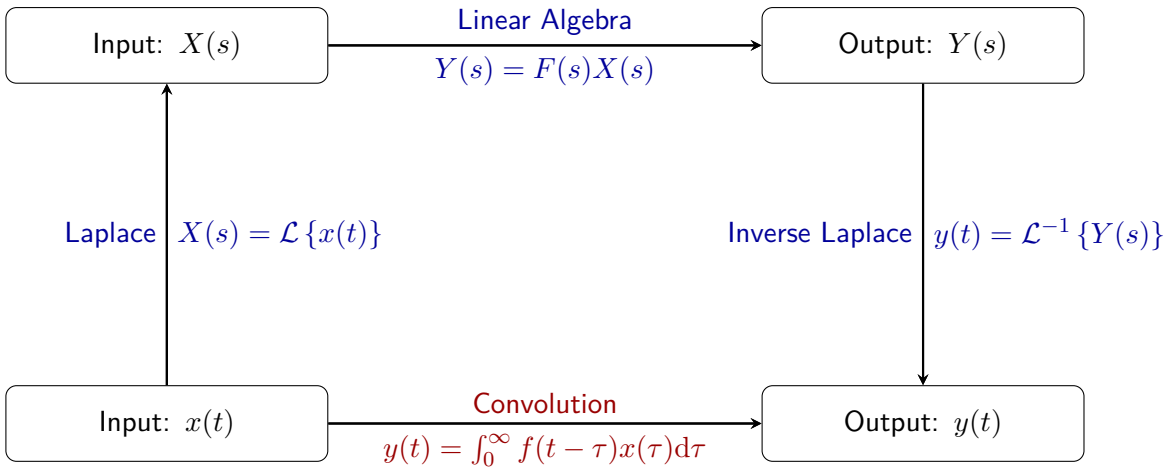


Figure 3.2: Two paths to describe a modification of an input signal $x(t)$ to an output signal $y(t)$ by as system described by $F(s) = \mathcal{L}\{f(t)\}$

3.1 Convolution

An alternative way of looking at a system is by representing it as a box with a function that transforms an input function to an output function. Let us define the situation: we take an input function denoted by $x(t)$ and an output function $y(t)$. The relation between the two is usually described by a differential equation. However we can also consider a function $f(t)$ and some operation that transforms $x(t)$ in $y(t)$ by means of $f(t)$. This is graphically presented in figure 3.3.

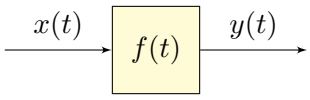


Figure 3.3: Function $f(t)$ relating input $x(t)$ and output $y(t)$ in the time domain.

This operation is called *convolution* and is defined as:

$$\begin{aligned} y(t) &\equiv f(t) \otimes x(t) \\ y(t) &\equiv \int_0^\infty f(t-\tau)x(\tau)\mathrm{d}\tau \end{aligned} \tag{3.1}$$

From the definition of convolutions we can derive the following properties:

- *Communtation:* $f \otimes g = g \otimes f$
- *linearity:* $f \otimes (g_1 + g_2) = f \otimes g_1 + f \otimes g_2$
- *Association:* $f \otimes (g \otimes h) = (f \otimes g) \otimes h$
- *Zero function:* $f \otimes 0 = 0 \otimes f = 0$

Using the Laplace route as indicated in figure 3.1 we will show in this chapter that the relation between the input and output in Laplace language is just a multiplication:

$$Y(s) = F(s)X(s) \tag{3.2}$$

This process is graphically presented in figure 3.4.

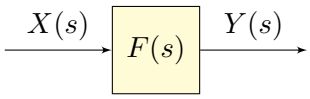


Figure 3.4: Function $F(s)$ relating input $X(s)$ and output $Y(s)$ in the Laplace domain.

Now we know the structure, we have to look for the mathematical formulations to relate $x(t) \leftrightarrow X(s)$, $y(t) \leftrightarrow Y(s)$ and $f(t) \leftrightarrow F(s)$. The relation between the two domains is done by the *Laplace Transform* end the *Inverse Laplace Transform*.
The functions $x(t)$ and $y(t)$ describe signals; input and output. The function $f(t)$ and its Laplace counterpart $F(s)$ represent a system. Later we will see that $F(s)$ is called the transfer function, to which a complete chapter is dedicated: chapter 4. The use of Transfer functions makes it possible to describe systems – by means of characteristics of the transfer functions – without having to deal with an input and an output function. The function $f(t)$ will later be identified as the impulse response of the system.
Before we dive into Laplace transforms we first introduce its generalisation: the *Integral Transform*.

3.2 Integral transforms

The Laplace transform is a transform of the family of integral transforms. These are characterized by the transformation of a function of some variable, in this case time t, to a function of a counterpart variable be means of an integral and a kernel function. The combination of the kernel function and the interval of integration define the actual transformation. The general form of a integral transform is:

$$g(s) = \int_a^b f(t)K(s,t)\mathrm{d}t \tag{3.3}$$

with $f(t)$ the function of the variable t that is mapped via the kernel $K(s,t)$ on the function $g(s)$ in the s domain. These integral transformations are linear and all have an inverse transformation. In fact the integral transformation and its inverse map one domain on a corresponding domain.
Examples of integral transforms that we will consider are the Laplace transform that maps the time domain t on the complex frequency domain s and the Fourier transform that maps the time domain t on the frequency domain ω.

3.3 The Laplace transform

Let's begin with the definition of the Laplace Transform. The Laplace transform is an integral transform with the Kernel given by:

$$K(s,t) = \mathrm{e}^{-st} \tag{3.4}$$

The time domain is restricted to $t \in [0,\infty]$ and the complex frequency domain is $s \in \mathbb{C}$. The operation of taking the Laplace transform of the function $f(t)$, is denoted as $\mathcal{L}\{f(t)\}$ and involves computing the following integral:

$$\mathcal{L}\{f(t)\} = F(s) = \int_{0^-}^{\infty} f(t)\,\mathrm{e}^{-st}\,\mathrm{d}t. \tag{3.5}$$

The resulting complex function $F(s)$ is called the Laplace transform of $f(t)$. Here s is a complex variable and the transform is said to exist for all values of s for which this integral converges. Since the time domain is only limited to positive times $t \in [0,\infty]$, in some cases the function $f(t)$ is

multiplied by the step function $u(t)$ to enforce this. The integral boundaries can be set to $[-\infty, \infty]$ in this case. The complex frequency s is in the complex domain $s \in \mathbb{C}$. Alternatively one could say that the functions in the time domain $f(t)$ are only defined for $t \geq 0$.
The lower limit in the integral in equation 3.5 is directly related to the discussion above on the addition of the step function and integrating over the whole domain of t. In order to include impulse and step functions at $t = 0$ we have to include the whole function and therefore a more appropriate formulation of $F(s)$ would be in terms of a limit:

$$F(s) = \lim_{\epsilon \uparrow 0} \int_{\epsilon}^{\infty} f(t)\, \mathrm{e}^{-st}\, \mathrm{d}t.$$

Now you know the background, we will drop the limit and the 0^- notation.

- $F(s)$ is independent of time since we integrate over time when computing the Laplace transform. If you ever compute a Laplace transform and your answer includes the variable t, you did something wrong!
- We don't usually plot $F(s)$. This would be quite awkward: because s and $F(s)$ are complex, each with a real and an imaginary component, such a plot is four-dimensional. Usually we only plot the position of the poles and zeroes of $F(s)$ in the s-plane.
- Since our purpose is to solve initial value problems, we don't really speak about the early childhood of $f(t)$ before $t = 0$. Here we made this explicit by including the step function $u(t)$, but it is also common practice to omit the $u(t)$ and to remember that it is there implicitly.

3.4 The inverse Laplace transform

There is a one-to-one mapping between a function $f(t)$ and its Laplace transform $F(s)$, and we can go back and forth between the time domain and the Laplace domain:

$$f(t) \leftrightarrow F(s)$$

Going from $F(s)$ back to $f(t)$ is called taking the inverse Laplace transform of $F(s)$. While it is possible to write down an explicit form for the inverse Laplace transform $\mathcal{L}^{-1}\{F(s)\}$, this is somewhat tricky because s is a complex number: there are many different paths over which to take an integral in the complex plane. This branch of mathematics is called complex analysis, and involves contour integration and calculation of residues. Just for completeness, the inverse Laplace Transform is given by the so called Bromwich integral:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi \mathrm{i}} \oint_{\gamma - \mathrm{i}\infty}^{\gamma + \mathrm{i}\infty} \mathrm{e}^{st} F(s) \mathrm{d}s \tag{3.6}$$

where γ is a vertical contour in the complex plane positioned so that all singularities of $F(s)$ are to the left of it.
In practice we do not compute inverse Laplace transforms using complex analysis. Instead, we rely on tables of standard transforms, based on the fact that if we know one way, we also know the way back. Due to the fact that the Laplace transform is a one-to-one mapping there is a one-to-one correspondence between pairs of functions in the two domains. We will build such a table as we proceed through the chapter. As far as we are concerned, if $F(s)$ is the Laplace transform of $f(t)$, then this must mean that $f(t)$ is the inverse Laplace transform of $F(s)$. You might have to do a little algebra to bring the function into the form of a function you find in the table to perform the transformation.

3.5 Examples

Let us work through a few concrete examples of Laplace transforms of signals, *i.e.* Laplace transforms of functions of time. The functions we use here will play an important role in characterizing systems at a later stage.

Example 3.5.1. *Trend function $x(t) = t$:*

$$\mathcal{L}\{t\} = X(s) = \int_{0^-}^{\infty} t\,\mathrm{e}^{-st}\,\mathrm{d}t = \left[\frac{\mathrm{e}^{-st}}{-s}\left(t - \frac{1}{-s}\right)\right]_{0^-}^{\infty} = \frac{1}{s^2}.$$

The integral converges, and thus the transform is defined, for all values of s that satisfy $\mathrm{Re}(s) > 0$. We do not do the full calculation here, but let's note in passing (and add to the table) that

$$\mathcal{L}\{t^n u(t)\} = \frac{n!}{s^{n+1}}.$$

Conversely, it follows that if we see a Laplace transform of the form $1/s^m$, it is linked to a function of the form $\propto t^{m-1}$ in the time domain. □

Example 3.5.2. *Exponential function: $x(t) = \mathrm{e}^{-at}$:*

$$\mathcal{L}\left\{\mathrm{e}^{-at}\right\} = \int_{0^-}^{\infty} \mathrm{e}^{-at}\,\mathrm{e}^{-st}\,\mathrm{d}t = \left[\frac{1}{-a-s}\,\mathrm{e}^{-(a+s)t}\right]_{0^-}^{\infty} = \frac{1}{s+a}.$$

Here the region of convergence is $\mathrm{Re}(s) > -a$. This one turns out to be particularly useful when solving ODEs. □

Example 3.5.3. *Unit-step function $u(t)$:*
This is the same calculation as example 3.5.2 if we set $a = 0$, so

$$\mathcal{L}\{u(t)\} = \mathcal{L}\{1\} = \int_{0^-}^{\infty} \mathrm{e}^{-st}\,\mathrm{d}t = \frac{1}{s}.$$

For higher values of s this resulting function goes to zero. □

Example 3.5.4. *Dirac-delta function $\delta(t)$:*

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty} \delta(t)\mathrm{e}^{-st}\mathrm{d}t = 1.$$

Note that the resulting function is a constant of 1 for every value of s. □

These examples are important, since many other functions can be related to these functions and therefore to their Laplace Transforms. These functions are also needed to illustrate some properties of the Laplace Transform.

3.6 Differentiation property of the Laplace transform

The differentiation property is a very important one and it will turn out to be the main reason why Laplace transforms are so useful in solving differential equations:

$$\mathcal{L}\left\{\frac{\mathrm{d}f}{\mathrm{d}t}\right\} = s\mathcal{L}\{f(t)\} - f(0^-). \tag{3.7}$$

The proof involves integration by parts:

$$\begin{aligned}\mathcal{L}\left\{\frac{\mathrm{d}f}{\mathrm{d}t}\right\} &= \int_0^\infty \mathrm{e}^{-st}\,\frac{\mathrm{d}f(t)}{\mathrm{d}t}\mathrm{d}t \\ &= \left[\mathrm{e}^{-st}\,f(t)\right]_{0^-}^{\infty} - \int_{0^-}^{\infty}(-s)\,\mathrm{e}^{-st}\,f(t)\mathrm{d}t \\ &= [0 - f(0^-)] + s\mathcal{L}\{f(t)\} \\ &= sF(s) - f(0^-).\end{aligned}$$

This property can be applied directly to examples 3.5.4 and 3.5.3 of the previous section.
Applying this relation recursively for higher-order derivatives with respect to t yields:

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f(t)}{\mathrm{d}t^n}\right\} = s^n F(s) - s^{(n-1)}\,f(t)|_{t=0^-} - s^{(n-2)}\,\frac{\mathrm{d}f(t)}{\mathrm{d}t}\bigg|_{t=0^-} - \ldots - \frac{\mathrm{d}^{(n-1)}f(t)}{\mathrm{d}t^{(n-1)}}\bigg|_{t=0^-}.$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^n f(t)}{\mathrm{d}t^n}\right\} = s^n F(s) - \sum_{i=0}^{n-1} s^{n-i-1}\,\frac{\mathrm{d}^{(i)}f(t)}{\mathrm{d}t^{(i)}}\bigg|_{t=0^-}$$

This characteristic gives us the tools to transform every derivative with respect to t to powers of the complex frequency s times the Laplace transform of the time functions and the initial conditions of the function and its derivatives. This transforms a differential equation to an algebraic equation to be solved. At the cost of a transformation we get an easier to solve algebraic equation.

3.7 Initial value problems

As mentioned in the previous section, the Laplace transform is a great help in solving differential equations, particularly the type we consider in this course: LTI ODE's. In order to get a non-zero output of a differential equation, we either set initial conditions that are different from the steady state values, or we us an input function (non-homogeneous differential equation). Use both an input function and initial conditions that deviate from the equilibrium values is also a possibility.
We will illustrate this with a few examples. We start with a simple one.

Example 3.7.1. *First-order differential equation with initial condition*
Suppose we want to solve the following initial value problem:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + 3y = 0;\ y(0) = 1$$

We can take the Laplace transform of both sides of the equation:

$$\begin{aligned}\mathcal{L}\left\{\frac{\mathrm{d}y}{\mathrm{d}t} + 3y\right\} &= \mathcal{L}\{0\} \\ [sY(s) - y(0)] + 3Y(s) &= 0 \\ Y(s) &= \frac{y(0)}{s+3} \\ y(t) &= y(0)\,\mathrm{e}^{-3t} = \mathrm{e}^{-3t}\end{aligned}$$

As a check we plug the solution back into the differential equation, which results in:

$$3\mathrm{e}^{-3t} - 3\mathrm{e}^{-3t} = 0$$

which is correct; checking for the initial value:

$$\mathrm{e}^{-3t}\Big|_{t=0} = 1$$

which is also correct. So we obtained the right solution.
This may look like more work at first, but if you are used to manipulating the transforms it's actually quite efficient since every step is applied systematically. It's also amusing that we never performed a differentiation operation, yet somehow we solved a differential equation. □

Let us look at a second order differential equation as a next step. We take an example to indicate that we have to work on a supporting technique to be able to make the transformation back to the time domain.

Example 3.7.2. *Second-order differential equation with initial conditions and input*
Solve the following initial value problem:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 5\frac{\mathrm{d}y}{\mathrm{d}t} + 6y(t) = u(t)\,; \qquad y(0) = 0,\ y'(0) = 1$$

where $u(t)$ is the step function and $\mathcal{L}\{u(t)\} = 1/s$. Taking the Laplace transform left and right gives us:

$$s^2Y(s) - sy(0) - y'(0) + 5sY(s) - 5y(0) + 6Y(s) = \frac{1}{s}$$

filling in the initial conditions and sorting terms gives us:

$$(s^2 + 5s + 6)Y(s) - 1 = \frac{1}{s}$$

Solving this equation for $Y(s)$ gives us:

$$Y(s) = \frac{1}{s^2 + 5s + 6}\frac{s+1}{s} = \frac{s+1}{s(s^2 + 5s + 6)}$$

For the transformation back to the time domain to get the complete output function $y(t)$, we have to take the inverse Laplace transform of $Y(s)$ as given above. Since it is not advisable to use the complex contour integration for the inverse Laplace transform, we need to transform the obtained expression in a form that can be looked up in the Laplace Transform table. So for the moment we let is rest at this point and come to it back after we have the tools to proceed. □

From the above example we have to work on a method that transforms the quotient of two polynomials in s in a form that can be looked up in a table. Since the LTI ODE's that we consider in general produce quotients of polynomials in s, this will be a valuable method to work on.

3.8 The inverse Laplace transforms of the quotient of two polynomials in s

When solving ODE's corresponding to LTI physical systems, we will typically generate expressions in the Laplace domain of the form

$$Y(s) = \frac{b_0 + b_1 s + \ldots + b_m s^m}{a_0 + a_1 s + \ldots + a_n s^n}\ (n \geq m). \tag{3.8}$$

How do we take the inverse transform to go back to the time domain, $y(t)$? It turns out that there is a very formulaic way to tackle this problem. First we need to factorize the denominator of $Y(s)$ such that it is in the form

$$Y(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{(s - p_1)(s - p_2)\cdots(s - p_n)}\ (n \geq m).$$

Here $p_1, p_2 \ldots p_n$ are the complex poles of the expression or, if you prefer, they are the roots of the polynomial in the denominator. *Note:* we use here the symbol p_i to distinguish the between the roots of the denominator polynomial that are the poles p_i and the roots of the numerator polynomial, that are the zeros z_i. We can now rewrite this expression in the form

$$Y(s) = \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2} + \cdots + \frac{k_n}{s - p_n}.$$

The coefficients $k_1, k_2 \ldots k_n$ are known as the "residues" of $Y(s)$ and we need to determine their values. Perhaps surprisingly, doing so is not hard. Suppose for definiteness that we want to determine the value of k_1. Then we note that

$$[(s-p_1)Y(s)]_{s=p_1} = \left[k_1 + \frac{k_2(s-p_1)}{(s-p_2)} + \cdots + \frac{k_n(s-p_1)}{(s-p_n)}\right]_{s=p_1} = k_1. \tag{3.9}$$

Finding this value is quite easy since

$$[(s-p_1)Y(s)]_{s=p_1} = \left[\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s-p_2)\ldots(s-p_n)}\right]_{s=p_1},$$

which we can compute by direct substitution of the actual numerical values of the a and p parameters. It looks complicated here, but you'll get the hang of it after doing a couple of exercises.
There is one caveat to the above derivation: we implicitly assumed in Eq. 3.9 that all of the poles have different values. If two poles have the same value, then things get slightly more complicated. Let's call the degenerate pole (that is, the pole whose value appears more than once) p_0. We're not judging the pole's moral compass by calling it degenerate, this is standard mathematical terminology. We will not go through a full derivation here but we will state the important results. For the (not so uncommon) case where two poles have the same value ($p_1 = p_2 = p_0$), the general form of the expansion is

$$Y(s) = \frac{k_1}{s-p_0} + \frac{k_2}{(s-p_0)^2} + \frac{k_3}{s-p_3} + \cdots + \frac{k_n}{s-p_n},$$

where

$$k_1 = \left[\frac{\mathrm{d}}{\mathrm{d}s}\left((s-p_0)^2 Y(s)\right)\right]_{s=p_0}$$

$$k_2 = \left[(s-p_0)^2 Y(s)\right]_{s=p_0}.$$

To see where those expressions come from, note that

$$\left[(s-p_0)^2 Y(s)\right]_{s=p_0} = \left[k_1(s-p_0) + k_2 + k_3\frac{(s-p_0)^2}{s-p_3} + \cdots + k_n\frac{(s-p_0)^2}{(s-p_n)}\right]_{s=p_0} = k_2$$

$$\begin{aligned}\left[\frac{\mathrm{d}}{\mathrm{d}s}\left((s-p_0)^2 Y(s)\right)\right]_{s=p_0} &= \left[\frac{\mathrm{d}}{\mathrm{d}s}\left(k_1(s-p_0) + k_2 + k_3\frac{(s-p_0)^2}{s-p_3} + \cdots + k_n\frac{(s-p_0)^2}{s-p_n}\right)\right]_{s=p_0} \\ &= \left[k_1 + 0 + 2k_3\frac{s-p_0}{s-p_3} - k_3\frac{(s-p_0)^2}{(s-p_3)^2} + \cdots + 2k_n\frac{s-p_0}{s-p_n} - k_n\frac{(s-p_0)^2}{(s-p_n)^2}\right]_{s=p_0} \\ &= k_1\end{aligned}$$

The most general form for an n-degenerate pole is given in the summary at the end of this chapter, but in practice it is rare to encounter anything more degenerate than a double pole.

Example 3.8.1. *Example of inverse Laplace Transform*
Suppose that we have the following Laplace transform:

$$Y(s) = \frac{5}{s^2 + 3s + 2}.$$

What is its inverse, $y(t)$? First we note that the denominator can be rewritten as

$$Y(s) = \frac{5}{(s+2)(s+1)}.$$

We rewrite this as

$$Y(s) = \frac{k_1}{s+2} + \frac{k_2}{s+1}$$

and compute

$$\begin{aligned} k_1 &= [(s+2)Y(s)]_{(s=-2)} = \left[\tfrac{5}{s+1}\right]_{(s=-2)} = -5 \\ k_2 &= [(s+1)Y(s)]_{(s=-1)} = \left[\tfrac{5}{s+2}\right]_{(s=-1)} = 5 \end{aligned}$$

so that

$$Y(s) = \frac{-5}{s+2} + \frac{5}{s+1}.$$

You can easily check that our algorithm is correct by recombining the two terms to recover the initial expression:

$$\begin{aligned} Y(s) &= \frac{-5}{s+2} + \frac{5}{s+1} \\ &= \frac{-5(s+1)+5(s+2)}{(s+2)(s+1)} \\ &= \frac{5}{(s+2)(s+1)} \end{aligned}$$

Taking the inverse transform using the result we calculated earlier is now very simple and we have

$$y(t) = -5\,\mathrm{e}^{-2t} + 5\,\mathrm{e}^{-t}\,.$$

Et voila!, Laplace would no doubt have exclaimed at this point. □

Now we have the technique available let us return to example 3.7.2 to complete the solution

Example 3.8.2. *Continuation of example 3.7.2*
We pick it up where we got stuck: taking the inverse Laplace transform of:

$$Y(s) = \frac{s+1}{s(s^2+5s+6)} = \frac{s+1}{s(s+2)(s+3)}$$

from which it is obvious the poles are $p_1 = 0$, $p_2 = -2$ and $p_3 = -3$, but we have one zero here too: $z_1 = -1$. Applying partial fraction decomposition we can write $Y(s)$ as:

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s+2} + \frac{k_3}{s+3}$$

and the corresponding solution:

$$y(t) = k_1 \mathrm{e}^{p_1 t} + k_2 \mathrm{e}^{p_2 t} + k_3 \mathrm{e}^{p_3 t}$$

with:

$$\begin{aligned} k_1 &= sY(s)|_{s=0} = \frac{1}{6} \\ k_2 &= (s+2)Y(s)|_{s=-2} = \frac{1}{2} \\ k_3 &= (s+3)Y(s)|_{s=-3} = -\frac{2}{3} \end{aligned}$$

Plugging the calculated values in the solution gives the final solution:

$$y(t) = \frac{1}{6} + \frac{1}{2}\mathrm{e}^{-2t} - \frac{2}{3}\mathrm{e}^{-3t}$$

Checking the solution by substitution it in the differential equation (left to the reader) gives the value 1, which is correct. Checking the initial conditions$y(0) = 0$ and $y'(0) = 1$:

$$y(t)|_{t=0} = \left(\frac{1}{6} + \frac{1}{2}\mathrm{e}^{-2t} - \frac{2}{3}\mathrm{e}^{-3t}\right)\Big|_{t=0} = 0$$
$$y'(t)|_{t=0} = \left(-\mathrm{e}^{-2t} + 2\mathrm{e}^{-3t}\right)\Big|_{t=0} = 1$$

which is also correct. □

Other methods of checking the results are based on the calculation of limits in both domains and relate them to each other. In order to apply those methods we need a further look in the properties of Laplace Transforms.

3.9 Properties of the Laplace transformation

The usefulness of the whole Laplace machinery relies on the fact that we can easily go back and forth between $f(t)$ and $F(s)$. However good our transform tables are, they cannot account for every combination of functions. Fortunately there are ways to manipulate known transforms to describe closely related functions, an example of which is the differentiation property that we employed above. Table 3.1 lists the most important of these properties as well as where each will be demonstrated.

Table 3.1: Some properties of the Laplace transform and where the demonstration of the property can be found.

#	**Property**	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	**Demo**
1	Linearity	$\alpha f_1(t) + \beta f_2(t)$	$\alpha F_1(s) + \beta F_2(s)$	E
2	Differentiation	$\frac{\mathrm{d}f(t)}{\mathrm{d}t}$	$sF(s) - f(0)$	R
3		$\frac{\mathrm{d}^n f(t)}{\mathrm{d}t^n}$	$s^n F(s) - \sum_{i=0}^{n-1} s^{n-i-1} \left.\frac{\mathrm{d}^{(i)} f(t)}{\mathrm{d}t^{(i)}}\right\|_{t=0^-}$	R
4	Integration	$\int_0^t f(t')\mathrm{d}t'$	$\frac{F(s)}{s}$	R
5	Time shift	$u(t-t_0)f(t-t_0)$	$\mathrm{e}^{-t_0 s}F(s)$	E
6	Frequency shift	$\mathrm{e}^{-at}f(t)$	$F(s+a)$	E
7	Initial value T	$f(0) = \lim_{s\to\infty} sF(s)$		R
8	Final value T	$f(\infty) = \lim_{s\to 0} sF(s)$		E
9	Convolution	$\mathcal{L}\{g(t) \otimes f(t)\}$	$G(s)F(s)$	R
10		$g(t)f(t)$	$\mathcal{L}^{-1}\{G(s) \otimes F(s)\}$	E

prop.2. See above in section 3.6

prop.3. See above in section 3.6

prop.4. By definition,

$$\mathcal{L}\left\{\int_0^t f(t')\mathrm{d}t'\right\} = \int_{0^-}^{\infty} \mathrm{e}^{-st}\left[\int_0^t f(t')\mathrm{d}t'\right]\mathrm{d}t$$

We can integrate this expression by parts to obtain

$$\begin{aligned}\mathcal{L}\left\{\int_0^t f(t')\mathrm{d}t'\right\} &= \left[\frac{\mathrm{e}^{-st}}{-s}\int_0^t f(t')\mathrm{d}t'\right]_0^{\infty} - \int_0^{\infty}\frac{\mathrm{e}^{-st}}{-s}f(t)\mathrm{d}t \\ &= 0 - 0 + \frac{1}{s}\int_0^{\infty}\mathrm{e}^{-st}\,f(t)\mathrm{d}t \\ &= \frac{F(s)}{s}.\end{aligned}$$

prop.7. Here we take the reverse approach:

$$\begin{aligned}\lim_{s\to\infty} sF(s) &= \lim_{s\to\infty}\left[sF(s) - f(0)\right] + f(0) \\ &= \lim_{s\to\infty}\mathcal{L}\left\{\frac{\mathrm{d}f}{\mathrm{d}t}\right\} + f(0) \\ &= \lim_{s\to\infty}\int_0^{\infty}\frac{\mathrm{d}f}{\mathrm{d}t}e^{-st}\mathrm{d}t + f(0) \\ &= 0 + f(0) \\ &= f(0).\end{aligned}$$

prop.9. In order to prove property 9, we use the time shift property (5). Since the functions $f(t)$ and $g(t)$ appear time-shifted in the convolution integral, we start with two Laplace transformed functions with different times in the transform: τ_1 and τ_2 that are related by:

$$t = \tau_1 + \tau_2$$

We get:

$$\begin{aligned}F(s) &= \int_0^{\infty}\mathrm{e}^{-s\tau_1}f(\tau_1)\mathrm{d}\tau_1 \\ G(s) &= \int_0^{\infty}\mathrm{e}^{-s\tau_2)}g(\tau_2)\mathrm{d}\tau_2 \\ &= \int_{\tau_1}^{\infty}\mathrm{e}^{-s(t-\tau_1}g(t-\tau_1)\mathrm{d}t \qquad \tau_1 \text{ is constant here, so: } \mathrm{d}\tau_2 = \mathrm{d}t \\ &= \mathrm{e}^{s\tau_1}\int_{\tau_1}^{\infty}\mathrm{e}^{-st}g(t-\tau_1)\mathrm{d}t\end{aligned}$$

The product of the two functions becomes:

$$\begin{aligned}F(s)G(s) &= \int_0^{\infty}\mathrm{e}^{-s\tau_1}f(\tau_1)\mathrm{e}^{s\tau_1}\int_{\tau_1}^{\infty}\mathrm{e}^{-st}g(t-\tau_1)\mathrm{d}t\mathrm{d}\tau_1 \\ &= \int_0^{\infty}f(\tau_1)\int_{\tau_1}^{\infty}\mathrm{e}^{-st}g(t-\tau_1)\mathrm{d}t\mathrm{d}\tau_1\end{aligned}$$

Assuming decent functions allows us to interchange the integrals. We have to take care of the integration limits though. In case we integrate first (inner integral) τ_1 over the interval $[0,\infty]$,

we also have to integrate t over the interval $[0, \infty]$, since we initially started the integration at τ_1 that can take the value zero now. So:

$$\begin{aligned} F(s)G(s) &= \int_0^\infty \mathrm{e}^{-st} \int_0^\infty f(\tau_1) g(t-\tau_1) \mathrm{d}\tau_1 \mathrm{d}t \\ &= \int_0^\infty \mathrm{e}^{-st} \left(g(t) \otimes f(t)\right) \mathrm{d}t \\ &= \mathcal{L}\left\{g(t) \otimes f(t)\right\} \end{aligned}$$

In order to check the limits for example 3.8.2 we get:

$$\begin{aligned} \lim_{t\to 0} y(t) = 0 \Leftrightarrow \lim_{s\to\infty} sY(s) = \lim_{s\to\infty} \frac{s+1}{s^2+5s+6} = 0 \\ \lim_{t\to\infty} y(t) = \frac{1}{6} \Leftrightarrow \lim_{s\to 0} sY(s) = \lim_{s\to 0} \frac{s+1}{s^2+5s+6} = \frac{1}{6} \end{aligned}$$

which shows a a good confirmation of the calculated values.

3.10 Solving coupled first-order differential equations

Due to the nature of the construction of models in the physics part of this course, which is strongly directed to the construction of a set of coupled first-order differential equations, we will mainly have to solve this type of sets of differential equations. Let us look at an example.

Example 3.10.1. *Solve a set of coupled ODE's with Laplace*
Suppose we want to solve the following initial value problem:

$$\begin{aligned} y'(t) + z(t) &= 0 \\ 3y(t) - z'(t) &= 6 \\ y(0) &= 0 \\ z(0) &= 0. \end{aligned}$$

We first take the Laplace transform of both sides of the equations:

$$sY(s) - y(0) + Z(s) = 0 \tag{3.10}$$
$$3Y(s) - sZ(s) + z(0) = 6/s. \tag{3.11}$$

Plugging in $y(0) = z(0) = 0$ and substituting the first equation into the second to eliminate $Z(s)$ then yields an algebraic equation in $Y(s)$ only:

$$3Y(s) + s^2 Y(s) = \frac{6}{s}. \tag{3.12}$$

Isolating $Y(s)$ then gives

$$Y(s) = \frac{6}{s(s^2+3)} = \frac{6}{s(s-\mathrm{j}\sqrt{3})(s+\mathrm{j}\sqrt{3})} \tag{3.13}$$

This function has three poles: $p_1 = 0$, $p_{2,3} = \pm\mathrm{j}\sqrt{3}$, and no zeros. Rewriting in the usual form applying partial fraction decomposition:

$$Y(s) = \frac{k_1}{s} + \frac{k_2}{s-\mathrm{j}\sqrt{3}} + \frac{k_3}{s+\mathrm{j}\sqrt{3}}$$

and computing the parameters:

$$\begin{aligned} k_1 &= sY(s)|_{s=0} = \frac{6}{(s-\mathrm{j}\sqrt{3})(s+\mathrm{j}\sqrt{3})} = 2 \\ k_2 &= (s-\mathrm{j}\sqrt{3})Y(s)\Big|_{s=\mathrm{j}\sqrt{3}} = \frac{6}{\mathrm{j}\sqrt{3}(\mathrm{j}\sqrt{3}+\mathrm{j}\sqrt{3})} = -1 \\ k_3 &= (s+\mathrm{j}\sqrt{3})\ Y(s)|_{s=-\mathrm{j}\sqrt{3}} = \frac{6}{-\mathrm{j}\sqrt{3}(-\mathrm{j}\sqrt{3}+-\mathrm{j}\sqrt{3})} = -1 \end{aligned}$$

gives us the form from which we can easily take the inverse Laplace transform to obtain the solution in the time domain:

$$y(t) = 2 - \mathrm{e}^{\mathrm{j}\sqrt{3}t} - \mathrm{e}^{-\mathrm{j}\sqrt{3}t} = 2 - 2\cos\sqrt{3}t.$$

The corresponding solution for $z(t)$ can be obtained from the differential equation directly:

$$z(t) = -y'(t) = -2\sqrt{3}\sin\sqrt{3}t.$$

Alternatively we could have eliminated $Y(s)$ of the equations, solve for $Z(s)$, inversely Laplace Transform the result to obtain $z(t)$ and use the differential equation to calculate $y(t)$. This should have given the same result. □

We can extend this example a bit further to apply Linear Algebra techniques and to prepare for a technique later in this course (reader part I): *State-Space* formalism.
Note that the two equations 3.10 and 3.11 that result form the Laplace Transform of the set of first-order differential equations – being a set of linear algebraic equations – can also be written in matrix form:

$$\begin{pmatrix} s & 1 \\ 3 & -s \end{pmatrix} \begin{pmatrix} Y(s) \\ Z(s) \end{pmatrix} = \begin{pmatrix} y(0) \\ \frac{6}{s} - z(0) \end{pmatrix} \tag{3.14}$$

This equation can be solved for $Y(s)$ and $Z(s)$ using techniques form Linear Algebra:

$$\begin{pmatrix} Y(s) \\ Z(s) \end{pmatrix} = \begin{pmatrix} s & 1 \\ 3 & -s \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ \frac{6}{s} \end{pmatrix} = \begin{pmatrix} \frac{s}{s^2+3} & \frac{1}{s^2+3} \\ \frac{3}{s^2+3} & -\frac{s}{s^2+3} \end{pmatrix} \begin{pmatrix} 0 \\ \frac{6}{s} \end{pmatrix} = \begin{pmatrix} \frac{6}{s(s^2+3)} \\ -\frac{6}{s^2+3} \end{pmatrix} \tag{3.15}$$

Where the initial conditions for $y(0)$ and $z(0)$ are substituted. The result is in agreement with equation 3.13. In the intermediate steps, the Laplace transform of the input signal is clearly recognized. To finalize the solutions to the differential equations, we have to taken the inverse Laplace transforms:

$$\begin{aligned} Y(s) &= \frac{6}{s\,(s^2+3)} && \Rightarrow y(t) = 2 - 2\cos\left(\sqrt{3}t\right) = 4\sin^2\left(\frac{\sqrt{3}t}{2}\right) \\ Z(s) &= -\frac{6}{s^2+3} && \Rightarrow z(t) = -2\sqrt{3}\sin\left(\sqrt{3}t\right) \end{aligned}$$

Although equation 3.15 is a specific equation as part of the solution to an example, the structure can be revealed from this. On the left side of the equal sign is a vector with the Laplace transforms of all variables in the problem; the dimension is equal to the order of the system. On the right side of the equal sign is inverse of the matrix containing the dynamics of the system by means of (powers of[1]) complex frequencies representing derivatives with respect to time. Inversion of the matrix contains the determinant of that matrix in the denominator; here this determinant has is expressed in s:

$$\mathrm{Det}\begin{pmatrix} s & 1 \\ 3 & -s \end{pmatrix} = -s^2 - 3$$

[1]Due to the fact that we treat only sets of coupled first-order differential equations, the highest power of s is equal to one and the lowest equal to zero.

as can be recognized in the calculated expression. The expression is a polynomial in s. The roots of this polynomial are also the poles $p_{2,3}$ of the output function $Y(s)$ in the Laplace domain. The extra pole that does not appear here $p_1 = 0$ comes directly from the input function $X(s)$ in the Laplace domain.
The vector on which the inverted matrix works, contains Laplace Transforms of input functions as well as initial conditions: these are the initiators of the dynamic behaviour of the system. It is obvious that if there are no inputs and all initial conditions are zero, this vector only contains zero's and the system remains in rest since the Laplace Transforms of all variables are zero. In generalized notation one could formulate the above as:

$$Y_i(s) = \sum_j \left\{ \mathbf{A}^{-1} \right\}_{ij} (X_j(s) + y_j(0)) \tag{3.16}$$

A few remarks can be made on the inverse of the matrix:

- The dimension of the matrix equals the order of the set of firs-order differential equations.
- Since the inverse of a matrix has the determinant in the denominator, all denominator polynomials in s are the same, unless one of the poles (roots of the denominator polynomial) is compensated by a zero (root of the numerator polynomial).
- The determinant, set to zero, is equal to the characteristic equation of any input-output equation derived from the set of equations.

The main result of this chapter is contained in equation 3.16, where the system dynamics from the homogeneous part of the differential equation is contained in the inverse of matrix $\mathbf{A}$, applied on the Laplace transform of the input function and the initial conditions. Compare this result with equation 3.2 in the beginning of this chapter: equation 3.16 reduces to equation 3.2, if we take a single input $x_j(t)$ and a single output $y_i(t)$. The corresponding matrix element $\{\mathbf{A}^{-1}\}_{ij}$ is the function $F(s)$.
In chapter 6 on State-Space models, we will obtain similar results.

Summaries

Summary Chapter 3a

Definitions:

- Definition of convolution of two functions $f(t)$ and $g(t)$:

$$f(t) \otimes g(t) \equiv \int_0^\infty f(t-\tau)g(\tau)\mathrm{d}\tau \tag{3.17}$$

- Definition of the Laplace Transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty \mathrm{e}^{-st} f(t)\mathrm{d}t \tag{3.18}$$

- Definition of the inverse Laplace Transform:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi\mathrm{i}} \oint_{\gamma-\mathrm{i}\infty}^{\gamma+\mathrm{i}\infty} \mathrm{e}^{st} F(s)\mathrm{d}s \tag{3.19}$$

- Inverse Laplace Transform using partial fraction decomposition:

$$X(s) = \frac{Q(s)}{P(s)} = \frac{b_m s^m + \ldots b_1 s + b_0}{a_n s^n + \ldots a_1 s + a_0} \qquad (m \leq n) \tag{3.20}$$

$$= \left(\sum_{i=1}^{n-l} \frac{r_i}{s-p_i}\right) + \frac{r_{n-l+1}}{s-p_0} + \ldots + \frac{r_n}{(s-p_0)^l} + K \text{ (for } l \text{ times } p_0) \tag{3.21}$$

$$r_i = (s-p_i)\left.\frac{Q(s)}{P(s)}\right|_{s=p_i \neq p_0}, \; i = 1, 2, \ldots, n-l \tag{3.22}$$

$$r_{n-k} = \frac{1}{k!}\frac{\mathrm{d}^k}{\mathrm{d}s^k}\left\{(s-p_0)^l \frac{Q(s)}{P(s)}\right\}\Big|_{s=p_0}, \; k = 0, 1, 2, \ldots, l-1 \tag{3.23}$$

$$x(t) = \left\{\sum_{i=1}^{n-l} r_i \mathrm{e}^{p_i t} + r_{n-l+1}\mathrm{e}^{p_0 t} + r_{n-l+2} t\mathrm{e}^{p_0 t} + \ldots + \frac{r_n}{(l-1)!} t^{l-1}\mathrm{e}^{p_0 t}\right\} u_s(t) + K\delta(t) \tag{3.24}$$

Summary Chapter 3b

Properties of Laplace Transforms

#	**Property**	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1	Linearity	$\alpha f_1(t) + \beta f_2(t)$	$\alpha F_1(s) + \beta F_2(s)$
2	Differentiation	$\dfrac{\mathrm{d}f(t)}{\mathrm{d}t}$	$sF(s) - f(0^-)$
3		$\dfrac{\mathrm{d}^n f(t)}{\mathrm{d}t^n}$	$s^n F(s) - \sum_{i=0}^{n-1} s^{n-i-1} \left.\dfrac{\mathrm{d}^{(i)} f(t)}{\mathrm{d}t^{(i)}}\right\vert_{t=0^-}$
4	Integration	$\int_0^t f(t')\mathrm{d}t'$	$\dfrac{F(s)}{s}$
5	Time shift	$u(t-t_0)f(t-t_0)$	$\mathrm{e}^{-t_0 s}F(s)$
6	Frequency shift	$\mathrm{e}^{-at}f(t)$	$F(s+a)$
7	Initial value T	$f(0) = \lim_{s\to\infty} sF(s)$	
8	Final value T	$f(\infty) = \lim_{s\to 0} sF(s)$	
9	Convolution	$g(t) \otimes f(t)$	$G(s)F(s)$
10		$g(t)f(t)$	$G(s) \otimes F(s)$

Summary Chapter 3c

Laplace Transforms

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t - t_0)$	e^{-st_0}
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\frac{1}{(n-1)!}t^{n-1}\mathrm{e}^{at}$	$\frac{1}{(s-a)^n}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$
$\mathrm{e}^{-\gamma t}\cos\omega t$	$\frac{s+\gamma}{(s+\gamma)^2+\omega^2}$
$\mathrm{e}^{-\gamma t}\sin\omega t$	$\frac{\omega}{(s+\gamma)^2+\omega^2}$

Problems

Problem 3.1. Laplace transforms
Validate the following Laplace transform pairs:

a. $\mathcal{L}\{\cos\omega t\} = \frac{s}{s^2+\omega^2}$

b. $\mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\} = \sin\omega t$

c. $\mathcal{L}\{\mathrm{e}^{-\gamma t}\cos\omega t\} = \frac{s+\gamma}{(s+\gamma)^2+\omega^2}$

d. $\mathcal{L}^{-1}\{\mathrm{e}^{-st_0}\} = \delta(t-t_0)\ \ (t_0 > 0)$

Problem 3.2. Initial value problems Solve the following equations for the given boundary conditions. You may need to compute new Laplace transforms along the way. In each case, substitute your answer back in to validate it.

a. $\frac{\mathrm{d}y}{\mathrm{d}t} - 3y = 0;\ y(0) = 1$

b. $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\frac{\mathrm{d}y}{\mathrm{d}t} + 6y = 0;\ y(0) = 2,\ y'(0) = 2$

c. $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 3\frac{\mathrm{d}y}{\mathrm{d}t} + 2y = 2;\ y(0) = 0,\ y'(0) = 0$

Problem 3.3. Properties of Laplace transforms
Demonstrate properties 1, 4, 5, 6 and 8 in table 3.1.

Problem 3.4. Coupled differential equations

Solve the following initial value problem,

$$\begin{aligned} y(t) + z'(t) - 3z(t) &= 0 \\ y''(t) &= 0 \end{aligned}$$

subject to the initial value conditions

a.

$$\begin{aligned} y(0) &= 0 \\ z(0) &= 0 \\ y'(0) &= 1 \end{aligned}$$

b.

$$\begin{aligned} y(0) &= 0 \\ z(0) &= \pi \\ y'(0) &= 1 \end{aligned}$$

Problem 3.5. Some more practice
Compute the inverse Laplace transform of the following expressions. You may need the properties.

a. $F(s) = \frac{10}{s^3+4s^2+9s+10} = \frac{10}{(s+2)(s^2+2s+5)}$

b. $F(s) = \frac{1}{(s+1)^4} + \frac{s-3}{(s-3)^2+6}$

c. $F(s) = \frac{\mathrm{e}^{-s}}{s}$

Problem 3.6. Dirac weirdness I: first-order system
Compute the solution to the following initial value problem for $x(t) = 0$, $x(t) = u(t)$ and $x(t) = \delta(t)$. Check whether your solutions satisfies the initial condition and explain any apparent discrepancy that you observe for the case $x(t) = \delta(t)$.

$$\begin{aligned} y'(t) + 3y(t) &= x(t) \\ y(0) &= 4 \end{aligned}$$

Problem 3.7. Dirac weirdness II: second-order system
Compute the solution to the following initial value problem for $x(t) = 0$ and $x(t) = \delta(t)$. Check whether your solution satisfies the initial condition and explain any apparent discrepancy that you observe for the case $x(t) = \delta(t)$.

$$\begin{aligned} y''(t) + 5y'(t) + 6y(t) &= x'(t) + 2x(t) \\ y(0) &= 4 \\ y'(0) &= 0. \end{aligned}$$

Problem 3.8. Double poles and unit step
Given the third order differential equation given by:

$$y^{(3)}(t) + 4y^{(2)}(t) + 5y^{(1)}(t) + 2y(t) = x(t)$$

with boundary conditions:

$$\begin{aligned} y^{(2)}(0) &= 1 \\ y^{(1)}(0) &= 1 \\ y(0) &= 0 \end{aligned}$$

The three solutions to the characteristic equation of the differential equation are: $s_i = \{-1, -1, -2\}$. Determine the unit-step response of this system, *i.e.* $x(t) = u(t)$, given the initial conditions.

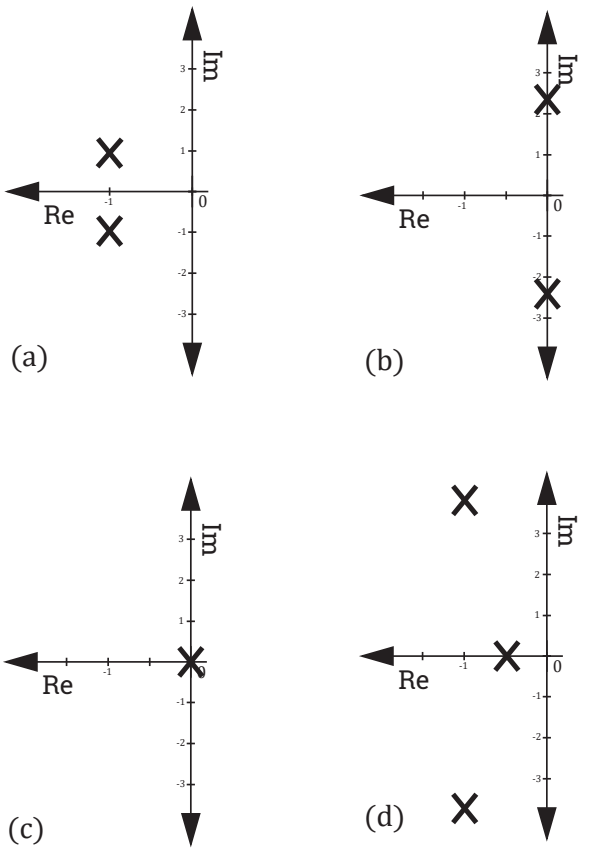


Figure 3.5: Four pole plots

Problem 3.9. Step response & Impulse response
In figure 3.5 the pole locations of four systems are given.
Each plot corresponds with a system. From the poles, given in the pole plots, the System can be written in the Laplace notations, with $Y(s)$ as the output representation and $X(s)$ as the input representation. Note that $X(s)$ is either the representation of a step function or that of an impulse function (Get these from the table).
Perform the following actions for the first three systems as represented by their poles (drop the last one (d); this is too much work):

a. Write down the output signal $Y(s)$ of each of the systems in the Laplace notation.

b. Apply partial fraction decomposition to rewrite the output signal $Y(s)$ in fractions.

c. Construct the output signal $y(t)$ in the time domain from these calculated fractions.

d. For each of the systems, make a sketch of the two output signals (step response and impulse response) in a single graph.

Problem 3.10. Step-response from poles of system
Verification of the responses in table 2.1 and 2.2
For the four items (a), (c), (d) and (f) in the tables, perform the following actions:

a. From the poles write down the differential equation.

b. Use Laplace Transform to find the step response of the system.

c. Verify your results with the graphs in table 2.2

d. Plot the step response of the system

e. Verify your results with the graphs in table 2.1

Problem 3.11. LRC-circuit permutations via Laplace
Consider the permutation of the LRC-circuits from problem 2.4.

a. Give the set of first-order differential equations for all of these three systems (Get them from your previous solution).

b. Transfer the set of first-order differential equations to the Laplace domain.

c. For all three systems give the solutions in the Laplace domain $Y(s)$ for both a unit-step input and a unit-impulse input.

d. Evaluate the time response of these three systems when the voltage is raised from 0 to $1\,\mathrm{V}$ at $t = 0\,\mathrm{s}$ using the Laplace technique.

e. Evaluate the time response of these three systems when an impulse with height $2\ \mathrm{V}$ at $t = 0\,\mathrm{s}$ is applied using the Laplace technique.

Problem 3.12. Solving an initial value problem *Question in test 2019*
We want to solve the following initial value problem,

$$y'' + y = x,$$

with $y(0) = 1$ and $y'(0) = 0$. The input function is $x(t) = \exp(-t)u(t)$.

a. (8 points). Determine $X(s) = \mathcal{L}\{x(t)\}$.

b. (10 points). Derive the Laplace transform of the left-hand side of the differential equation, and give the expression for $Y(s) = \mathcal{L}\{y(t)\}$ in terms of $X(s)$ and s.

c. (12 points). Perform a partial fraction expansion for $Y(s)$, and determine the inverse Laplace transform to get $y(t)$.

Problem 3.13. Property 10 from the Laplace transform
Prove property 10, the convolution theorem from the Laplace Transform property table 3.1.

Chapter 4

Transfer functions

4.1 Introducing the transfer function

A transfer function describes the transfer of an input signal to an output signal of a system in the complex frequency domain. For a single-input single-output system (SISO) there is one function, derived from the input-output differential equation. In a multi-input multi-output system (MIMO), that is most likely represented by a set of first-order differential equations, there are multiple transfer functions, combining every input-output combination. For a system with m inputs and n outputs, the number of transfer functions is the product of m and n.

4.1.1 SISO transfer function

For a SISO transfer function of an LTI system we start with a general input-output differential equation as given in equation 1.2:

$$\begin{aligned} a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \ldots + a_1 y^{(1)}(t) + a_0 y(t) \\ = b_m x^{(m)}(t) + b_{m-1} x^{(m-1)}(t) + \ldots + b_1 x^{(1)}(t) + b_0 x(t). \end{aligned} \tag{4.1}$$

Here $y^{(k)}$ is the kth derivative of y, as usual.
From this input-output equation we now derive the so-called transfer function by applying a Laplace Transform. Here we are solely interested in the response of the system to an input signal to produce an output. In particular, we are not interested in the start-up behaviour that is determined by the initial conditions. We therefore apply a Laplace transform on the differential equation assuming all initial conditions ($t = 0$) to be equal to zero.
The Laplace transform is then:

$$a_n s^n Y(s) + \ldots + a_1 s Y(s) + a_0 Y(s) = b_m s^m X(s) + \ldots + b_1 s X(s) + b_0 X(s), \tag{4.2}$$

with $X(s)$ and $Y(s)$ the Laplace transforms of $x(t)$ and $y(t)$, respectively. Although the integers m and n are in principle independent, physically valid systems require $m \leq n$ in order to have a causal description, that is, one in which the output at a certain time is not dependent on an input at a later time; we can not use information from the future[1]. The system transfer function is defined as the ratio of the Laplace transforms of the output and input in the Laplace domain. This gives:

$$T(s) \equiv \frac{Y(s)}{X(s)} = \frac{b_m s^m + \ldots + b_1 s + b_0}{a_n s^n + \ldots + a_1 s + a_0} \tag{4.3}$$

For a LTI-system this is the ratio of two polynomials in s with $m+1$terms in de numerator and $n+1$ terms in the denominator. For each value of s the transfer function is a complex number with a real

[1] Mathematically there is no objection.

and imaginary part, or equivalently a magnitude $|T(s)|$ and a phase $\angle T(s)$. These latter components can be readily interpreted as the gain (often called attenuation if the gain is smaller than 1) and the change in phase of a sinusoid input signal at the frequency ω with $s = j\omega$.
The transfer function can be graphically represented by a block diagram as depicted in figure 4.1.

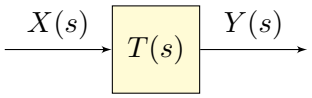


Figure 4.1: Transfer function relating input and output.

The use of block diagrams allows to characterize parts of a system and build the complete system of connecting various subsystem blocks. As we will see this highly simplifies the evaluation of the transfer function of a complete system. In section 4.3 we will construct a system from sub-systems with each having their own transfer function by means of block-diagram algebra.

4.1.2 MIMO transferfunctions

In a MIMO system consisting of a set of n first-order differential equations we take also a number of n inputs in order to have a general form, with the exception that every differential has one single input (In the most general case, all linear combinations of the input functions could contribute to every first-order differential equation.). In this case the set of equations is:

$$\begin{aligned}
y_1'(t) + \sum_{i=1}^{n} a_{1i} y_i(t) &= \sum_{j=1}^{m} b_{1j} X_j(s) \\
y_2'(t) + \sum_{i=1}^{n} a_{2i} y_i(t) &= \sum_{j=1}^{m} b_{2j} X_j(s) \\
&\vdots \\
y_n'(t) + \sum_{i=1}^{n} a_{ni} y_i(t) &= \sum_{j=1}^{m} b_{nj} X_j(s)
\end{aligned} \tag{4.4}$$

Apply the Laplace Transform on this set of equations and setting all initial conditions to zero we obtain:

$$\begin{aligned}
sY_1(s) + \sum_{i=1}^{n} a_{1i} Y_i(s) &= \sum_{j=1}^{m} b_{1j} X_j(s) \\
sY_2(s) + \sum_{i=1}^{n} a_{2i} Y_i(s) &= \sum_{j=1}^{m} b_{2j} X_j(s) \\
&\vdots \\
sY_n(s) + \sum_{i=1}^{n} a_{ni} Y_i(s) &= \sum_{j=1}^{m} b_{nj} X_j(s)
\end{aligned}$$

Since this is a linear set of equations that have to be solved, we can rewrite this in matrix notation:

$$\begin{pmatrix} s+a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & s+a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & s+a_{nn} \end{pmatrix} \begin{pmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_n(s) \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \\ \vdots \\ X_m(s) \end{pmatrix} \tag{4.5}$$

This matrix equation can be solved for $Y_i(s)$:

$$\begin{pmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_n(s) \end{pmatrix} = \begin{pmatrix} s+a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & s+a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & s+a_{nn} \end{pmatrix}^{-1} \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \\ \vdots \\ X_m(s) \end{pmatrix} \tag{4.6}$$

or more simple in matrix notation:

$$\vec{Y}(s) = (s\mathbb{I} + \mathbb{A})^{-1}\,\mathbb{B}\vec{X}(s)$$

Note: The dimensions of the matrices $\mathbb{A}$, $\mathbb{I}$ and $\mathbb{B}$ are respectively $n \times n$, $n \times n$ and $n \times m$. The number of inputs (m) is not necessary equal to the number of outputs (n).
From this equation we can obtain expressions for all $Y_i(s)$ in terms of the coefficients and all inputs:

$$Y_i(s) = \sum_{j=1}^{n} \left\{ (\mathbf{A} + s\mathbf{I})^{-1} \right\}_{ij} \sum_{k=1}^{m} b_{jk} X_k(s) = \sum_{k=1}^{m} \sum_{j=1}^{n} \left\{ (\mathbf{A} + s\mathbf{I})^{-1} \right\}_{ij} b_{jk} X_k(s) \tag{4.7}$$

This is a nice result, since We can relate an arbitrary output $Y_i(s)$ to and arbitrary $X_j(s)$ to get the transfer function relating input to the output; the transfer functions are the elements of the matrix $\mathbb{T}(s)$ given by:

$$\mathbb{T}(s) = (s\mathbb{I} + \mathbb{A})^{-1}\,\mathbb{B} \tag{4.8}$$

From this equation we see that the elements of the inverse matrix are by definition the transfer functions, *i.e.* by calculating the inverse matrix we obtain a matrix of transfer functions of every combination between an input and an output.
Taking the inverse of the matrix results in a reshuffling of the elements, calculation of minors, here and there multiplying with -1, AND a division of all element by the determinant of the matrix, which is a polynomial in s of the order n. This polynomial determines the dynamics of the system. This denominator polynomial is also the characteristic equation of the input-output differential equation. Both SISO systems as well as MIMO systems result in transfer functions of the kind as presented in equation 4.3: consisting of the the quotient of two polynomials in the complex frequency s. Therefore we need to dive a bit deeper in the properties of these transfer functions.

Example 4.1.1. *MIMO system in matrix notation.*
We consider the set of first-order differential equations given by:

$$y_1'(t) = -4y_1(t) - 2y_2(t) + x_1(t)$$
$$y_2'(t) = -y_1(t) - 3y_2(t) + 2x_2(t)$$

Apply Laplace, with initial conditions set to zero we get:

$$sY_1(s) + 4Y_1(s) + 2Y_2(s) = X_1(s)$$
$$sY_2(s) + Y_1(s) + 3Y_2(s) = 2X_2(s)$$

Which becomes in matrix notation:

$$\begin{pmatrix} s+4 & 2 \\ 1 & s+3 \end{pmatrix} \begin{pmatrix} Y_1(s) \\ Y_2(s) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \end{pmatrix} \tag{4.9}$$

From which we can solve for the Laplace transform of the output:

$$\begin{aligned}\begin{pmatrix} Y_1(s) \\ Y_2(s) \end{pmatrix} &= \begin{pmatrix} s+4 & 2 \\ 1 & s+3 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \end{pmatrix} \\ &= \begin{pmatrix} \frac{s+3}{s^2+7s+10} & -\frac{4}{s^2+7s+10} \\ -\frac{1}{s^2+7s+10} & \frac{2s+8}{s^2+7s+10} \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \end{pmatrix} \end{aligned} \tag{4.10}$$

For completeness we write down the determinant equation, that is equal to the characteristic equation for calculating the poles of the system:

$$s^2 + 7s + 10 = 0$$

with the roots: $s_1 = -2$ and $s_2 = -5$. Note that two of the four transfer functions also have a zero: $z_1 = -3$ and $z_2 = -4$. □

4.2 Properties of System Transfer Function

4.2.1 Limiting behaviour

In order to work with transfer functions we first take a closer look at the limits. The fist limit to consider is to let s go to zero in equation 4.3. According to table 3.1 this is the Final value:

$$\lim_{s\to 0} T(s) = \lim_{s\to 0} \frac{b_m s^m + \ldots + b_1 s + b_0}{a_n s^n + \ldots + a_1 s + a_0} = \frac{b_0}{a_0} \tag{4.11}$$

under the condition that $a_0 \neq 0$. The term Final value indicates that this is the value the system will achieve after infinite time. This number is sometimes referred to as the static gain, the amplification of a signal with frequency zero, i.e. a DC signal or the result of a step response at large time.
In case the transfer function has $a_0 = 0$, there are two options:

$b_0 = 0$ In this situation the limit depends on the lowest power in s of the numerator polynomial with respect to the lowest power in s of the denominator polynomial, e.g. if both $b_1 and a_1$ not equal zero, the limit will become $\frac{b_1}{a_1}$.

$b_0 \neq 0$ In this situation the mathematical limit is not finite. The actual system will go to its limits not incorporated in the mathematical expression.

Similarly we have the high complex frequency limit, also known as the initial value:

$$\lim_{s\to\infty} T(s) = \begin{cases} 0 & n > m \\ \frac{b_m}{a_n} & n = m \end{cases} \tag{4.12}$$

We do not have to consider the condition $n < m$, since that would correspond to a non-causal system. Table 3.1 links the initial value to the value at time 0.

4.2.2 Zeros and Poles

Since the transfer function of a system can be written as a quotient of two polynomials, the characteristics of a transfer function is derived from the characteristics of these polynomials. So let us define a general transfer function as the ratio of two polynomials:

$$T(s) = \frac{B(s)}{A(s)} \tag{4.13}$$

where $B(s)$ is a polynomial of order m with coefficients b_i and $A(s)$ is a polynomial of order n with coefficients a_i:

$$A(s) = a_n s^n + \ldots + a_1 s + a_0, \tag{4.14}$$

$$B(s) = b_m s^m + \ldots + b_1 s + b_0. \tag{4.15}$$

We still keep the condition for a causal system: $m \leq n$.
Since we have a numerator polynomial $B(s)$ and a denominator polynomial $A(s)$ the effect on the characteristics of the transfer function is different. Most prominent effects are expected for situations where either the numerator is equal to zero, or the denominator is equal to zero. Therefore we factorize both polynomials. This is equivalent to finding the roots of both polynomials.
Knowing the roots, we can rewrite the polynomial $B(s)$ in the form:

$$B(s) = b_m(s - z_1)(s - z_2)\ldots(s - z_m), \tag{4.16}$$

where $z_1, z_2, \ldots, z_m$ are the zeros of the transfer function. Note that, for each of the zeros, the transfer function becomes zero, unless the roots of the polynomial $B(s)$ corresponds with roots from the denominator polynomial $A(s)$:

$$B(s)|_{s=z_i} = 0 \; \forall \; i = 1, 2, \ldots, m. \tag{4.17}$$

The roots of the polynomial in the denominator of the transfer function, $A(s)$, are called the poles of the system transfer function. The transfer function of an nth-order system has n poles, which can also be real or complex numbers, and the polynomial $A(s)$ can similarly be written as:

$$A(s) = a_n(s - p_1)(s - p_2)\ldots(s - p_n), \tag{4.18}$$

where the poles are solutions of the equation

$$A(s)|_{s=p_i} = 0 \; \forall \; i = 1, 2, \ldots, n. \tag{4.19}$$

Since both the coefficients of the numerator polynomial $B(s)$ as well as the coefficients of the denominator polynomial $A(s)$ are real numbers, *i.e.* $\{a_i, b_j\} \in \mathbb{R}$, the roots are real-valued or pairs of complex-valued with their conjugate counter parts.
Note that the form of the denominator polynomial $A(s)$ is the same as the form of the characteristic equation of an nth-order linear system, equation 1.9. Therefore the system poles are the same as the roots of the system characteristic equation. It can thus be implied that the poles of the system transfer function provide complete information about the inherent dynamic characteristics of the system. Specifically, each real pole corresponds to a time constant and each pair of complex-conjugate poles corresponds to a natural frequency and a damping ratio of the system.
In conclusion, we write the transfer function its factorized form:

$$T(s) = \frac{b_m(s - z_1)(s - z_2)\ldots(s - z_m)}{a_n(s - p_1)(s - p_2)\ldots(s - p_n)} \tag{4.20}$$

Where z_i and p_i are the zero's and poles respectively of the input-output transfer function.
A few concluding remarks to this section:

- In general the denominator polynomial is the same for all transfer functions of all input-output combinations, so showing the same dynamics of the system; Time constants τ_i, natural frequencies $\omega_{n,i}$ and damping factors ζ_i are related to the poles p_i of the system.
- From equation 4.20 is is easily seen that poles can be compensated by zeros for every $p_i = z_j$. Which reduces the order of the system.
- Computers are very skilled in manipulating transferfunctions and writing them either as a coefficient of polynomials, eq. 4.3 or as the ratio of the product of zeros and poles, eq. 4.20. **Learn how to use a computer to manipulate transfer functions. See appendices D, E and F how to this.**

4.3 Transfer-function block-diagram algebra

The transfer functions is a very strong concept. It can connect various subsystems with each their own transfer function to construct a system with an overall transfer function. The output of each subsystem can be the input of a next subsystem, or outputs of various subsystems can be combined to create a new input signal. Also feedback is easily depicted with blocks representing transfer functions. This all starts with the already introduced pictorial transfer function in figure 4.1, which related the related the output $Y(s)$ in the (complex) frequency domain to the input $X(s)$ in the (complex) frequency domain through the transfer function $T(s)$.

$$Y(s) = T(s)X(s) \tag{4.21}$$

This is the building block for constructing more complicated networks of transfer functions.
The basis operations to combine transfer functions are addition, subtraction (variant of addition) and multiplication. The former two require specific addition and subtraction nodes, whereas multiplication appears from a sequential structure. One might consider a gain element ($Y(s) = GX(s)$, where G is a constant) as a separate block in a network. It can however also be incorporated into a more complex transfer function. In this document, we do not treat gain as a separate block.
Note that constructing transfer functions of simple blocks or more complicated network structures follows the same rules. You can combine transfer functions at any level.

Rule 1: Combining blocks in series (cascaded). For two blocks of transfer functions $T_1(s)$ and $T_2(s)$ in series, as shown in Fig. 4.2, the equivalent transfer function is

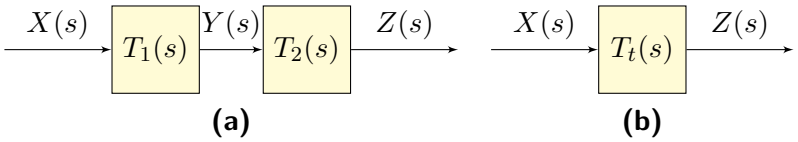


Figure 4.2: Rule 1: The sequence of two blocks (a), and the equivalent single-block representation (b).

$$T_t(s) = \frac{Z(s)}{X(s)} = \frac{Y(s)}{X(s)}\frac{Z(s)}{Y(s)} = T_1(s)T_2(s). \tag{4.22}$$

In general, the equivalent transfer function for n blocks in series is

$$T_{ser}(s) = \prod_{i=1}^{n} T_i(s) \tag{4.23}$$

That is, we only need to *multiply the magnitudes* and *add the phases* of the individual transfer functions to compute the overall transfer function.

Rule 2: Blocks in parallel.

$$T_t(s) = \frac{Z(s)}{X(s)} = \frac{Y_1(s) + Y_2(s)}{X(s)} = \frac{X(s)T_1(s) + X(s)T_2(s)}{X(s)} = T_1(s) + T_2(s) \tag{4.24}$$

In general, the equivalent transfer function for n blocks in parallel is

$$T_{par}(s) = \sum_{i=1}^{n} T_i(s) \tag{4.25}$$

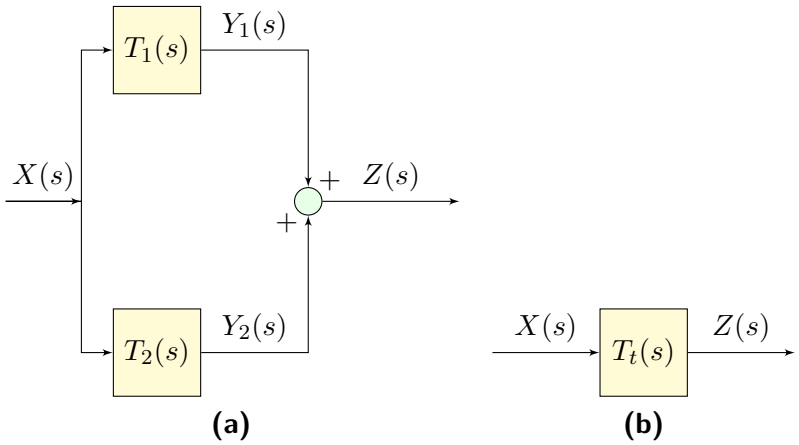


Figure 4.3: Rule 2: Two blocks in parallel (a) and the equivalent single-block representation (b).

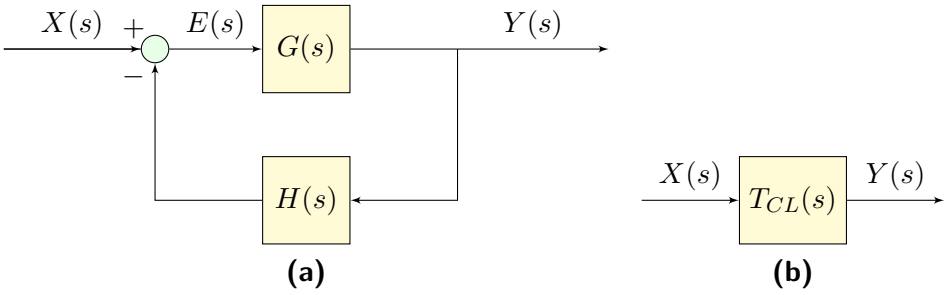


Figure 4.4: Rule 3: Feedback system (a) and the equivalent single-block representation (b).

Rule 3: Combining blocks in a feedback system. In a feedback system, the system output variable $Y(s)$ is transferred through a feedback transfer function $H(s)$ back to the input of the system where it is subtracted from the input signal, thus closing what is called a negative-feedback loop. Figure 4.4 shows a block diagram of a closed-loop feedback system with the summator having a positive and negative sign to indicate negative feedback. The output variable can be expressed as

$$\begin{aligned} Y(s) &= E(s)G(s) \\ &= [X(s) - H(s)Y(s)]G(s) \\ &= X(s)G(s) - H(s)Y(s)G(s). \end{aligned} \tag{4.26}$$

Note that in writing down this expression, we start from the back, *i.e.* $Y(s)$ and gradually work through the system to the front. Hence the equivalent transfer function of a closed-loop negative feedback system is

$$T_{CL}(s) = \frac{Y(s)}{X(s} = \frac{G(s)}{1 + G(s)H(s)} \tag{4.27}$$

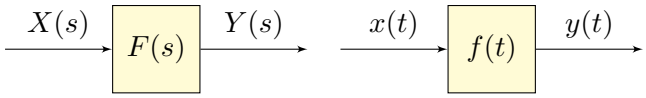
.

Of course also a positive feedback can be constructed. In that case the closed loop transfer function is given by:

$$T_{CL_{positive}}(s) = \frac{Y(s)}{X(s)} = \frac{G(s)}{1 - G(s)H(s)} \tag{4.28}$$

4.4 Time representation of a system function

As we defined in section3.1 we could represent a system in the frequency domain as well as in the time domain.



(a) *frequency domain* (b) *time domain*

Figure 4.5: Frequency response and time response of a system to relate the system functions $F(s)$ and $f(t)$

In this chapter we concentrated on the frequency domain, where the system function is the transfer function, relating the output to the input:

$$Y(s) = F(s)X(s)$$

Referring to section 3.1 the equivalent relation in the time domain is:

$$y(t) = \int_0^\infty f(t-\tau)x(\tau)\mathrm{d}\tau$$

The relation between the two system function is therefore:

$$F(s) = \mathcal{L}\{f(t)\}$$
$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

This can easily be seen from the fact that the Laplace transform of the impulse response is equal to one. In formula's:

$$\begin{aligned} y(t) &= \int_0^\infty f(t-\tau)x(t)\mathrm{d}\tau \\ &= \int_0^\infty f(t-\tau)\delta(t)\mathrm{d}\tau \\ &= f(t) \\ &= \mathcal{L}^{-1}\{F(s)\} \end{aligned}$$

or from the frequency perspective:

$$\begin{aligned} Y(s) &= F(s)X(s) \\ &= F(s)1 \\ &= \mathcal{L}\{f(t)\} \end{aligned}$$

Both methods are used to calculate output from an input and a given system. Although the procedure of the convolution method looks complicated it is often used in discrete-time systems.
Before we dive further in the use of transfer functions and the analysis of systems, we first look at the meaning of the complex frequency domain s.

4.5 Frequency Analysis

The response of linear systems to sinusoidal inputs forms the basis of an extensive body of theory dealing with the modelling and analysis of dynamic systems. This theory was developed first in the

field of communications (telephone and radio) and later became extended and then widely used in the design and development of automatic control systems. The theory is useful not only in determining the sinusoidal response of a system but also in specifying performance requirements (as in sound system components or in a radar-tracking system), in finding the responses to other periodic inputs (square wave, sawtooth wave, etc.), and for predicting the stability of feedback control systems (amplifiers, regulators, and automatic controllers). A graphical portrayal of the input and output sinusoids of a typical linear system is shown in Fig. 4.6.

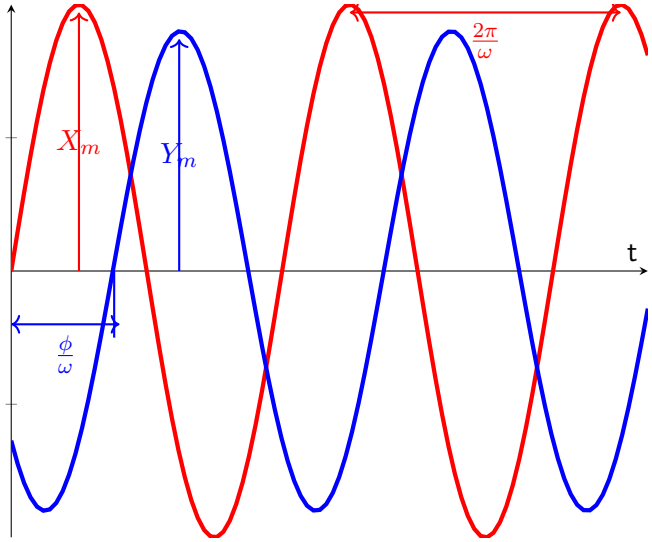


Figure 4.6: Sinusoidal input ($x(t)$) and output ($y(t)$) waveforms for a linear dynamic system.

Recall that the response of a linear system to an input signal $x(t)$ is the sum of the homogeneous (free) and particular (forced) solutions: $y(t) = y_h(t) + y_p(t)$. Here the homogeneous part of the response of a stable system is of minor interest (because it usually dies away soon and, in a linear system, strictly independent of the particular solution). After an initial start of a system at $t = 0$ when the system is set in motion, the longest characteristic time or the smallest characteristic frequency determines the presence of the homogeneous solution in the output. This time span we call the transient time: τ_t. After this time the output is dominated by the input signal modified by the system as given by the transfer function.
Since the transfer function is relating the output $Y(s)$ of a system to the input of a system $X(s)$ in the complex frequency domain we will dive a bit deeper in the meaning of the complex frequency.
The Laplace Transform (and the inverse Laplace Transform) relates two domains: the time domain $t \in [0, \infty]$ and the complex frequency domain $s \in \mathbb{C}$. The origin of the relation is in the kernel for the Laplace Transform $K(s,t) = e^{-st}$. The complex frequency can be written in a real and an imaginary part:

$$s = \sigma + \mathrm{j}\omega \tag{4.29}$$

with σ and ω real numbers. Due to the fact that the complex frequency s appears in the solution as e^{st} we can write this as:

$$e^{-st} = e^{-(\sigma+\mathrm{j}\omega)t} = e^{-\sigma t}e^{-\mathrm{j}\omega t} \tag{4.30}$$

where the first part is a growing or decaying factor and the second part represents an oscillation (together with its complex conjugate). Later we will see that $K(\omega, t) = e^{-\mathrm{j}\omega t}$ is the kernel for the Fourier Transform that relates the time domain $t \in [-\infty, \infty]$ and the real frequency domain $\omega \in [-\infty, \infty]$.
The difference between the Laplace Transform and the Fourier Transform is mentioned here to point at the effect of the $e^{-\sigma t}$-term, that enforces convergence of the transformation integrals in the case

of Laplace (of course this only works for positive values of σ). This makes the Laplace Transform more practical for engineering type of applications.
Now we know this relation, we can use it in the interpretation of the transfer function. The transfer function of a LTI system can be expressed as the quotient of two polynomials in the complex frequency s. For an input-output relation, we are not interested in the effect of the initial conditions, *i.e.* the homogeneous solution of the differential equation as was argument above. We are only interested how the output relates to an input only. From Fourier we will learn that a transformation from the time domain to the frequency domain is a complete mapping, without loosing information.
So if we Calculate the output $Y(s) = T(s)X(s)$, for an input

$$x(t) = e^{-\mathrm{j}\omega t} \quad \Leftrightarrow \quad X(s)|_{\sigma=0} = X(\mathrm{j}\omega) \tag{4.31}$$

with all initial conditions set to zero we can also set $\sigma = 0$, corresponding to an harmonic input signal with constant amplitude (no decay or growth). In the same way we can substitute $s = \mathrm{j}\omega$ in the transfer function:

$$T(s) \rightarrow T(\mathrm{j}\omega) \equiv H(\omega) \tag{4.32}$$

to obtain a function of ω only.
Here we defined the transfer function $H(\omega)$ as function of ω. $H(\omega)$ is a complex valued function, *i.e.* it returns a complex value for every real value of ω. For the interpretation of this it is most convenient to rewrite this in two factors:

$$H(\omega) = r(\omega)e^{j\phi(\omega)} \tag{4.33}$$

one containing the amplitude $r(\omega)$ and one containing the phase $\phi(\omega)$.
Since $Y(s) = T(s)X(s) \leftrightarrow Y(\omega) = H(\omega)X(\omega)$ this means that every pure harmonic signal as an input is multiplied by $H(\omega)$ consisting of a frequency dependent real multiplication factor $r(\omega)$ and a frequency dependent phase factor $\phi(\omega)$. The phase factor appears in an exponential $e^{\mathrm{j}\phi(\omega)}$; Note that the magnitude of this exponential factor is equal to one. This means that the original harmonic input is multiplied by a real gain factor $r(\omega)$ and phase shifted by a real phase $\phi(\omega)$. Both gain and phase are frequency dependent.
For the moment we only need to know that we can decompose any input signal in its frequency components by means of a Fourier Transform and compose the output signal from the modified frequency components by means of the inverse Fourier Transform. This will become clear in the chapter 9 on Fourier Transforms. We will focus now on the description of the system by means of the transfer function $H(\omega)$.
So we now know that for LTI systems, the transfer function results in a complex number for every value of the frequency ω, and that this complex number is represented by an amplitude $r(\omega)$ and a phase $\phi(\omega)$. Mathematically there is not so much to say about his, however various tools are developed in a graphical representation of the transfer function. We will consider two representations here: The Bode-plot and the Nyquist plot.

Summaries

Summary Chapter 4-I

Transfer functions

- Transfer function of an LTI system is the quotient of two polynomials:

$$T(s) = \frac{B(s)}{A(s)} \tag{4.34}$$

with the polynomials:

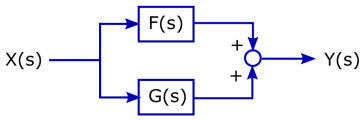
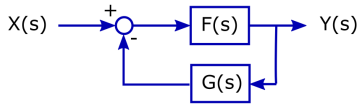
$$B(s) = b_m s^m + b_{m-1} s^{m-1} + \ldots + b_1 s + b_0 \tag{4.35}$$
$$A(s) = a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0 \tag{4.36}$$

with $(m \leq n)$.

- the transfer function can also be written as:

$$T(s) = \frac{(s - s_m)(s - s_{m-1}) \ldots (s - s_1)}{(s - s'_n)(s - s'_{n-1}) \ldots (s - s'_1)} \tag{4.37}$$

Block algebra

Diagram	Operation	Equation
	Product	$H(s) = F(s)G(s)$
	Sum	$H(s) = F(s) + G(s)$
	Feed-back	$H(s) = \frac{F(s)}{1 + F(s)G(s)}$

Frequency analysis and Bode-plots

- Figure 5.5 is part of this summary. The Bode-plots correspond to the systems for which the step responses are shown in figure 2.12
- The quality factor of a second-order system is related to the damping factor and is defined as:

$$Q \equiv \frac{1}{2\zeta} \tag{4.38}$$

- The width (full-width-half-maximum) of the absolute value of the transfer function $|H(\omega)|$ is also related to the damping factor:

$$\Gamma_\omega = \Delta\omega = \frac{\omega_n}{Q} = 2\zeta\omega_n \tag{4.39}$$

Problems

Problem 4.1. Transfer function of first-order system
Consider the differential equation given by:

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} + 3y(t) = x(t)$$

a. Determine the transfer function $T(s)$ of this system, relating the output $Y(s)$ to the input $X(s)$.

b. Find the pole of the transfer function.

c. Express the transfer function in the variable ω by applying a pure harmonic input:

$$x(t) = A\mathrm{e}^{\mathrm{j}\omega t}$$

d. What is the value of the transfer function – expressed in magnitude and phase – for the following values of ω

(i) $\omega \to \infty$
(ii) $\omega \to 0$
(iii) $\omega = 3$.

Problem 4.2. Transfer function of a second order system
Consider the differential equation given by:

$$\frac{\mathrm{d}^2y(t)}{\mathrm{d}t^2} + \frac{1}{2}\frac{\mathrm{d}y(t)}{\mathrm{d}t} + 4y(t) = x(t)$$

a. Determine the transfer function $T(s)$ of this system, relating the output $Y(s)$ to the input $X(s)$.

b. find the poles of the transfer function.

c. Express the transfer function in the variable ω by applying a pure harmonic input:

$$x(t) = A\mathrm{e}^{\mathrm{j}\omega t}$$

d. What is the value of the transfer function – expressed in magnitude and phase – for the following values of ω

(i) $\omega \to \infty$
(ii) $\omega \to 0$

e. For what value of ω is the phase value $-\pi/2$?

f. For what value of ω has the transfer function its maximum magnitude?

g. Compare the values of the last two problems and comment on the result.

Problem 4.3. Transfer function algebra 1
Consider the system that is graphically presented in figure 4.7.

a. Determine the total transfer function of the system, relating the output $Y(s)$ to the input $X(s)$.

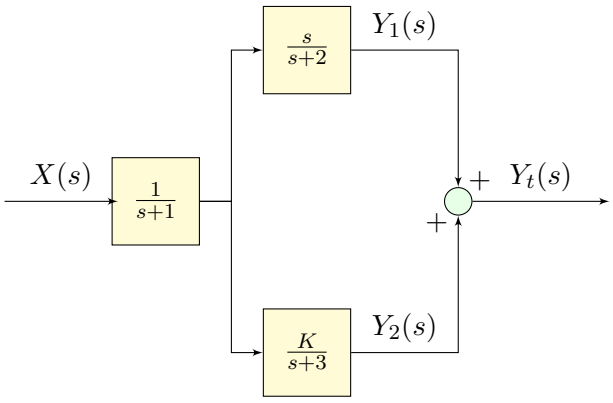


Figure 4.7: System to determine the overall transfer function.

b. Express the total transfer function as a function of ω and determine the magnitude and phase value for:

   (i) $\omega \to 0$.

   (ii) $\omega \to \infty$.

Problem 4.4. Transfer function algebra 2

Consider the system that is graphically presented in figure 4.8.

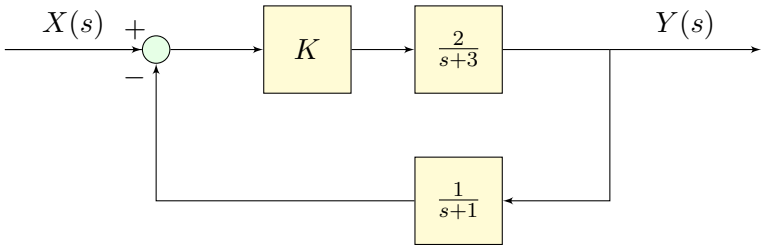


Figure 4.8: System to determine the overall transfer function.

a. Determine the total transfer function of the system, relating the output $Y(s)$ to the input $X(s)$.

b. Express the total transfer function as a function of ω and determine the magnitude and phase value for:

   (i) $\omega \to 0$.

   (ii) $\omega \to \infty$.

Problem 4.5. Transfer function algebra 3

Consider a system that is graphically presented in figure 4.9. This systems has two input signals: the regular input $X_I(s)$ and a disturbance input $X_D(s)$. Both inputs have their own "flow" through the system. Transfer function $P(s)$ describes the process, $M(s)$ the transfer function of the measurement; $D(s)$ is the part of the system that is only seen by the distortion signal, and $C(s)$ is the controller. The controller is the part of the system that can be designed by the operator to optimize the process. In this problem it has a given transfer function.

The transfer functions of the various blocks in figure 4.9 are defined as:

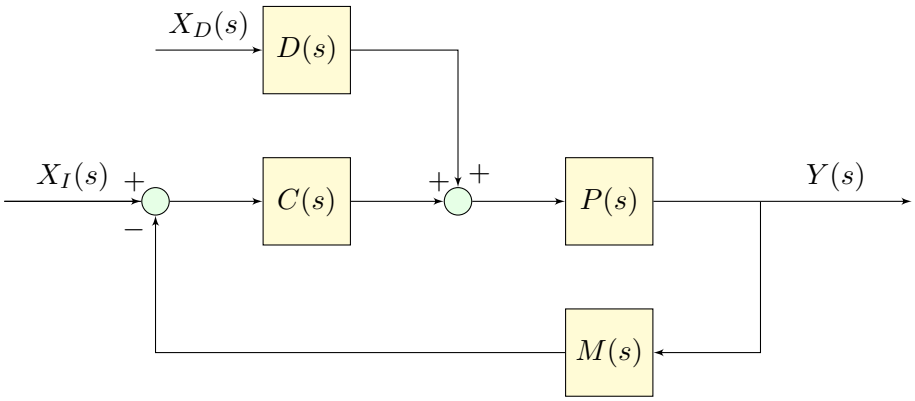


Figure 4.9: System with two inputs to determine the transfer functions.

- The process transfer function: $P(s) = \dfrac{2}{s^2 + 2s + 4}$
- The measurement transfer function: $M(s) = \dfrac{1}{s+1}$
- The disturbance input transfer function: $D(s) = \dfrac{5}{s+5}$
- The controller transfer function: $C(s) = \dfrac{s+1}{s}$

The open-loop output signal is defined as the signal just before it is fed back into the system.

With the given system and the transfer functions of the parts of the system given, answer the following questions.

a. Construct the open-loop transfer function of this system.

b. Construct the closed-loop transfer function $T_{CL-I}(s)$ for the regular input $X_I(s)$.

c. Construct the closed-loop transfer function $T_{CL-D}(s)$ for the disturbance input $X_D(s)$.

d. Calculate the poles and zeros of the closed-loop transfer function $T_{CL-I}(s)$.

e. Calculate the poles and zeros of the closed-loop transfer function $T_{CL-D}(s)$.

Problem 4.6. Feedback control system
In figure 4.10 a control system can be found. The dashed line is an extra feedback loop to enhance the performances of the control system, called the inner feedback loop. First we will look to the system without the inner feedback loop.

a. Determine the transfer function $T_1(s)$ of this system, relating the output $Y(s)$ to the input $X(s)$, without the inner feedback loop $M(s)$.

b. Determine the transfer function $T_2(s)$ of this system, relating the output $Y(s)$ to the input $X(s)$, with the inner feedback loop $M(s)$ added.

c. Determine the transfer function $T_d(s)$ of this system, relating the output $Y(s)$ to the disturbance $D(s)$, with the inner feedback loop $M(s)$ added.

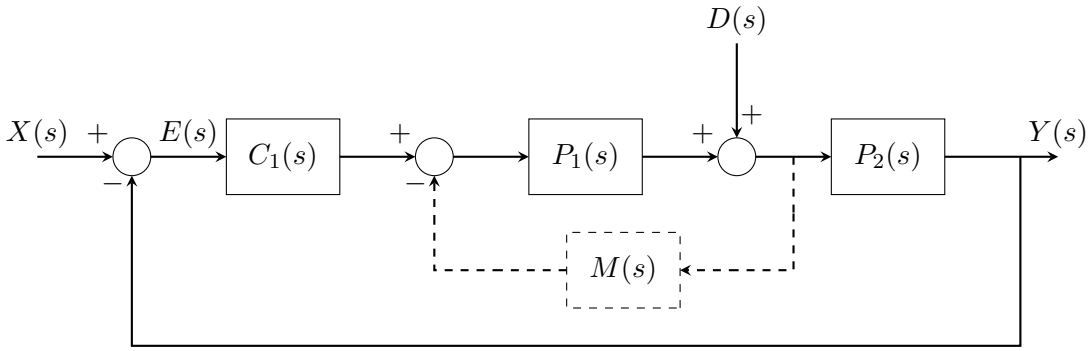


Figure 4.10: Block diagram of control system

Chapter 5

System response in the Frequency domain

In the previous chapter dynamic systems initially described by differential equations were described in transfer functions as the result of a Laplace Transform of the differential equations. The transfer function is an input-output function in the complex frequency domain, that transfers the input to an output.
In this chapter we focus on the interpretation of the transfer function of a system in the complex frequency domain. This as a complementary view on dynamic behaviour of systems via the time response.

5.1 The magnitude and phase of a transfer function

As we have derived in the previous chapter, a transfer function in the complex frequency domain can be interpreted as a amplitude transfer and a phase change by the system. By the substitution of $s \to \mathrm{j}\omega$ in the transfer function $T(s) \to T(\mathrm{j}\omega) \equiv H(\omega)$, we obtain a function that is a function of a real number, giving a complex number:

$$H(\omega) = r(\omega)e^{\mathrm{j}\phi(\omega)} \tag{5.1}$$

This complex number is written in polar coordinates, which separates the amplitude transfer and the phase transfer in two distinct functions.
This achievement can be used to our benefit in a sequence of systems, where the input of system 2 is the output of system 1 is represented by the transfer function:

$$T(s) = \frac{Y_2(s)}{X_1(s)} = \frac{Y_2(s)}{X_2(s)}\frac{Y_1(s)}{X_1(s)} = T_2(s)T_1(s)$$

where we used that $X_2(s) = Y_1(s)$.
Converting to the frequency description $s \to \mathrm{j}\omega$ and expressing the complex number in polar coordinates gives:

$$H(\omega) = r_1(\omega)e^{\mathrm{j}\phi_1(\omega)}r_2(\omega)e^{\mathrm{j}\phi_2(\omega)} = r_1(\omega)r_2(\omega)e^{\mathrm{j}(\phi_1(\omega)+\phi_2(\omega))}$$

From this expression we see that the total phase change of an harmonic signal through both systems is equal to the sum of the phases:

$$\phi_t(\omega) = \phi_1(\omega) + \phi_2(\omega) \tag{5.2}$$

at that frequency. The total amplitude multiplication factor is:

$$|H(\omega)| = |r_1(\omega)r_2(\omega)| = |r_1(\omega)||r_2(\omega)| \tag{5.3}$$

The result is two procedures to calculate the magnitude change and phase change respectively of the combined transfer function. The combined magnitude is constructed via a *multiplication* of the individual magnitude transfers, whereas the combined phase is constructed via the *sum* of the individual phase transfers. Since both the magnitude and the phase are functions of ω the construction of two transfer functions has to be done for a continuous span of frequencies.

5.1.1 The dB-scale for magnitude

Traditionally in system and control engineering one does look at powers instead of amplitudes, so the power transfer is equal to:

$$|H(\omega)|^2 = |r_1(\omega)r_2(\omega)|^2 = |r_1(\omega)|^2|r_2(\omega)|^2$$

Since adding is more easy then multiplication (certainly in the pre-computer era), engineers used the 10-log – logarithm with base 10 – of this quantity giving:

$$|H(\omega)|_B = \log|H(\omega)|^2 = 2\log|H(\omega)|$$

Although this quantity is dimensionless, it is given the dimension Bell or abbreviated [B]. Like with other units, this definition does not always give nice numbers, so it became common practice to express the power transfer ratio in units of deci-Bell's dB: 1 Bell $=$ 10 deci-Bell. Therefore the quantity to be calculated is:

$$|H(\omega)|_{dB} = 20\log|H(\omega)| = 20\log(r(\omega)) \tag{5.4}$$

in units of [dB]
Just to mention some practical implications: for

- $|H(\omega)| = 1 \Rightarrow |H(\omega)|_{dB} = 0[dB]$
- $|H(\omega)| < 1 \Rightarrow |H(\omega)|_{dB} < 0[dB]$ damping
- $|H(\omega)| > 1 \Rightarrow |H(\omega)|_{dB} > 0[dB]$ amplification

In units of dB, the multiplication of two transfer functions becomes an addition of the magnitudes:

$$|H_t(\omega)|_{dB} = 20\log|H_1(\omega)| + 20\log|H_2(\omega)| = 20(\log(r_1(\omega)) + \log(r_2(\omega))) \tag{5.5}$$

In the Bode plots, the transfer function in dB and the phase in degrees or radians can be added per frequency. So calculation of transfer function of a sequence of two systems addition of two graphs, one for the power in dB and one for the phase.
There is one aspect of Bode plots we did not yet mention that makes Bode plots even more useful: plot also the independent frequency axis on a logarithmic scale. The relevance of this becomes immediate clear for a very simple first-order transfer function.

Example 5.1.1. *Bode Plot of a first-order system*
Let us consider the Transfer function:

$$T(s) = \frac{1}{s+1} \;\Leftrightarrow\; H(\omega) = \frac{1}{\mathrm{j}\omega + 1} = \frac{-\mathrm{j}\omega + 1}{\omega^2 + 1}$$

Where we used the method of multiplying numerator and denominator with the complex conjugate of the denominator to make the denominator real-valued. This transfer function has no zero's and one pole at $s = -1$. Calculating the limits:

$$\lim_{s\to 0} T(s) = 1$$
$$\lim_{s\to\infty} T(s) \approx \lim_{s\to\infty} \frac{1}{\mathrm{j}\omega} = -\lim_{\epsilon\to 0} \mathrm{j}\epsilon = 0$$

Calculating the magnitude and the phase:

$$r(\omega) = \sqrt{\frac{1}{\omega^2+1}}$$
$$\phi(\omega) = \arctan\left(\frac{-\omega}{1}\right)$$

This leads to the limits of: $\phi(0) = 0$ and $\phi(\infty) = -\pi/2$ for the phase and $r(0) = 1 \Rightarrow 0$ dB and $r(\infty) = 0 \Rightarrow -\infty$ dB for the magnitude. For the magnitude we also want the trend at low values of ω and high values of ω

$$|H(\omega)|_{dB} = 10\log\left(\frac{1}{\omega^2+1}\right)$$

For small values of ω this is constant 0 dB, since ω is small compared to 1. For large values of ω we neglect 1 compared to ω to obtain:

$$|H(\omega)|_{dB} = 10\log\left(\frac{1}{\omega^2}\right) = 10\log\left(\omega^{-2}\right) = -20\log(\omega)$$

So if we plot $|H(\omega)|_{dB}$ as a function of $\log(\omega)$ we get a straight line with slope -2; see figure 5.1. The slope of -2 can be recognized in the figure as -6 db/octave or -20 dB/decade. Where octave stands for doubling of the frequency and decade for a multiple of 10 to indicate the change in magnitude per unit of frequency ratio. The limits of the phase are also clearly observed and consistent with the calculated values: 0 and $-\pi/2$.

□

Now we have seen an example of a graphical representation of the transfer function, we take a more systematic approach on Bode-plots in the next section.

5.2 Bode plot

Thanks to the pioneering work of H.W. Bode[1] on feedback amplifier design, the use of logarithmic charts to portray the magnitude and phase characteristics has led to the development of so-called Bode diagrams. These diagrams are now widely used in describing the dynamic performance of linear systems. The idea is that the logarithm of the power ratio (output over input) and phase difference are plotted against frequency. This helps with visualization when designing LTI-system, as the power ratio in dB and phase differences can simply be added when combining sequences of systems.

5.2.1 Constructing a Bode-plot

The Bode plot consists of two plots as function of frequency: ω on a logarithmic scale. One is the $20\log(|H(\omega)|)$, representing the power of the signal in dB (Decibel or deci-Bell). The second one is the phase as function of ω. The origin of these Bode-plot dates from before the existence of computers.

[1] H.W. Bode, Network Analysis and Feedback Amplifier Design (Van Nostrand, Princeton, NJ, 1945).

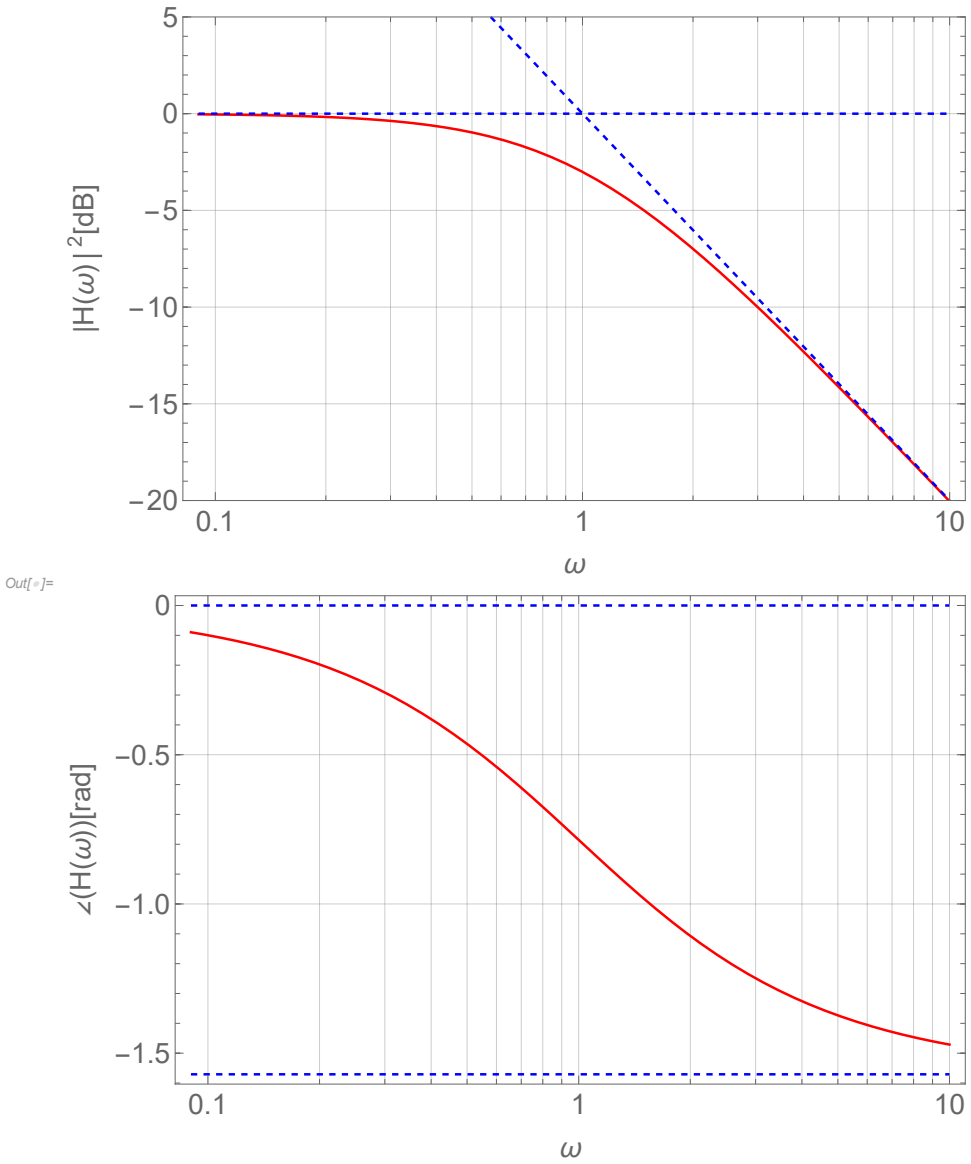


Figure 5.1: Bode plot of a first-order system as given in the example.

A Bode plot consists of two plots for the amplitude and phase of $T(s)$ as a function of frequency. The amplitude axis is either logarithmic ($\log T$) or quasi-logarithmic (decibels), whereas the phase axis is linear. Our emphasis here is on the use of Bode diagrams for analysing systems. However, these diagrams are also used in portraying the results of frequency-response measurements that one obtains experimentally by applying an input sinusoid and measuring the resultant power ratio and phase difference of the output with respect to the input, for a range of applied input frequencies. The procedure for preparing a set of Bode diagram curves analytically is as follows:

- Step 1. Determine the system transfer function $T(s)$.
- Step 2. Convert $T(s)$ to the sinusoidal transfer function $H(\omega)$ by by substitution of $s = \mathrm{j}\omega$.
- Step 3. Develop expressions for the magnitude, $|H(\omega)|$, and phase angle, $\phi_T(\omega) = \angle H(omega) = \arctan(H(\omega))$.
- Step 4. Plot $20 \log |T(\omega)|$ versus $\log \omega$ and ϕ_T versus $\log \omega$.

Remark: In Bode-plots, the decibel scale, which is linear, is sometimes instead shown as a $\log$-scale of the ratio of amplitudes. This can be convenient for reading back values by inspection. However, strictly speaking, this is not a Bode-plot.
In example 5.1.1we already showed that asymptotic behaviour is important in frequency analysis. This is the behaviour for low frequencies $\omega \to 0$, which corresponds to $\log \omega \to -\infty$, and the behaviour for high frequencies, $\omega \to \infty \Rightarrow \log \omega \to \infty$. In many cases this results in linear behaviour of the amplitude part of the Bode plot. Horizontal lines indicate constant frequency response. Straight lines with a slope are often expressed in $\#\,\mathrm{dB/oct.}$, or $\#\,\mathrm{dB/dec.}$, denoting a gain or attenuation as a number of decibels per octave (factor of two in frequency) or as a number of decibels per decade (factor of ten in the frequency). Usually these slopes are multiples of 6, which relates to the order in s or ω of the asymptotic behaviour or the transfer function. For example value of $-6\,\mathrm{dB/oct.}$ corresponds to a factor two damping in amplitude per frequency doubling ($-20 \log 2 = -6$), which in turn corresponds to an order of s in the numerator. Orders of s in the denominator correspond to negative slopes,; orders of s in the numerator correspond to positive slopes. In the same way, the phase changer per order in s or ω is $\pi/2$ or 90° for orders in s or ω in the numerator. Orders of s in the denominator instead correspond to negative phase shifts.
We illustrate this these characteristics for first-order and second-order systems.

5.2.2 Bode diagram of a first-order system.

Let us prepare Bode diagram curves for the first-order system described by the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{y}{\tau} = \frac{kx}{\tau} \tag{5.6}$$

where $x(t)$ and $y(t)$ are the input and output, respectively.

- Step 1. Transforming to the s domain yields

  $$sY(s) + \frac{Y(s)}{\tau} = \frac{kX(s)}{\tau} \tag{5.7}$$

  The system transfer function is

  $$T(s) = \frac{Y(s)}{X(s)} = \frac{k}{\tau s + 1} \tag{5.8}$$

- Step 2 The sinusoidal transfer function is

  $$T(\mathrm{j}\omega) = T(s)|_{s=\mathrm{j}\omega} = H(\omega) = \frac{k}{\mathrm{j}\omega\tau + 1} \tag{5.9}$$

- Step 3. To simplify the process of obtaining expressions for $|H(\omega)|$ and $\phi(\omega)$, note that $H(\omega)$ can be presented as a ratio of complex functions $N(\omega)$ and $D(\omega)$:

  $$H(\omega) = \frac{N(\omega)}{D(\omega)} = \frac{r_N(\omega)\mathrm{e}^{\mathrm{j}\phi_N(\omega)}}{r_D(\omega)\mathrm{e}^{\mathrm{j}\phi_D(\omega)}} = \frac{r_N(\omega)}{r_D(\omega)}\mathrm{e}^{j(\phi_N(\omega)-\phi_D(\omega))} \tag{5.10}$$

  Realize that the magnitude of the exponential is equal to one, due to the pure imaginary character of the argument of the exponential. In this case, $r_N(\omega) = k$ and $r_D(\omega) = |\mathrm{j}\omega\tau + 1|$, so that

  $$|H(\omega)| = \frac{k}{|\mathrm{j}\omega\tau + 1|} = \frac{k}{\sqrt{1 + \omega^2\tau^2}} \tag{5.11}$$

  and the phase angle ϕ_T is given by

  $$\phi_T = \phi_N - \phi_D = \arctan(0/k) - \arctan(\omega\tau/1) = 0 - \arctan(\omega\tau) \tag{5.12}$$

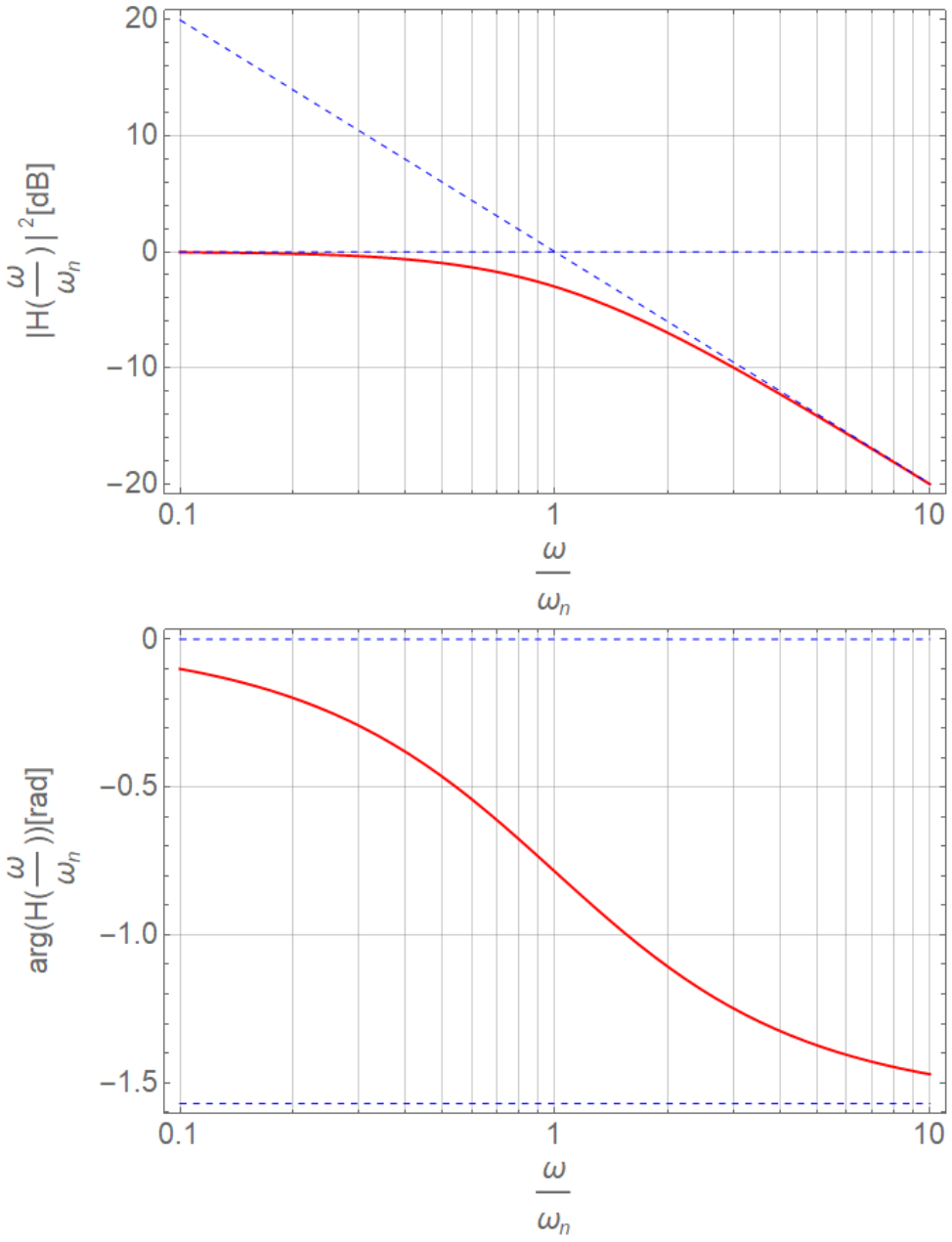


Figure 5.2: Amplitude and phase curves for the Bode diagram of the first order system. The frequency axis is normalized to the inverse of the time constant $1/\tau_n$ expressed as ω_n. The asymptotic behaviour is indicated by the blue dashed lines.

- Step 4. The Bode plot curves for magnitude and phase can now be plotted with calculations based on the expressions for $|H(\omega)|$ and $\phi(\omega)$ developed in Step 3. The resulting magnitude and phase curves for this first-order system are shown in figure 5.2.

  In a preliminary system analysis, it is often sufficient to approximate the amplitude and phase curves by use of simple straight-line asymptotes, which are very easily sketched by hand. These straight-line asymptotes are included in figure 5.2. The first asymptote for the magnitude curve corresponds to the case in which ω is very small, approaching zero. Using the Equation for the magnitude as $\omega \to 0$ yields

$$\lim_{\omega \to 0} [\log H(\omega)] = \lim_{\omega \to 0} \left[\log \frac{k}{\sqrt{1+\omega^2\tau^2}} \right] = \log(k) \tag{5.13}$$

  The second asymptote for the amplitude curve is based on the case in which ω is very large.

$$\lim_{\omega \to \infty} [\log H(\omega)] = \lim_{\omega \to \infty} \left[\log \frac{k}{\sqrt{1+\omega^2\tau^2}} \right] = \log(k) - \log(\omega\tau) \tag{5.14}$$

  Hence, at high frequencies,

$$\log(H(\omega)) = \log(k) - \log(\tau) - \log(\omega) \tag{5.15}$$

  Note that the slope of the second asymptote on $\log - \log$ paper is -1, which corresponds to $-6\,\mathrm{dB/oct.}$ on the decibel scale. The two asymptotes described by Eqs. 5.13 and 5.15 thus constitute the asymptotic magnitude versus frequency characteristic. The maximum error incurred by use of the asymptotic approximation occurs at the cut-off frequency where $\omega = 1/\tau$, i.e., where the two asymptotes intersect – sometimes also referred to as the "corner frequency".

  The asymptotic approximation for the phase angle curve consists of three straight lines. The first, for very low frequency, is given by

$$\lim_{\omega \to 0} [\arctan(\omega\tau)] = -\arctan(0) = 0 \tag{5.16}$$

  The second is a straight-line tangent to the phase angle curve at its cut-off frequency, having a slope given by

$$\frac{\mathrm{d}\phi_T(\omega)}{\mathrm{d}\log(\omega)} = \frac{\mathrm{d}\phi_T(\omega)}{\mathrm{d}\omega} \frac{\mathrm{d}\omega}{\mathrm{d}\log(\omega)} = -\frac{\omega\tau}{1+\omega^2\tau^2} \tag{5.17}$$

  Thus, at the cut-off frequency $\omega = 1/\tau$, the phase angle is $-\pi/4$ or -45° and the slope is given by

$$\frac{\mathrm{d}\phi_T(\omega)}{\mathrm{d}\log(\omega)} = -\frac{1}{2}\ln(10). \tag{5.18}$$

  The third asymptotic approximation occurs at very large values of ω, where

$$\lim_{\omega \to \infty} [\arctan(\omega\tau)] = -\arctan(\infty) = -\frac{\pi}{2}. \tag{5.19}$$

  From the graph in figure 5.2 we observe that frequencies larger than the natural frequency ω_n (in the discipline of filters also called cut-off frequency ω_c) are increasingly suppressed with increasing frequency. In terms of filtering we call this a low-pass filter. Frequencies below the natural frequency or cut-off frequencies are unaltered. We will treat filtering in more detail in chapter 10.7. We note that for a first-order system, the cut-off frequency is given by: $\omega_c = \omega_n = 1/\tau$.

5.2.3 Bode plot of a second-order system

The general equation for second order system is:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\zeta\omega_n\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_n^2 y = \omega_n^2 kx. \tag{5.20}$$

In this ω_n is called the natural frequency and ζ the damping ratio as introduced in chapter 2.4 and k is the static gain, the ratio between output and input signal in a stationary situation. The Bode diagram of this function is again made by applying the various steps:

- Step 1. The system transfer function is

$$T(s) = \frac{Y(s)}{X(s)} = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{5.21}$$

- Step 2. The sinusoidal transfer function is

$$H(\omega) = \frac{k\omega_n^2}{(\mathrm{j}\omega)^2 + 2\zeta\omega_n\mathrm{j}\omega + \omega_n^2} = \frac{k}{(1-(\omega/\omega_n)^2) + \mathrm{j}2\zeta(\omega/\omega_n)} \tag{5.22}$$

- Step 3. Develop expression for $|H(\omega)|$ and $\phi(\omega)$. First the magnitude is

$$|H(\omega)| = \frac{k}{\sqrt{[1-(\omega/\omega_n)^2]^2 + 4\zeta^2(\omega/\omega_n)^2}} \tag{5.23}$$

  The phase angle is:

$$\phi(\omega) = -\arctan\frac{2\zeta(\omega/\omega_n)}{(1-(\omega/\omega_n)^2)} \tag{5.24}$$

- Step 4. Prepare the Bode plot by graphing the magnitude and the phase. In figure 5.3 the frequency-response of a second-order system with a natural frequency $\omega_n = 1.0$ rad/s, a damping ratio $\zeta = 0.25$ and a static gain of $k = 1$ is presented. Note that the static gain has no influence on the phase and does not influence the shape of magnitude plot, it merely changes the values on the y-axis. From a graph like figure 5.3 the static gain can be read off from the value of the magnitude for $\omega \to 0$. (In a Bode diagram a change in static gain by a factor k implies you add $20\log(k)$ to the magnitude. For $k = 1$ the value for $\omega \to 0$ is $20\log(1) = 0$). Unless specified the static gain is assumed to be one.

  Some values of the magnitude and phase are presented in table 5.1.

Table 5.1: Frequency response power, magnitude and phase angle for a second order system with $\omega_n = 1$, $\zeta = 0.25$ and $k = 1$.

Frequency (rad/s)	**Power** (dB)	**Magnitude**	**Phase** (rad)	**Phase** (deg)
0.1	0.076	1.01	-0.05	-2.9
0.2	0.31	1.04	-0.10	-5.9
1.0	6.0	2.0	$\pi/2$	-90
5.0	-28	0.04	-3.04	-174
10.0	-40	0.01	-3.09	-177

For three frequency values ω the time response of this system is given in figure 5.4. One can clearly observe that over the frequency range a phase lag (negative values!) is developed. At the natural frequency $\omega = \omega_n$ the phase lag is $-\pi/2$. For $\omega \to \infty$ the phase lag goes to $-\pi$.

Again we see that the phase lag is roughly $-\pi/2$ for every order in the denominator polynomial of the transfer function and $+\pi/2$ for ever order in the numerator polynomial of the transfer function. In this second order system this results in a total phase lag of $-\pi$.

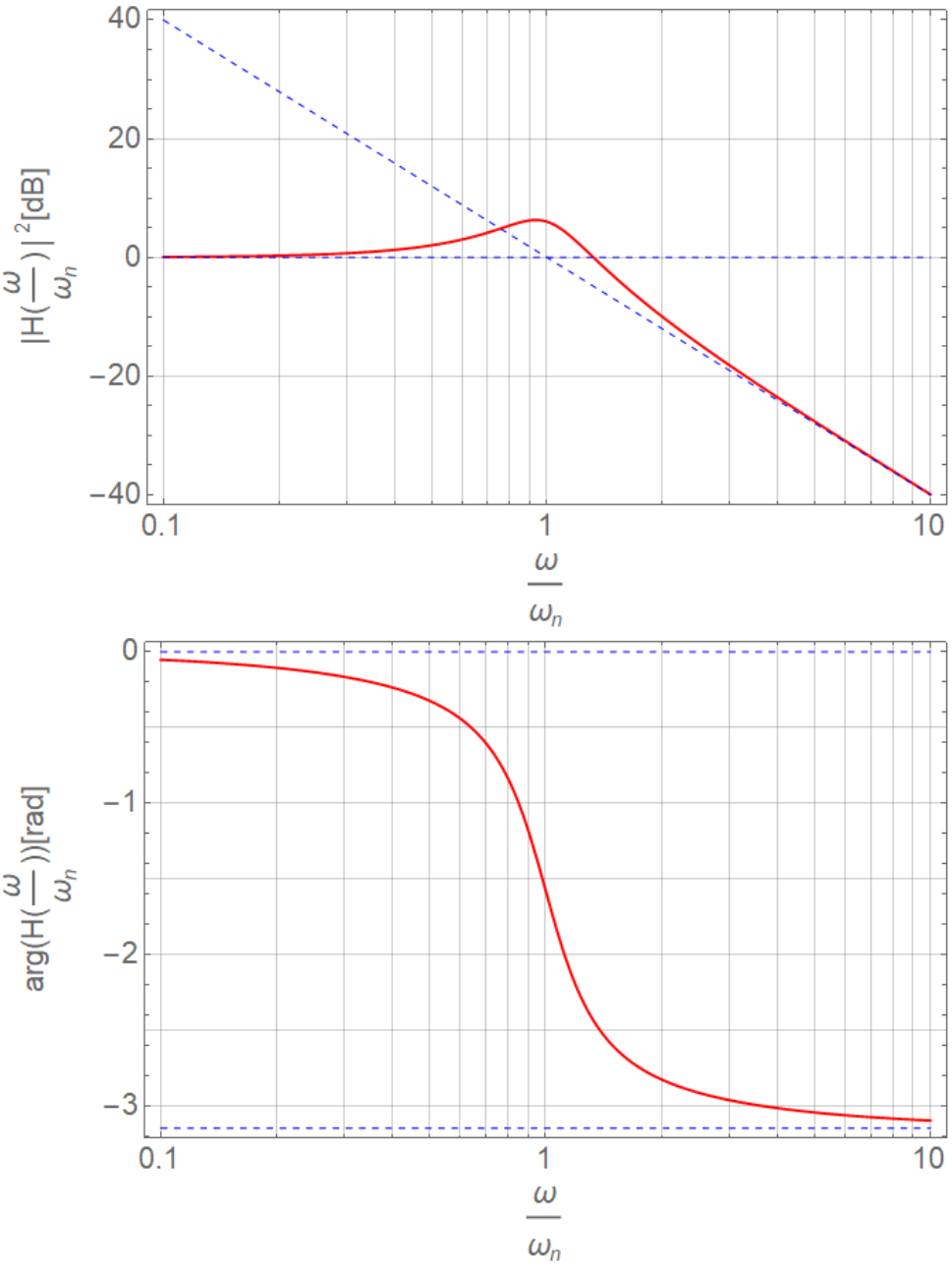


Figure 5.3: Bode diagram of second order low pass system. The frequency axis is normalized to the natural frequency ω_n, static gain is 1. The asymptotic behaviour is indicated by the blue dashed lines.

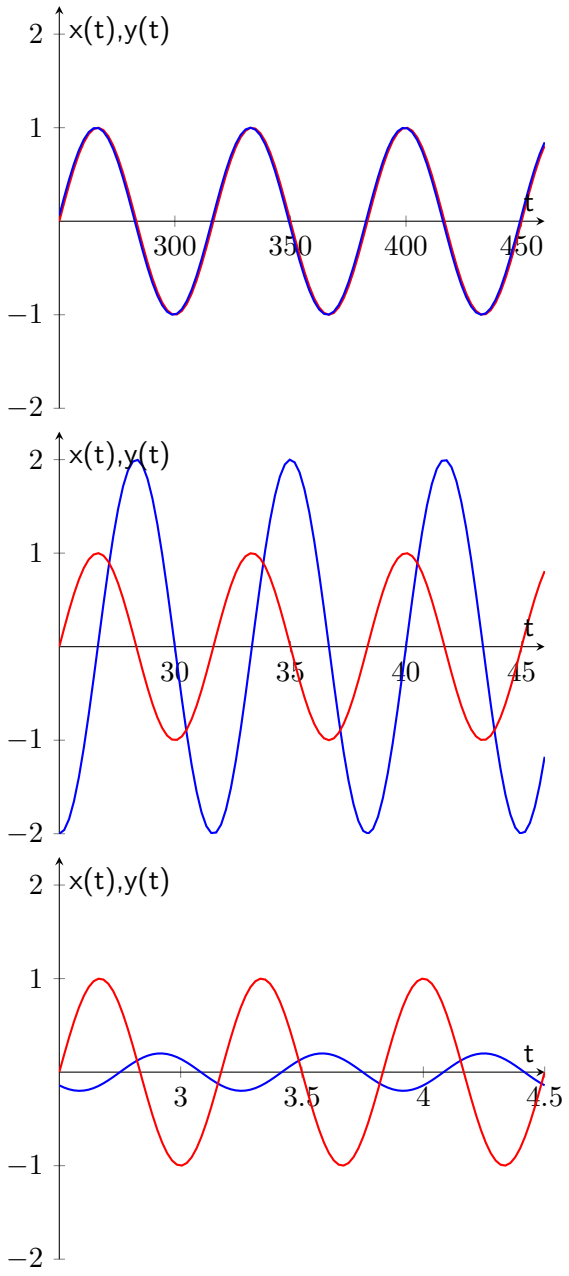


Figure 5.4: Input $x(t)$ (red) and output $y(t)$ (blue) signals when the driving frequency is lower ($0.1\omega_n$) (top), at (ω_n) (middle) or above ($10\omega_n$) (bottom) the natural frequency. Note that the phase shifts shows a lag of 0, $\pi/2$ and π for the three driving frequencies ($0.1\omega_n$, ω_n and $10\omega_n$) respectively.

5.2.4 Resonance and Damping in a second-order system

In a second-order system two parameters determine the response of the system. These can either be the poles, the two of the three coefficients in the differential equation (the third one is just a scaling), or the the more meaningful derived parameters ω_n and ζ. In fact the we can reduce the characteristics of second-order system to one parameter if we normalize the frequency axis to the natural frequency.
Plotting the Bode-plots on a normalized frequency axis and the static gain set to leafs us one parameter to vary in the system: The damping factor ζ. In figure 5.5 the Bode plots for a second-order systems for various values for the damping factor are presented.
The distinction between under-damped and over-damped systems is clear. For $\zeta < 1$, a peak around the natural frequency is observed. This graph should be compared to the time response given in figure 2.12. For the under-damped system the time response exhibits oscillations with a larger magnitude at a particular frequency, which corresponds to the peak frequency in the Bode diagram.
The Bode diagram of this low-pass second-order system shows a few characteristics:

a. The strength of the transmission in inversely related to the damping factor.

b. The slopes of the magnitude for high frequencies is the same and is twice as large as for a first order system. If the limit for $\omega \to \infty$ is taken it is found that the slope of the asymptote is $-12\,\mathrm{dB/oct.}$ or -2 on the $\log - \log$ plot. The maximum slope for the asymptote is $-6n\,\mathrm{dB/oct.}$ for an n-order system.

c. The total phase change over the frequency range is $-\pi$ or $-180°$, also twice as large as for a first-order system. Actually the maximum phase difference for an n-order system is $n\pi/2$.

d. The phase change at the natural frequency is independent of ζ and equals to $-\pi/2$, half the phase change of the whole frequency span.

e. The rate of the phase change at the natural frequency is higher for smaller ζ.

f. The frequency at which the maximum magnitude occurs ω_m is slightly below the natural frequency for underdamped systems. By setting the derivative of the magnitude with ω to zero the position of this maximum can be evaluated. Some tedious mathematics can be avoided with the observation that a maximum in the magnitude occurs a the same position as the minimum in $\left[(1-(\omega/\omega_n))^2\right]^2 + 4\zeta^2(\omega/\omega_n)^2$. The position of the maximum of the magnitude is

$$\omega_m = \omega_n\sqrt{1-2\zeta^2} \tag{5.25}$$

This effect can be seen quite clearly in figure 5.5. Note that $\omega_m \neq \omega_d$; the former one $\omega_m = \omega_n\sqrt{1-2\zeta^2}$ is the the maximum of the resonance curve, obtained from a frequency sweep on the system, the latter one $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the frequency observed in the impulse response.

g. The width of the resonance peak is smaller for smaller values of ζ.

The strength of the transmission is actually a good measure for the determination of the damping factor. Because the actual height of a peak is considered for this, the static gain has to be taken into account. If the value of the magnitude at the resonance peak ω_m is used, the magnitude at at this frequency is:

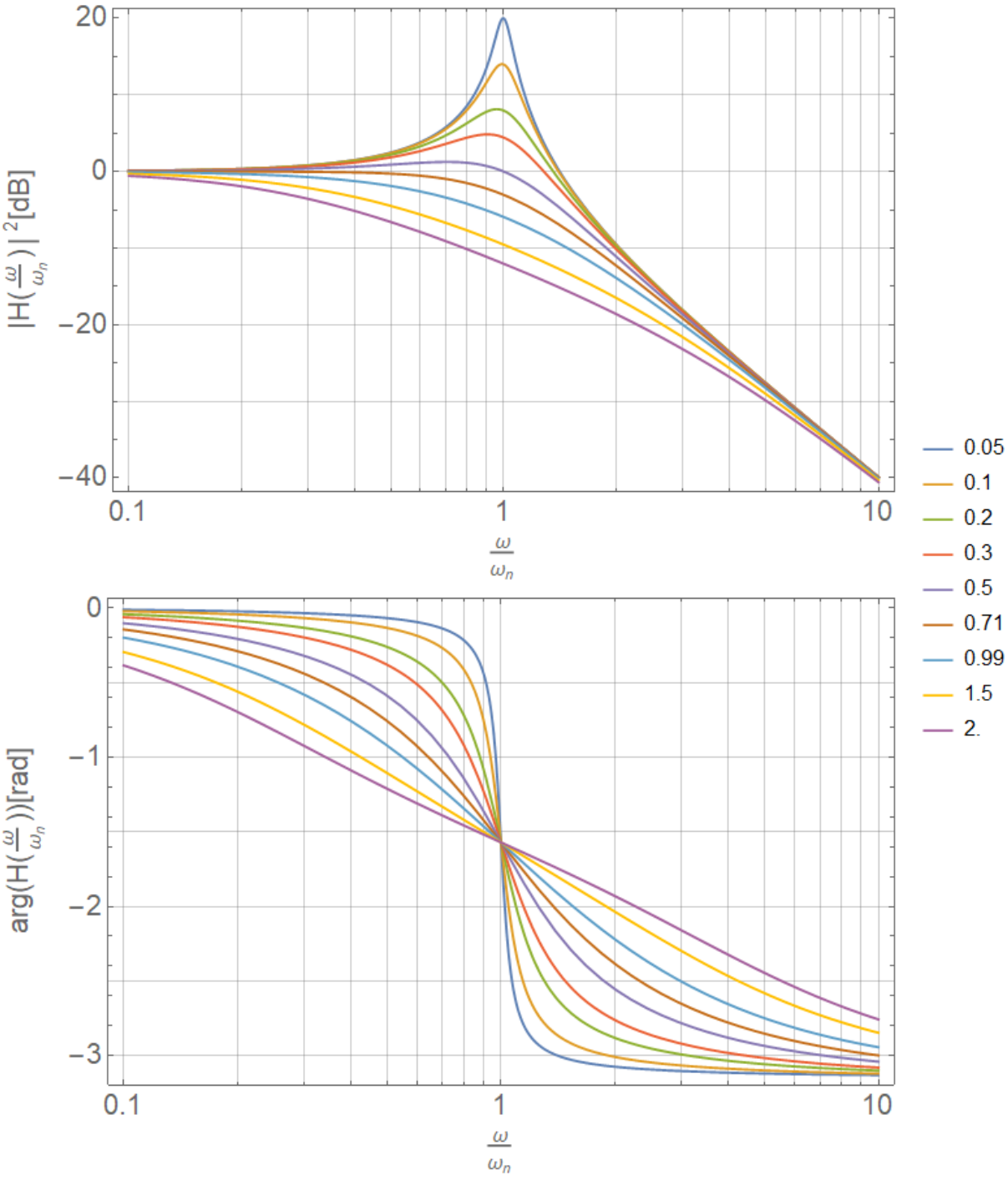


Figure 5.5: Bode diagram (magnitude and phase) of a second order system for various damping factors. The frequency is normalized.

$$
\begin{aligned}
|H(\omega_m)| &= \frac{k}{\sqrt{\left[1-(1-2\zeta^2)\right]^2 + 4\zeta^2(1-2\zeta^2)}} \\
&= \frac{k}{\sqrt{(2\zeta^2)^2 + 4\zeta^2(1-2\zeta^2)}} \\
&= \frac{k}{2\zeta\sqrt{1-\zeta^2}}
\end{aligned} \tag{5.26}
$$

For $\zeta \ll 1$ this reduces to $H(\omega_m) \approx k/(2\zeta)$. The ratio of the height of the resonance peak and the static gain is called the Quality factor, defined as:

$$
Q \equiv \frac{1}{2\zeta} \tag{5.27}
$$

In a Bode diagram the logarithm is depicted: $20\log(k) - 20\log(2\zeta)$ and the Quality factor is the difference in strength between the value for ω is zero and the resonance peak. The width of the resonance peak is also related to the quality factor Q:

$$
\Delta\omega = \frac{\omega_n}{Q} \tag{5.28}
$$

where $\Delta\omega$ is taken at the $-6\,\mathrm{dB}$ points with respect to the peak. As such the Quality factor expresses both the width and the relative height of a resonance peak, an important factor if you want to tune to your rive frequency to a resonant system. The physical interpretation of the quality factor is the ratio of energy stored in the system over the energy dissipated per cycle; in formula:

$$
Q = 2\pi \frac{\text{Energy stored in the system}}{\text{Energy dissipated per cycle}} = \omega \frac{\text{Energy stored}}{\text{Power loss}} \tag{5.29}
$$

Figure 5.5 shows the magnitude for several underdamped systems. For $\zeta = 0.05$, $Q = 10$ and a peak $20\,\mathrm{dB}$ can be read from this graph when the height of the peak is compared to the low frequency strength.
Now we have all ingredients for the Bode plots and combination of transfer functions, we can perform the calculations in a graphical way:

5.2.5 Bode diagrams of higher order systems

The construction of Bode diagrams of systems of an order higher than two can be performed by braking up the transfer function into fragments of first and second order that are then added together. This can always be done on the basis of formula 4.20. First-order fragments can be used for real poles and zeros, whereas second-order systems can be used for the complex and complex-conjugate pairs of poles or zeros.
The total transfer function is the product of all fragments. This translates to an addition of the amplitudes in a logarithmic plot, and an addition of the phases in a linear plot (which is a Bode plot). This characteristic is what makes the use of Bode diagrams so powerful in system analysis.

Example 5.2.1. *Adding Bode-plots*
In figure 5.7 the addition of two transfer functions in a Bode plot is illustrated with an example.
In this example two transfer functions:

$$
\begin{aligned}
T_1(s) &= \frac{2s+1}{2s^2+4s+8} \\
T_2(s) &= \frac{2}{2s^2+s+2}
\end{aligned}
$$

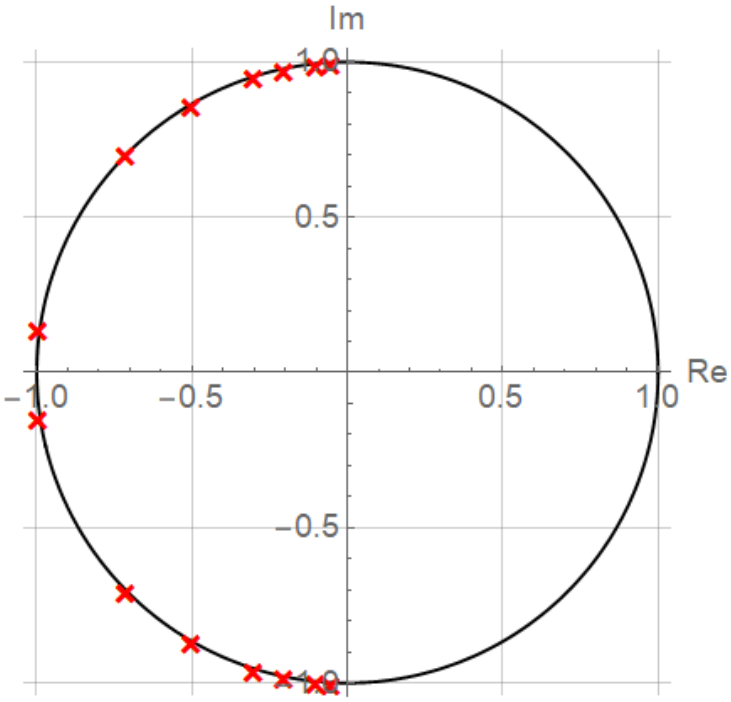


Figure 5.6: Poles of a second order system for various damping factors plotted in the complex plane. The poles correspond to the graphs in the Bode plot of figure 5.5 (limited to values $\zeta \leq 1$). Note that the poles are in complex conjugate pairs.

are put in cascade to get the total transfer function which is the product of the two individual transfer functions:

$$T_t(s) = T_1(s)T_2(s) = \frac{2s+1}{2s^4+5s^3+12s^2+8s+8}$$

In the bode diagram this results in the addition of the blue and the green curves to obtain the red curve, that represent the three transfer functions. This holds for every frequency. A check on the limits for $\omega \to 0$ on the magnitude gives:

$$|T_t(0)|_{\,\mathrm{dB}} = |T_1(0)T_2(0)|_{\,\mathrm{dB}} = 20\log\left(\frac{1}{8}\right) + 20\log(1) = 20\log\left(\frac{1}{8}\right) \approx -18.1 \ \mathrm{dB}$$

This number is verified with the graph. The phase for $\omega \to 0$ gives:

$$\angle T_t(0) = \angle T_1(0) + \angle T_2(0) = 0 + 0 = 0$$

which is also verified with the values in the graph. Now e check the ω-dependence for large values of ω. The dominant behaviour of the total transfer function for $\omega \gg 1$ becomes:

$$H_t(\omega) = H_1(\omega)H_2(\omega) \approx \frac{2\mathrm{j}\omega}{-2\omega^2}\frac{2}{-2\omega^2} = \frac{\mathrm{j}\omega}{\omega^4} = \frac{1\mathrm{j}}{\omega^3}$$

In units of $\mathrm{dB/oct}$ this is $-6 \ \mathrm{dB/oct} - 12 \ \mathrm{dB/oct}$ which gives a total of $-18 \ \mathrm{db/oct}$ which is effectively a third-order system.
The corresponding phase angels for $\omega \to \infty$ are:

$$\angle T_t(\infty) = \angle T_1(\infty) + \angle T_2(\infty) = -\frac{\pi}{2} - \pi = -\frac{3\pi}{2}$$

All these values can be verified in the Bode plot in figure 5.7.

□

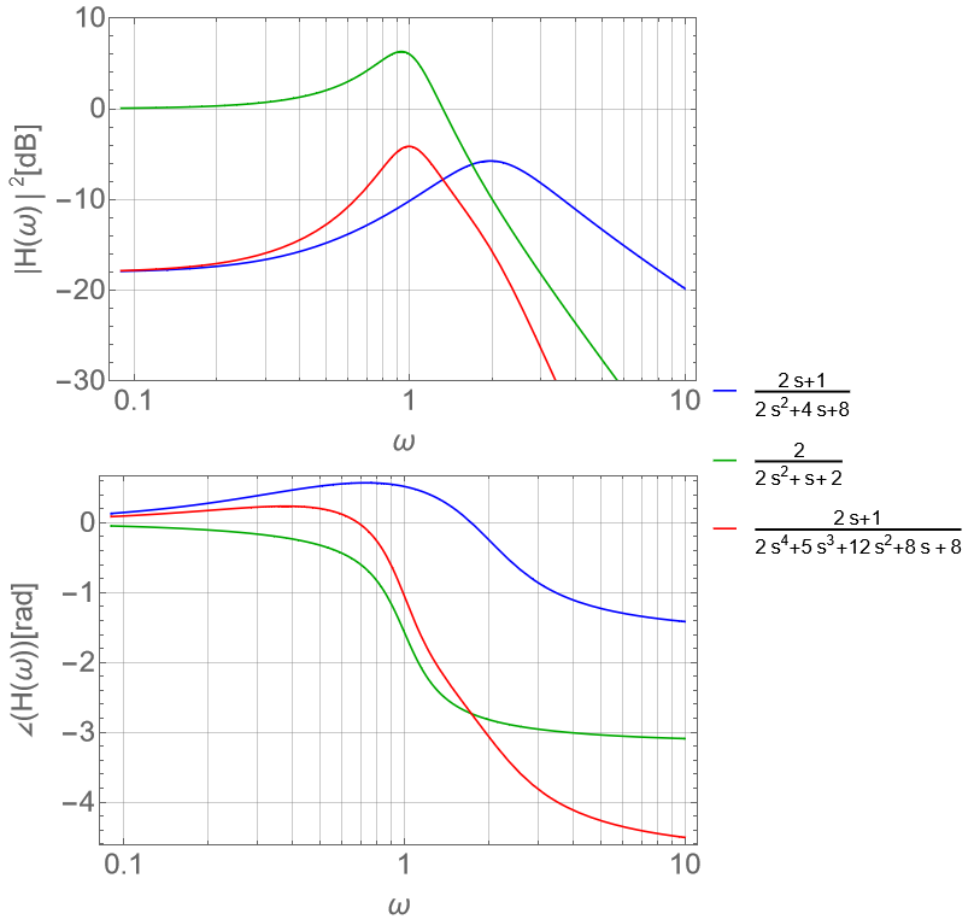


Figure 5.7: Cascade of two transfer functions $T_t(s) = T_1(s)T_2(s)$ results in adding of the Bode plots, both for the magnitude as well as for the phase.

5.2.6 Characteristic times and frequencies in Bode plots

The fact that we can write a transfer function as the quotient of two polynomials in s with real coefficients; see equation 3.8 or 4.20:

$$T(s) = \frac{\sum_{i=0}^{m} b_i s^i}{\sum_{j=0}^{n} a_j s^j} \tag{5.30}$$

with $m \leq n$, gives us the opportunity to factor the two polynomials:

$$T(s) = \prod_{i=1}^{m}(s - z_i) \prod_{j=1}^{n}(s - p_j)^{-1} \tag{5.31}$$

with z_i the zeroes of the transfer function and p_j the poles of the transfer function. These zeroes and poles contain the characteristics of the system. From the analysis of first-order and second-order system we know that there are two options: either poles and zeroes are real and can be associated with inverse time constants, or the poles and zeroes come in complex conjugated pairs, that can be associated with a natural frequencies and corresponding damping factors. We define the time constants as:

$$\tau_{z,i} \equiv -\frac{1}{z_i}$$

$$\tau_{p,j} \equiv -\frac{1}{p_j}$$

being the characteristic times of the system, and

$$\omega_{z,n,i} \equiv \sqrt{z_i z_i^*}$$
$$\omega_{p,n,j} \equiv \sqrt{p_j p_j^*}$$

being the characteristic natural frequencies of the system with the corresponding damping factors:

$$\zeta_{z,i} \equiv -\frac{z_i + z_i^*}{2\sqrt{z_i z_i^*}}$$
$$\zeta_{p,j} \equiv -\frac{p_j + p_j^*}{2\sqrt{p_j p_j^*}}$$

Now we have defined the characteristic times, frequencies and damping factors, we can identify these in the Bode plots. The natural frequencies can be found on the frequency axis in the Bode plot and give a maximum in the magnitude plot for a pole and a minimum for a zero. The corresponding damping factors determine the hight/depth and width of the peak. The change in slope is $\pm 12\ \mathrm{dB/oct}$; where the positive sign is for zeroes and the negative sign for poles. The characteristic times can be found as the inverse of the times on the ω axis of the Bode plot. At these locations we find a change in slope of the curve with an amount of $\pm 6\ \mathrm{dB/oct}$, where the plus sign is for a zero and the minus sign for a pole. In the phase plot of the Bode plot, the phase change is $+\pi/2$ for a single zero and $-\pi/2$ for a single pole. These values become $+\pi$ for a natural frequency related to zeroes and $-\pi$ at a natural frequency related to poles. For both resonances holds that the the smaller the damping factor ζ the faster the phase change at the natural frequency.

5.2.7 Measuring the transfer function

The use of Bode plots can also be useful in the identification of unknown systems. By probing a system with a suitable signal we have a means to identify the system in terms of the transfer function, and therefore the input-output differential equation.
A straightforward way to measure a Bode plot is of course to measure the magnitude and the phase difference between the input and output at various frequencies. Another method is to use an input function $x(t)$ that yields an output signal that echoes the transfer function

$$T(s) = \frac{Y(s)}{X(s)} \tag{5.32}$$

If the input is chosen such that $X(s) = 1$, the Laplace transform of the output signal $y(t)$ is simply equal to the actual transfer function $T(s)$. The inverse Laplace transform of 1 is the impulse function, $\delta(t)$ and thus the response to an impulse function input provides the transfer function in a straightforward manner. This is a very elegant result. The reason for this, which we will explore in detail in Part 3, is that the impulse function can be regarded as a summation of sinusoids of all frequencies with equal weight. In practice any impulse generated in the lab is not infinitely sharp in time and therefore only contains frequencies up to its inverse duration.

5.3 Nyquist plot

– Optional in part I – Essential in part II –
Another way to visualize the transfer function is via a so called Nyquist plot. The Nyquist plot is parametric plot of the transfer function in the complex plane with the variable ω as parameter. It plots the points $(\mathrm{Re}(H(\omega)), \mathrm{Im}(H(\omega)))$ as a function of ω in the complex plane. In figure 5.8 an example of a Nyquist is given for the second order system (in Laplace notation):

$$T(s) = \frac{1}{s^2 + 2s + 1}$$

Note that the transfer function starts in $H(\omega)|_{\omega=0} = 1$ and end in $H(\omega)|_{\omega=\infty} = 0$. Note that the parametric curve approaches zero from the negative real axis, due to the fact that:

$$H(\omega) \approx -\frac{1}{\omega^2} \quad \omega \gg 1$$

This corresponds to a phase dependence starting from $\phi(0) = 0$ to $\phi(\infty) = -\pi$.

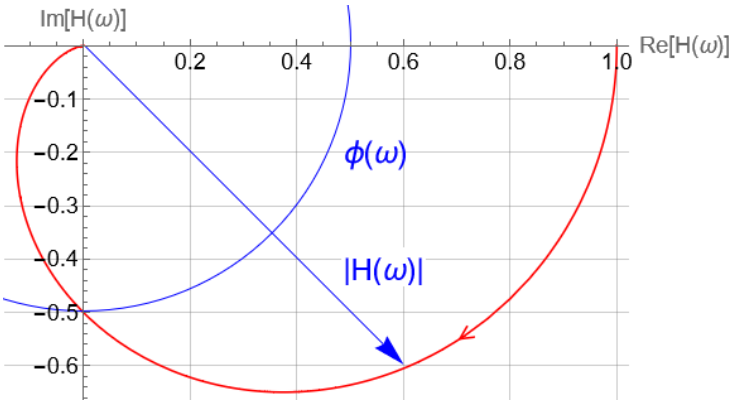


Figure 5.8: Nyquist of a second order system with annotation of magnitude and phase of the transfer function.

An way to look at the Nyquist plots is by imagining a vector from the origin to a position on the parametric curve, as indicated in blue in the figure. The length of the vector represents the magnitude of the transfer function for that specific value of ω, where as its angle with the positive real axis represents the phase shift for that specific value of ω.
In this specific case, the phase angle was set to $\phi(\omega) = -\pi/4$. The corresponding value of ω of that phase angle can be found by solving for equal real and imaginary values but opposite signs of the transfer function:

$$T(s) = \frac{1}{s^2 + 2s + 1} \;\Rightarrow\; H(\omega) = \frac{(1-\omega^2) - 2\omega \mathrm{j}}{\omega^4 + 2\omega^2 + 1}$$

This is true for a value of $\omega = -1 + \sqrt{2}$. Plugging this value back in the transfer function gives:

$$H(\omega)|_{\omega=-1+\sqrt{2}} = \frac{1}{4}(1-\mathrm{j})\left(1+\sqrt{2}\right) \approx 0.60(1-\mathrm{j})$$

Which gives a magnitude of $|H(\omega)| \approx 0.85$, which is equivalent with $-1.4\,\mathrm{dB}$.
Verify these numbers in figure 5.8.

The Nyquist plot does not add much at this moment, however it becomes important in the treatment of feed-back systems and the tuning of them. It is a convenient indication for the stability of the stability of a feed-back system.
For completeness lets look at an example of a Nyquist plot for a combination of two transfer functions.

Example 5.3.1. *Continuation of example 5.2.1*
At this point we pick up example 5.2.1 to draw the Nyquist plots of these three transfer functions. In figure 5.9 the three Nyquist plots of example are presented.
The limit behaviour can be directly compared from the values of the example. In order to compare the plots with the transfer functions, we express the transfer functions in the sum of a real and

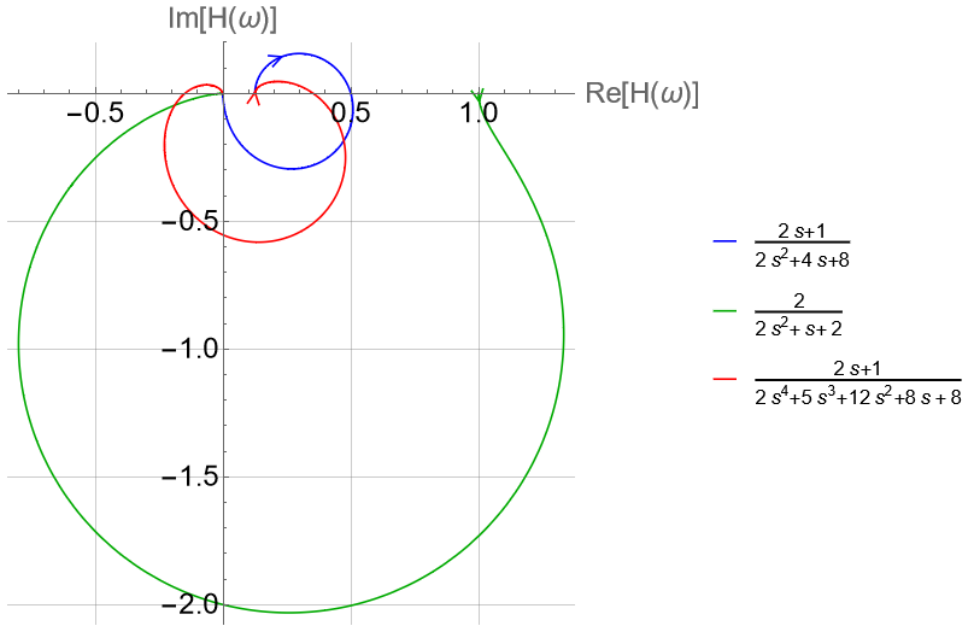


Figure 5.9: Nyquist of a cascade of two systems from example 5.2.1. The third one (red) is the cascade of the first two: blue and green.

imaginary part to be able to check the axes crossings of the curves:

$$H_1(\omega) = \frac{6\omega^2+8}{4\omega^4-16\omega^2+64} + \frac{-4\omega^3+12\omega}{4\omega^4-16\omega^2+64}\mathrm{j}$$
$$H_2(\omega) = \frac{-4\omega^2+4}{4\omega^4-7\omega^2+4} + \frac{-2\omega}{4\omega^4-7\omega^2+4}\mathrm{j}$$
$$H_t(\omega) = \frac{-8\omega^4+4\omega^2+8}{4\omega^8-23\omega^6+96\omega^4-128\omega^2+64} + \frac{4\omega^5-19\omega^3+8\omega}{4\omega^8-23\omega^6+96\omega^4-128\omega^2+64}\mathrm{j}$$

Crossings of the real axis are obtained by setting the imaginary part of the transfer functions to zero; leading to:

$$-4\omega^3+12\omega = 0 \;\Rightarrow\; \omega = 0 \;\vee\; \omega = \pm\sqrt{3}$$
$$-2\omega = 0 \;\Rightarrow\; \omega = 0$$
$$4\omega^5-19\omega^3+8\omega = 0 \;\Rightarrow\; \omega = 0 \;\vee\; \omega = \pm 0.68 \;\vee\; \omega = \pm 2.07$$

Substituting these values back in the transfer function gives us the values of intersection of the real axis:

$$H_1(0) = \frac{1}{8} \qquad H_1(\sqrt{3}) = \frac{1}{2}$$
$$H_2(0) = 1$$
$$H_t(0) = \frac{1}{8} \qquad H_t(0.68) = 0.35 \qquad H_t(2.07) = -0.15$$

Crossings of the imaginary axis are obtained by setting the real part of the transfer functions to zero; leading to:

$$6\omega^2+8 = 0 \;\Rightarrow\; \omega = \{\,\} \quad \textit{no axis crossings!}$$
$$-4\omega^2+4 = 0 \;\Rightarrow\; \omega = \pm 1$$
$$-8\omega^4-4\omega^2+8 = \pm 1.13$$

Substituting these values back in the transfer function gives us the values of intersection of the real axis:

$$\begin{aligned} H_1(\ldots) &= \{\ \} \quad \textit{no axis crossings!} \\ H_2(1) &= -2\mathrm{j} \\ H_t(1.13) &= -0.55\mathrm{j} \end{aligned}$$

All these values can be found in figure 5.9. From these checks we conclude that the graphs are good graphical representations of the transfer functions in the complex plane.
A small note to the calculations above: Although negative values and complex values of ω are obtained in solving the equations, these are no real solutions and are therefore ignored in finding the axes crossings.

□

5.4 Numerical evaluation and graphical representations of transfer functions

The transfer function has reformulated the mathematical model of an LTI system from an Ordinary Differential Equation with constant coefficients into an algebraic, frequency-dependent formula. Such a formula can be easily evaluated and programming languages have extensive toolboxes for the manipulation (addition, multiplication etc.) and evaluation of transfer functions (Bode diagram, step response, arbitrary response, pole derivation). Since we are working with three kind of programs, reference to useful functions are given for:

MATLAB Table D.1 and D.3

PYTHON Table E.1

MATHEMATICA Table F.3

In these programmes, transfer function objects can be defined that can be visualized and on which various functions can be applied.

Summaries

Summary Chapter 5

Frequency analysis and Bode-plots

- Figure 5.5 is part of this summary. The Bode-plots correspond to the systems for which the step responses are shown in figure 2.12
- The quality factor of a second-order system is related to the damping factor and is defined as:

$$Q \equiv \frac{1}{2\zeta} \tag{5.33}$$

- The width (full-width-half-maximum) of the absolute value of the transfer function $|H(\omega)|$ is also related to the damping factor:

$$\Gamma_\omega = \Delta\omega = \frac{\omega_n}{Q} = 2\zeta\omega_n \tag{5.34}$$

Problems

Problem 5.1. Bode plot
Figure 5.10 shows the response of a second order system.

a. How does the plot show that this is a second order system.

b. Determine the natural frequency and the damping ratio of this system.

c. Derive the transfer function of this system. Verify your answer.

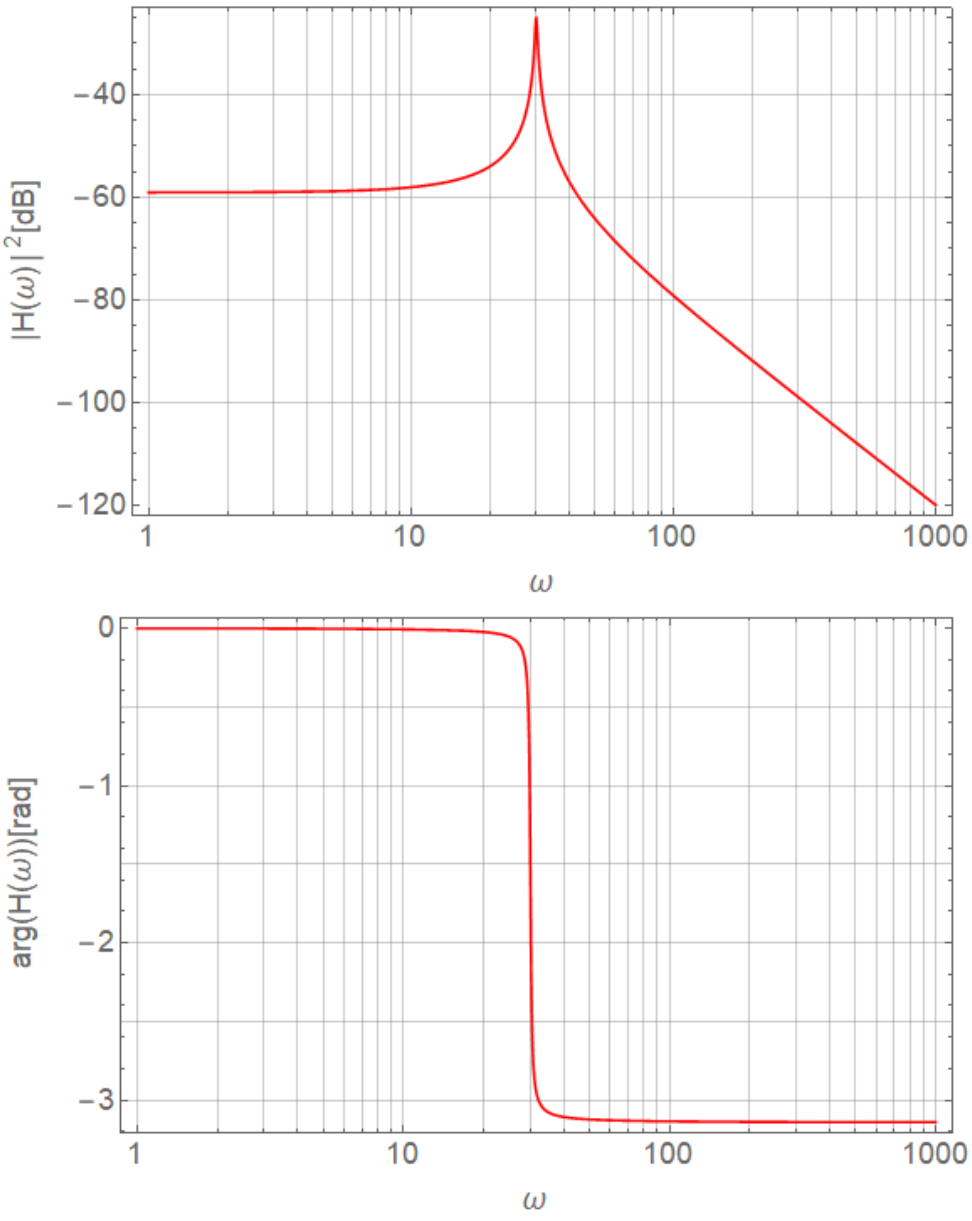


Figure 5.10: Bode plot of second order system.

Problem 5.2. LCR-permutation *(0:45h)*
In question 2.4 from chapter 2 you derived the input-output equation of three LCR networks.

a. Derive the transfer function of these networks.

b. Plot the Bode diagrams of these networks. (You can also use your computer to do this)

c. Plot the step response of these networks. (You can also use your computer to do this)

d. Describe the difference and similarities of these networks.

Problem 5.3. Feedback system
Consider a negative feedback system as depicted in fig. 4.4 and described in eq. 4.27 with subsystem G and H with

$$G(s) = \frac{s+1}{s^2+4s+3}$$

and

$$H(s) = \frac{1}{s+1}$$

a. Derive the closed loop transfer function of the negative feedback system in two ways, by hand and with a computer. (Hint: in general the programs MATLAB, MATHEMATICA and PYTHON do not provide the most compact answer. All programs have their own method to reduce the answer tot the most compact form. The programs also provide methods to represent the transfer function in zero–pole representation, which might be a more convenient result for interpretation.).

b. Derive the poles of G, H and T_{CL}.

c. Plot the Bode diagrams of the feedback network.

Problem 5.4. Cascade of three first-order systems
Consider a cascade of three equal first order systems (which means they are placed behind each other), of which the transfer function of a single first order system is given by:

$$G_1(s) = \frac{1}{s+1}$$

a. Determine the total transfer function of the cascaded circuit named G_3.

b. Determine the poles of the transfer functions.

c. Construct the response of this system on a unit-step input.

d. Consider a negative feedback of this cascaded circuit with $H = 1$ and $G = KG_3$, with K a real gain parameter. Make a schematic drawing of the system.

e. Derive the transfer function of the negative feedback system T_{CL}, with K as a parameter.

f. For the values $K = 1, 8, 27$ calculate the poles of the transfer function. What can be said about the behaviour of these systems as a function of K?

g. Give the range of K-values for which the system has a asymptotically stable output.

h. Give the step response for K=1.

i. Give the response of the system for $K = 8$ as the result of a 1 second long sawtooth pulse (signal increases linearly from 0 to $0.5\,\mathrm{s}$ to $1\,\mathrm{V}$ and decreases again linearly from 0.5 to $1\,\mathrm{s}$ to $0\,\mathrm{V}$.

Problem 5.5. Transfer function calculations with the computer
For Problems 5.2 to 5.4 define the transfer functions in either MATLAB, MATHEMATICA or PYTHON and determine:

a. The poles and zeros of the transfer function.

b. The Step-response of the system.

c. The Impulse-response of the system.

d. The Bode plot of the system.

Hint: use the appendices and files supplied on Canvas for specified information on the functionality of the various programs.

Problem 5.6. Location of characteristics in Bode plots
For example 5.2.1 write the transfer function in the form of equation 5.31, determine all poles and zeros en translate those to characteristic times, frequencies and damping factors. Indicate these quantities in the Bode plots of figure 5.7, both in the magnitude as well as in the phase plot.

Chapter 6

State-Space representations

In the previous chapters, a systematic approach for modelling systems that are made up of energy storing and dissipating elements was introduced. This approach starts with the identification of the constituting elements and linking them together by applying conservation and continuity. The basic result is a set of first order differential equations which through mathematical manipulation can be combined to a single input-output equation. Laplace transforms can be used to limit the amount of manipulation. The motivation for this effort is that analysis of the input-output equation, especially the homogeneous part, provides detailed information on the system response. The same information can however be computed with less manipulation effort using the State Model formalism that will be outlined in this chapter. This formalism is not only the basis for efficient simulation routines of dynamical systems, but can also stress the importance of the co-called State-variables, variables whose values at any given time provide detailed information about the system studied and how it will evolve in time.
In order to make this chapter more readable, a few notation conventions are made.

- All variables are functions of time, so the time dependence is dropped in the notation: $q(t) \rightarrow q$.
- Time derivatives with respect to time are represented with a dot above the variable: $\mathrm{d}q/\mathrm{d}t \rightarrow \dot{q}$.
- The various state variables in a system, including both input and output variables, can be collected in a state vector in which the elements are the state variables: $\{q_i\} \rightarrow \mathbf{q}$.

In order to be consistent with the rest of the document, we use the notation x_i for input variables, y_i for output variables and q_i for state variables.

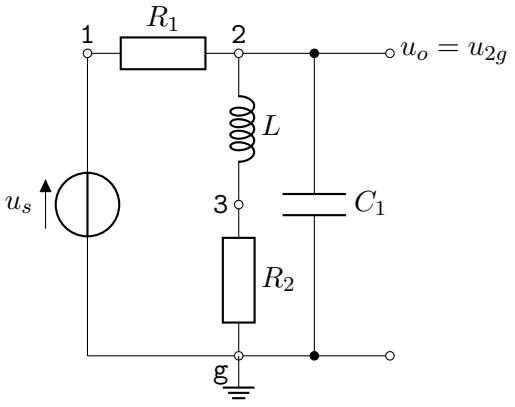


Figure 6.1: LCR network with inductor, capacitor and two resistors.

6.1 Matrix representation

Instead of deriving a single input-output equation for a physical model, it is more conveniant to collect a set of first-order differential equations that are coupled. From the *Models* cource we take the sytem presented in figure 6.1. The capacitor and the inductor are the only two components that have a differenitial relation between the voltage and the current. The variables in the differential are the variables that determine the state of the system: $u_{2,g}$ and i_L. The set of two equations can be written in a form where the differentail terms are only functions of the variables in the differentials and input variables. This leads to:

$$C\frac{\mathrm{d}u_{2,g}}{\mathrm{d}t} = \frac{1}{R_1}\left[u_s - u_{2,g}\right] - i_L \tag{6.1}$$

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = u_{2,g} - R_2 i_L. \tag{6.2}$$

This is a set of coupled first order differential equations. Besides the variables $u_{2,g}$ and i_L obviously also the external input u_s enters the set of equations. For analysis it is beneficial to have the derivatives explicit as a function of the rest, so we divide the two equations by C and L respectively to obtain:

$$\frac{\mathrm{d}u_{2,g}}{\mathrm{d}t} = -\frac{1}{R_1 C}\,u_{2,g} - \frac{1}{C}\,i_L + \frac{1}{R_1 C}\,u_s \tag{6.3}$$

$$\frac{\mathrm{d}i_L}{\mathrm{d}t} = \frac{1}{L}\,u_{2,g} - \frac{R_2}{L}\,i_L. \tag{6.4}$$

This set of equations can be formulated in a matrix notation. The first step is the introduction of the state vector $\mathbf{x}$ that incorporates $u_{2,g}$ and i_L, the variables that at any moment in time define the state of the system, i.e. how much energy is in the network and stored in which component, how much power is dissipated in each resistor, what is the voltage over the inductor and what is the current through the resistor etc. Of course obtaining the actual values of all these requires a calculation but they can always be expressed in terms of $u_{2,g}$, i_L, the input voltage and parameter values of the components involved.
Let q

$$\mathbf{q} = \begin{pmatrix} u_{2,g} \\ i_L \end{pmatrix}$$

and let $\mathbf{x}$ denote the input, $x = u_s$. Then (6.3)–(6.4) is of the form

$$\dot{\mathbf{q}} = f(\mathbf{q}, \mathbf{x}, t) \tag{6.5}$$

for some vector-valued function f. The right-hand side of (6.3)–(6.4) does not depend explicitly on time so we could have skipped the third argument of f and written instead $\dot{\mathbf{q}} = f(\mathbf{q}, \mathbf{x})$. Often we are not interested in the entire state $\mathbf{q}$ but only in some of its components or some combination of $\mathbf{x}$ and $\mathbf{q}$, for example to evaluate the voltage drop over resistor 1. This output signal $\mathbf{y}$ that we are interested in can be written in the form

$$\mathbf{y} = h(\mathbf{q}, \mathbf{x}, t) \tag{6.6}$$

for some known function h. We call $\mathbf{y}$ the *output* of the system. Note that although the value of $\mathbf{y}$ may be determined by only the value of $u_{2,g}$ the full knowledge of $\mathbf{q}$ at any time is required as a result of the connection between the various ODE's that together determine the function f.
In our example the state vector has two components. In general $\mathbf{q}$ can be a vector with many components. With these vectorial definitions the set of first order differential equations can be rewritten in a matrix notation:

$$\begin{pmatrix} \frac{\mathrm{d}u_{2,g}}{\mathrm{d}t} \\ \frac{\mathrm{d}i_L}{\mathrm{d}t} \end{pmatrix} = \begin{pmatrix} \frac{-1}{R_1 C} & \frac{-1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{pmatrix} \begin{pmatrix} u_{2,g} \\ i_L \end{pmatrix} + \begin{pmatrix} \frac{1}{R_1 C} \\ 0 \end{pmatrix} u_s.$$

or, more succinctly,

$$\dot{\mathbf{q}} = \mathbb{A}\mathbf{q} + \mathbb{B}\mathbf{x}$$

with

$$\mathbf{q} = \begin{pmatrix} u_{2,g} \\ i_L \end{pmatrix}, \qquad \mathbf{x} = u_s \qquad \mathbb{A} = \begin{pmatrix} \frac{-1}{R_1 C} & \frac{-1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{pmatrix}, \qquad \mathbb{B} = \begin{pmatrix} \frac{1}{R_1 C} \\ 0 \end{pmatrix}$$

The matrix $\mathbb{A}$ is square with as many rows and columns as $\mathbf{q}$ has entries. Here $\mathbb{B}$ is a matrix as well. Typically it is a column vector for only one input signal. Suppose we are only interested in $y = u_{2,g}$. This is linear in state (and input) and can be written in matrix notation as

$$\mathbf{y} = \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{q}.$$

6.2 General formulation of the state-space model

The most general form of state representation considered in this course is

$$\dot{\mathbf{q}} = f(\mathbf{q}, \mathbf{x}, t)$$
$$\mathbf{y} = h(\mathbf{q}, \mathbf{x}, t).$$

Time-invariant systems that today behave the same as they did yesterday and as they will tomorrow, the explicit time dependence can be left out, which gives.

$$\dot{\mathbf{q}} = f(\mathbf{q}, \mathbf{x})$$
$$\mathbf{y} = h(\mathbf{q}, \mathbf{x}).$$

In case the functional's f and h are linear functions, these equations can be mapped to a Linear Algebra (or matrix) formulation, then we can write, as we did in our previous LCR-network, a matrix representation, called a State Model

$$\dot{\mathbf{q}} = \mathbb{A}\mathbf{q} + \mathbb{B}\mathbf{x} \tag{6.7}$$
$$\mathbf{y} = \mathbb{C}\mathbf{q} + \mathbb{D}\mathbf{x} \tag{6.8}$$

with matrices $\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}$ where the elements are constant (i.e. they do not depend on time).
The virtue of the State Model is that it simplifies calculations. The only requirement on the model of an LTI system (Linear, Time-invariant and Causal) is a formulation as a set of first order differential equations. All other processes like the evaluation of the characteristic polynomial, the derivation of the input-output equation (which is the weighted addition and subtraction of the various first order DE's) etc. can be described as matrix manipulations known from Linear Algebra.
Before we solve the state-space equations we first look for the solution in steady state.

6.2.1 Steady state solution

The stationary state solution of the state space equations for a given steady state input signal $\mathbf{x_{ss}}$ is found be setting all time derivatives to zero: The first of these is characteristics is the stationary situation at a given input signal $\mathbf{x_{ss}}$. I any stationary situation, all time derivatives are zero:

$$\frac{\mathrm{d}\mathbf{q}}{\mathrm{d}t} = 0 = \mathbb{A}\mathbf{q} + \mathbb{B}\mathbf{x_{ss}}.$$

In the stationary situation the state vector $\mathbf{q}$ has the value $\mathbf{q_{ss}}$

$$\mathbb{A}\mathbf{q_{ss}} = -\mathbb{B}\mathbf{x_{ss}}$$

From this the value of the state vector can be evaluated applying Linear Algebra

$$\mathbf{q_{ss}} = -\mathbb{A}^{-1}\mathbb{B}\mathbf{x_{ss}}.$$

Substitution of this result in the output equation gives us the steady state values of the output:

$$\mathbf{y}_{ss} = -\mathbb{C}\mathbb{A}^{-1}\mathbb{B}\mathbf{x}_{ss} + \mathbb{D}\mathbf{x}_{ss} = \left(\mathbb{D} - \mathbb{C}\mathbb{A}^{-1}\mathbb{B}\right)\mathbf{x}_{ss} \tag{6.9}$$

The steady state output can be calculated directly from the state-space matrices applied on the steady state input.

6.3 Solve State-Space equations: Laplace transform

For finding the solutions to the state space equations we use the technique we already know: Apply the Laplace Transform on these equations to obtain algebraic equations. The fact that the state space equations are written in a matrix form, makes it even more easy.
In section 4.1.2 we have already seen that the Laplace transform was applied on a set of first-order differential equations. The resulting set of linear equations was subsequently written in matrix form to be solved. The solutions obtained solutions are the transfer functions.
The approach here is not very different, however there are a few subtle differences. Starting point is writing the set of first order differential equations in matrix form and subsequently apply the Laplace transform on the obtained matrix equation. Here we find the first significant difference in approach: due to the fact that we use a kind of intermediate variables: the state space variables. The state variables give a direct view on the state of the system, unlike the output variables. Due to this separation between state variables and output variables, we need an additional set of linear equations that calculate the output variables from these state variables. This extra step pays back itself in the interpretation of the results: the state variables give complete information about the state of the system, including its dynamics.
From the State space model the set of Transfer functions can be directly derived in the form of a matrix. The Laplace transform of the State-Space equations 6.8 (we set all initial conditions to zero; otherwise an extra term $+\,\mathbf{q}|_0$ comes into the equation on the right-hand side) gives:

$$s\mathbf{Q} = \mathbb{A}\mathbf{Q} + \mathbb{B}\mathbf{X} \tag{6.10}$$
$$\mathbf{Y} = \mathbb{C}\mathbf{Q} + \mathbb{D}\mathbf{X} \tag{6.11}$$

The first equation we rewrite to:

$$(s\mathbb{I} - \mathbb{A})\mathbf{Q} = \mathbb{B}\mathbf{X}$$

From this equation we can extract $\mathbf{Q}$ and substitute the result in the output equation in Laplace form:

$$\mathbf{Q} = (s\mathbb{I} - \mathbb{A})^{-1}\mathbb{B}\mathbf{X}$$
$$\mathbf{Y} = \mathbb{C}(s\mathbb{I} - \mathbb{A})^{-1}\mathbb{B}\mathbf{X} + \mathbb{D}\mathbf{X}$$

The general transfer function is defined as

$$\mathbb{G}(s) = \mathbb{C}(s\mathbb{I} - \mathbb{A})^{-1}\mathbb{B} + \mathbb{D} \tag{6.12}$$

In fact $\mathbb{G}(s)$ is a matrix of transfer functions; the elements of the matrix are transfer functions for every input-output combination. Therefore we could not write $\mathbb{G}(s)$ in terms of the input and output vectors, however we can formulate it for a individual element from the matrix:

$$G_{ij} = \{\mathbb{G}(s)\}_{i,j} = \frac{Y_i(s)}{X_j(s)}$$

which are in fact the individual transfer functions between input $X_j(s)$ and output $Y_i(s)$.
We can make a more direct comparison with the transfer function calculation by assuming that the output variables are the same as the state-space variables:

$$\mathbf{Y} = \mathbb{I}\mathbf{Q} = \mathbf{Q}$$

Calculating the transfer functions in this situation gives:

$$\mathbb{G}(s) = (s\mathbb{I} - \mathbb{A})^{-1}\mathbb{B}$$

which is equivalent with equation 4.8; strictly speaking for the case: $\mathbf{A} = -\mathbb{A}$, which has as a consequence that the eigenvalues are equal in magnitude, however with opposite signs for the two matrices. The eigenvectors are the same for $\mathbf{A}$ and $\mathbb{A}$.
This formulation is a matrix equation and does hold for a single input, single output (SISO) system, where it reduces to a single transfer function as we have seen in chapter 4. Now we have a more general form of this single equation. It directly describes the Transfer function for a Multiple Input Multiple Output (MIMO) system with each matrix element the Transfer function coupling the specific input-output given by the row-column index of the specific matrix element.
Let us illustrate this with an example.

Example 6.3.1. *Application to a second order two-input two-output system*
To illustrate the effect of equation 6.12, we illustrate this on an two-input, two-output MIMO-system. We start from a given state-space system and calculate the matrix of transfer functions, that relates the inputs to the outputs.
Consider the following set of State-space equations:

$$\begin{pmatrix}\dot{q}_1\\ \dot{q}_2\end{pmatrix} = \begin{pmatrix}-1 & -2\\ 3 & -2\end{pmatrix}\begin{pmatrix}q_1\\ q_2\end{pmatrix} + \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \tag{6.13}$$
$$\begin{pmatrix}y_1\\ y_2\end{pmatrix} = \begin{pmatrix}2 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}q_1\\ q_2\end{pmatrix} + \begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \tag{6.14}$$

First we calculate the steady state solution for a steady state input $\mathbf{x}_{ss}$ by applying equation 6.9:

$$\begin{pmatrix}y_{1,ss}\\ y_{2,ss}\end{pmatrix} = \begin{pmatrix}\frac{1}{2} & -\frac{3}{8}\\ \frac{1}{2} & \frac{1}{8}\end{pmatrix}\begin{pmatrix}x_{1,ss}\\ x_{2,ss}\end{pmatrix}$$

In order to derive the characteristics from the states-space equations we apply equation 6.12 to write this state-space model set of four transfer functions; every combination of an input with an output gives an individual transfer function. The best representation is a matrix form:

$$G(s) = \begin{pmatrix} \dfrac{3s+4}{s^2+3s+8} & \dfrac{s-3}{s^2+3s+8} \\ \dfrac{s+4}{s^2+3s+8} & \dfrac{s+1}{s^2+3s+8} \end{pmatrix} \tag{6.15}$$

This matrix transfers the input vector $\mathbf{x}$ to the output vector $\mathbf{y}$:

$$\begin{aligned} \mathbf{y} &= G(s)\mathbf{x} \\ \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} &= \begin{pmatrix} \dfrac{3s+4}{s^2+3s+8} & \dfrac{s-3}{s^2+3s+8} \\ \dfrac{s+4}{s^2+3s+8} & \dfrac{s+1}{s^2+3s+8} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \end{aligned} \tag{6.16}$$

Note that the denominator of all four transfer functions is the same. This is not surprising, since the dynamics of the system is determined by the eigenvalues of the $\mathbb{A}$-matrix. These eigenvalues correspond to the poles of the transfer function. The eigenvalues of the $\mathbb{A}$-matrix, and therefore also the poles, are:

$$s_{1,2} = -\frac{3}{2} \pm \frac{1}{2}\sqrt{23}\mathrm{j} \tag{6.17}$$

The verification of this is left as an exercise.
Note that the steady state solution can also be found from the transfer function matrix by taking the limit for s to zero and apply this on the steady state input $\mathbf{x}_{ss}$:

$$\mathbf{y}_{ss} = \lim_{s\to 0} \begin{pmatrix} \dfrac{3s+4}{s^2+3s+8} & \dfrac{s-3}{s^2+3s+8} \\ \dfrac{s+4}{s^2+3s+8} & \dfrac{s+1}{s^2+3s+8} \end{pmatrix} \mathbf{x}_{ss} = \begin{pmatrix} \frac{1}{2} & -\frac{3}{8} \\ \frac{1}{2} & \frac{1}{8} \end{pmatrix} \mathbf{x}_{ss}$$

which is also obvious from the equivalent of setting all time-derivatives to zero is equivalent with setting all complex frequencies s set to zero; property 2 and 3 of the Laplace transform; see table 3.1.
A Bode plot of the transfer functions is given in figure 6.2
From the results presented in figure 6.2 it is obvious that the dynamics of the system is reflected in all four transfer functions. The location of the resonance, the width of the resonance (related to the damping factor) and the limit behaviour are all very similar. This is not surprising since all dynamics is contained in the $\mathbb{A}$-matrix.

□

Although these calculations can be done by hand, in almost all cases you leave this to the computer. The programmes MATLAB, PYTHON (library SCIPY.SIGNAL) and MATHEMATICA have functions to make this transformation; see appendices D, E and F respectively.

6.4 Relating characteristics

Although we know that the state-space formalism is in fact nothing more than the rewriting of a set of first-order linear time-invariant differential equation in matrix form. However in doing so, we can immediately apply the toolbox of linear algebra in the analysis of these systems.
So let us look at the state space equations from a different perspective. Assume we have no input and the system is set into motion by means of initial conditions. In that case, the dynamic par of the equation becomes:

$$\dot{\mathbf{q}}(t) = \mathbb{A}\mathbf{q}(t) \tag{6.18}$$

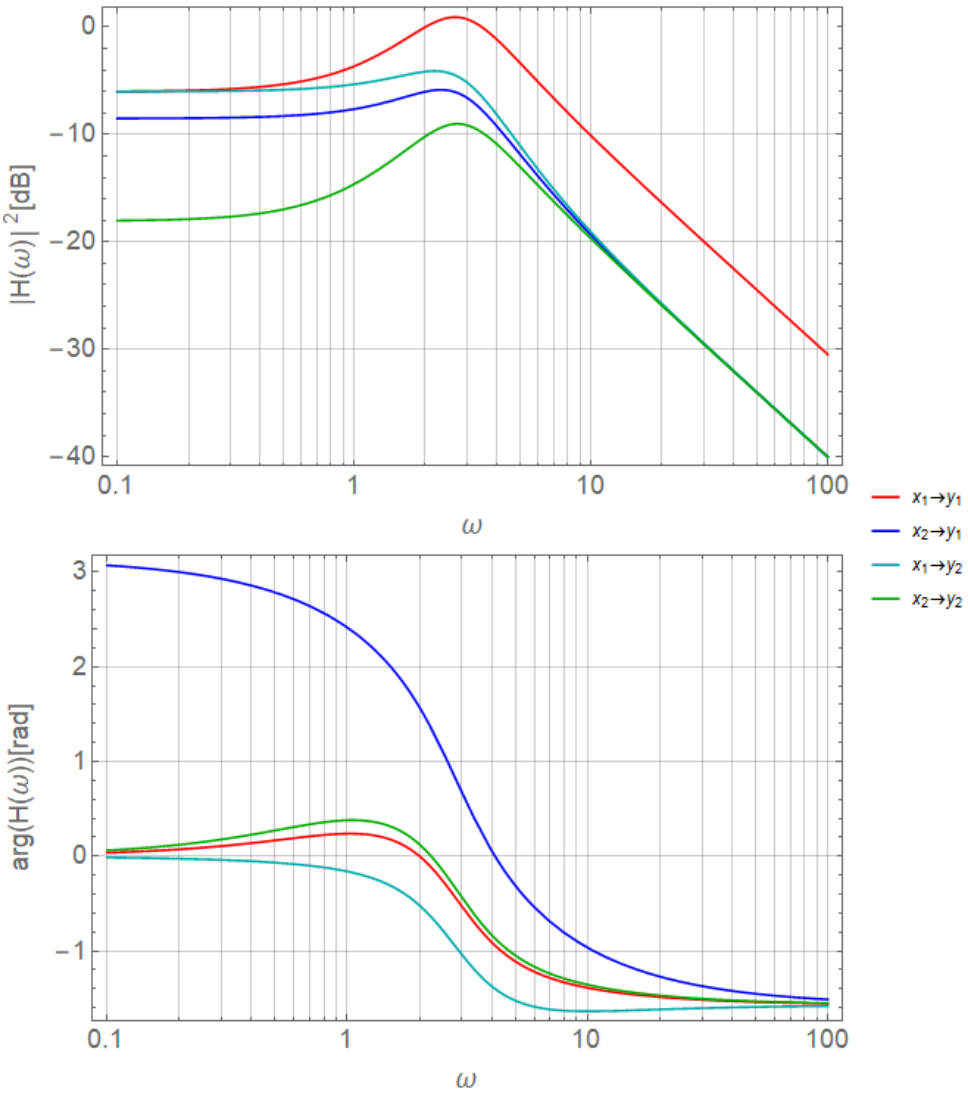


Figure 6.2: Bode-plots of the transfer functions from equation 6.15.

Applying the Laplace transform gives us:

$$s\mathbf{Y}(s) - \mathbf{q}(0) = \mathbb{A}\mathbf{Y}(s) \tag{6.19}$$

which results in:

$$\mathbf{Y}(s) = (s\mathbb{I} - \mathbb{A})^{-1}\,\mathbf{q}(0) \tag{6.20}$$

Calculating the inverse matrix of $(s\mathbb{I} - \mathbb{A})$ involves the determinant of that matrix, which becomes the eigenvalue equation of matrix $\mathbb{A}$ if we set this determinant to zero:

$$|s\mathbb{I} - \mathbb{A}| = 0 \tag{6.21}$$

Since this determinant appears in the denominator of the solution for $\mathbf{Y}(s)$, it gives us the poles of the transfer function as we have seen in section 4.1.2.
At this point we can draw a very strong conclusion: *The poles of the transfer function are the same as the eigenvalues of the $\mathbb{A}$-matrix*. we can make it even stronger: the poles of the transfer function, the eigenvalues of the $\mathbb{A}$-matrix and the roots of the characteristic equation of the input-output differential equation, are all the same.
So, as we already announced, the matrix formalism pays off: by writing the model in terms of coupled first-order differential equations and transfer them to a matrix form, gives us a short route to the dynamic characteristics of the system by calculating the eigenvalues of the $\mathbb{A}$-matrix.

6.5 Simulation of response using the state representation

In chapter 2 the numerical evaluation of a time response was introduced. The generalization to a matrix method of this scheme is quite natural and is already implicitly present in the formulation in that chapter. Certainly for LTI systems this is rather convenient as the state model matrices are constant. In that case the State model and the transfer function are just different representations. An important difference is that a transfer function usually represents a Single Input Single Ouput (SISO) system, wheras the State model provides a simple extension to Multiple Input Multiple Output (MIMO) models. Of course transfer function can also be defined for MIMO systems, defining a transfer function for each combination of input and output. Note however that inspection of eq. 6.12 indicates that all the transfer functions for a MIMO system have the same denominator, determined by the eigenvalues of the dynamic matrix $\mathbb{A}$. All kind of tools for defining a model within a computational programme as touched upon in chapter 4, Bode plot, Step response etc. are accessible with the formulation of the system response as a State Model.
For non-linear systems, the matrices are no longer constant, but will depend on the actual value of the state vector. This implies that tools from linear algebra to characterize a system will no longer be available. However the general formulation to evaluate a time response in the time domain (not using Laplace transform etc.) described in chapter 2 still allow evaluation of the response. The other possibility is to represent the non-linear system with a linear equivalent that has a comparable behaviour within the oprational range of the model. An example of the latter would be the behaviour of a car limited to a speed of $95 - 105\,\mathrm{km/h}$. In this range the effect of the wind speed at $100\,\mathrm{km/h}$ can well be used as an average.

Summaries

Summary Chapter 6

- State-space equations for inputs variables given by $\mathbf{x}$, output variables given by $\mathbf{y}$ and the state variables of the system by $\mathbf{q}$:

$$\frac{\mathrm{d}\mathbf{q}}{\mathrm{d}t} = \mathbb{A}\mathbf{q} + \mathbb{B}\mathbf{x} \tag{6.22}$$

$$\mathbf{y} = \mathbb{C}\mathbf{q} + \mathbb{D}\mathbf{x} \tag{6.23}$$

- The eigenvalues of the $\mathbb{A}$-matrix are the roots of the characteristic equation and the poles of the transfer function. Therefore all the dynamics of a system is in the $\mathbb{A}$-matrix.

$$\text{Det}|s\mathbb{I} - \mathbb{A}| = 0 \tag{6.24}$$

- The stationary solution is the situation in which all time derivatives are zero. With $\mathbf{x_{ss}}$ the value of the input variables in the stationary situation, the values of the state variables are at this point given by:

$$\mathbf{y}_{ss} = \left(\mathbb{D} - \mathbb{C}\mathbb{A}^{-1}\mathbb{B}\right)\mathbf{x}_{ss} \tag{6.25}$$

- Transfer function matrix is calculated from the Laplace transform and some linear algebra:

$$\mathbb{G}(s) = \mathbb{C}(s\mathbb{I} - \mathbb{A})^{-1}\mathbb{B} + \mathbb{D} \tag{6.26}$$

- Input-output ($\mathbf{X}(s) \rightarrow \mathbf{Y}(s)$; $\{x_i\} \rightarrow \{y_j\}$) relation in Laplace domain:

$$\mathbf{Y}(s) = \mathbb{G}(s)\mathbf{X}(s) \tag{6.27}$$

Problems

A-type problems

Problem 6.1. State-Space
Consider the following set of first order differential equations:

$$x'(t) + 3x(t) = u(t) \tag{6.28}$$
$$y'(t) + 2y(t) = \frac{3}{2}x(t) \tag{6.29}$$

where $u(t)$ is an input signal. The output for this system is $y(t)$.

a. Derive the State-Space model and write this in matrix notation.

b. Determine the characteristic equation from the A-matrix.

c. Derive the transfer function from the State-Space model.

d. Derive the transfer function model from the input-output equation, derived from the two second order differential equations.

e. Compare the results and identify the role of the characteristic equation.

Problem 6.2. LCR-permutation
In question 2.4 from chapter 2 you derived the input-output equation of 3 LCR networks.

a. Write a state model for each of these three networks. This includes the output equation.

b. Determine the eigenvalues of these three networks and compare the results.

B-type problems

Problem 6.3. State-Space
Consider the following set of differential equations:

$$x'(t) + ax(t) = by(t) + u_1(t) \tag{6.30}$$
$$y'(t) + cy(t) = -bx(t) + u_2(t) \tag{6.31}$$

In this description, $u_1(t)$ and $u_2(t)$ are the input signals. The system has two outputs given by:

$$z_1(t) = x(t) + y(t)$$
$$z_2(t) = x(t) - y(t)$$

a. Set-up the state-space model for this system and identify the four matrices.

b. Derive the characteristic equation for this system and relate the coefficients to the natural frequency ω_n and the damping factor ζ, according to equation 2.36.

c. Calculating the eigenvalues of the A-matrix. How do these relate to the poles of the system.

d. Give the general solution of the set of homogeneous equations, neglecting initial conditions.

Problem 6.4. Computer model
Consider problem 6.3.
In a computer model of the system, the characteristics and behaviour of the system is explored for four sets of coefficient values:

set 1: $\{a = 1,\, b = 1,\, c = 1\}$

set 2: $\{a = 2,\, b = 0,\, c = 2\}$

set 3: $\{a = \frac{1}{4},\, b = 1,\, c = \frac{1}{3}\}$

set 4: $\{a = 1,\, b = \sqrt{5},\, c = -1\}$

a. Make a computer model of the system in which you can easily set and change the parameter values. This model should be able to produce the answers to the questions below.

For all four sets of parameters calculate and produce the answers to the questions:

b. Make a State-Space representation of the system.

c. Calculate the eigenvalues.

d. Calculate the transfer functions.

e. Make the Bode plots and plot the traces of one set in one graph, *i.e.* four traces.

f. Calculate the step response for the four systems and plot the result for each input-output combination in one graph. (set the input that does not have the step input to zero).

g. Reflect on the results you obtained.

C-type problems

Problem 6.5. Electrical network
The electrical network depicted in figure 6.3 has electrical components with values $C_1 = C_2 = 0.1\,\mu\text{F}$, $L = 2\,\mu\text{H}$ and $R_2 = 1\,\Omega$. The value R_1 can be set between 0.5 and $8\,\Omega$. At $t = 0$s the input voltage is changed from 1 to 2V.

a. Derive the set of first order differential equations for this network. The voltage u_s is the input signal.

b. Write the set of State-equations in matrix form, with u_s as input and u_{R_2} and i_L as output variable.

c. Determine the characteristic properties of this system.

d. Use the characteristics of the system, in combination with the boundary conditions and the given step input, to construct the resulting output signal.

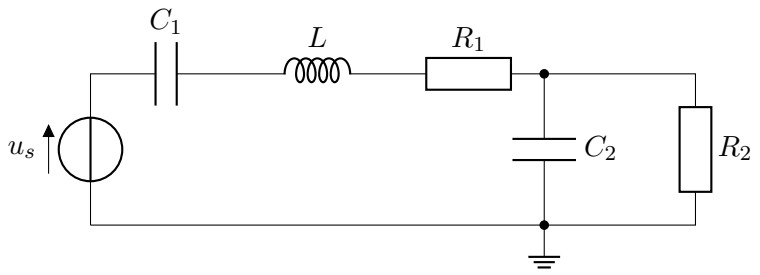


Figure 6.3: Electrical network with two resistors, two capacitors and an inductor.

Problem 6.6. State-Space *Question test 2019 (it only tests State Space knowledge, no mechanical or electrical modelling required)*
Consider the following set of differential equations:

$$y'(t) + 2y(t) = z(t) \tag{6.32}$$
$$z'(t) + 3z(t) = -y(t) + x(t) \tag{6.33}$$

In this description, $x(t)$ is the input and $z(t)$ is the output.

a. Set-up the state-space model for this system and identify the four matrices.

b. Derive the characteristic equation for this system.

c. Find the poles of this system.

Part II

Signal Analysis, Signal Processing and Filter Design

Chapter 7

Prologue to the Signals section

7.1 Part I: Module 4

In part I of this reader, scheduled in Module 4, the Laplace Transform was defined and used. This was introduced primarily as a mathematical tool to solve sets of linear differential equations. It was also used to construct the transfer functions of physical systems, giving qualitative insight into their behaviour. In particular, the transfer function yields the frequency characteristics of a system which can be visualized using Bode plots. Most calculations took place in the "Laplace domain" or complex frequency space. In the Laplace description, the system and the signals (input and output) are both treated as functions in the same domain: The product of an input function and the transfer function results in the output function. In part II and part III, the systems and signals are more or less separated and the applications are heading in different directions. Where part II is concentrated on the constructions of models of physical systems, part III is more concentrated on the signals. The systems in this part are primarily to modify signals: The systems in this part are filters that have to be designed according to design criteria that follow from requirements to enhance the signals.

7.2 Part III: Module 5

In part III, scheduled in Module 5, we concentrate on signals. Characterisation of signals is an important aspect, but also the conditioning of signals. We start with continuous signals, but the main emphasis will be on discrete signals, since nowadays almost all signals are treated digitally.
A new transformation is introduced for continuous signals: the *Fourier Transform* (FT or CFT). This at first looks quite similar to the *Laplace Transform*, however it will be used to treat the more fundamental relation between the time domain and the frequency domain.
Before we discuss the Fourier Transform, however, we will start by introducing the so-called *Fourier Series*, a special case of Fourier Transformations that applies to periodic functions. This is easier to visualize than the Fourier Transform and provides a conceptual picture of the concept of frequency build up of a time signal.
In the next step, we shift our attention to discrete signals, which consist of a series of discrete samples and represent the core of digital signal processing. The step to discrete time and signals is relatively easily made by introducing the Discrete Fourier Transform (DFT). Before this switch to Discrete Fourier Transforms is treated, a classification of Signals and Systems and their characteristics in the time and frequency domains is made as an intermezzo. A parallel between continuous signals and discrete signals is drawn as a basis to define the Discrete Fourier Transform as a counterpart of the Continuous Fourier Transform, and later the z-Transform as a discrete counterpart to the continuous Laplace Transform.
The various properties of the Discrete Fourier Transform are explored in the Filter design chapters, in particular the Finite-Impulse Response (FIR) filters. As an introduction to this chapter, first the concept of time domain filtering is introduced as a way to design systems that modify signals.

As an accompanying tool for filter design, windowing is briefly treated after the FIR-filter section to improve the reliability and applicability of the Discrete Fourier Transform.
We then proceed with two chapters on the z-Transform, its definition, characteristics and applications, and the design of Infinite-Impulse Response (IIR) filters. The z-Transform is the discrete counterpart of the continuous Laplace transform. Just like Laplace is used to solve a set of linear differential equations (continuous time description) and calculate (continuous time) Transfer functions, the z-Transform is used to solve sets of difference equations and calculate discrete-time Transfer functions. In the IIR-filter chapter these techniques are also used to design and characterise powerful filters for processing discrete-time signals.

Chapter 8

Fourier Series

In this chapter we will describe how a periodic signal $f(t)$ can be described as a sum of sinusoids. As always, we assume that we are dealing with well behaved signals: mostly smooth, with an occasional step-like discontinuity or impulse component.

8.1 Periodic functions

A *periodic* function $f(t)$ with period T_0 is defined mathematically by the relation

$$f(t) = f(t + mT_0) \qquad \forall\, m \in \mathbb{Z}. \tag{8.1}$$

The frequency associated with this periodicity is

$$f_0 = \frac{1}{T_0} \tag{8.2}$$

and the corresponding angular frequency is

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}. \tag{8.3}$$

Example 8.1.1. Modulo function
A trivial example of a periodic functions is *e.g.*

$$f(t) = \sin \omega t$$

which has a periodicity of $2\pi/\omega$. A function that is only defined on an interval that is repeated can be defined using the *modulo* function *e.g.*:

$$f(t) = 1 - \mathrm{e}^{-\mathsf{mod}(t,T_0)} \tag{8.4}$$

where only the part of the function on the interval $[0, T_0)$ is used to define the repetitive function on the full time range: $[-\infty, \infty]$. □

8.2 The Fourier series

A periodic function $f(t)$ can be represented by its Fourier series,

$$f(t) = \sum_{n=-\infty}^{\infty} F_n \mathrm{e}^{\mathrm{j}n\omega_0 t}. \tag{8.5}$$

In this equation:

- The coefficients F_n are complex numbers known as *Fourier coefficients*. Different functions $f(t)$ are represented by unique sets of values for these coefficients.
- The complex exponential has the usual meaning can be written in the linear combination of cosines and sines, $\mathrm{e}^{\mathrm{j}n\omega_0 t} = \cos(n\omega_0 t) + \mathrm{j}\sin(n\omega_0 t)$.
- The different terms in the sum have names:
  - The $n = 0$ term has no time dependence since $\mathrm{e}^{\mathrm{j}0} = 1$. This term is therefore often called the *constant term* or the *DC component*. As we will see from the expression for F_n below, its value is simply equal to the average of $f(t)$.
  - The $n = 1$ term is known as the *fundamental component* or just *the fundamental*.
  - For $|n| > 1$, the *n*th term in the sum is known as the *n*th harmonic of the function.

Calculating the coefficients F_n for a given function $f(t)$ requires evaluating an integral:

$$F_n = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} f(t)\,\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t. \tag{8.6}$$

Here t_0 is an arbitrary constant: since the argument of the integral is periodic with period T_0, it does not matter where we start integrating. However you are advised to take a suitable value for t_0 when you have the freedom.
Demonstration:
We substitute the definition of $f(t)$ into the expression for F_n, yielding

$$\begin{aligned}
\frac{1}{T_0}\int_{t_0}^{t_0+T_0} f(t)\,\mathrm{e}^{\mathrm{j}n\omega_0 t}\mathrm{d}t &= \frac{1}{T_0}\int_{t_0}^{t_0+T_0}\left[\sum_{m=-\infty}^{\infty} F_m \mathrm{e}^{\mathrm{j}m\omega_0 t}\right]\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t \\
&= \frac{1}{T_0}\sum_{m=-\infty}^{\infty} F_m \int_{t_0}^{t_0+T_0} \mathrm{e}^{\mathrm{j}m\omega_0 t}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t \\
&= \frac{1}{T_0}\sum_{m=-\infty}^{\infty} F_m \int_{t_0}^{t_0+T_0} \mathrm{e}^{\mathrm{j}(m-n)\omega_0 t}\mathrm{d}t \\
&= \frac{1}{T_0}\sum_{m=-\infty}^{\infty} F_m T_0 \delta_{m,n} \\
&= F_n
\end{aligned} \tag{8.7}$$

Note that here we used as a shorthand the so-called Kronecker delta function, defined as

$$\delta_{m,n} \equiv \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases} \tag{8.8}$$

You may want to take a moment to convince yourself that the integral in the third last line indeed simplifies to a Kronecker delta function. This can be done by inspection without the need to whip out your integral table.

Example 8.2.1. *Fourier Series decomposition of a block wave*
Consider the periodic square wave function defined as

$$f(t) = \begin{cases} C & 0 < t \leq T/2 \\ 0 & T/2 < t \leq T \end{cases}$$

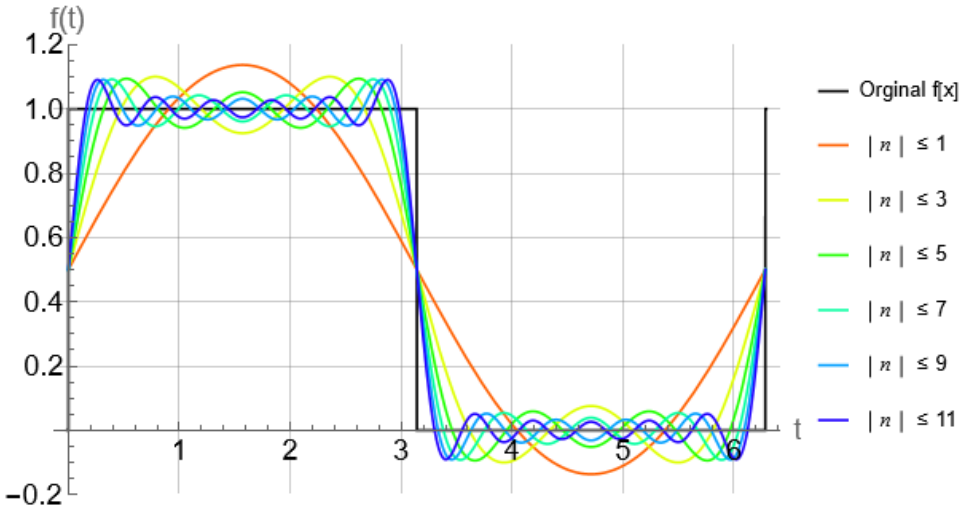


Figure 8.1: Representation of a square wave with a sum of sinus functions; the n here indicates the minimum and maximum element contributing to the plotted function: $F_i \; i \in [-n, n]$

Let's express it as a Fourier series. Picking $t_0 = 0$ for convenience, the Fourier coefficients are given by

$$\begin{aligned}
F_n &= \frac{1}{T}\int_0^T f(t)\,\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t \\
&= \frac{1}{T}\int_0^{\frac{T}{2}} C\mathrm{e}^{-\frac{\mathrm{j}2\pi nt}{T}}\mathrm{d}t + \frac{1}{T}\int_{\frac{T}{2}}^T 0\mathrm{e}^{-\frac{\mathrm{j}2\pi nt}{T}}\mathrm{d}t \\
&= \frac{1}{T}C\frac{T}{-\mathrm{j}2\pi n}\left[\mathrm{e}^{-\frac{\mathrm{j}2\pi nt}{T}}\right]_0^{\frac{T}{2}} \qquad (for\ n \neq 0) \\
&= \frac{\mathrm{j}C}{2\pi n}\left(\mathrm{e}^{-\mathrm{j}\pi n} - 1\right) \\
&= \begin{cases} \frac{-\mathrm{j}C}{n\pi} & n = \pm 1, \pm 3, \pm 5, \ldots \\ 0 & n = \pm 2, \pm 4, \pm 6, \ldots \\ \frac{C}{2} & n = 0 \end{cases}
\end{aligned} \tag{8.9}$$

Here F_0 for $n = 0$ was obtained by performing that integral separately. To write out the series explicitly in a more elegant form without complex numbers, we note that the terms with positive and negative index n can be grouped:

$$\begin{aligned}
f(t) &= \frac{C}{2} - \frac{\mathrm{j}C}{\pi}\mathrm{e}^{\frac{\mathrm{j}2\pi t}{T_0}} + \frac{\mathrm{j}C}{\pi}\mathrm{e}^{-\frac{\mathrm{j}2\pi t}{T_0}} - \frac{\mathrm{j}C}{3\pi}\mathrm{e}^{\frac{\mathrm{j}6\pi t}{T_0}} + \frac{\mathrm{j}C}{3\pi}\mathrm{e}^{-\frac{\mathrm{j}6\pi t}{T_0}} + \ldots \\
&= \frac{C}{2} + \frac{2C}{\pi}\sin\frac{2\pi t}{T_0} + \frac{2C}{3\pi}\sin\frac{6\pi t}{T_0} + \frac{2C}{5\pi}\sin\frac{10\pi t}{T_0} + \ldots \\
&= \frac{C}{2} + \frac{2C}{\pi}\sum_{q=0}^{\infty}\frac{1}{2q+1}\sin\frac{2\pi(2q+1)t}{T_0}.
\end{aligned} \tag{8.10}$$

The result only contains sine functions, apart from the DC component, which makes sense, since the source function also is an odd function with a DC offset.[1]

□

[1]The classification "odd" and "even" functions refers to the odd and even harmonics; cosine is a symmetric function and relates to the "even" coefficients, whereas the sine is an anti-symmetric function and relates to the "odd" coefficients.

While an infinite number of terms is required to reproduce the square wave function perfectly, we get an increasingly good approximation as we add more terms. This is illustrated in figure 8.2 for a block wave that is symmetric around zero in this case.

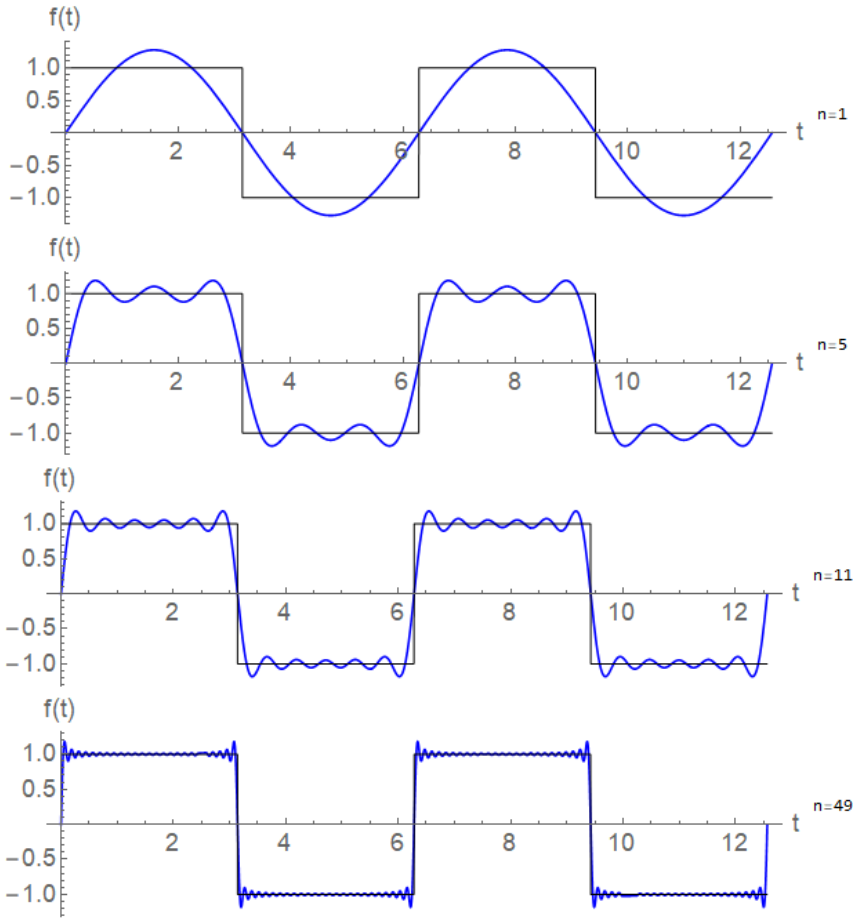


Figure 8.2: Representation of square wave by 1, 5, 11 and 49 sinus functions.

Miscellaneous observations on Fourier series

- The definition of the Fourier series of Eq. 8.5 is not unique. You may encounter other variants in different books, articles, etc. One particularly common form you may encounter is

$$f(t) = B_0 + \sum_{n=1}^{\infty} B_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} C_n \sin(n\omega_0 t). \tag{8.11}$$

  A particular feature of this representation is that $C_n = 0$ for all *n* when $f(t)$ is symmetric around $t = 0$ (that is, when $f(t)$ is even, $f(-t) = f(t)$). Similarly, $B_n = 0$ for all *n* when $f(t)$ is antisymmetric around $t = 0$ (that is, when $f(t)$ is odd, $f(-t) = -f(t)$). We will work out in the exercises how to interconvert between the representations of Eq. 8.5 and Eq. 8.11.

- Another way to think about the coefficients F_n is that they consist of an amplitude, $|F_n|$, and a phase, φ_n , such that

$$F_n = |F_n| e^{j\varphi_n} \tag{8.12}$$

  and

$$f(t) = \sum_{n=-\infty}^{\infty} |F_n| e^{j(n\omega_0 t + \varphi_n)}. \tag{8.13}$$

This way of representing the coefficients is reminiscent of a Bode plot and is particularly popular when plotting experimental data.

- The Fourier series can also be used to describe any (well-behaved) function in the interval t_0 to $t_0 + T_0$ when one does not care about what happens outside this interval[2].
- The expression for F_n requires integrating over an interval of length T_0. Because the function is periodic, the information contained in the time interval of length T_0 is independent of the starting point of the integration. Therefore the most general for for F_n equation 8.6 is used:

$$F_n = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} f(t)\,\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t. \tag{8.14}$$

  In most of the cases it is most convenient to choose $t_0 = 0$ as is done in equation 8.9 Here t_0 is any constant we want to choose. This is often convenient when doing calculations because we are free to pick the integration integral that simplifies the problem.
- The lowest frequency in the representation is directly related to the time interval T_0:

$$\omega_0 = \frac{2\pi}{T_0}$$

  This might be important in case you limit yourself to finite time interval of which you think a signal is repetitive: you will not find frequencies lower than the one related to $T - 0$.
- Note that the number of Fourier coefficient is infinite in general. This means that for a representation of a signal in the frequency domain we have to calculate an infinite number of coefficients. As illustrated in example 8.2.1 a limited number of the lowest Fourier components is sufficient to get a reasonable representation of the original signal. This means that the Fourier series can be truncated to limit the calculations at the cost of a reduced accuracy. This is often done using Fourier Series representation of signals.

A few words about convergence

Due to the increasing frequency of the higher Fourier components, the magnitude will decrease, even for functions with a limited number of discontinuities.

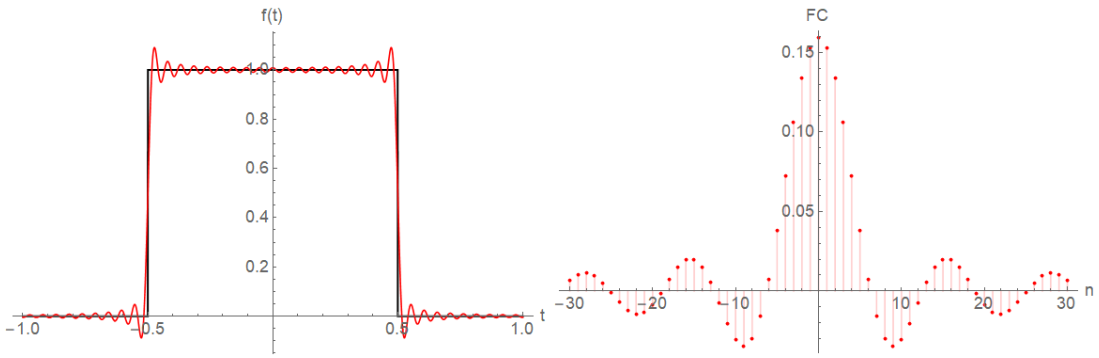


Figure 8.3: Representation of a block function by its Fourier components: the magnitude of the coefficients converges to zero if n goes to infinity. Even if we include a lot of terms in the series, oscillations remain near discontinuities in $f(t)$.

We will not attempt to formally prove the convergence of the Fourier series, but it is worth noting a few qualitative points.

[2] In fact, Fourier did not initially develop the series to describe periodic functions at all. He was trying to model the cooling of a metal rod of finite length.

- In the representation of the signal in terms of an infinite sum of Fourier components, there is a need to convergence to obtain a finite result. In general this means that the contribution of higher-order terms decrease with the order, or the series is limited by itself: $F_n = 0 \; \forall \; n > n_{\max}$.
- As a direct result of the above, only a few terms are often sufficient to obtain a fairly good approximation for a smooth function.
- For a functions with a discontinuity, some oscillatory behaviour is observed near the discontinuity. This is illustrated for a square pulse in Figure 8.3. Adding more terms makes the step sharper but does *not* remove the oscillations very close to the discontinuity.
- At a discontinuity, the series converges to the average of the values on either side of the discontinuity.

8.2.1 Real-valued functions

In real life we most often deal with cases where $f(t)$ is a real-valued function such as, for example an electrical current, the height of waves in the sea, etc. Then the negative-frequency Fourier components of the function are simply the complex conjugate of the corresponding positive frequencies: $F_{-n} = F_n^*$. To see this, note that:

$$\begin{aligned} F_{-n} &= \frac{1}{T}\int_0^T f(t)\, \mathrm{e}^{-\mathrm{j}(-n)\omega_0 t}\mathrm{d}t \\ &= \frac{1}{T}\int_0^T f(t)\, \mathrm{e}^{+\mathrm{j}n\omega_0 t}\mathrm{d}t \\ &= \frac{1}{T}\int_0^T f(t)\, \left(\mathrm{e}^{-\mathrm{j}n\omega_0 t}\right)^* \mathrm{d}t \\ &= \left(\frac{1}{T}\int_0^T f(t)\, \mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t\right)^* \\ &= F_n^*. \end{aligned} \tag{8.15}$$

It is therefore common to plot F_n only for positive frequencies when dealing with real-valued signals since the negative frequency components do not contain additional information. Note also that our specific example of a square wave (Eq. 8.9) explicitly satisfies this condition; this is why we were able to group the terms so as to obtain a series of real sinusoids (Eq. 8.10).

8.2.2 Two alternative ways to look at Fourier Series

This two alternative ways of looking at Fourier series is just for completion and is not a part that you should master for your course. However it might help in building a consistent picture of the domain of Signals.

8.2.2.1 Projection on a basis

The calculation of the Fourier coefficients can also be interpreted as a projection of a function on a basis of eigenfunctions. The projection is the function equivalent of a dot-product in function space and is defined as:

$$F_n = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} f(t)\, \mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t. \tag{8.16}$$

with $\omega_0 = 2\pi/T_0$. This is equivalent with the Fourier Series decomposition, defined in equation 8.6. The eigenfunctions are given by:

$$\phi_n(t) = \mathrm{e}^{-\mathrm{j}n\omega_0 t} \qquad \forall \qquad n \in \mathbb{Z} \tag{8.17}$$

The function $f(t)$ is the sum of all components, weighted with the calculated coefficients:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n \phi_n(t) \tag{8.18}$$

In case the range of eigenfunction for reconstruction is limited to a maximum $n = N$, the reconstruction is an approximation.

8.2.2.2 Integral Transform

The structure of the integral in equation 8.6 can also be seen as an integral transform as defined in section 3.2, relating two variables, in this particular situation t and n in the kernel function:

$$K(t,n) = \mathrm{e}^{-\mathrm{j}n\omega_0 t} \qquad t \in \mathbb{R} \ \wedge \ n \in \mathbb{Z} \tag{8.19}$$

and the corresponding Integral Transform:

$$F[n] = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} f\,(t)\, K(t,n) \mathrm{d}t. \tag{8.20}$$

with again $\omega_0 = 2\pi/T_0$. Note that this Integral Transform maps a Real-variable domain on an Integer-variable domain and visa versa.
We do not go further in detail, however we will come back to this description when we treat the Fourier Transform.

8.2.3 Energy, power and Parseval's theorem

Most of the functions representing signals are A-type or T-type variables, or can easily related to one of the two. Consequence is that the absolute-squared value of this variable is proportional to the power of that signal:

$$P_f(t) \propto |f(t)|^2 \tag{8.21}$$

from which we can calculate the energy E per fundamental period T_0 as an integral over time of the power:

$$E \propto \int_{t_0}^{t_0+T_0} |f\,(t)\,|^2 \mathrm{d}t. \tag{8.22}$$

As an example in electronics the instantaneous power dissipated across a resistor R is given by $V^2\,(t)\,/R$ or $I^2\,(t)\,R$, where $V\,(t)$ and $I\,(t)$ are the voltage across and the current through the resistor, respectively. Such a quadratic dependence similarly occurs in mechanics, optics, electromagnetism, quantum mechanics, etc. In the context of a specific physical problem, however, the definition above will not have the right physical units and it is necessary to add the appropriate physical constants to the expression to obtain units of energy (for example, adding a factor R if the signal one is dealing with is a current $I\,(t)$).
Because a periodic function $p\,(t)$ does not decay to zero for $t \to \pm\infty$, its energy is always infinite (apart from the trivial case $p(t) = 0$).
In case we normalize the calculated energy over a time interval T_0 to the time interval, we obtain the *average power*, $\hat{P}$:

$$\hat{P} \propto \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |f\,(t)\,|^2 \mathrm{d}t. \tag{8.23}$$

Again, the integral has the same value irrespective of the choice of the arbitrary constant t_0 due to periodicity. Because periodic signals have a well-defined average power, they are also called *power signals*.

Let us re-express the power $\hat{P}$, in terms of Fourier coefficients:

$$\begin{aligned}
\hat{P} &\propto \frac{1}{T_0}\int_{t_0}^{t_0+T_0} |f(t)|^2 \mathrm{d}t \\
&\propto \frac{1}{T_0}\int_{t_0}^{t_0+T_0} \left(\sum_{n=-\infty}^{\infty} F_n \mathrm{e}^{\mathrm{j}n\omega_0 t}\right)\left(\sum_{m=-\infty}^{\infty} F_m \mathrm{e}^{\mathrm{j}m\omega_0 t}\right)^* \mathrm{d}t \\
&\propto \frac{1}{T_0}\sum_{n=-\infty}^{\infty} F_n \sum_{m=-\infty}^{\infty} F_m^* \int_{t_0}^{t_0+T_0} \left(\mathrm{e}^{\mathrm{j}(n-m)\omega_0 t}\right) \mathrm{d}t \\
&\propto \frac{1}{T_0}\sum_{n=-\infty}^{\infty} F_n \sum_{m=-\infty}^{\infty} F_m^* T_0 \delta_{m,n} \\
&\propto \sum_{n=-\infty}^{\infty} |F_n|^2. \qquad (8.24)
\end{aligned}$$

This result is known as Parseval's theorem for periodic functions. It tells us that each harmonic n carries with it a well-defined amount of power that is proportional to $|F_n|^2$, and the total power in the signal is the sum of the contributions from the individual harmonics. Representing a signal by a Fourier series is thus not a mere mathematical exercise – there is also a physical meaning attached to the amplitude of the harmonics. (This also motivates some obscure-sounding engineering lingo like "the carrier wave is transmitted at 10 milliwatts").

Summaries

Summary Chapter 8

Key concepts on Fourier series:

- A periodic function $f(t)$ with period T_0 (i.e. $f(t+T_0) = f(T_0)$) can be expressed as an infinite series of complex exponentials,

$$f(t) = \sum_{n=-\infty}^{\infty} F_n \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

- The value of the Fourier coefficients is given by

$$F_n = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} f(t) \mathrm{e}^{-\mathrm{j}n\omega_0 t} \mathrm{d}t$$

- The average power in a signal can be attributed to specific frequency components:

$$\hat{P} \propto \frac{1}{T_0} \int_{t_0}^{t_0+T_0} |f(t)|^2 \, \mathrm{d}t = \sum_{n=\infty}^{\infty} |F_n|^2$$

Problems

Problem 8.1. Any duty cycle

a. Determine the Fourier Series of

$$f_\alpha(t) = \begin{cases} 1 \text{ for } t \in [0, \alpha] \\ 0 \text{ for } t \in [\alpha, 2] \end{cases}$$

and repeating itself.

b. What happens if $\alpha = \frac{1}{2},\ 1, \frac{3}{2}$ and 2?

Problem 8.2. Fourier series
Sketch a graph of each of the following periodic functions (period T), paying attention to any symmetry that may exist. Then express each function as a Fourier series:

a. The rectangular square wave:

$$f(t) = \begin{cases} +1, & |t| < T/4 \\ -1, & T/4 \leq |t| < T/2 \end{cases}$$

b. The sawtooth wave:

$$f(t) = 2t/T, |t| < T/2$$

c. The comb function, which consists of a periodic train of Dirac delta functions with period T:

$$f(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Problem 8.3. Sum of cosines

a. Determine Fourier Series of

$$f(t) = \cos(\pi t) + 2\cos(2\pi t)$$

b. Can you avoid calculation? Which "implicit" property are you actually using?

Problem 8.4. Voltage rectification
Consider the function $V_{in}(t) = 325 \sin 100\pi t$, where t is in units of seconds and the amplitude is in volts, that describes the voltage measured at a power outlet.

a. Determine the coefficients F_n for the Fourier series of $V_{in}(t)$.

b. Sketch your answer to part (a) as a plot of the amplitude of F_n versus frequency f in units of Hertz (as opposed to angular frequency ω).

Now imagine that we use a clever circuit to transform the input signal into a new signal $V'_{out}(t)$ that is the absolute value of the input voltage: $V_{out}(t) = |V_{in}(t)| = 325\,|\sin 100\pi t|$

c. Using a plot similar to that of part (b), make a qualitative prediction as to what the Fourier series of $V_{out}(t)$ will look like. In particular, predict which frequencies will have a non-zero amplitude.

d. Calculate the coefficients F_n for the Fourier series of $V_{out}(t)$.

e. What could be the use for such a circuit? Hint: odds are you own several variants.

Problem 8.5. Another form of Fourier Series

a. Determine Fourier Series of

$$f(t) = \mathrm{e}^{-\mathsf{mod}(2t,\frac{1}{2})}.$$

b. Find the coefficients in the series

$$f(t) = B_0 + \sum_{n=1}^{+\infty} B_n \cos(n\omega_0 t) + \sum_{n=1}^{+\infty} C_n \sin(n\omega_0 t)$$

Problem 8.6. Alternate representation of the Fourier series
We stated above that the Fourier series is sometimes written in the form

$$f(t) = B_0 + \sum_{n=1}^{\infty} B_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} C_n \sin(n\omega_0 t)$$

instead of our complex exponential notation. Derive expressions for B_0, B_n and C_n in terms of the complex exponential Fourier coefficients, F_n.

Problem 8.7. Fourier Series and reusing known results

a. Determine the Fourier Series of

$$f_A(t) = \begin{cases} \frac{1}{2} \text{ for } t \in [0,1) \\ -\frac{1}{2} \text{ for } t \in [1,2) \end{cases}$$

and repeating itself. If F_n is real or purely imaginary, discuss its cause.

b. Determine the Fourier Series of

$$f_B(t) = \begin{cases} 1 \text{ for } t \in [-\frac{1}{2},\frac{1}{2}) \\ 0 \text{ for } t \in [\frac{1}{2},\frac{3}{2}) \end{cases}$$

and repeating itself. If F_n is real or purely imaginary, discuss its cause.

For questions c) and d) we define

$$f_\alpha(t) = \begin{cases} 1 \text{ for } t \in [0,\alpha] \\ 0 \text{ for } t \in [\alpha,2] \end{cases}$$

c. What is the relationship between $f_A(t)$ and $f_\alpha(t)$? Use this relationship to find the Fourier Series of $f_A(t)$.

d. What is the relationship between $f_B(t)$ and $f_\alpha(t)$? Use this relationship to find the Fourier Series of $f_B(t)$.

Problem 8.8. e^{-t} and plotting

a. Determine Fourier Series of

$$f(t) = \mathrm{e}^{-\mathsf{mod}(t,1)},$$

where $\mathsf{mod}(a,b)$ maps a to $\tilde{a} \in [0,b)$ such that $a = \tilde{a} + kb$ for some $k \in \mathbb{Z}$.

b. We know that if $f(t)$ is real valued $F_{-n} = F_n^*$ holds. Use this fact to show that

$$F_{-n}e^{-j\omega_0 nt} + F_n e^{j\omega_0 nt} = 2|F_n|\cos(\omega_0 nt + \arg(F_n))$$

holds in general.

c. Make plots of $f(t)$ for $t \in [-2, +2]$ for $\sum_{n=-10}^{10} F_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$ and $\sum_{n=-50}^{50} F_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$.

d. What is the average power of the $n = 0$ term of the Fourier Series of $f(t)$?

Problem 8.9. Ministry of Silly Periods (an homage to Monty Python)
We've shown that the Fourier coefficients of the periodic square wave function defined as

$$f(t) = \begin{cases} C & 0 < t \leq T/2 \\ 0 & T/2 < t \leq T \end{cases}$$

are

$$F_n = \begin{cases} \frac{-\mathrm{j}C}{n\pi} & n = 1, 3, 5, \ldots \\ 0 & n = 2, 4, 6, \ldots \\ \frac{C}{2} & n = 0 \end{cases} \tag{8.25}$$

Suppose that we decide to represent $f(t)$ using a Fourier series and pick a time interval of $2T$ instead of T for the period. While this is not the most intuitive choice, $f(t)$ is periodic with period $2T$ as well, so it must be a valid option.

a. Evaluate the Fourier coefficients F_n in this case.

b. The coefficients F_n have different values when using periods of T or $2T$. Explain why the two results are nonetheless fully consistent with each other.

Problem 8.10. (non-)Matching intervals and frequencies
Consider a continuous-time signal that contains two harmonic frequencies:

$$x(t) = \sin\left(\frac{2\pi t}{\tau_a}\right) + \sin\left(\frac{2\pi t}{\tau_b}\right)$$

with the two time constants $\tau_a = 1$ s and $\tau_b = 1.5$ s. The signal has to be represented by a Fourier Series.

a. What is an appropriate period T_0 to pick for the evaluation of the Fourier coefficients?

b. Explain what would happen if we picked $T_0 = \tau_b$.

Problem 8.11. Checking Parseval's theorem During the lecture we derived the Fourier series for a square wave function,

$$f(t) = \begin{cases} C, & 0 < t \leq T/2 \\ 0, & T/2 < t \leq T \end{cases}$$

a. Calculate the power for this function starting from the definition, $P = \frac{1}{T}\int_0^T |f(t)|^2 \, \mathrm{d}t$.

b. Calculate the power again, but this time using the Fourier domain expression, $P = \sum_{n=-\infty}^{\infty} |F_n|^2$. *Hint: you may have to look up a table of infinite series.*

c. Is Parseval's theorem satisfied in this specific instance?

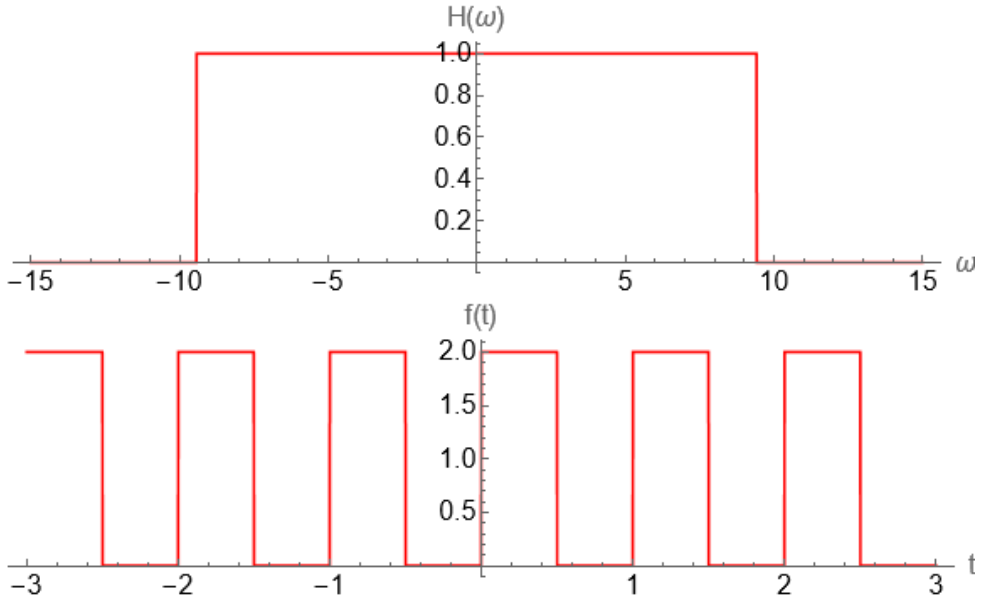


Figure 8.4: Top: Ideal filter. Bottom: 50 % duty cycle square wave.

Problem 8.12. The perfect filter

Consider a filter circuit whose transfer function $H(\omega)$ has only a real component as shown in graph 8.4:

Suppose we input a 50 % duty cycle square wave to this circuit and measure the output signal. Calculate the input and output average power. *Note: a transfer function with a perfectly sharp discontinuity as in this exercise is actually impossible to realize using analog electronic circuits. But we can build close enough approximations when needed.*

Problem 8.13. Truncating periodic functions

Plot the Fourier Series of $f_A(t)$ on $[-4, +4]$ of the following finite sums:

a. $\sum_{n=-1}^{+1} F_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$

b. $\sum_{n=-5}^{+5} F_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$

c. $\sum_{n=-25}^{+25} F_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$

Use $f_A(t) = -\frac{1}{2}\mathrm{Sign}(\mathrm{mod}(t, 2) - 1)$ and for a real valued function $f(t)$

$$F_{-n}e^{-j\omega_0 nt} + F_n e^{j\omega_0 nt} = 2|F_n| \cos(\omega_0 nt + \arg(F_n))$$

Problem 8.14. Total harmonic distortion

A perfect amplifier reproduces the shape of the input signal at its output. In particular, if an amplifier is perfectly linear, a sinusoidal input will yield a sinusoidal output at the same frequency. Any non-linearity will instead lead to the presence of harmonics in the output signal. This provides a convenient way of quantifying the linearity of an amplifier via a quantity called the *total harmonic distortion* (THD). This is defined as the ratio of the power in the harmonics of the output signal divided by the power in the fundamental component of the output when a perfect sinusoid is applied at the input:

$$THD = \frac{\sum_{n=2}^{\infty}(|F_n|^2 + |F_{-}n|^2)}{(|F_1|^2 + |F_{-}1|^2)}$$

It is most commonly expressed as a percentage. As illustration, consider an amplifier whose output in response to a 500 Hz sinusoidal input is

$$f(t) = \begin{cases} 1 - 4t^2, & -0.5 < t < 0.5 \\ 4(t-1)^2 - 1, & 0.5 < t < 1.5 \end{cases}$$

a. Make a sketch this output function.

b. Compute the Fourier coefficients of f(t).

   Note: the integrals get a little heavy if you do not make use of symmetry.

c. Calculate the THD of the amplifier.

   Hint: You can solve this last question by brute force and evaluate an infinite sum, but there is a simpler, equally valid way to go about it.

In case you wonder whether your answer is big or small, here's an example taken from the website of Cambridge Audio, a rather respected maker of audio equipment. Note the implications that (1) the amount of distortion depends on frequency and (2) the THD may depend on volume. How this affects human perception and our enjoyment of music is very subjective and the subject of countless reviews! *Full disclosure: The author owns an older Cambridge model but wasn't compensated for this product placement ad.*



Figure 8.5: Technical specifications THD: $<$ 0.001% 1 kHz at 80% of rated power 0.01% 20 Hz - 20 kHz at 80% of rated power.

Problem 8.15. Gibb's phenomenon

Using a Fourier Series, a square wave can be reconstructed. It is obvious that you need an infinity number of Fourier elements to completely reconstruct the square wave.
The square wave is mathematically given by:

$$f(x) = \begin{cases} 1 & |x \mod 2\pi| \leq \pi \\ -1 & \textit{else} \end{cases} \tag{8.26}$$

In the reconstruction of a square wave a remarkable effect shows up: The overshoot of the maximum level as well as the minimum level is always 9 % of the amplitude. This is independent of the number of Fourier coefficients used in the reconstructed signal. This phenomenon is called the "Gibbs"-phenomenon[3].
The idea of this problem is to verify this phenomenon empirically.

[3]Yes this is the Josiah W. Gibbs from the thermodynamics theory. In fact the phenomena was first discovered by Henry Wilbraham

a. Determine the value of the coefficients of the Fourier series that represents the square wave as given above.

b. Set-up a reconstruction of a square wave as a sum of a Fourier series in which you can set the number of elements as a parameter.

c. Make a function to determine the maximum value of the sum of Fourier elements.

d. Verify the "Gibbs"-phenomenon for a few values of the number of elements in the series.

e. Illustrate the results with a plot.

Chapter 9

Fourier Transforms

9.1 Introduction

In Part I of this reader in section 3.2 the Integral Transform was introduced as a transformation of signals between two variable domains that are connected via a Kernel function. In section 3.3 this transformation was defined as the Laplace transformation, useful to solve differential equations and to describe LTI-systems by transfer functions. The Kernel function used in the Laplace Transform is:

$$K(s,t) = \mathrm{e}^{-st} \tag{9.1}$$

In the same manner we will define the Fourier Transform. The Fourier Transform is a generalisation of the decomposition of a function in its Fourier Series by means of calculating its Fourier components.

9.2 Definition of the Fourier Transform

The Fourier Transform is defined via the Integral Transform with the Kernel function:

$$K(\omega,t) = \mathrm{e}^{-\mathrm{j}\omega t} \tag{9.2}$$

in the integral:

$$\mathcal{F}\{f(t)\} = F(\omega) \equiv \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t. \tag{9.3}$$

The arguments t and ω are both real valued over the full range in $\mathbb{R}$, which means that ω can also have negative values. A formal name for $F(\omega)$ is the *spectral density function*, but more commonly it is just called the Fourier Transform of $f(t)$.
Both $f(t)$ and $F(\omega)$ can be complex valued, although in signal processing $f(t)$ is usually a real-valued function (a condition we maintain in this reader) while $F(\omega)$ is in general a complex valued function. We most often speak about signals such that t has units of time and ω is an angular frequency (thus with physical units of inverse time). In this case, $F(\omega)$ is often said to be the representation in *frequency space*. But in some cases, for example in quantum mechanics and electrodynamics, one also encounters situations where the independent variable has units of distance and the "Fourier space" variable consequently has units of inverse distance. A common notation in that case is to label the functions $f(x)$ and $F(k)$. The more general terms *Fourier space* or *reciprocal space* are then often employed instead of frequency space.
It is also possible to take the Fourier transform in multiple dimensions simultaneously. This is done in image manipulation software, for example. But here we will focus on single-valued signals[1].

[1] In relativistic electrodynamics, space-time has four dimensions and so does the corresponding reciprocal space!

One important difference with the Laplace Transform is that the variable s is complex and $F(s)$ is defined over a two-dimensional domain in the complex s-plane. In the Fourier domain, the Fourier Transform $F(\omega)$, which is a complex-valued function in general, is only defined for real values of its argument ω. This makes it possible to plot $F(\omega)$, either by plotting its real and imaginary parts or its amplitude and phase as a function of ω here[2].
Another key difference between the two transforms is that their integration boundaries are different. Whereas we only cared about positive real values of time $t \in [0^-, \infty]$ when taking the Laplace Transform (or added a step function, $u(t) = 0 \; \forall \; t < 0$, as a reminder) the range for t when taking the Fourier transform extends over the entire Real interval and includes negative values: $t \in [-\infty, \infty]$. One could relate this to the definition of the Fourier Series (equation 8.6) where we take the limit for $t_0 \to -\infty$ and $t_0 + T_0 \to \infty$. With this action the smallest frequency of the series is set to zero and the frequency step evolved to a continuous frequency variable; $n\omega_0 \to \omega$ since $\omega_0 \to 0 \wedge n \to \infty$.

9.3 Definition of the inverse Fourier Transform

Like the situation for the Laplace transform, also for the Fourier Transform there exists an inverse Fourier Transform. This one-to-one transformation relates a Fourier pair of functions in the time-domain to that in the frequency-domain:

$$f(t) \leftrightarrow F(\omega). \tag{9.4}$$

Contrary to Laplace, explicitly writing down the inverse operation is straightforward and rather similar to the forward operation:

$$\mathcal{F}^{-1}\left\{F(\omega)\right\} = f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\left(\omega\right) \mathrm{e}^{+\mathrm{j}\omega t} \mathrm{d}\omega. \tag{9.5}$$

and therefore easier to calculate; no complex contour integration involved here.
Like in other inverse Integral Transforms, the inverse of the Kernel function is used. The integral boundaries are the same: $\omega \in [\infty, \infty]$ and since the variable of integration ω is a Real variable, we can use standard integration over a Real variable as opposed to the complex contour integration in the inverse Laplace Transform.
Note however that, although the time function $f(t)$ is in general a real-valued function[3], its Fourier Transform $F(\omega)$ is in general complex-valued.
Note that the inverse Fourier Transform has a pre-factor of $1/(2\pi)$, which is needed since ω and t are not exactly each others inverse, and need that factor 2π: $\omega = 2\pi/t$. The need for this correction factor can lead to some enjoyable/frustrating confusion: The location of the correction factor 2π is not defined; it depends on the specific domain in which you use the Fourier Transforms where you put the factor 2π. The definition used here is common for signal processing. The pure mathematicians distribute the correction over both transforms: both transforms get the same pre-factor $1/\sqrt{2\pi}$ to make the transformations symmetric. To make it even more confusing, the in some domains, the sign in the exponent is swapped between the regular and inverse Fourier Transform. See MathWorld for a clear discussion on this and indication of the definitions used in the various disciplines[4].
We will calculate several examples of the Fourier and inverse Fourier transform during the course. In practice, however, one usually still looks them up in a table just like any integral. Every transform that we calculate can be found in the table available in the summary at the end of this chapter or derived from these functions by applying properties of Fourier Transforms.

[2] Do not confuse this with the trick we used with transfer functions where we substituted $s \to \mathrm{j}\omega$ to make the Laplace Transform a function of a single real variable ω

[3] Not strictly, however in all descriptions of real signals we consider only real-valued functions for $f(t)$.

[4] Here you find also the justification for choosing your `FourierParamters`-setting in MATHEMATICA for the correct results in signal processing.

Example 9.3.1. *Fourier Transform of a square pulse*
Let us consider a single square pulse of amplitude A and duration τ centred around $t = 0$, as shown in figure 9.1:

$$f(t) = \begin{cases} A & |t| < \tau/2 \\ 0 & \text{otherwise} \end{cases}$$

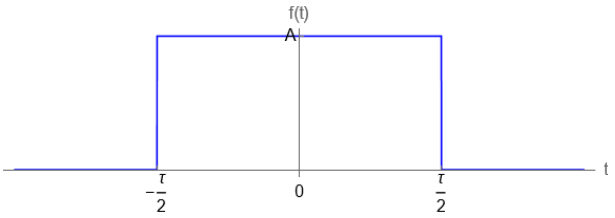


Figure 9.1: A single square pulse function

Its Fourier transform is then

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t)\, \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t \\ &= \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t \\ &= \frac{A}{-\mathrm{j}\omega} \mathrm{e}^{-\mathrm{j}\omega t} \Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \qquad (\omega \neq 0) \\ &= \frac{2A}{\omega} \sin\left(\frac{\omega\tau}{2}\right) \\ &= \tau A \,\mathrm{sinc}\left(\frac{\omega\tau}{2}\right). \end{aligned} \tag{9.6}$$

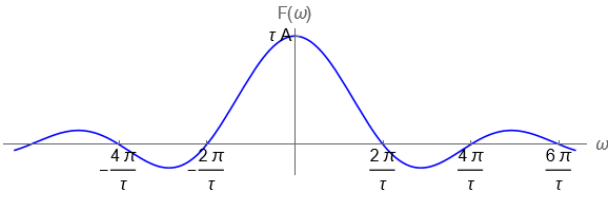


Figure 9.2: The sinc fucntion, the Fourier transform of a square pulse function.

In the last step we used the definition

$$\mathrm{sinc}(x) \equiv \frac{\sin x}{x} \quad \text{with} \quad \mathrm{sinc}(0) \equiv 1.$$

The value at $x = 0$ is defined to be the limiting value from L'Hôpital's rule, $\mathrm{sinc}(0) = 1$. For $\omega = 0$ we can calculate the integral to find $F(0) = \tau A$. This conveniently matches the defined value of $\mathrm{sinc}(0) \equiv 1$. Our expression for $F(\omega)$ is therefore valid for all ω. The sinc-function is depicted in figure 9.2.

□

Note that the width of the oscillations in $F(\omega)$ is proportional to $1/\tau$. Reducing τ by a factor of two, causes $F(\omega)$ to be stretched by a factor of two along the frequency axis. We will quickly learn that this is a general feature of Fourier transform pairs: functions that vary rapidly in the time domain are

broad in the frequency domain, while slowly varying functions in time tend to predominantly contain low frequencies.[5]

Example 9.3.2. *Fourier Transform of the impulse function*
Let us look at the Fourier transform of the impulse function:

$$F(\omega) = \int_{-\infty}^{\infty} \delta(t)\, \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t = \mathrm{e}^{-\mathrm{j}\omega 0} = 1. \tag{9.7}$$

That is, the function contains all frequencies with equal weight. This is consistent with the qualitative idea that, since the function is infinitely sharp in time at the singularity, it must contain contributions from arbitrarily high frequencies. Applying an inverse Fourier transform to $F(\omega)$ yields a new, alternative definition of the impulse function:

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\, \mathrm{e}^{\mathrm{j}\omega t} \mathrm{d}\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{j}\omega t} \mathrm{d}\omega. \tag{9.8}$$

We will use this result later on.
In the same spirit, let us look at a function described by an impulse function in the frequency domain, $F(\omega) = 2\pi\delta(\omega - \omega_*)$. In the time domain this function is given by

$$\mathcal{F}^{-1}\{2\pi\delta(\omega - \omega_*)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_*)\, \mathrm{e}^{\mathrm{j}\omega t} \mathrm{d}\omega = \mathrm{e}^{\mathrm{j}\omega_* t}. \tag{9.9}$$

That is, the inverse Fourier transform of an impulse function in the frequency domain is a pure sinusoidal in the time domain.

□

9.4 Convergence of the Fourier Transform

As with all Integral Transforms we need to be cautious about the convergence of the integrals. We will not attempt to establish the conditions necessary for the existence of the Fourier transform. It is however comforting to know that a sufficient condition for convergence for "normal" functions encountered in the real world (that is, mostly continuous and devoid of divergences) is that the energy E of the function be finite. We also deal with some cases of signals that do not satisfy this condition and still have a well-defined Fourier Transform, most importantly impulse functions and periodic functions, but these are exceptions to the norm.

9.5 Properties of Fourier Transforms

A very convenient feature of transforms is that once you know the mathematical form of a particular Fourier pair, you can easily deduce that for similar functions that are related by a simple transformation. Here we state and motivate some of the more important relations and characteristics. These are, with some exceptions, reminiscent of the properties of Laplace transforms, which is not surprising given the fact that these are two of the same family.
The collection of properties will be collected in a table at the end of the chapter.

[5] A quick aside: If this reminds you of the uncertaity principle in quantum mechanics, you are not wrong! Although you have not (yet) seen it expressed in this language in quantum mechanics, the wavefunction $\psi(x)$ and the momentum distribution function $\phi(p)$ a Fourier transform pair. In the wave formulation of quantum mechanics, the uncertainty principle is simply a consequence of the mathematics of Fourier transforms.

9.5.1 Real functions

The positive and negative frequency components are trivially related in the case of a real signal in the time-domain:

$$F(-\omega) = F^*(\omega) \tag{9.10}$$

This is why it is conventional to plot only the positive frequency range when dealing with real-valued signals: it contains all information.
Demonstration:

$$\begin{aligned} F(-\omega) &= \int_{-\infty}^{\infty} f(t)\, \mathrm{e}^{-\mathrm{j}(-\omega)t} \mathrm{d}t \\ &= \int_{-\infty}^{\infty} f(t) \left(\mathrm{e}^{-\mathrm{j}\omega t}\right)^* \mathrm{d}t \\ &= \left(\int_{-\infty}^{\infty} f(t)\, \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t\right)^* \\ &= F^*(\omega)\,. \end{aligned} \tag{9.11}$$

This holds for all real-valued time signals.

9.5.2 Reciprocal scaling

Above we saw that the width in frequency of the Fourier transform of a square pulse is inversely proportional to the width of the pulse in the time-domain, and vice versa. This result is not limited to the block pulse and sinc combination but actually holds for *all* Fourier pairs:

$$\mathcal{F}\{f(\alpha t)\} = \frac{1}{\alpha} F\left(\frac{\omega}{\alpha}\right) \qquad \forall\, \alpha > 0. \tag{9.12}$$

The actual proportionality constant depends on the functions and the definitions of the width of a function. There are many ways you can define a width of a function.
The reciprocal scaling holds for all Fourier pairs.
Demonstration:

$$\mathcal{F}\{f(\alpha t)\} = \int_{-\infty}^{\infty} f(\alpha t)\, \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t = \int_{-\infty}^{\infty} f(t')\, \mathrm{e}^{-\mathrm{j}\left(\frac{\omega}{\alpha}\right)t'} \frac{1}{\alpha} \mathrm{d}t' = \frac{1}{\alpha} F\left(\frac{\omega}{\alpha}\right).$$

9.5.3 Time shift (delay)

Shifting a function in time does not affect the *amplitude* of the Fourier function: after all, the function still contains the same energy with the same distribution between frequency components. It does however affect the phase:

$$\mathcal{F}\{f(t - t_0)\} = F(\omega)\, \mathrm{e}^{-\mathrm{j}\omega t_0}. \tag{9.13}$$

Demonstration:

$$\mathcal{F}\{f(t - t_0)\} = \int_{-\infty}^{\infty} f(t - t_0)\, \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t = \int_{-\infty}^{\infty} f(t')\, \mathrm{e}^{-\mathrm{j}\omega\left(t' + t_0\right)} \mathrm{d}t' = F(\omega)\, \mathrm{e}^{-\mathrm{j}\omega t_0}.$$

Since we only consider real-valued time signals and ω and t_0 are real, we conclude that a time shift results in a phase shift in the frequency domain.

9.5.4 Frequency shifting (modulation)

This is just the complementary relation to the one above:

$$\mathcal{F}\left\{f(t)\,\mathrm{e}^{\mathrm{j}\omega_0 t}\right\} = \int_{-\infty}^{\infty} f(t)\,\mathrm{e}^{-\mathrm{j}\omega t+\mathrm{j}\omega_0 t}\mathrm{d}t = F(\omega-\omega_0)\,.$$

This result is extremely useful from a technological point of view, as it allows shifting a function to a different frequency range without otherwise affecting its contents. This is the basis for transmission of signals via radio waves and optical fibres, where ω_0 is the frequency of the so-called carrier wave or of the transmitted light, respectively.

9.5.5 Differentiation

Differentiation in the time-domain can directly be translated to multiplication by $\mathrm{j}\omega$ per order of differentiation in the frequency-domain.

$$\mathcal{F}\left\{\frac{\mathrm{d}}{\mathrm{d}t}f(t)\right\} = \mathrm{j}\omega F(\omega) \tag{9.14}$$

Demonstration:

$$\frac{\mathrm{d}}{\mathrm{d}t}f(t) = \frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\,\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\left(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{\mathrm{j}\omega t}\right)\mathrm{d}\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\mathrm{j}\omega F(\omega)\right]\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega$$

It also follows that further orders of differentiation simple lead to extra factors of $\mathrm{j}\omega$:

$$\mathcal{F}\left\{\frac{\mathrm{d}^n}{\mathrm{d}t^n}f(t)\right\} = (\mathrm{j}\omega)^n F(\omega)\,. \tag{9.15}$$

This is useful when dealing with differential equations, since differentiation is replaced with a much simpler multiplication in Fourier space (just as for the Laplace transform).

9.5.6 Duality

The Fourier transform and its inverse are almost the same operation. It is not surprising therefore that transform pairs are almost the same if you swap t and ω. If

$$\mathcal{F}\{f(t)\} = F(\omega) \tag{9.16}$$

then

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)\,. \tag{9.17}$$

Demonstration
This one is a bit more abstract and difficult to wrap one's head around because of its apparent simplicity, but remember that $f(x)$ and $F(x)$ are just different mathematical functions until one specifies what x is:

$$\mathcal{F}\{F(t)\} = \int_{-\infty}^{\infty} F(t)\,\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = 2\pi\frac{1}{2\pi}\int_{-\infty}^{\infty} F(t)\,\mathrm{e}^{\mathrm{j}(-\omega)t}\mathrm{d}t = 2\pi f(-\omega)\,.$$

What this means in practice is that if you have a transform pair $f(t) \leftrightarrow F(\omega)$ then, apart from sign changes and/or complex conjugates, the reciprocal is true: $F(\omega) \leftrightarrow f(t)$.

9.5.7 Symmetry

It is sometimes convenient to think of symmetry when performing a Fourier transform. Using Euler's formula, $\exp(-\mathrm{j}\omega t) = \cos(\omega t) - \mathrm{j}\sin(\omega t)$, we can rewrite the definition of the Fourier transform as

$$F(\omega) \equiv \int_{-\infty}^{\infty} f(t)\cos(\omega t)\mathrm{d}t - \mathrm{j}\int_{-\infty}^{\infty} f(t)\sin(\omega t)\mathrm{d}t. \tag{9.18}$$

Now consider a function $f(t)$ which is symmetric around $t = 0$ (*i.e.*, $f(-t) = f(t)$). Because the factor $\sin(\omega t)$ is anti-symmetric around $t = 0$ ($\sin(-\omega t) = -\sin(\omega t)$ and $\cos(-\omega t) = \cos(\omega t)$), the product $f(t)\sin(\omega t)$ is also antisymmetric and the integral is exactly zero. This means that $F(\omega)$ is real-valued and can be determined by taking only the $\cos$ integral instead of dealing with integrals of complex exponentials. For functions that are anti-symmetric (*i.e.* $f(-t) = -f(t)$) leads a function of $F(\omega)$ that is imaginary-values and can be determined by taking only the $\sin$ integral.

9.6 Fourier Transform of periodic functions → Fourier Series

In case we imply a periodicity on a function that is Fourier Transformed, we should get the equivalence of a Fourier Series. In treating Fourier Series we considered a finite interval of a function that repeated itself infinite times; with the condition in mathematical form:

$$f(t) = f(t + nT_0)$$

In order to explore the consequences of this periodicity for the Fourier transform of such function $f(t)$ we can take the Fourier Transform of this condition:

$$\mathcal{F}\{f(t)\} = \mathcal{F}\{f(t - T_0)\}$$

with this expression we can use the time-shift property of Fourier Transforms which yields:

$$F(\omega) = F(\omega)\mathrm{e}^{-\mathrm{j}\omega T_0}. \tag{9.19}$$

This condition can be fulfilled in two ways: either $F(\omega) = 0$ or the exponent is equal to one. The latter can only be fulfilled in case the angular frequency satisfies:

$$\mathrm{e}^{-\mathrm{j}\omega T_0} = 1 \;\Rightarrow\; \omega_n T_0 = 2\pi n \;\Rightarrow\; \omega_n = n\omega_0 \;\forall\; n \in \mathbb{Z}. \tag{9.20}$$

with $\omega_0 = 2\pi/T_0$ the fundamental frequency corresponding to the periodicity of the initial function. That is, $F(\omega)$ is always zero except at discrete frequencies $n\omega_0$. This is why we were able to express periodic functions in terms of Fourier series, that explicitly contain those frequencies that are harmonics of ω_0, leading to a(n infinite) discrete set of harmonic functions, being the Fourier Series of the periodic function.

9.7 Convolution property of the Fourier Transform

Consider two functions $x(t)$ and $h(t)$. The *convolution* between two functions is defined as

$$[x \otimes h](t) = x(t) \otimes h(t) \equiv \int_{-\infty}^{\infty} x(\tau)h(t - \tau)\mathrm{d}\tau. \tag{9.21}$$

Not that this definition of convolutions deviates from the definition of convolution in the Laplace domain by the integration interval. There is however a relation between the two.
We will explore in more detail below the physical meaning of convolution.

In the frequency domain, this expression corresponds to

$$\begin{aligned}
\mathcal{F}\left\{x(t) \otimes h(t)\right\}(\omega) &= \mathcal{F}\left\{x(t) \otimes h(t)\right\} \\
&= \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau \right\} \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t \\
&= \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega') \mathrm{e}^{\mathrm{j}\omega'\tau} \mathrm{d}\omega' \right] \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega'') \mathrm{e}^{\mathrm{j}\omega''(t-\tau)} \mathrm{d}\omega'' \right] \mathrm{d}\tau \right\} \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t \\
&= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(\omega') H(\omega'') \mathrm{e}^{\mathrm{j}(\omega'-\omega'')\tau} \mathrm{e}^{\mathrm{j}(\omega''-\omega)t} \mathrm{d}t \mathrm{d}\tau \mathrm{d}\omega' \mathrm{d}\omega'' \\
&= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(\omega') H(\omega'') \mathrm{e}^{\mathrm{j}(\omega'-\omega'')\tau} 2\pi\delta(\omega''-\omega) \mathrm{d}\tau \mathrm{d}\omega' \mathrm{d}\omega'' \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(\omega') H(\omega) \mathrm{e}^{\mathrm{j}(\omega'-\omega)\tau} \mathrm{d}\tau \mathrm{d}\omega' \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega') H(\omega) 2\pi\delta(\omega'-\omega) \mathrm{d}\omega' \\
&= X(\omega)\, H(\omega) \qquad (9.22)
\end{aligned}$$

Thus, convolution in the time domain corresponds to multiplication in the Fourier domain.
While we will not perform extensive manipulations of the convolution, we note here some identities of this operation for reference.

$$f \otimes h = h \otimes f \qquad (9.23)$$
$$f \otimes (g \otimes h) = (f \otimes g) \otimes h \qquad (9.24)$$
$$f \otimes (g + h) = f \otimes g + f \otimes h \qquad (9.25)$$
$$(f \otimes h)' = f' \otimes h + f \otimes h'$$

In fact the convolution in the Laplace domain and the Fourier domain are not equal. There are subtle differences. We don't address them here since we use a special class of signals. For finite length and real-values signals, these differences are suppressed.

9.7.1 Interpretation of convolution

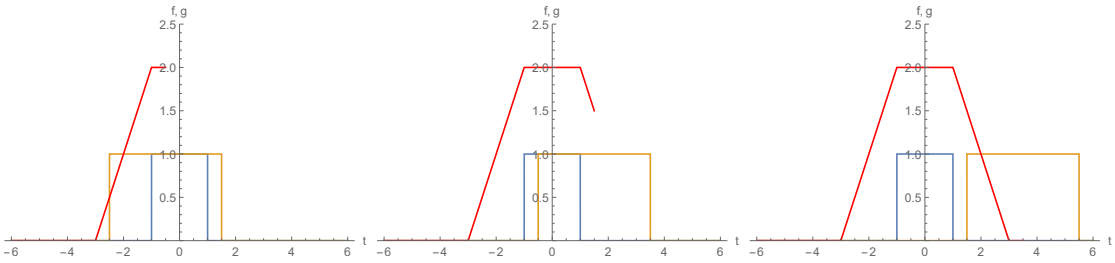


Figure 9.3: Convolution step-wise illustrated. Convolution between the yellow and blue block pulse results in the red curve and is given for three values of t: the time shift between the two functions.

In figure 9.3 the principle of convolution of two block pulses (blue and yellow) is illustrated. The blue block signal is fixed. The orange block signal is shifted with respect the the blue signal by an amount of τ ranging from $-\infty$ to ∞. This parameter is the independent variable for the red curve on the horizontal axes. For each time shift between the two block pulses, the overlap integral is calculated and added to the resulting curve in red. In general, evaulating the integral may involve several separate cases, as illustrated in the figure. The first graph shows a full overlap between the

two pulses. In the second graph the overlap is only partial, resulting in a decrease in the value of the convolution. In the third graph, the pulses do not overlap anymore, resulting in a function value of zero. This illustration was done with symmetric pulses. For a complete understanding of the concept of convolution, you should repeat the exercise with asymmetric pulses, especially if you do another one where you flipped one of the pulses, you get the full picture.

Example 9.7.1. Impulse response
As an example we consider the impulse response since it plays an important role in the applications of filters on signals. Remember that the impulse response is the time-domain equivalent of the Transfer function in the Laplace domain of the system.
Consider a system characterized by the transfer function $H(s)$ and let us denote the input with $x(t)$ and output with $y(t)$ or with their Laplace equivalents: $X(s)$ and $Y(s)$ respectively. Using the impulse function as an input, *i.e.* $x(t) = \delta(t)$ or, equivalently, $X(s) = 1$. Then the output is

$$y(t) = \mathcal{L}^{-1}\{H(s)X(s)\} = \mathcal{L}^{-1}\{H(s)\} = h(t). \tag{9.26}$$

Using the time representation of the system, we can also calculate the result of the output using the convolution:

$$y(t) = x(t) \otimes h(t) = \int_{-\infty}^{\infty} \delta(\tau)h(t-\tau)\mathrm{d}\tau = h(t). \tag{9.27}$$

which gives us the same result. Thus $h(t)$ and $H(s)$ are two representations of the same system and contain the same information of the system, this means that $h(t)$, just like $H(s)$, represents a complete characterization of the system. Indeed, Eq. 9.22 indicates that an alternative way of solving a LTI differential equation is to follow the following steps:

a. Evaluate $h(t)$. This can be done either by taking the inverse Laplace transform of $H(s)$, or solving the LTI ODE for an input $x(t) = \delta(t)$ to get the output $y(t) = h(t)$ using either classic or Laplace Transform methods.

b. Armed with $h(t)$, convolve it with the input function $x(t)$ in order to obtain the output function $y(t)$.

□

The method in this example will become important in the manipulation of digital signals. For continuous systems and signals this method is not the most efficient way to calculate the output of a system from an given input function in the time domain. This method is however important in the application of filters and digital signal processing in general, since a numerical implementation on sampled data is relatively easy and efficient to apply.

9.8 Energy and Parseval's theorem

Using the same arguments as we applied in the previous chapter on Fourier Series, we can associate the power of a signal to the absolute-square of the signal:

$$P_f(t) \propto |f(t)|^2$$

where we took the proportionality constant equal to one in this case, for simplicity. From this power function we can calculate the energy E in the signal by integrating over time; to obtain the total energy we have to integrate over the full domain, thus:

$$E \propto \int_{-\infty}^{\infty} |f(t)|^2 \mathrm{d}t \tag{9.28}$$

for functions $f(t)$ for which this integral converges (you should mostly be thinking of smooth functions that have a finite amplitude and decay to zero for $t \to \pm\infty$). Applying the Fourier Transform to the function $f(t)$ in the energy expression, we can formulate the total energy in terms of the Fourier Transformed functions $F(\omega)$:

$$\begin{aligned} E \propto \int_{-\infty}^{\infty} |f(t)|^2 \mathrm{d}t &= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \mathrm{e}^{\mathrm{j}\omega t} \mathrm{d}\omega \right) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega') \mathrm{e}^{\mathrm{j}\omega' t} \mathrm{d}\omega' \right)^* \mathrm{d}t \\ &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} F(\omega) \int_{-\infty}^{\infty} F^*(\omega') \left(\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{j}((\omega-\omega')t} \mathrm{d}t \right) \mathrm{d}\omega' \mathrm{d}\omega \\ &= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} F(\omega) \int_{-\infty}^{\infty} F^*(\omega') \left(2\pi\delta\left(\omega - \omega'\right) \right) \mathrm{d}\omega' \mathrm{d}\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) F^*(\omega) \mathrm{d}\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 \mathrm{d}\omega. \end{aligned} \tag{9.29}$$

This is the generalization of Parseval's theorem to energy signals. Each frequency interval carries with it a well-defined amount of energy. For example, the statement "the pulse carries one μJ of energy in the 10-120 kH range" has a precise physical meaning. Note that this also implies, combined with Eq. 9.7, that the function $\delta(t)$ carries infinite energy. Although the integral of a delta function is equal to one, the integral of the square of the delta function is infinite. This can be seen if you go to the other Fourier domain: a sine stretches over the full domain from $[-\infty, \infty]$ resulting in a total energy of infinity. Since both integrals (time-domain and frequency-domain) should give the same number, they are both infinite. This can be shown by the fact that the integral of the square of the delta function can be approximated by:

$$\int_{-\infty}^{\infty} \delta^2(t) \mathrm{d}t \approx \lim_{\epsilon \to 0} \frac{1}{\epsilon} \frac{1}{\epsilon} \epsilon = \infty$$

This picture is consistent with the integral of the delta function:

$$\int_{-\infty}^{\infty} \delta(t) \mathrm{d}t \approx \lim_{\epsilon \to 0} \frac{1}{\epsilon} \epsilon = 1$$

Summaries

Summary Chapter 9a

Main concepts in Fourier transforms:

- A function $f(t)$ can also be described in the frequency domain by its spectral density function $F(\omega)$:

$$f(t) \longleftrightarrow F(\omega)$$

- Switching between these domains require computing an integral:

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}dt \;\; f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega$$

- Definition of convolution:

$$[x \otimes g](t) = x(t) \otimes g(t) \equiv \int_{-\infty}^{\infty} x(\tau)g(t-\tau)\mathrm{d}\tau. \tag{9.30}$$

- Parseval's theorem extends to nonperiodic energy signals in the following form:

$$E \propto \int_{-\infty}^{\infty} |f(t)|^2 \,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2 \,\mathrm{d}\omega$$

Summary Chapter 9b

Fourier Transform Properties:

#	Property	$f(t) = \mathcal{F}^{-1}\{F(\omega)\}$	$F(\omega) = \mathcal{F}\{f(t)\}$
1	Real Functions		$F(-\omega) = F^*(\omega)$
2	Linearity	$\alpha f_1(t) + \beta f_2(t)$	$\alpha F_1(\omega) + \beta F_2(\omega)$
3	Reciprocal Scaling	$f(\alpha t)$	$\frac{1}{\alpha} F\left(\frac{\omega}{\alpha}\right)\ \forall\ \alpha > 0$
4	Differentiation	$\frac{\mathrm{d}}{\mathrm{d}t} f\,(t)$	$\mathrm{j}\omega F\,(\omega)$
6	Time Shift	$f(t - t_0)$	$F(\omega)\mathrm{e}^{-\mathrm{j}\omega t_0}$
7	Frequency Shift	$f(t)\mathrm{e}^{\mathrm{j}\omega_0 t}$	$F(\omega - \omega_0)$
8	Convolution	$f(t) \otimes g(t)$	$F(\omega)G(\omega)$
9		$f(t)g(t)$	$\frac{1}{2\pi} F(\omega) \otimes G(\omega)$
10	Parseval's theorem	$\int_{-\infty}^{\infty} \lvert f(t)\rvert^2\,\mathrm{d}t$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} \lvert F(\omega)\rvert^2\,\mathrm{d}\omega$

Summary Chapter 9c

Fourier Transform pairs:

$f(t) = \mathcal{F}^{-1}\{F(\omega)\}$	$F(\omega) = \mathcal{F}\{f(t)\}$
$\delta(t)$	1
$\mathrm{e}^{\mathrm{j}\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$A \sin \omega_0 t$	$-\mathrm{j}\pi A\delta(\omega - \omega_0) + \mathrm{j}\pi A\delta(\omega + \omega_0)$
$= \begin{cases} A & \lvert t\rvert < \tau/2 \\ 0 & \textit{otherwise} \end{cases}$	$\tau A \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$
$= \begin{cases} A\mathrm{e}^{-t/\tau} & t > 0 \\ 0 & \textit{otherwise} \end{cases}$	$\frac{\tau A}{1 + \mathrm{j}\omega\tau}$
$= \begin{cases} 1 - \lvert t\rvert/\tau & -\tau t < \tau \\ 0 & \textit{otherwise} \end{cases}$	$\tau \operatorname{sinc}^2\left(\frac{\omega\tau}{2}\right)$
$\frac{A}{\pi}\frac{a}{a^2 + t^2}$	$A\mathrm{e}^{-a\lvert\omega\rvert}$

Problems

Problem 9.1. Definition, table and properties

a. Assume a is real and $a > 0$. Calculate $F(\omega)$ of

$$f(t) = \mathrm{e}^{-at}\, u(t)$$

Plot $|F(\omega)|$ and $\arg(F(\omega))$. Comment on the shape of the two graphs.

b. Assume a is real and $a > 0$. Calculate $F(\omega)$ of

$$f(t) = \mathrm{e}^{-a|t|}$$

c. Compare the results of a) and b).

d. Find $F(\omega)$ of (using table, known properties, etc.)

$$f(t) = 2\sin(40(t-7)) + \mathrm{e}^{-3t}\, u(t) \otimes \mathrm{e}^{-|t|}$$

For every step, explicitly mention which property you are using.

e. Find the Fourier transform of $\mathrm{e}^{-3t}\, u(t) \otimes \mathrm{e}^{-|t|}$ using the definition of the Fourier transform directly.

f. Use duality to find $F(\omega)$ of

$$f(t) = \frac{3}{1+t^2}$$

Problem 9.2. Fourier Transforms
Sketch a graph of each of the following functions, paying attention to any symmetry that may exist. Then compute its Fourier transform $F(\omega)$ and sketch the result (note that a Fourier transform can give a complex valued function, in which case you should explicitly plot the real and imaginary components).

a. A pure sine function:

$$f(t) = A\sin(\omega t)$$

b. An exponentially decaying step:

$$f(t) = \begin{cases} A\mathrm{e}^{-|t|/\tau} & t > 0 \\ 0 & elsewhere \end{cases}$$

c. A triangular pulse with amplitude A and duration 2τ:

$$f(t) = \begin{cases} 1 - |t|/\tau & -\tau < t < \tau \\ 0 & elsewhere \end{cases}$$

Problem 9.3. Inverse Fourier transforms
Sketch a graph of each of the following Fourier transform functions, paying attention to any symmetry that may exist. Then compute the corresponding time-domain function $f(t)$.

a. A shifted delta function:

$$F(\omega) = 2\pi\delta(\omega - \omega_0)$$

b. A symmetric exponential decay:

$$F(\omega) = A\mathrm{e}^{-a|\omega|}$$

Problem 9.4. Calculation versus table and properties

a. Find the Fourier transform $F(\omega)$ by using its definition of $f(t) = \begin{cases} 2, \ t \ \in [-4, 0] \\ -2, \ t \in [0, 4] \\ 0 \text{ otherwise} \end{cases}$

b. Redo a), but now write $f(t)$ as a rect function and use a table. Verify it matches the answer of the previous question.

c. Redo a), but now write $f(t)$ as a difference of time shifted $u(t)$ functions and use a property and a table. Verify it matches the answer of the previous question.

Problem 9.5. A third interpretation of the FT

a. When you inspect a Fourier transform pair table, you will see the entry

$$\mathrm{e}^{\mathrm{j}\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$$

What does this Fourier transform pair tell you?

b. We can generalize this by using the inverse FT formula,

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega)\,\mathrm{e}^{\mathrm{j}\omega t}\,d\omega.$$

What does the Fourier transform tell you about $f(t)$? It might help to interpret the integral as a (Riemann) sum with many terms.

c. Consider the Fourier transform of $f(t) = e^{-|t|}$. What does the graph of $F(\omega)$ tell you about $f(t)$?

d. When you inspect a Fourier transform properties table, you will find the frequency shift property

$$f(t)\,\mathrm{e}^{\mathrm{j}\omega_0 t} \leftrightarrow F(\omega - \omega_0)$$

Give an intuitive explanation why this must be true. Use your answer to b) in your explanation.

Problem 9.6. Filter and Parseval

a. We filter $f(t) = \mathrm{e}^{-t}\,u(t)$ with a filter with transfer function $H(\omega) = \dfrac{1}{1 + \mathrm{j}\omega}$ yielding output $y(t)$. Find the Fourier transform $Y(\omega)$.

b. Determine $\displaystyle\int_{-\infty}^{+\infty} |y(t)|^2 dt$ twice using Parseval. Use Mathematica or Wolfram Alpha to evaluate the frequency domain integral.

Problem 9.7. Energy and filtering

We have discussed that the Dirac delta function has infinite energy as it contains all frequency components. If a delta function is used at the input to a low-pass filter, however, the high frequencies will be suppressed. Consider for example a simple RC circuit whose transfer function is given by

$$G(\omega) = \frac{1}{1 + \mathrm{j}\omega RC}$$

a. If the input of this circuit is $\delta(t)$, what is the energy of the output signal?

b. We can define the bandwidth B of our filter as the frequency at which the transmitted power is halved. What is B for our RC filter?

c. What happens if we decrease the time constant RC of the circuit by using different resistors and/or capacitors?

Notes: In real life no system can respond infinitely fast and high enough frequencies are always suppressed such that infinite energies do not show up in the output signal. The Dirac delta function is just a very convenient mathematical idealization introduced by, you guessed it, one Mr. Dirac to describe a pulse that is way faster than what the system can handle. He was working on the foundations of quantum mechanics, but his mathematical trick was later picked up in many fields of science and technology.

Problem 9.8. AM radio

In Amplitude Modulated (AM) transmission, a low-frequency signal $f(t)$ such as the sound of your voice is multiplied with a high-frequency sinusoidal $A\cos(\omega_0 t)$ to yield the transmitted signal f_{trans}:

$$f_{trans} = Af(t)\cos(\omega_0 t)$$

Another way to think about it is that $f(t)$ modulates the amplitude of the periodic wave $A\cos(\omega_0 t)$, hence the AM label. The resulting signal is transmitted as electromagnetic radiation emanating from an antenna.

a. Suppose that the sketch in figure 9.4 represents the spectral density function $F(\omega)$ of your broadcast:

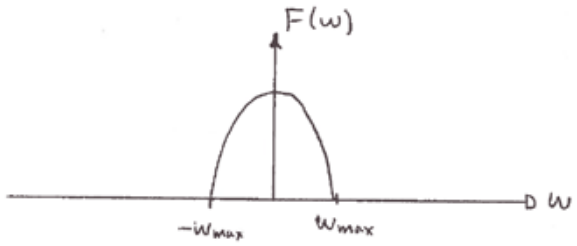


Figure 9.4: Spectral density function of broadcast.

Here ω_{max} is the maximum angular frequency present in the signal. We take $F(\omega)$ to be a real function for convenience in plotting (none of the discussion below is changed by the presence of a complex component, which does basically the same thing as the real component). Typically f_{max} is of order 10 kHz while f_0 is of order 1 MHz, so you can assume $\omega_{max} << \omega_0$. Make a sketch similar to the above of the Fourier transform of the radio signal being sent out.

b. Upon reception by your radio's antenna, the signal is again multiplied by $\cos(\omega_0 t)$ (choosing the right frequency ω_0 is what you're doing when tuning to a station). Make a sketch of the spectral density function of the demodulated signal (that is, the transmitted signal after it was multiplied by $\cos(\omega_0 t)$).

c. Is there any reason to use a low-pass filter to kill any bits of the signal that we might not want? Explain your reasoning.

d. There are many different radio stations broadcasting at the same time. Make a rough sketch of what the electromagnetic spectral density function looks like when that is the case. e. You

sometimes hear in the news about "spectrum being auctioned" or mobile phone companies "buying spectrum". Cell phones don't use AM protocols but the idea is pretty much the same. What do you think is being bought (by companies) or auctioned (by governments)?

Philosophical musing: the saying goes that "time is money". But bandwidth is money too! Somehow the reciprocity relation does not apply to money...

Problem 9.9. Fourier transforms

Sketch a graph of each of the following functions, then compute its Fourier transform $F(\omega)$. You can often make use of the properties of Fourier transforms to simplify the calculation.

a.

$$f(t) = \frac{t}{|t|} \mathrm{e}^{-b|t|}$$

In the limit $b \longrightarrow 0$ this is the signum function, $f(t) = sign(t)$. We could not evaluate the Fourier transform of $sign(t)$ directly, however, because it is not absolutely integrable.

b. The two rectangular pulses sketched in figure 9.5.

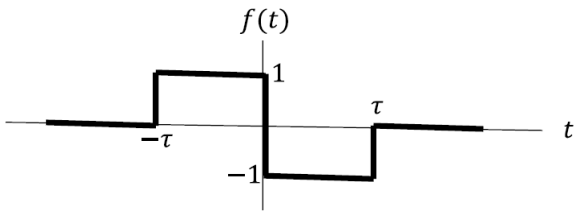


Figure 9.5: Two rectangular pulses

c. The modulated pulse function,

$$f(t) = \begin{cases} A\cos(\omega_0 t) & |t| < \tau/2 \\ 0 & elsewhere \end{cases}$$

d. Determine the complex-valued constant k required to make the following Fourier transform pair relation hold:

$$\mathcal{F}\{\mathrm{e}^{\mathrm{j}\pi t} f(t - \frac{1}{2})\} = k\mathrm{e}^{-\frac{\mathrm{j}\omega}{2}} F(\omega - \pi)$$

Problem 9.10. Fourier Transforms of basic functions

a. Rebuild the Fourier Transforms of the functions from the table in the summary.

b. Make plots of the Fourier pairs

Problem 9.11. Role of phase in spectra

In order to illustrate the role of the phase in the Fourier Tranfomations, you have to look at a signal given by a specific function:

$$f(t) = \begin{cases} \mathrm{e}^{-\frac{t}{\tau}} & t > 0 \\ 0 & \textit{otherwise} \end{cases}$$

a. Make a Fourier Transform of the function.

b. Plot the real part and the imaginary part in one plot.

c. Take the absolute value of the Fourier Transform and plot this in a separate plot.

d. Use the calculated absolute value function and take the inverse Fourier Transform.

e. Plot the result and compare with the original signal.

Problem 9.12. Convolution two blocks

a. Define $f(t) = 1$ for $t \in [-1, 1]$ and 0 otherwise. Calculate

$$g(t) = f(t) \otimes f(t) = \int_{-\infty}^{+\infty} f(\tau) f(t - \tau) d\tau$$

Hint: how does the definition impact the integral and what can you say about $f(\tau)f(t - \tau)$?

b. Find $g(t)$ through $G(\omega)$.

Problem 9.13. Convolution
Find the convolution of the rectangular pulse $f_1(t)$ and the triangular pulse $f_2(t)$ shown in figure 9.6 (hint: there are 5 different regimes).

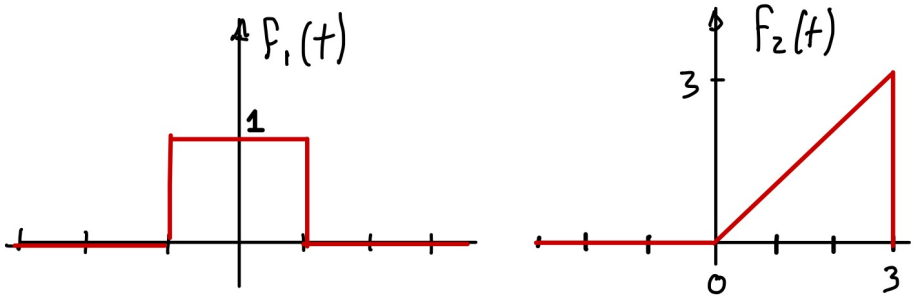


Figure 9.6: Definition of functions for exercise 20.1.

Problem 9.14. Convolution and Fourier
Determine the Fourier transform of the signal $g(t) = u(t)\exp(-t) \otimes u(t)\exp(-2t)$, where $u(t)$ is the unit step function, by

a. Performing the convolution first and then taking the Fourier transform of the result.

b. Taking the Fourier transform of each term and then multiplying.

Problem 9.15. Equivalent representations of convolution
Show that the two representations

$$[x \otimes g](t) =\equiv \int_{-\infty}^{\infty} x(\tau) g(t - \tau) \mathrm{d}\tau \tag{9.31}$$

and

$$[g \otimes x](t) =\equiv \int_{-\infty}^{\infty} g(\tau) x(t - \tau) \mathrm{d}\tau. \tag{9.32}$$

are equivalent.

Problem 9.16. rectangular, sinc and Heisenberg demystified
The `rect` function is defined as

$$\text{rect}(t) = \begin{cases} 1, \ t \in [-\frac{1}{2}, +\frac{1}{2}] \\ 0 \text{ otherwise} \end{cases}$$

We have

$$\frac{1}{\tau}\text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \text{sinc}\left(\frac{\omega\tau}{2}\right) \text{ or } \text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau\,\text{sinc}\left(\frac{\omega\tau}{2}\right)$$

This Fourier transform pair enables us to study two interesting limit cases.

a. Select the expression on the left. What happens if $\lim_{\tau\to 0}$?

b. Select the expression on the right. What happens if $\lim_{\tau\to+\infty}$?

c. One of the cornerstones of quantum physics is Heisenberg's uncertainty principle that is "a fundamental limit to the accuracy with which the values for certain pairs of physical quantities of a particle can be predicted from initial conditions" as explained in `https://en.wikipedia.org/wiki/Uncertainty_principle`. This article uses position and momentum, but another lesser known version involves energy and time. In the latter case, the principle reads

$$\sigma_E \sigma_t \geq \frac{\hbar}{2}.$$

Here, the $\sigma_{\ldots}$ denotes the standard deviation in the measurement of $\cdots$ and $\hbar$ is Planck's constant. This constant relates energy E of a photon to its frequency ω through $E = \hbar\omega$. Questions a) and b) demonstrate that a signal cannot be concentrated both in time and frequency, or as an inequality:

$$\sigma_\omega \sigma_t \geq \frac{1}{2}.$$

Show that Heisenberg's uncertainty relation $\sigma_E\sigma_t \geq \frac{\hbar}{2}$ is a fundamental property. N.B. It can be shown that the minimum $\frac{1}{2}$ is obtained by Gaussian shaped distributions.

Problem 9.17. Frequency convolution
Show that

$$\mathcal{F}\{x(t)g(t)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(u)H(\omega - u)\mathrm{d}u. \tag{9.33}$$

Problem 9.18. Convolution Theorem for Fourier Transform
Consider two signals given by their mathematical descriptions:

$$f(t) = \mathrm{e}^{-t^2}$$
$$g(t) = \sin 7\pi t$$

a. Show that the Fourier Transform of the convolution of the two signals is equal to the product of the Fourier Transforms of the individual signals.

$$\mathcal{F}\{f(t) \otimes g(t)\} = \mathcal{F}\{f(t)\} \cdot \mathcal{F}\{g(t)\}$$

b. Make Graphical representations of the intermediate steps.

Problem 9.19. Convolution Theorem for Laplace Transform
Consider two systems given by their transfer functions:

$$H(s) = \frac{1}{s^2 + 9}$$
$$G(s) = \frac{1}{s + 4}$$

where ω_0 is a constant.

a. Show that the inverse Laplace Transform of the product of the two functions is equal to the convolution of the two inverse Laplace Transforms of the individual functions for the given transfer functions.

$$\mathcal{L}^{-1}\{H(s) \cdot G(s)\} = \mathcal{L}^{-1}\{H(s)\} \otimes \mathcal{L}^{-1}\{G(s)\}$$

b. Make a graphical representation of your results.

Chapter 10

Digital Signals and Systems

Entering the digital domain of Signals

Up to this point we treated mainly continuous signals. We did some simulations of systems with signals, however this was all a description of continuous signals. From here onward we will concentrate on digital signals. Nowadays almost all signals are treated digitally, since it is far more efficient en flexible. The digital domain has many parallel descriptions with the analogue domain, like difference equations as oppose to differential equations and z-Transforms as oppose to Laplace transforms, however there are also differences that lead to specific artefacts like aliasing.

10.1 Introduction

Digital Signal Processing (DSP) is the manipulation of a digital signals in all forms: time signals, scientific data, images. Examples of manipulations include increasing the yellow color intensity in an image to better observe yellow stained features, enhancing speech in a noisy background by amplifying the high frequencies as used to be done in the communication with astronauts, combining signals from different telescopes to enhance the resolution of an image of the universe or in an MRI scan. These examples share that a digitized signal is manipulated by a computer in a designed manner to achieve a specific objective. A lot of these applications used to be carried out using analogue electronics. DSP has however become dominant in the last decades. Pushed by the flexibility and low cost of digital electronics, high frequency sampling techniques and computation have evolved at an enormous rate. Furthermore, doing everything digitally means that the manipulation process can easily be adjusted in software without re-designing and replacing costly hardware.

Nowadays measured signals are digitized with a minimum of analogue treatment. Typically, only a (simple) low-pass filter needs to be applied before the signal is fed into an ADC (analogue to digital converter). The motivation will become clear later. Although the world of Analogue-to-digital converters is a very interesting one, we do not address this topic in this course: we will treat it as a black box that does its work. Once the signal is digitized, the world is open for tweaking and manipulation. This can be done with a general-purpose computer, but more commonly it is carried out by a dedicated, programmable DSP chip. Once the signal has been processed, it is either stored in this form or sent back into the analogue world again via a DAC (digital to analogue converter). Filtering the output may be needed in cases where the sharp edges from digital signals have a detrimental effect. The process is schematically illustrated in figure 10.1.

The remainder of the course will be mainly about digital signal processing, in particular for the design of filters. Since sampling and digitalisation are of central importance to the complete process, however, we first focus on these steps in this chapter.

In order to make a clear distinction and reduce confusion, we will use a clear notation for digital

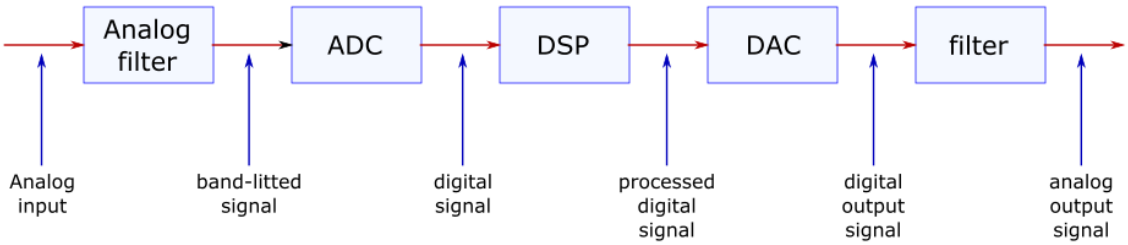


Figure 10.1: Illustration of a Digital Signal Processing procedure.

signals. The digital equivalent of a continuous time-function is a time-series:

$$x(t) \leftrightarrow x[n]$$

We use round brackets for continuous-time functions and square brackets for discrete time-series.

10.2 Discrete and time-sampled signals

In the process of converting a continuous signal to a digital signal we distinguish two functional processes in the ADC that are important: time-sampling and discretisation. The first - time sampling - is the process of taking regular samples of the signal values. This is the main process we focus on in this reader. The second - dicretisation - is the process of translating the signal value to a numerical representation that can be stored in a computer. This latter process, which might be very important in the actual process of digitising an analogue signal will hardly be addressed in this reader. In the next paragraph we highlight very shortly a few important concepts of discretisation before we focus on time-sampling.

10.2.1 Discretisation

The discretisation process in an ADC can be done in various ways, however in the end it narrows down to representing the signal value by a digital number. Important in this process is the number representation often expressed in the number of bits available to store the number. Values of 8, 12, 16, 24, 32 and even higher bit numbers are used. These bit numbers do not determine all the characteristics, also the voltage range of the ADC is important. The bit number n determines the number of voltage levels: 2^n, where as the voltage range together with the number of voltage levels determines the resolution of the ADC in ΔV. As an example: a 12-bit ADC with a voltage range of -5 v to +5 V has a resolution of $\Delta V = 4.9\,\mathrm{mV}$. Here one bit is used for the sign of the number, so effectively the range is divided by 2^{11} steps.
Unless otherwise stated, for the rest of the reader we do not care about the discretisation process neither on the representation of the discrete signal values. In practice you might run into limitations of the discretesation process, however in modern computers, the representation of the numbers in memory is never a problem.

10.2.2 Time-sampling

The time-sampling of a signal is regulated by a (stable) clock signal. On a regular basis a sample of the signal is digitized and stored. That is, instead of dealing with the full functional form of a continuous signal $x\,(t)$, we sample it at a fixed time interval T_s and record a series of discrete values

$$x\,[n] = x\,(nT_s)\,, \tag{10.1}$$

where n is an integer. By convention we usually take *n* in the range $n = 0, 1, 2, \ldots, N-1$, where N is the total number of data points in the sampled data set. This sampling of an analogue signal is illustrated in figure 10.2. Some remarks:

- The sampling interval T_s is related to the sampling frequency f_s by

$$f_s = \frac{1}{T_s}. \tag{10.2}$$

  In order to use these quantities in transfer functions it is more convenient to use the cyclic frequency ω_s:

$$\omega_s = 2\pi f_s = \frac{2\pi}{T_s} \tag{10.3}$$

- As in earlier chapters, the quantities f and ω are often both denoted by "frequency", even though they differ by a factor 2π, as defined above. This might get very confusing. In general one uses f in case the frequency is directly related to time, and not to a process, *e.g.* the sampling frequency f_s. ω is more commonly related to processes described by a transfer function $H(\omega)$. As always there is no "right" or "wrong" choice, merely conventions. Some authors, for example, prefer instead to work only with f and write down explicitly all the factors of 2π that then arise when dealing with complex exponentials.

- The discrete function $x[n]$ is just a one-dimensional array of numbers; it contains no intrinsic time information. To be able to express $x[n]$ in the time domain with a time axis, one must additionally know the sampling interval T_s.

- $x\,[n]$ can in general be a complex function, but unless stated explicitly otherwise we will assume that it is as a real-valued function.

- During the process of digital sampling, the value of $x[n]$ also becomes quantized. In conventional 16-bit sampling, for example, the allowed range of input signals is subdivided into 2^{16} equally spaced levels. We will not deal further with the consequences of this quantization, but it can become relevant when detecting low signal levels that may correspond to only a few quantized levels.

10.3 Sampling a periodic signal

Suppose that $x(t)$ is a periodic signal with period T. This signal is sampled with period T_s. Will the resulting digital data $x[n]$ be periodic in n?
In order for $x[n]$ to be periodic, both $x(t)$ and $x[n]$ must repeat after a certain number of periods,

$$pT = qT_s, \qquad p, q = 1, 2, \ldots, \tag{10.4}$$

or, equivalently,

$$T_s = \frac{p}{q}T, \qquad p, q = 1, 2, \ldots \tag{10.5}$$

If this condition is not satisfied, then the digitized trace is *not* periodic, which can be rather counter intuitive. When the condition is satisfied, the period of $x[n]$ is q samples, where $q = pT/T_s$.

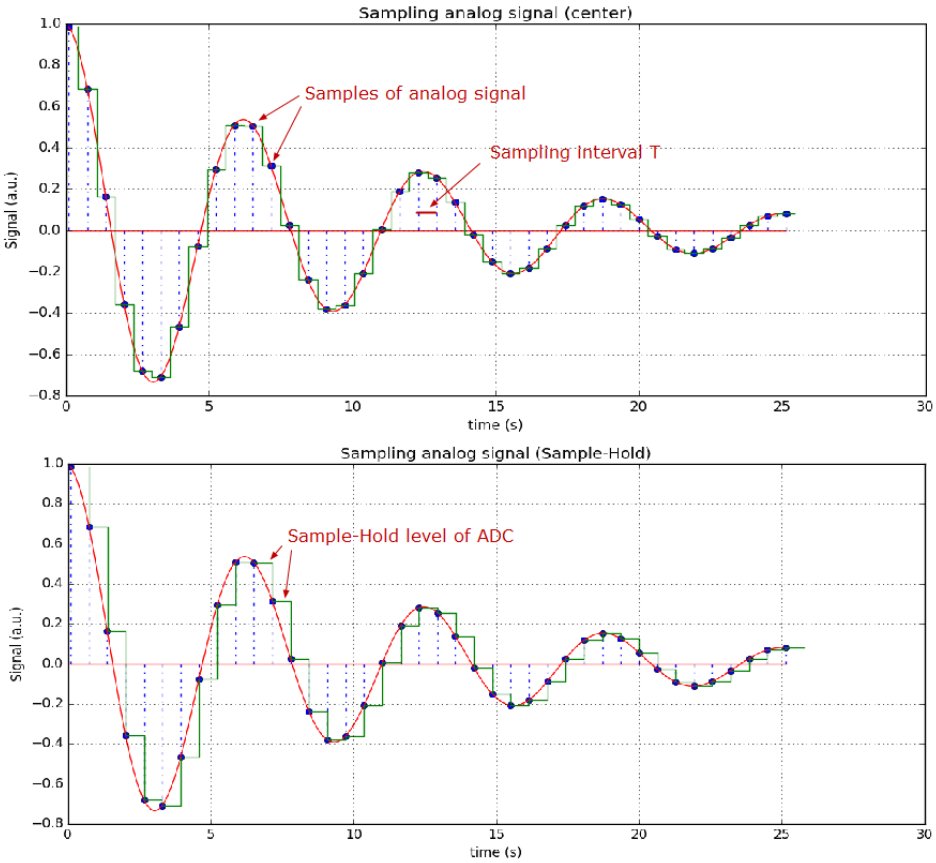


Figure 10.2: Sampling of an analogue signal with a sample and hold scheme.

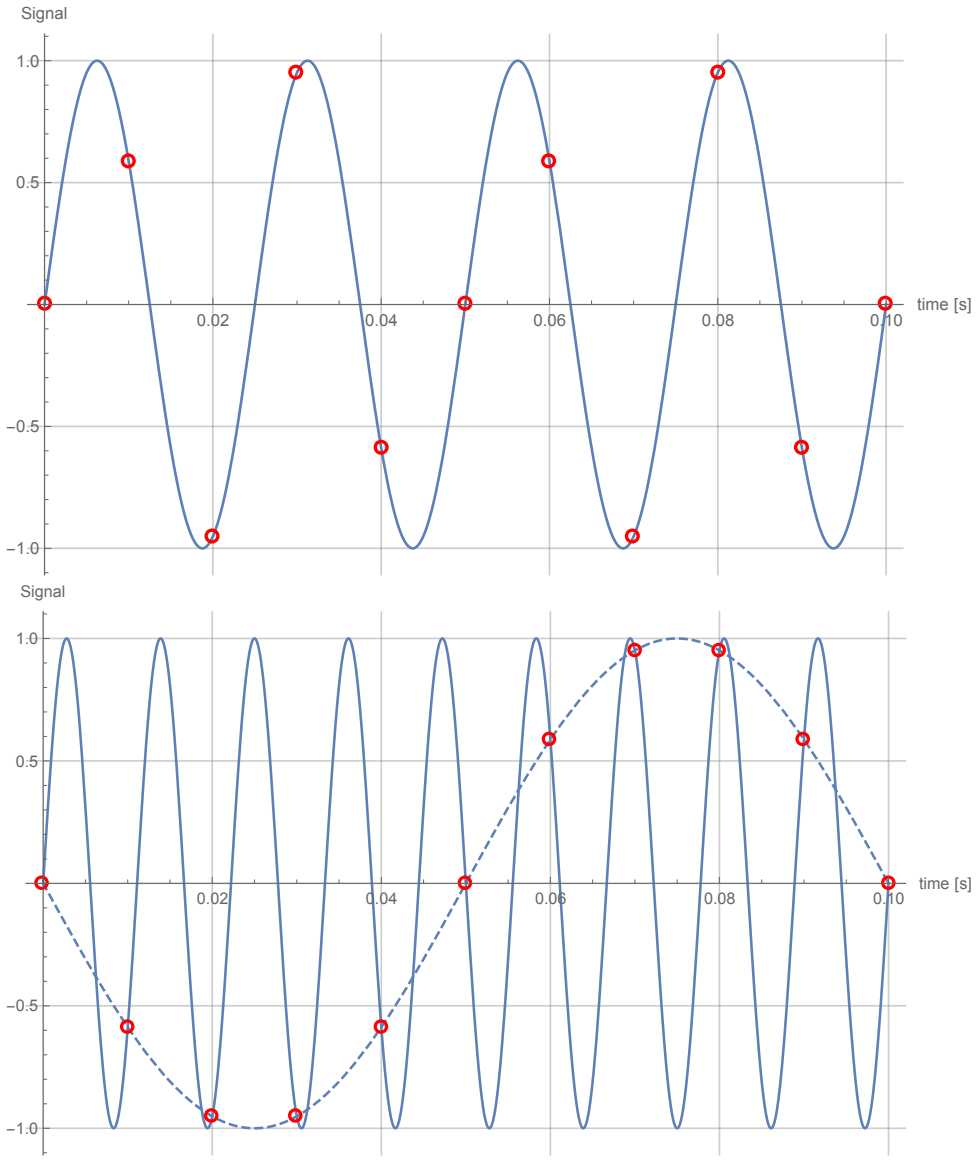


Figure 10.3: Sampling of a 40Hz sinus (top) and a 90 Hz sinus with a sampling frequency of 100Hz.

10.4 Aliasing

The choice of the sampling time puts restrictions on the frequency content of the analog input signal $x(t)$ if we want the digitized version $x[n]$ to be a faithful representation. It is obvious that one will lose or badly corrupt all information of signals with a frequency close to or above $1/T_s = f_s$, the sampling frequency. The form this corruption takes is depicted in fig. 10.3, where two sinusoidal signals are sampled at intervals of 0.01 s ($f_s = 100$ Hz). For a 40 Hz sinusoid analog signal (top panel), the samples (red circles) still more or less mimick a 40 Hz sinusoid, albeit in a choppy manner. On the other hand, when one samples a 90 Hz sinusoidal signal (bottom panel), the resulting samples look very much like a 10 Hz sinusoid. The sampling frequency of 100 Hz is clearly unsuitable for distinguishing between a 10 Hz and a 90 Hz signal.
We will see in detail in the next chapter that only frequencies below half the sampling frequency ($f_s/2$, known as the *Nyquist frequency*) can be sampled correctly. For this reason, the analogue input to a DSP system is usually filtered prior to digitization so as to retain frequency components below the Nyquist frequency but eliminate all frequencies larger than $f_s/2$. Such a filter is known as an *anti-aliasing filter*. In fig. 10.1, this role is played by the analog filter at the input of the circuit.

10.5 A simple DSP system

Now we have the continuous signal sampled and digitised in the computer, we need tools to manipulate the digital data. The digital data appears in lists: so called time-series. How do we manipulate these time-series? We need a system that transfers an input signal to an output signal that is modified in a way that meets the specifications we set. First we have to represent the input signal. A digital signal is just a list, or array, of numbers, $x[n]$. We employ square brackets to indicate that $x[n]$ is a discrete rather than a continuous function. An alternative to this list of numbers an algorithmic description of the $x[n]$ as a function of n can be given, *e.g.*:

$$x[n] = \mathrm{e}^{-an}u[n]$$

where a is a constant and $u[n]$ is the unit step function. In fact this definition defines the infinite list of numbers.
What the sequence of numbers represent here is irrelevant for the implementation of the algorithm that works on this time-series: they could represent a sound amplitude, a position or a temperature. The time-series does not contain any information off time. This information is usually provided separately in the form of a sampling frequency[1].
To get an impression of what Digital Signal Processing (DSP)/indexDSP is all about, we first consider a simple example: a moving average filter. It is not difficult to imagine that applying a moving average algorithm on a signal will suppress sharp peaks present in the signal. A system that suppressing sharp peaks can be regarded as a low-pass filter.
A very simple algorithm that does the moving averaging is one that replaces every data point in the list by the average of itself and its two neighbours. In mathematical form this can be written as:

$$y[n] = \frac{1}{3}\left(x[n-1] + x[n] + x[n+1]\right). \tag{10.6}$$

Intuitively, we can imagine that this so-called 3-point average algorithm smooths out fast variations and can thus be used to suppress high frequency components in the time-series.
The sort of equations such as equation 10.6 are called difference equations. These are the discrete equivalent of differential equations for the continuous domain. Equation 10.6 is a very simple example of a difference equation; more complex cases will be treated in chapter 11.1.

[1] Time-series are sampled at equally spaced intervals in time, but this information is usually not contained in the array of numbers itself.

At this point we introduce a new concept: real time processing. In our discussion of Fourier transforms, we assumed that a complete trace $f(t)$ was available for mathematical manipulation. Indeed, this was necessary to evaluate the Fourier transform. Examples of applications where this is the case are post-processing of stored recordings or images so as to filter out unwanted features. Most control systems, on the other hand, must deal with a signal as it comes in and respond as fast as possible. In continuous systems we identified this as causal systems: the manipulation of data can only be done on signal values from the past.
The 3-point average filter defined by Equation 10.6 lacks this very important characteristic of a real time algorithm: it requires knowledge of the future (more specifically, it requires the value $x[n+1]$ to calculate $y[n]$). This is a non-causal algorithm. Fortunately this can be easily amended with a simple time shift, mathematically formulated as

$$y[n] = \frac{1}{3}(x[n-2] + x[n-1] + x[n]). \tag{10.7}$$

This implementation ensures the causal behaviour required for a real-time application: the newly calculated output at step n only depends on inputs at step n or earlier. However, this shift comes at a price: the whole output will now have a time delay of one time step. It is easy to understand that more complicated filtering usually requires larger time shifts.
The essence of digital filtering is that the actual filter implementation takes place in the time domain, making real time applications possible. An alternative method for non-real time applications, filtering in the frequency domain, may seem a viable alternative but in practice it requires several non-trivial actions:

a. Fourier Transform to the frequency domain

b. Removal of the unwanted part of the spectrum

c. Transformation back to the time domain

These 3 steps seem easier than they actually are, especially the second step. In the frequency domain one has both an amplitude and a phase of the signal. Both amplitude and phase have to be manipulated simultaneously to insure that the inverse transform leads to a real-valued signal again. If the phase is not treated properly, it quickly leads to a *signal in – garbage out* signal processor. For this reason, DSP computations are nearly always carried out in the time domain. The frequency domain still plans an important role as a visualization tool during design, however.

10.6 Difference equations - signal manipulation

The digital signal manipulation part that of the system is described by difference equations. As we did in part 1 of this reader we limit the treatment to Linear Time-Invariant (LTI) systems. Continuous LTI systems are characterized by a differential equations that determine the relationship between an input $x(t)$ and the corresponding output $y(t)$. When we sample a signal, we sample its values at equally spaced time intervals T_s:

$$y[n] = y(t = nT_s) \tag{10.8}$$

But how do we translate these differential equations into the digital world? Doing so involves a certain degree of approximation, in the sense that we start from the definition of a derivative,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \lim_{\Delta t \to 0} \frac{y\,(t + \Delta t) - y\,(t)}{\Delta t}, \tag{10.9}$$

but instead of taking the limit $T_s \to 0$ we use the sampling interval, T_s , as our 'limit':

$$\frac{\mathrm{d}y}{\mathrm{d}t} \approx \frac{\Delta y}{\Delta t} = \frac{y\,[n] - y\,[n-1]}{T_s}. \tag{10.10}$$

This is a pragmatic choice since T_s is the smallest time interval captured by the discretization process. Higher derivatives follow the same principle, for example

$$\begin{aligned} y''[n] &\approx \frac{1}{T_s}\left(y'[n] - y'[n-1]\right) \\ &\approx \frac{1}{T_s}\left(\frac{y[n]-y[n-1]}{T_s} - \frac{y[n-1]-y[n-2]}{T_s}\right) \\ &= \frac{y[n]-2y[n-1]+y[n-2]}{T_s^2}. \end{aligned}$$

Note that it might seem more logical to estimate the second derivative in a symmetric way around n by using $y''[n] = (y[n+1] - 2y[n] + y[n-1])/T_s^2$. This expression however violates causality and therefore cannot be used in real-time systems.
For a Linear Time-Invariant (LTI) system, the total description of the system with one input and one output is a linear combination of differences (derivatives) of the input signal and the output signal. In general we can formulate a LTI difference equation in the form:

$$y[n] = \sum_{i=0}^{N} b_i x\left[n-i\right] \; + \; \sum_{j=1}^{M} a_j y\left[n-j\right] \tag{10.11}$$

Important to note that the output $y[n]$ is only dependent on previous elements of the input and output signal, which makes this a causal system.
This equation can also be transformed to an equivalent of a Transfer function as we did with continuous systems by means of the Laplace Transform, however we need the the z-Transform technique for discrete systems. This will be treated in chapter 13.

Example 10.6.1. *An example of a second order differential equation*
Consider the differential equation

$$y''\left(t\right) + by'\left(t\right) + cy\left(t\right) = kx\left(t\right). \tag{10.12}$$

Applying the discretization principles from above, we have

$$\frac{y\left[n\right] - 2y\left[n-1\right] + y\left[n-2\right]}{T_s^2} + b\frac{y\left[n\right] - y\left[n-1\right]}{T_s} + cy\left[n\right] = kx\left[n\right] \tag{10.13}$$

or, grouping terms,

$$\left\{\frac{1}{T_s^2} + \frac{b}{T_s} + c\right\} y\left[n\right] + \left\{-\frac{2}{T_s^2} - \frac{b}{T_s}\right\} y\left[n-1\right] + \left\{\frac{1}{T_s^2}\right\} y\left[n-2\right] = kx\left[n\right]. \tag{10.14}$$

We can rewrite this as

$$y[n] = b_1 x[n] - a_1 y[n-1] - a_2 y[n-2] \tag{10.15}$$

where the coefficients are given by:

$$b_1 = \frac{kT_s^2}{cT_s^2 + bT_s + 1} \qquad a_1 = \frac{-2 - bT_s}{cT_s^2 + bT_s + 1} \qquad a_2 = \frac{1}{cT_s^2 + bT_s + 1}$$

□

Note that

a. Two initial value conditions $y[-1]$ and $y[-2]$ are necessary to be able to calculate $y[0]$. In analogy to continuous differential equations, we refer to such a difference equation as being of the second order. Not surprisingly, a continuous nth-order differential equation leads to a nth-order difference equation upon discretisation.

b. The output time-series is calculated numerically by iterating Equation 10.15. That is, $y[0]$ is first calculated using the initial value conditions. Then, $y[1]$ can be calculated using the known values of $y[-1]$, $y[0]$ and $x[1]$. Then, $y[2]$ can be calculated using the known values of $y[0]$, $y[1]$ and $x[2]$ *etc.*.

In the next section we address the difference equations is a practical manner to define discrete time-domain filtering. A full treatment of difference equations will be given in chapter 13 on z-Transforms. The relation between difference equations and the z-Transform is the digital equivalent of the relation between differential equations and the Laplace Transform. The approach in solving and analysing difference equations via a z-Transform is the same as we used the Laplace Transform to solve differential equations. These techniques are important in the frequency behaviour of the transfer functions.

10.7 Time-domain filtering

We saw above that signals can be manipulated in the time domain by means of difference equations. The algorithmic approach is very powerful if we have a time-series and a definition of the filter in terms of a difference equation.
In the example above on the moving-average filter we defined an algorithm where the output was only dependent on the (past) input signal values, in the most general form for an LTI system this is represented by equation10.11.

10.7.1 FIR and IIR filters: the concept

The digital filtering treated in this course covers so-called Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters on 1D signals. This is prerequisite knowledge for more advanced applications such as image processing and adaptive filters. The difference between FIR filters and IIR filters can directly be derived from the difference equation describing the filter. For a FIR filter the output $y[n]$ is formed by a limited addition of components of the input signal $x[n]$ where the weight coefficients b_i determine the characteristic of the filter:

$$y[n] = \sum_{i=0}^{N-1} b_i x[n-i] \tag{10.16}$$

Causality is assured as only values of $x[n-i]$ that are in the past ($i \geq 0$) are used and the filter can be applied as a real-time filter at the cost of some delay.
For an IIR filter, the output $y[n]$ at time n depends on both the input signal $x[n]$ and previous values of the output $y[n]$:

$$y[n] = \sum_{i=0}^{N-1} b_i x\left[n-i\right] \; + \; \sum_{j=1}^{M-1} a_j y\left[n-j\right] \tag{10.17}$$

In principle, each $y[n]$ on the right side can be recursively related to a sequence of $x[n]$ only. This usually leads to a series of infinite length, hence the name Infinite Impulse Response. This implies that the complete history of the input is taken into account in specifying the output.
The implementation of FIR-filters differs from that of IIR-filters. FIR-filters are dominantly applied in the time domain by means of the convolution method (see below in section 10.7.2), whereas the IIR-filters are dominantly applied in the time domain by means of recursive algorithms.
For a given amount of computing power available, IIR-filters tend to be far more efficient than FIR-filters. But FIR-filters are more straightforward to design and provide the best route to visualise and understand what we are doing. We will therefore start with FIR-filters in chapter 12 after treatment of the Discrete Fourier Transform in chapter 11.1.

Before we go to the real filter design in coming chapters let us first get some experience in using a difference equation to apply a filter action on a signal: the moving average filter revisited. Once you got the idea, you can explore this further and develop a basis for digital signal processing.

Example 10.7.1. The Moving-Average filter
Let us illustrate the method of section 10.5, in a moving-average filter in which a number of values of the input signal are added with the same weight. This is mathematically formulated as:

$$y[n] = \frac{1}{M} \sum_{i=0}^{M-1} x[n-i] \tag{10.18}$$

The evaluation speed of this moving average filter can be made very fast: for any length M, it can be re-programmed as a recursive filter with only 3 coefficients: the new value for $y[n]$ is the old value $y[n-1]$ plus the new input sample $x[n]$ and kicking out the input value $M+1$ samples in the past, $x[n-M-1]$. Weighing by the number of average samples M, this yields in mathematical notation

$$y[n] = y[n-1] + \frac{x[n] - x[n-M]}{M} \tag{10.19}$$

Note that although eq. 10.19 hides itself as a recursive filter (IIR class, where the output depends on earlier values of the output) it is still a FIR filter since it can be written out in terms of M input samples, as per Eq. 10.18.
On problem at the end of the chapter is devoted to apply this filter to a signal to explore its effect. What this means for the frequency behaviour of the filter will be treated in the FIR-filters section, after you learned the working of the Discrete Fourier Transform, and can relate the (discrete) time-domain and frequency-domain.

□

10.7.2 Discrete Linear Convolution

For the moment let us assume we know the impulse response of a (filter) system $h[n]$. With this impulse response we can calculate the response of the filter to an arbitrary input signal $x[n]$. Every element in the input signal will give an impulse response, however delayed by successive steps according to the sequence of elements in in the input signal. Since we only consider linear systems, all contributions to a certain time sample in the output signal can be added. In fact this is described in the definition of a discrete convolution below. The convolution for a time series $x[n]$ and a filter impulse response array $h[n]$ is defined analogous tot the continuous definition in equation 3.1 as:

$$y[n] \equiv \sum_{m=-\infty}^{\infty} x[m]h[n-m] \tag{10.20}$$

In real-world situations, the impulse response is limited to a certain number of elements, which makes the calculation of the output signal via the convolution doable. Note: this is the definition for a linear convolution in the discrete domain. This convolution definition is related to the application of a known impulse response of the system.
In section 11.3 we will introduce the circular convolution for application of a filter to a signal designed with the frequency sampling method. That definition of convolution is related to the Discrete Fourier Transform (DFT).

10.8 Cross-correlation and Auto-correlation

At this point it is good to introduce the concept of correlation. De mathematical definition of correlation (in the discrete domain) looks very similar to the definition of convolution: the correlation

between two discrete signals $x[n]$ and $y[n]$ is defined as:

$$R_{x,y}[n] \equiv \sum_{m=-\infty}^{\infty} x[m]y[m-n] \tag{10.21}$$

The actual operation that is this equation represents is a measure of likeliness for all subsequent delays between the two signals. In case $x[n]$ and $y[n]$ are different signals, $R_{x,y}$ is called *cross-correlation* and in case the correlation is calculated with its own signal $R_{x,x}$ it is called *auto-correlation*. Also here in practice we do not need to run over an infinite number of calculations, since the signals are limited in length in practice.
Correlation is a good measure to compare signals and to quantify the similarity between the two signals. In order to get an impression off the use of correlations think of a signal that is sent out, reflected by some object and received back at the sender; *i.e.* the principle of a radar. The calculation of the cross-correlation between the transmitted signal and the received signal will show a peak in the cross-correlation signal at a time that is equal to the travelling-time of the signal form the transmitter to the the object and back to the receiver. Knowing the travelling speed of the signal will give you the distance of the object with respect to the transmitter/receiver.
Beside this application there are many other applications of correlations in relation to signals: at any instance you want to find similarities between signals or within a signal.
Not that the mathematical definitions of correlation an convolution look very similar; one could even think that it is more or less the same. The major difference is that in correlation the time orientation is the same for both signals, whereas for convolution, the time orientation of one of the signals is reversed. This is clear in the index of the second signal in the operation: *i.e.* $[n-m]$ for convolution and $[m-n]$ for correlation. An immediate consequence of this is that if one of the signals is symmetric, both convolution and correlation will give the same resulting signal. The concept of both definitions is completely different.

Before we address the quantitative properties of digital filters and filter design, we introduce the discrete equivalent of the Fourier Transform: The Discrete Fourier Transform in next chapter.

Summaries

Summary Chapter 10

Sampling and discrete signals

- The relation between a continuous-time and a discrete-time signal is that for the discrete-time signal time is represented by the number n of the nth time interval T_s:

$$f_d[n] = f_a(nT_s) \tag{10.22}$$

- Sampling frequency is denoted by f_s or ω_s:

$$f_s = \frac{1}{T_s} \quad \Rightarrow \quad \omega_s = 2\pi f_s = \frac{2\pi}{T_s} \tag{10.23}$$

- Aliasing appears for frequencies in the signal larger than half the sampling frequency, $f_s/2$.
- Aliasing can be suppressed by applying a low-pass filter, with a cut-off frequency f_c below half the sampling frequency, in the analogue time signal before the digital sampling process.
- Equivalent of a first derivative in a difference equation:

$$\frac{\mathrm{d}x}{\mathrm{d}t} \approx \frac{\Delta x}{\Delta t} = \frac{x\,[n] - x\,[n-1]}{T_s}. \tag{10.24}$$

- Equivalent of a second derivative in a difference equation:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} \approx \frac{x[n] - 2x[n-1] + x[n-2]}{T_s^2}. \tag{10.25}$$

- Application of a difference equation: the moving-average filter (M is the number of time-samples in the average):

$$y[n] = \frac{1}{M}\sum_{i=0}^{M-1} x[n-i] \tag{10.26}$$

- Definition of Convolution of signal $x[n]$ and impulse response $h[n]$:

$$y[n] = \sum_{m=-M}^{M} x[m]h[n-m] \tag{10.27}$$

- Definition of Correlation between signal $x[n]$ and $y[n]$:

$$R_{x,y}[n] = \sum_{m=-M}^{M} x[m]y[m-n] \tag{10.28}$$

Problems

Problem 10.1. Aliasing: looking outside
You are sitting in a classroom and at $t = 0$ you quickly look outside and you see a bus. You repeat this every 20 minutes and every time you see the bus. What can you say about the schedule of the bus?

Problem 10.2. Sampling of a signal
In many modern signal recorders, *e.g.* digital oscilloscope, audio recorder ect, an analogue signal is fed into an Analogue-Digital-Converter – in short: ADC. This process can be mimicked by the taking of values from an analytical function and put the numbers in a list.
So take the function:

$$f(t) = \sin\left(\frac{2\pi}{T}t\right) \tag{10.29}$$

where the repetition time of this periodic signal is given by T, that we set to the value of 1 s.
From this analytical signal – equivalent for analogue signal – we take the values (samples) every Δt, and put these values in a list. You have to repeat this process for various values of Δt and plot the result in a graph. Take the number of points (samples) equal to $N = 20$. For every Δt also give the total time span of your list of numbers from the signal.
Use the following values for Δt:

a. $\Delta t = 0.1$ s

b. $\Delta t = 0.2$ s

c. $\Delta t = 0.6$ s

d. $\Delta t = 1.1$ s

e. $\Delta t = 2.4$ s

Explain the results you obtained, and reflect on the observed periodicity of the sampled signal in relation to the given analytical function.

Problem 10.3. Periodicity and aliasing

a. You sample $x(t) = \sin(t)$ every $T_s = 1$ms. Is $x[n]$ periodic? If so, what is its period?

b. You sample $x(t) = \sin(2\pi t)$ every $T_s = 4$ms. Is $x[n]$ periodic? If so, what is its period?

c. You sample $x(t) = \sin(t)$ every $T_s = 1$s. Give a formula for $x[n]$. Do you observe aliasing?

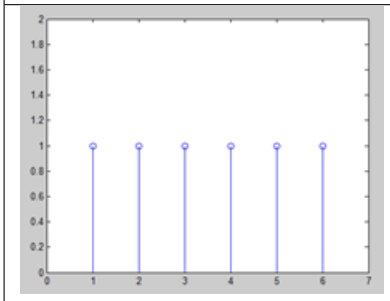
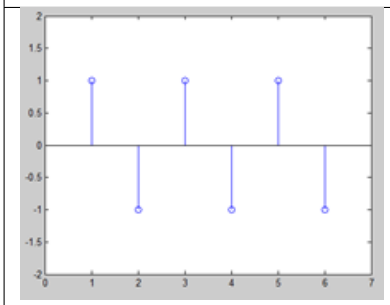
d. You sample $x(t) = \sin(t)$ every $T_s = \pi$s. Give a formula for $x[n]$. Do you observe aliasing?

e. You sample $x(t) = \sin(t)$ every $T_s = 6$s. Give a formula for $x[n]$. Based on your sampling frequency, predict the frequency of $x(t)$ based on $x[n]$ and the assumption that there is no aliasing.

Problem 10.4. Digital transfer functions
To get an impression of the effect of a simple difference equation, sketch the result of the difference equations on the two prototype signals (a DC signal and one fluctuating at highest frequency possible, f_s) and describe the filter behaviour of the two digital filters presented in table 10.1.

Problem 10.5. Auto- and cross-correlation Consider the sequences $x[n] = \cos(\frac{\pi}{3}n)$ and $y[n] = \cos(\frac{\pi}{6}n)$ for $n = 0, \cdots, 11$.

Table 10.1: Two examples of filters, given the difference equations working on two different signals.

Difference equation→ Signal ↓	y[n]=x[n]+x[n-1]	y[n]=x[n]-x[n-1]
	sketch the response	sketch the response
	sketch the response	sketch the response

a. Calculate the autocorrelation $r_x[\ell] = \sum_{n=-\infty}^{\infty} x[n]x^*[n-\ell]$ of $x[n]$. Use MATLAB or PYTHON . Make a stem plot such that the graph plots the middle element of the autocorrelation at $\ell = 0$.

b. Why is the value at lag $\ell = 0$ the largest? Also, what is the physical interpretation of this value? Finally, what do you notice?

c. Calculate the crosscorrelations $r_{xy}[\ell] = \sum_{n=-\infty}^{\infty} x[n]y^*[n-\ell]$ and $r_{yx}[\ell]$ of $x[n]$ and $y[n]$. Use MATLAB or PYTHON . Make stem plots such that the graph plots the middle element at $\ell = 0$. What do you notice?

Problem 10.6. DC component in cross-correlation Given is $x(t) = \cos(\pi t)$. You sample $x(t)$ every $T = 0.2$s, yielding $x[n]$.

a. Take $n = 0, \cdots, 29$. Calculate the autocorrelation of $x[n]$ and make a stem plot.

b. Define $x_1[n] = x[n] + 10$ and $x_2[n] = x[n] + 20$. Calculate the cross-correlation of $x_1[n]$ and $x_2[n]$. Explain the shape.

c. Why is it beneficial to remove the DC component of a signal before you calculate cross-correlation?

Problem 10.7. Integrator with feedback Consider the discrete system

$$y[n] = x[n] + y[n-1]$$

a. Apply this filter to $x[n] = u[n]$, for $n = 0, \cdots, 9$ and make a stem plot. Use MATLAB or PYTHON . Write the system behaviour as

$$y[n] - y[n-1] = x[n]$$

b. Why do we call this system an integrator?

Problem 10.8. LR**-circuit** Consider the LR series circuit shown in figure 10.4 whose transfer function is given by $H(\omega) = \dfrac{1}{1 + \mathrm{j}\omega\frac{L}{R}}$.

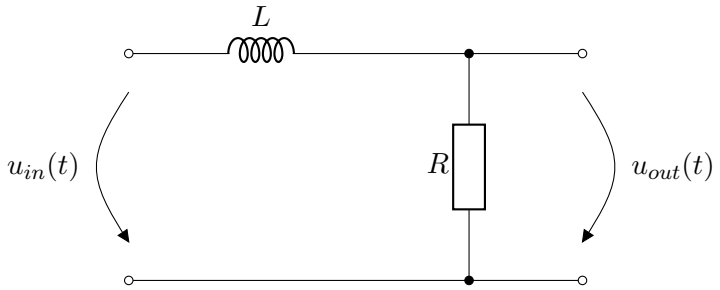


Figure 10.4: A LR low-pass filter.

a. If $u_{in}(t)$ is the input voltage and $u_{out}(t)$ the output voltage over R, show that

$$\frac{L}{R}u'_{out}(t) + u_{out}(t) = u_{in}(t)$$

b. Convert this to a linear difference equation.

Chapter 11

Discrete Fourier Transform

Just like a continuous function $x(t)$ that can be represented as a continuous function $F(\omega)$ in the Fourier domain, a discrete signal $x[n]$ can be represented by a discrete Fourier-domain function $X[k]$, where k is an integer variable.
Parallel to what we learned from the continuous time and frequency domains and the relations between them, we will explore equivalent relations in the discrete domain. It is important to master these relations, since we will need them in the design of digital filters and analysis of digital signals.

11.1 Discrete Fourier Transform

For a finite time-series with N time samples, sampled at a sampling frequency of f_s, the lowest frequency not being zero, retrievable from this time series is:

$$\omega_l = \frac{2\pi f_s}{N}$$

The highest frequency that can be retrieved from this time-series is:

$$\omega_u = \frac{1}{2} 2\pi f_s$$

Dividing the positive frequency range in $N/2$ equidistant frequency intervals we obtain (here we only consider the positive frequencies):

$$\Delta\omega = \frac{\omega_u}{N/2} = \frac{2\pi f_s}{N} = \omega_l$$

In terms of frequency content of a signal is this the range of frequencies that we can use to represent a discrete signal. So we have to project the signal on these components.
In terms of a Discrete Fourier Transformation (DFT) of $x[n]$ we can write this as:

$$X\left[k\right] = \mathcal{F}\left\{x\left[n\right]\right\} = \sum_{n=0}^{N-1} x\left[n\right] \mathrm{e}^{-\mathrm{j}2\pi kn/N}. \tag{11.1}$$

Here $k = 0, 1, 2, \ldots, N-1$ is the frequency index of the corresponding frequency components $\mathrm{e}^{-\mathrm{j}2\pi k/N}$.
Unsurprisingly, the definition of the inverse DFT:

$$x\left[n\right] = \mathcal{F}^{-1}\left\{X\left[k\right]\right\} = \frac{1}{N} \sum_{k=0}^{N-1} X\left[k\right] \mathrm{e}^{+\mathrm{j}2\pi nk/N}. \tag{11.2}$$

The factor of $1/N$ is needed for normalization. A demonstration of the complementary function of the DFT and the inverse DFT is addressed in the exercises.

The index n of the time-series is the discrete equivalent of time:

$$t = nT_s \qquad n \in [0, N)$$

where t is the time and T_s is the sampling time related to the sample frequency $f_s = 1/T_s$. The index k of the frequency-series is the discrete equivalent of frequency:

$$\omega_k = \frac{k}{N}\frac{2\pi}{T_s} \qquad k \in [0, N)$$

Here the subscript k indicates that ω_k is discrete and therefore a list of numbers. Since the time-series and frequency-series are just lists of numbers, the index relates these numbers to the time and frequency respectively. The complementary lists are of equal length. In the domain of signal analysis and processing, the frequency is often used in a normalized representation to be able to quickly apply analyses and calculations to other data-series or data-series with other parameters. A normalisation that is frequently used is to normalise ω_k to the sample frequency f_s:

$$\Omega_k = \omega_k T_s = \frac{2\pi k}{N} \qquad \Omega \in [0, 2\pi)$$

This abstract formalism can be illustrated by means of a mapping of the frequency axis on the unit circle. Remember that the maximum frequency contained in a time-series is half the sample frequency: $f_s/2$; you need a minimum of two data points to produce an alternating signal. At first you would say that the minimum frequency is zero, however from the continuous Fourier analysis we learned that also negative frequencies need to be present in the formulation. Using this hint we would suggest that the frequency span of a sampled signal is from $-f_s/2$ to $f_s/2$. This frequency span is projected on the normalised frequency axis: $[-\pi, +\pi)$. In fact is is not critical how you chose the range of 2π normalised frequency span as we will show somewhat later.
The advantage of the formulation is that the normalised frequency Ω can be treated as a continuous variable, in which the discrete series is substituted when the actual data is needed. This requires a small adjustment in the formulation of the Discrete Fourier Transforms (back and forth):

$$X[\Omega] = \sum_{n=0}^{N-1} x[n]\mathrm{e}^{-\mathrm{j}\Omega n} \tag{11.3}$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X[\Omega]\mathrm{e}^{\mathrm{j}\Omega n}\mathrm{d}\Omega \tag{11.4}$$

Note that the integral is taken over a whole unit circle, and therefore the exact boundaries are not that important as long as they do not overlap.
This formulation makes it also a bit easier to show that these transformations are each others inverse:

$$\begin{aligned}
x[n] &= \frac{1}{2\pi}\int_{2\pi} X[\Omega]\mathrm{e}^{\mathrm{j}\Omega n}\mathrm{d}\Omega \\
&= \frac{1}{2\pi}\int_{2\pi}\sum_{n'=0}^{N-1} x[n']\mathrm{e}^{-\mathrm{j}\Omega n'}\mathrm{e}^{\mathrm{j}\Omega n}\mathrm{d}\Omega \\
&= \frac{1}{2\pi}\sum_{n'=0}^{N-1} x[n']\int_{2\pi}\mathrm{e}^{-\mathrm{j}\Omega n'}\mathrm{e}^{\mathrm{j}\Omega n}\mathrm{d}\Omega \\
&= \frac{1}{2\pi}\sum_{n'=0}^{N-1} x[n']\int_{2\pi}\mathrm{e}^{\mathrm{j}\Omega(n-n')}\mathrm{d}\Omega \\
&= \frac{1}{2\pi}\sum_{n'=0}^{N-1} x[n']2\pi\delta_{n,n'} \\
&= x[n]
\end{aligned}$$

Here we used the fact that both n and n' are integers and so the integral above is zero, unless $n = n'$ holds. In the latter case the exponent is equal to one for all values of Ω. The integral evaluates to zero in case $n \neq n'$ since the magnitude of the exponent is always one, but the phases are equally distributed over the full range $[0, 2\pi)$.

11.1.1 Frequency span and Aliasing

As we already showed in the previous chapter on sampling, aliasing is an effect of that can appear in the sampling of continuous signals. In case a frequency component in the signal is higher than half the sample frequency, this component appears at a lower frequency is illustrated in figure 10.3, where a 90 Hz signal looks like a 10 Hz signal in the sampled data (the sampling frequency is $f_2 = 100$ Hz in this situation. The 40 Hz signal of the top graph in the same figure shows the correct frequency though.

This aliasing effect can be explored using the Discrete Fourier Transform. For this illustration we define a frequency-series of N elements, *i.e.* $k = 0, 1, 2, \ldots, N-1$. Assume that the list of elements contains one frequency component at k_0, all other elements are zero. This index k_0 can be chosen freely. When we map the frequency axis on the unit circle (for convenience we choose the range$[0, 2\pi)$ for now. This results in an angle Ω_0 somewhere on the unit circle. Now we change this frequency to $k_1 = k_0 + mN$, resulting in a frequency that is larger than the sample frequency. The mapping on the unit circle puts the normalized frequency Ω_1 exactly on Ω_0 for every value of m. Mathematically this can be shown:

$$\begin{aligned} X[k_0 + mN] &= \sum_{n=0}^{N-1} x[n] \mathrm{e}^{-\mathrm{j}2\pi(k_0+mN)n/N} \\ &= \sum_{n=0}^{N-1} x[n] \mathrm{e}^{-\mathrm{j}2\pi k_0 n/N} \mathrm{e}^{-\mathrm{j}2\pi mn} \\ &= X[k_0] \end{aligned}$$

here we made use of the fact that both m and n are integers, leading to the second exponent to be equal to one.

With this exercise we justified that all frequencies are confined to the unit circle, all higher frequencies beyond the 2π range are mapped in this range. This principle is depicted in figure 11.1.

Using knowledge from the continuous Fourier Transform, and indicated by the observed aliasing effect we can make a further reduction. From the characteristics of the continuous Fourier Transform, we learned that for real-valued time-functions we obtained the relation:

$$F(-\omega) = F^*(\omega)$$

Translated to the discrete frequency this means that for real-valued time-series the frequency components $X[k]$ come in pairs.

$$X[N-k] = X[k]^* \tag{11.5}$$

This means that we only have to look at one half of the unit circle, since the other half is dependent on that. So all information of real-valued time-series is contained in one half of the unit circle. Needless to say, we have to incorporate the content of the second part in a proper way in our results: we can not simply ignore this part.

Let us use this condition in the inverse Discrete Fourier Transform to check the implications. For this we derive

$$
\begin{aligned}
x[n] &= \frac{1}{N}\sum_{k=0}^{N-1} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} \\
&= \frac{1}{N}\sum_{k=0}^{N/2} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} + \frac{1}{N}\sum_{k=N/2+1}^{N-1} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} \\
&= \frac{1}{N}\sum_{k=0}^{N/2} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} + \frac{1}{N}\sum_{k'=1}^{N/2} X[N-k']\mathrm{e}^{\mathrm{j}2\pi (N-k')n/N} \\
&= \frac{1}{N}\sum_{k=0}^{N/2} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} + \frac{1}{N}\sum_{k'=1}^{N/2} X[N-k']\mathrm{e}^{\mathrm{j}2\pi n}\mathrm{e}^{-\mathrm{j}2\pi k'n/N} \\
&= \frac{1}{N}\sum_{k=0}^{N/2} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} + \frac{1}{N}\sum_{k'=1}^{N/2} X[N-k']\mathrm{e}^{-\mathrm{j}2\pi k'n/N} \\
&= \frac{X[0]}{N} + \frac{1}{N}\sum_{k=1}^{N/2} X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} + \frac{1}{N}\sum_{k=1}^{N/2} X[k]^*\mathrm{e}^{-\mathrm{j}2\pi kn/N}
\end{aligned}
$$

where we used:

$$\mathrm{e}^{\mathrm{j}2\pi n} = 1 \qquad \text{and} \qquad X[N-k'] = X[k]^*$$

So both parts of the frequency array have the same content. The implementation of this is a bit tricky, due to the fact that there is a difference in numbering of the elements in a frequency-series with even or odd number of elements.
The consequence of the real-values time-series on the frequency components is depicted in figure 11.1. Now we have explored the conditions and possibilities of the frequency-series we can concentrate on the interpretation of these numbers. Due to the characteristics we only have to concentrate on the first half part of the list. These give the frequency components of the transformed signal. The lowest frequency is zero and the step size in frequency is half the the sample frequency divided by N equivalent to exactly one cycle in the whole time-series: $f_0 = 1/(NT_s) = f_s/N$; the maximum frequency is $f_s/2$. Which is consistent with the fact that all information is contained in $N/2$ elements.

Mapping of the frequency axis on a circle

The normalisation of the frequency axis for continuous frequency Ω in the discrete frequency domain can be made more clear if we map this axis on a circle. This is illustrated in figure 11.2.
From this figure it is immediately clear that the range of Ω is $[-\pi, \pi)$ or $[0, \pi)$. This figure illustrates also the aliasing effect, since frequencies larger than 2π or f_s map in the same domain due to the cyclic nature of the circular axis. The fact that this already occurs for frequencies larger that π or $f_s/2$ has its origin in the fact that for real time-series, the frequency axis is symmetric around 0 for the range $\Omega \in [-\pi, \pi)$. In the figure also the k-index is indicated for an odd number of elements in the frequency series: $K = 23$ in this case. The corresponding normalized frequencies Ω_k are homogeneously spread over the circle resulting in $\Delta\Omega = 2\pi/23$. Note that the the frequency corresponding to the highest $k = (K-1)/2$ value is positioned $-\frac{1}{2}\Delta\Omega$ from π.

Example 11.1.1. *Peak in spectrum show up in other places too.*
Consider the discrete function, sampled in N samples:

$$x[n] = \cos\left(\frac{6\pi n}{N}\right) = \frac{1}{2}\left(\mathrm{e}^{\mathrm{j}\frac{6\pi n}{N}} + \mathrm{e}^{-\mathrm{j}\frac{6\pi n}{N}}\right). \tag{11.6}$$

By inspection (and as addressed in detail in the exercises), one can easily see that $X[k]$ has peaks at $k = \pm 3$. However, when plotting $X[k]$ over the range $k = 0, 1, 2, \ldots, N-1$, the peak at $k = -3$

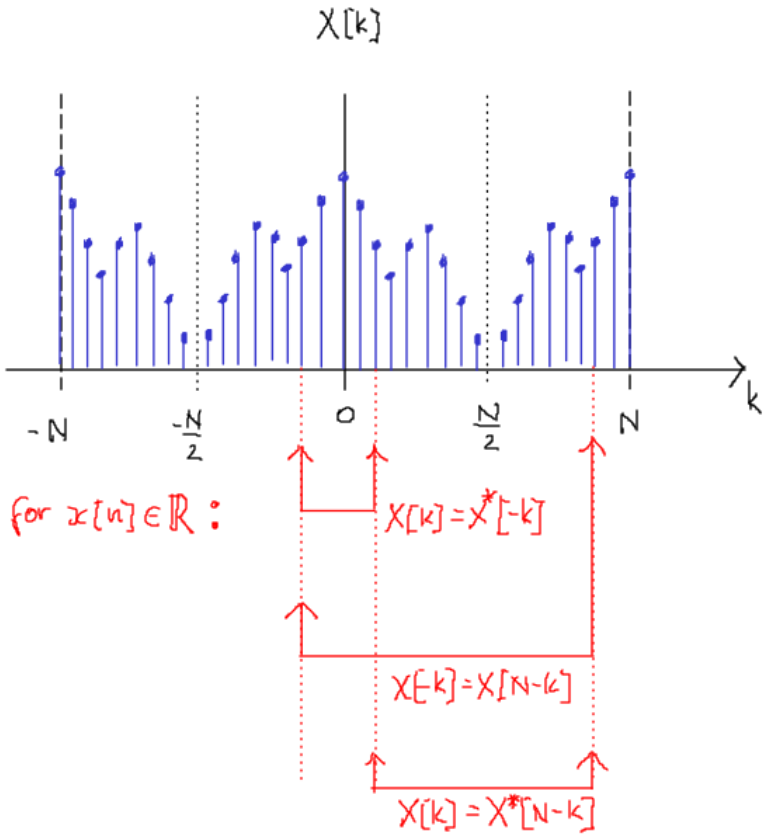


Figure 11.1: The negative frequencies $-N/2 \leq k \leq 0$ are also mapped unto the region $N/2 \leq k \leq N$.

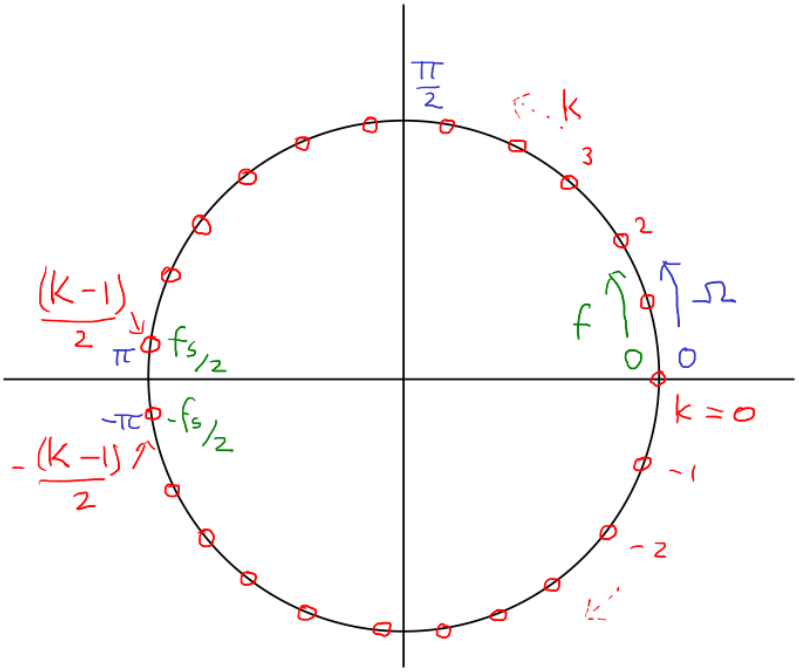


Figure 11.2: Frequency axis mapped on a circle to illustrate the normalisation and make aliasing intuitive. See text for further explanations.

is represented by its copy at $k = N - 3$ of $k = -3$, which might seem like a high frequency. This is illustrated in Figure 11.3.

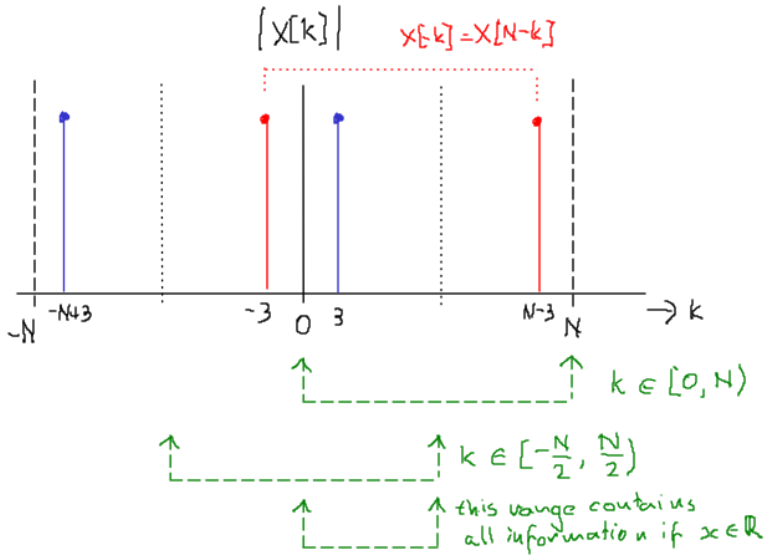


Figure 11.3: Fourier components appear in multiple places on the frequency axis!

□

Short reflection digitizing signals in terms of frequency content

Note that $f_s/2$ determines the maximum frequency that can be found in a signal, and N determines the resolution of the frequency axis in the analysis. In case the digital signal is the result of sampling, the original analogue signal could have contained frequencies above half the sample frequency. Since these frequencies degrade the digitized signal, you want to avoid these frequencies in your system. Referring to the schematic overview of a DSP-process you notice an analogue filter in front of the ADC, that takes care of the sampling. This filter is usual a low-pass filter with a cut-off set close to half the sample frequency, with the task to filter out the disturbing frequencies above $f_s/2$, which would otherwise show up all over the spectrum. This is the only place where you need an analogue filter since it makes no sense to filter after the sampling process since these disturbing frequencies are not distinguishable any more after the sampling process.
The fact that we work only with finite-length time-series means that there is a limit on the lowest frequency too in the digitised signals. In general this is not a problem.
Due to the symmetry of the Fourier Transformation process, one might think that shorter times can be brought into the time-series by extending the frequency-series (we know that intervals are appearing repetitively). This is however not the case unless the sampling frequency is changed. Otherwise, one would add information to the signal, which can not be the case.
Similar arguments hold for resolution in the time as well as in the frequency domain.

As soon as the time-series (with T_s and N) or the frequency-series (with f_s and N) are determined, no mathematical (or other) trick can add information to the signals. In the design of digital filters,

you will explore this field, due to the fact that you have to optimize your design for efficiency.

11.2 Properties of the Discrete Fourier Transform

All of the properties of the continuous Fourier Transform have parallels in the Discrete Fourier Transform. We tend to use these less, however, since a lot of computations involving the DFT are done numerically. Two properties will prove very relevant and are proven here. The other properties are presented in the summaries at the end of this chapter. Some of them will come back in the problems to be proven by you.

11.2.1 Real functions

One that we already used above but is repeated here for completeness. For a real functions $x[n]$,

$$X[-k] = X^*[k]. \tag{11.7}$$

Combined with the periodicity of the DFT, this also means that

$$X[N-k] = X^*[k]. \tag{11.8}$$

11.2.2 Time shift property of the DFT

A general result that will be very useful to describe difference equations is the time shift property:

$$\mathcal{F}\{x[n+a]\} = X[\Omega]\mathrm{e}^{-\mathrm{j}\Omega a} \tag{11.9}$$

The effect of a time-shift is a phase change in the frequency domain, as was also the result in the continuous Fourier situation.

11.3 Convolution related to DFT

In paragraph 10.7.2 the convolution in the discrete domain is defined to calculate the output response of an LTI system via a convolution of the input signal with the impulse response of the difference equation.
For the use of convolution in combination with the Discrete Fourier Transform we use a definition that is slightly altered. This has to do with the fact that the whole frequency span in the frequency domain is limited to a finite domain, *i.e.* $\Omega \in (\pi, \pi]$, represented by a finite list of filter coefficients. For a filter length of M elements, the definition of the convolutions becomes:

$$f[n] \otimes g[n] = \{f \otimes g\}[n] \equiv \sum_{m=0}^{M-1} f[m]g[n-m]_M \tag{11.10}$$

The subscript M on list variable not only determines the length of the filter, but is also an indication that the list can be circular permuted when applied in the convolution. In fact it is equivalent with the representation of the Fourier components as presented in figure 11.3. Also from the definition of the convolution it is clear that a circular permutation of the list of filter coefficients does not have an effect on the result, since the sum runs over all elements M for every delay $n - m$ between the two signals. Unlike the system representation by its impulse response obtained from the difference equations, the time response obtained from the inverse Discrete Fourier Transform of a spectrum does not contain absolute time information. In chapter 12 an illustration of these effects will be given.
The implementation of the convolution in digital signal processing is quite easy, but many program environments have special functions for these kind of processes that are even more easy to use with the benefit of being more efficient. An additional advantage is that you do not have to care about the bookkeeping of the shifts in the arrays.
As already mentioned in section 9.7.1 the convolution method is often used with digital signals when filters are applied to a signal. This in contrast with methods used with analogue signals. We will see that it plays a dominant role in the application of digital filters, specific the FIR-filters. FIR stands for Finite-Impulse-Response, which immediately reveals the motivation for this choice as opposed to IIR-filters, which stands for Infinite-Impulse-Response filters. The difference between these two types of filters and explanation why the convolution method is dominantly used for FIR-filters will be treated in the corresponding chapters.
Also for the Digital Fourier Transform we have a convolution theorem:

$$f[n] \otimes g[n] = F[k]G[k] \tag{11.11}$$

$$f[n]g[n] = \frac{1}{N} F[k] \otimes G[k] \tag{11.12}$$

Note that in case $F[k]$ and $G[k]$ are formulated as function of the independent variable Ω, the normalisation factor in front of the last term becomes: $1/N \rightarrow 1/(2\pi)$.

11.4 The Fast Fourier Transform (FFT)

Computing $X[k]$ requires N^2-multiplications of complex numbers, which can get computationally costly when N gets large. It turns out that there exists a clever algorithm whose computational cost scales as $N \log N$ instead of N^2. If you think that's not so impressive, note that it saves you a factor of 100,000 in computing time if N is of the order of one million; it gets more extreme when you consider the amount of data in your typical 1-hour CD or 2-hour DVD. Strictly speaking this algorithm requires that the data contain $N = 2^M$ samples, but there are workarounds that allow

dealing with data of any length while reaping most of the benefits. This algorithm is known as the Fast Fourier Transform (FFT), and is the standard function for calculating DFT's numerically[1].
As useful a practical tool as the FFT is, however, it's good to keep one thing in mind: computing a FFT is the same as computing a DFT, just faster. The FFT is not a different transform, it's a tool to get to the answer faster. This is often misunderstood. Nowadays, this method is hidden in most of the implementations of DFT in computational programmes or libraries, so you don't have to worry about modifying your data to make the calculation efficient and scale frequency or time axes accordingly[2]. Modern algorithms for DFT and iDFT return frequency-series and time-series respectively with the same number of elements and a straight forward scaling as we used in this chapter.

[1]To some extent you have to know some of the artefacts of the FFT implementation in the software that you use, since the results might depend on the exact implementation method. While we will not discuss the idiosyncrasies of the different variants in detail, in practice it is wise to consult the help functions of your software to get details on the implementation.

[2]Only a few years ago, you had to extend your time-series to give it 2^M elements, perform the transformation and use an appropriate scaling too. This could give you a lot of trouble in the construction of the discrete spectrum of a signal

Summaries

Summary Chapter 11.1a

Main concepts in discrete Fourier transforms:

- The definition of the discrete Fourier transform:

$$X\left[k\right]=\mathcal{F}\left\{x\left[n\right]\right\}\equiv\sum_{n=0}^{N-1}x\left[n\right]\mathrm{e}^{-\mathrm{j}2\pi kn/N}. \tag{11.13}$$

  where $k=0,1,2,\ldots,N-1$ represents the (discrete) frequency of each component in the transform.

- The corresponding inverse transform is given by:

$$x[n]=\mathcal{F}^{-1}\left\{X[k]\right\}=\frac{1}{N}\sum_{k=0}^{N-1}X[k]\mathrm{e}^{\mathrm{j}2\pi kn/N} \tag{11.14}$$

- Relation between Normalized (to the sample frequency f_s) frequency $\Omega\in[0,2\pi)$ or $\Omega\in[-\pi,\pi)$ and frequency number $k\in[0,N-1]$:

$$\Omega_k=\frac{2\pi}{N}k \tag{11.15}$$

  with N the number of samples.

- Semi-continuous formulation of the DFT:

$$X[\Omega]=\sum_{n=0}^{N-1}x[n]\mathrm{e}^{-\mathrm{j}\Omega n} \tag{11.16}$$

$$x[n]=\frac{1}{2\pi}\int_{2\pi}X[\Omega]\mathrm{e}^{\mathrm{j}\Omega n}\mathrm{d}\Omega \tag{11.17}$$

- Convolution:

$$f[n]\otimes g[n]=F[k]G[k] \tag{11.18}$$

$$f[n]g[n]=\frac{1}{N}F[k]\otimes G[k] \tag{11.19}$$

$$f[n]\otimes g[n]=\{f\otimes g\}[n]\equiv\sum_{m=0}^{M-1}f[m]g[n-m]_M \tag{11.20}$$

Summary Chapter 11.1b

Discrete Fourier Transform Properties:

#	Property	$x[n] = \mathcal{F}^{-1}\{X[k]\}$	$X[k] = \mathcal{F}\{x[n]\}$
1	Real Functions		$X[-k] = X^*[k]$
2	Linearity	$\alpha x_1[n] + \beta x_2[n]$	$\alpha X_1[k] + \beta X_2[k]$
3	Reciprocal Scaling	$x[\alpha n]$	$\frac{1}{\alpha} X\left[\frac{k}{\alpha}\right] \ \forall\ \alpha > 0$
4	Differentiation	$x[n] - x[n-1]$	$\left(1 - \mathrm{e}^{-\mathrm{j}2\pi k/N}\right) X[k]$
6	Time Shift	$x[n - n_1]$	$X[k]\mathrm{e}^{-\mathrm{j}2\pi k n_1/N}$
7	Frequency Shift	$x[n]\mathrm{e}^{\mathrm{j}2\pi k_1 n/N}$	$X[k - k_1]$
8	Convolution	$f[n] \otimes g[n]$	$F[k]G[k]$
9		$f[n]g[n]$	$\frac{1}{N} F[k] \otimes G[k]$
10	Parseval's theorem	$\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$	$\frac{1}{N} \sum_{k=-\infty}^{\infty} \lvert X[k]\rvert^2$

Summary Chapter 11.1c

Discrete Fourier Transform pairs:

$x[n] = \mathcal{F}^{-1}\{X[k]\}$	$X[k] = \mathcal{F}\{x[n]\}$
$\delta[n]$	1
$\mathrm{e}^{\mathrm{j}2\pi k_1 n/N}$	$2\pi\delta[k - k_1]$
$\cos\left(\frac{2\pi k_1 n}{N}\right)$	$\frac{N}{2}\left(\delta[k - k_1] + \delta[k + k_1]\right)$
$\sin\left(\frac{2\pi k_1 n}{N}\right)$	$-\mathrm{j}\frac{N}{2}\left(\delta[k - k_1] - \delta[k + k_1]\right)$
$x[n] = 1$	$N\delta[k]$
$a^n u[n]\ (\lvert a\rvert < 1)$	$\frac{1}{1 - a\mathrm{e}^{-\mathrm{j}2\pi k/N}}$

Problems

Problem 11.1. DFT calculation, interpretation and a computer

a. Consider the sequence $x_1[n] = \cos(\pi n) = (-1)^n$ for $n = 0, \cdots 3$. Calculate $X_1[k]$. Comment on your answer.

b. Consider the sequence $x_2[n] = \cos(\pi n) = (-1)^n$ for $n = 0, \cdots 127$. Calculate $X_2[k]$. Comment on your answer.

c. Verify your answers using MATLAB or PYTHON .

Problem 11.2. Sampling and DFT relations
Consider two digital recordings on a Compact Disk (CD) with durations of $2\,\mathrm{s}$ and $20\,\mathrm{s}$, respectively. The recordings are sampled at a sample frequency of $f_s = 44.1\,\mathrm{kHz}$.

a. How many samples make up the recordings?

b. What are the highest frequencies that can be encoded in these recordings?

Next we want to Fourier Transform this recording digitally.

c. How many discrete frequency components k do you get after a Fourier transform of these signals?

d. What i the frequency resolution of these discrete frequency components in units of Hz?

For digital signals it is more practical to use the normalized frequency Ω.

e. Give an expression for the normalisation.

f. What is the domain of Ω, and how does it relate to the non-normalised Fourier domain?

Problem 11.3. DFT on a short time-series
Depending on your preferences on computer support, use either MATHEMATICA, PYTHON or MATLAB, apply the digital Fourier Transform on the time-series given below. Assume a sampling frequency of $f_s = 1.0\,\mathrm{Hz}$.

c. For the data sequence $\{1, -1, 1, -1, 1, -1, 1, -1\}$.

d. For the data sequence $\{1, -1, 1, -1, 1, -1, 1\}$.

e. Comment on the obtained results.

Problem 11.4. Computing the DFT
Compute the discrete Fourier transform of the (discrete) function

$$x[n] = \sin(2\pi an/N) \tag{11.21}$$

defined over the interval $n = 1, 2, \ldots, N-1$. You can take a to be an integer (what does this mean in terms of the function $x[n]$?). Sketch the result over the interval $k = 0, 1, 2, \ldots, N$ and $k = -N/2, \ldots, N/2$.

Problem 11.5. Zero padding Consider a sequence $x[n]$ of length N. We add $(r-1)N$ zeroes at the end of $x[n]$ ("zero padding"), obtaining sequence $y[n]$ of length rN. Feel free to answer b) first.

a. How can we retrieve $X[k]$ from $Y[k]$?

b. Pick $x[n] = 1, 2, 3, 4$ and $r = 2$ and use FFT to demonstrate a).

Problem 11.6. Up-sampling Consider a sequence $x[n]$ of length N. After each sample we insert $(r-1)$ zeroes, obtaining sequence $y[n]$ of length rN. This operation is called up-sampling and is one of the fundamental operations in digital signal processing.

a. How can we retrieve $X[k]$ from $Y[k]$?

b. Explore for which values of k your previous result holds and why?

c. Pick $x[n] = 1, 2, 3, 4$ and $r = 2$ and use FFT to demonstrate a).

Problem 11.7. Visualising the harmonics

a. Show that that for a real function $x[n]$, it follows that its DFT $X[k]$ obeys the property $X[\frac{N}{2} + p] = X^*[\frac{N}{2} - p]$, where p is an integer and N is even.

b. The symmetry you derived in part a) may appear somewhat enigmatic. Why would a higher frequency component echo a lower frequency? To see why, make sketches of the real and imaginary parts of the following harmonics $\exp(\mathrm{j}2\pi kn/N)$ as a function n for $N = 16$ and the following values of k (Instead of one exponential you can also take a linear combination of two exponentials to form a $\sin$ or $\cos$ to illustrate the effect.):

(i) $k = 0$
(ii) $k = N$
(iii) $k = 1$
(iv) $k = 2$
(v) $k = N/2$
(vi) $k = N - 1$
(vii) $k = N - 2$

Note: this can be done either by manipulating the trigonometric functions algebraically or using a computer.

Problem 11.8. Spectral leakage Consider the sequence $x[n] = \cos(\frac{\pi}{2}n)$ of infinite length. Since we cannot use most infinitely long sequences we study finite sequences instead. Define

a. $x_{16}[n] = x[n]$, length 16.

b. $x_{17}[n] = x[n]$, length 17.

c. $x_{18}[n] = x[n]$, length 18.

d. $x_{19}[n] = x[n]$, length 19.

e. $x_{20}[n] = x[n]$, length 20.

Calculate and plot the DFT of each of these sequences. The effect that you observe is spectral leakage. Explain what is happening, and include the length 16/20 cases as well.

Problem 11.9. Discrete Fourier Transform
Consider the function $x(t) = \cos(2\pi f_0 t)$. Now suppose that we sample $x(t)$ at a frequency f_s for N data points.

a. What is the condition for which f_0 will correspond exactly to one of the harmonics k of the DFT $X[k]$? Express your answer in terms of f_0, f_s and N.

b. This part pretty much requires the use of a computer. We have shown before that the continuous Fourier transform of $x(t)$ is simply

$$X(\omega) = \pi\delta(\omega - 2\pi f_0) + \pi\delta(\omega + 2\pi f_0).$$

Apart from the special case discussed in part a), the position of the delta function does not correspond to one of the harmonics of the DFT. What happens then in the DFT? To illustrate this, consider the case where N = 100 and $f_s = 100$ Hz. Compute and plot $X[k]$ for $f_0 = 10Hz$, 10.25 Hz, 10.5 Hz, 10.75 Hz and 11 Hz. You may plot the real and imaginary parts separately or just the magnitude $|X[k]|$.

Note: This leaves you with a code that allows you with only minor adjustments to crank out $X[k]$ for any discrete function $x[n]$. Discrete Fourier transforms may have their drawbacks, not least the risk of aliasing, but they are unparalleled in the (relative) ease with which they can be computed.

Problem 11.10. Validating the definition
The discrete Fourier transform (DFT) is defined as

$$X[k] = \mathcal{F}\{x[n]\} = \sum_{n=0}^{N-1} x[n]\mathrm{e}^{-\mathrm{j}2\pi kn/N}$$

while its inverse is

$$x\left[n\right] = \mathcal{F}^{-1}\left\{X\left[k\right]\right\} = \frac{1}{N}\sum_{k=0}^{N-1} X\left[k\right]\mathrm{e}^{+\mathrm{j}2\pi nk/N}.$$

Challenge this definition by checking whether $\mathcal{F}^{-1}\{\mathcal{F}\{x[n]\}\} = x[n]$.

Problem 11.11. Some properties of the DFT
Consider the sequences $x[n] = \mathrm{e}^{\mathrm{j}\frac{\pi}{2}n}$ and $y[n] = (-1)^n$ for $n = 0, \cdots, 3$.

a. Calculate $X[k]$ and $Y[k]$.

b. Calculate $z[n] = x[n] \otimes y[n]$.

c. Find $Z[k]$ from (1) $z[n]$ and from (2) $X[k]$ and $Y[k]$ and verify matching answers.

d. Use Parseval to calculate energy of $x[n]$ in sample domain and in the frequency domain.

e. Calculate $w[n] = \mathrm{e}^{\mathrm{j}\frac{\pi}{2}n}\, x[n]$. Calculate $W[k]$ from $w[n]$.

f. Derive $W[k]$ from $X[k]$ and verify matching answers.

Problem 11.12. More properties and a computer Consider the sequences $x[n] = 3 + (-1)^n$ for $n = 0, \cdots, 3$ and $y[n] = 1 + 2(-1)^n$ for $n = 0, \cdots, 3$.

a. Calculate $X[k]$ and $Y[k]$.

b. Calculate $z[n] = x[n]y[n]$.

c. Find $Z[k]$ from (1) $z[n]$ and from (2) $X[k]$ and $Y[k]$ and verify matching answers.

d. Use MATLAB or PYTHON to calculate a)-c).

Problem 11.13. Convolution
Consider two signals represented by two number series given by:

$s_1 = \{0, 0, 1, 1, 1, 0, 0\}$
$s_2 = \{0, 0, 1, 1, 0, 0, 0\}$

a. Calculate the convolution of the two series.

b. Note that many programs have options on how to deal with the ends of the series. Explore these options.

c. Make graphical representations of the results.

Chapter 12

FIR filters

12.1 Introduction

Filters are used to shape signals to the desire of the user. In the section 10.7 we have seen that applying a difference equation on a signal can perform a filtering on the signal in terms of suppressing frequencies or enhancing frequencies in the signal. The result of filtering is described in terms of effect in the frequency domain of the signal.
A great advantage of FIR filters is that they can easily be designed to have a linear phase response. Different frequency components get delayed by the same amount of time (that is, by the same number of samples), preserving the shape of the signal in time. Especially for audio signals this is important as a non-linear phase response distorts the signal shape in time and the ear is very sensitive to this. Of course audio is not the only domain where you can benefit from the phase-preservation characteristics of a FIR filter. Detection and analysis of signals consisting of discrete pulses or steps, is another field where this characteristic is important.
Since the main purpose of filtering is to modify the signal according to the requirements set by the user, we need some means to design filters. Since one method of designing FIR filters is closely related to the Discrete Fourier Transform and its inverse, we start with the design of FIR filters.
A FIR filter is characterized by a difference equation in which the output value $y[n]$ is determined only by a combination of a finite number of elements from the input signal $x[n]$:

$$y[n] = \sum_{i=0}^{K-1} b_i \; x[n-i] \tag{12.1}$$

We use here capital K for the number of filter elements. This is also called the kernel of the filter. In case we know the coefficients b_i of the filter, we can apply the filter to an input signal $x[n]$. The output only depends on previous samples of the input signal, *i.e.* a causal system.
So now we can focus on the technique to determine the coefficients b_i of the filter. For this we need the convolution property of the Discrete Fourier Transform: equation 11.12:

$$\begin{aligned} f[n] \otimes g[n] &= F[k]G[k] \\ f[n]g[n] &= \frac{1}{N} F[k] \otimes G[k] \end{aligned}$$

The first of the two equations gives us the motivation to design a filter in the frequency domain, perform an inverse DFT and apply the result in a convolution on the input signal. For an input signal $x[n]$ and an output signal $y[n]$ and a Filter described by $H[k]$ the convolution would read:

$$y[n] = x[n] \otimes h[n] \qquad \Rightarrow \qquad Y[k] = X[k]H[k] \tag{12.2}$$

Which makes $H[k]$ the Transfer function of the filter: every frequency component of the input signal is multiplied with a (complex) factor. If $H[k]$ is the transfer function of the filter, then $h[n]$ is

something like an impulse response of the filter[1] In order to make this work we need a trick, that is presented in the next section where we treat an ideal low-pass filter.
In order to make a link with the filter coefficients b_i in equation 12.1 we explicitly write out the convolution:

$$y[n] = x[n] \otimes h[n] = \sum_{i=0}^{K-1} x[i]h[n-i]_K \tag{12.3}$$

Which shows that the filter coefficients are equal to the elements in impulse response of the filter[2]. This method is applied from the back: first determine the frequency content of the signal by calculating $X[k]$ via a DFT from $x[n]$. Subsequently determine the frequency characteristics $H[k]$ of the filter. Apply the inverse transformed filter function $h[n]$ via a convolution on the original input signal. This is the design process in rough sketched form and it needs some tweaks to make it really work.

12.1.1 Frequency-sampling or Time-sampling

In the actual realisation of the FIR filter there are roughly two methods to construct the actual list of filter coefficients in the design process: the *frequency-sampling* method and the *time-sampling* method.

FS: In the frequency-sampling method a sketch or function is sampled in the frequency-domain to get a list and subsequently transformed to the time-domain via an inverse Discrete Fourier Transform. The resulting list is the list of filter coefficients. The resolution, and therefore the number of filter coefficients, is determined in the frequency-domain.

TS: In the time-sampling method a sketch needs to to be transformed to a continuous function that is transformed via an inverse Fourier Transform to a function in the time-domain. This time function is subsequently sampled to obtain the filter coefficients in the time-domain. The choice of number of coefficients and resolution is a bit harder is this case.

Both methods are used and find in implementations in filter-design procedures and toolboxes in scientific software programs. To some extend it is a matter of taste which method you prefer. The frequency-sampling method is more transparent and direct.

We will start with an illustrative example of an ideal low-pass FIR-filter constructed via the frequency-sampling method to explore this method and its consequences.

12.2 Ideal low-pass FIR-filter

Let us try to rationally construct a filter with a desired frequency response. To illustrate this process we consider the ideal low pass filter. Its transfer function is a rectangular function, with a value of 1 (perfect transmission) below some cut-off frequency Ω_c (normalized frequency) and a magnitude 0 (no transmission) above Ω_c. This is illustrated in Figure 12.1.
Up to this point, the frequency-sampling method and the time-sampling method are the same. Now we will first continue with the frequency-sampling method to construct the filter.

[1] We should be a bit careful here, since we face the same difference between the continuous Fourier Transform and the Laplace Transform. The inverse Laplace transform of the Transfer function results in the impulse response, not the inverse Fourier Transform. Strictly we need to use the inverse z-Transform here.

[2] It might look as if here the order of the convolution arguments is changed, however $x[n] \otimes h[n] = h[n] \otimes x[n]$ holds; This notation is more in line with the circular application of the convolutions that is bounded to the sum of the smallest list, being that of the filter.

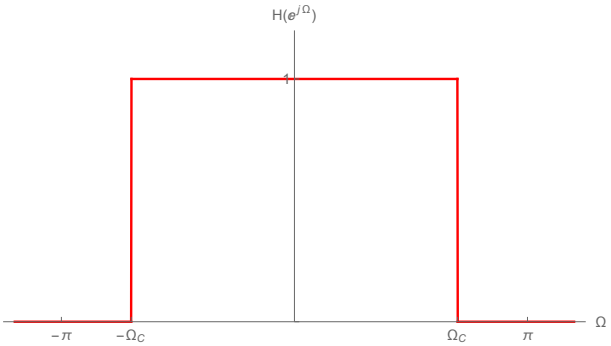


Figure 12.1: A sketch of the transfer function of an ideal low-pass filter with cut-off frequency Ω_c.

12.2.1 Frequency-sampling design of the low-pass FIR-filter

Starting from the continuous function as presented in figure 12.1 we need to sample this function. Note that we need a full range $\Omega \in [-\pi, \pi)$ for the frequency axis (alternatively one could chose $\Omega \in [0, 2\pi)$). For a discrete spectrum with K samples in the normalised range or 2π the cut-off frequency is associated with the index k_c via:

$$k_c = \text{Floor}\left(\frac{K\Omega_c}{2\pi}\right) \tag{12.4}$$

Note that $\Omega_c \in \mathbb{R}$ and $k_c \in \mathbb{Z}$ so we take the highest value for $k_c \leq \Omega_c$. Since we did choose $\Omega \in [-\pi, \pi)$, this means that for this filter we construct an array that consists of values of one for the index k with the condition $-k_c \leq k \leq k_c$ and the rest $-K/2 < k < -k_c$ and $k_c < k < K/2$ filled with zeros. In figure 12.2 this is presented for $K = 59$ and $k_c = 7 \Leftrightarrow \Omega_c = 0.8$ for half of the k-values $k \in [0, 29]$. Since the filter coefficients for negative k-values are equal to the ones for the positive values of k all information is in this part. Note that this implies that it is convenient to chose an odd number of coefficients for the filter in order to have a filter element at index $k = 0$ that is equivalent with $\Omega = 0$ in the continuous domain[3]. The resolution in terms of Ω is equal to $\Delta\Omega = 0.106$ for

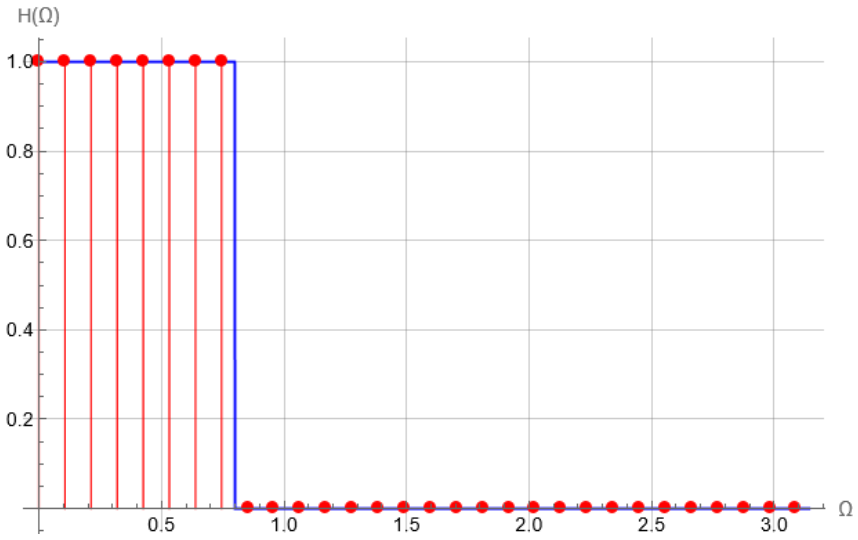


Figure 12.2: Combination of the continuous graph of the filter as a function of Ω and the discrete values with index k of the filter for $K = 59$, $\Omega_c = 0.8$ and $k_c = 7$. See text for further explanations.

[3]Since there are many ways to chose the filter coefficients and many implementations in various codes, you have to be careful, read the help file and do some test to check the working of filters during a filter design.

this number of elements $K = 59$. Note that the endpoint $k = 29$ does not correspond to $\Omega = \pi$, however has a value of $\Delta\Omega/2$ below π at $\Omega = 3.09$. These details are important - special with a low number of filter coefficients - in order to start with a symmetric distribution of filter elements around zero, where the cyclic nature of the filter coefficients is not disturbed.
In order to obtain the filter coefficients in the time-domain the inverse Discrete Fourier Transform is applied on the sampled block function.
The resulting coefficients in the time-domain $h[n]$ for this situation is shown in figure 12.3. A few

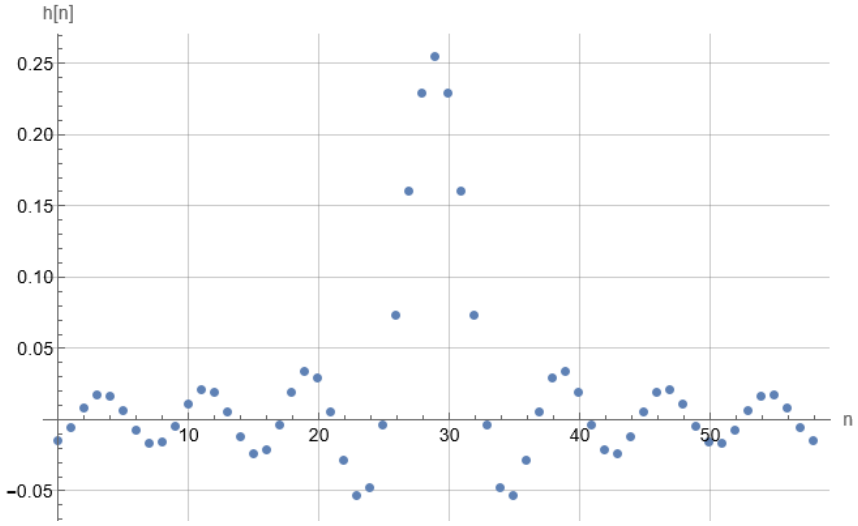


Figure 12.3: The resulting time-domain filter coefficients of the low-pass filter, $h[n]$, with $K = 59$ elements.

things have to be mentioned on the results:

- There are in total $K = 59$ filter elements.
- The coefficients presented in figure 12.3 are indexed starting with $k = 0$ denoting the first coefficient and $k = K - 1 = 58$ the final coefficient.
- The coefficient with index $k = (K - 1)/2 = 29$ is the center of the time-series.
- The mirror symmetry in the filter coefficients array can clearly be seen.
- The width of the filter in the time domain is inversely proportional to the width of the filter in the time-domain: compare figures 12.2 and 12.3. This is a property of the Fourier Transform as we know.
- The shape of the graph looks like a sampled sinc-function that is truncated at the edges. A close comparison of the filter coefficients with a sinc-function shows that it is *not* a sinc-function; It is a sum of cosines that are aligned at $k = 29$ in this case. It is however, not a surprise that it looks like a sinc-function, since in the limit for an infinite number of coefficients, it will approach a sinc-function.

The truncation of the sinc-like-function will result in some unwanted artefacts in the filtered signal. A way to suppress these artefacts, is to force the filter coefficients to zero near the edges. This can be achieved by multiplying the filter list with a gradually changing weighing function approaching zero at the edges, a method that is called windowing and is treated in chapter 12.6.

12.2.2 Time-sampling design of the low-pass FIR-filter

From the continuous Fourier Transform pairs we know that the inverse Fourier Transform of a block is a sinc-function. In the time-sampling method, also called *windowing method*, first the inverse Fourier Transform is calculated to obtain the continuous time-domain representation of the filter:

$$h(t) = \frac{\Omega_c}{\pi}\,\mathrm{sinc}(\Omega_c t) \tag{12.5}$$

This sinc-function has to be sampled to obtain the time-domain filter coefficients. From the previous method we know the number of filter coefficients: $K = 59$. We also know the time resolution which has to match the sampling time $T_s = 1/f_s$. Consequently, the full range of the list is equivalent with KT_s. Combining this information we can directly calculate the time-domain filter coefficients via:

$$h[n] = \frac{\Omega_c}{\pi}\,\mathrm{sinc}\left(\frac{\Omega_c n}{2\pi}\right) \tag{12.6}$$

A few additional remarks have to be made on this method and its results (all but the last bullitpoints hold here):

- In this method a sinc-function is sampled, so the coefficients differ with those from the other method.
- There is also a subtle difference in the use of the cut-off frequency Ω_c between the two methods. In the latter method, the Ω_c is the cut-off frequency that determines the shape of the sinc-function, whereas in the former method Ω_c is indication within a spread of $\Delta\Omega = 2\pi/K$ for the cut-off frequency.
- The fact that the cut-off frequency is used in different ways in both methods has consequences in the frequency responses of the two filters. This can easily be observed from the Discrete Fourier Transforms of the filter coefficients. The frequency-domain filter coefficients of the former method are exactly the filter coefficients we started with and those are located exactly on the initial block function. The effect is illustrated below in the next section.
- For the time-sampling method we used the number of filter coefficients from the frequency-sampling method. In more general cases you have to start from scratch and then the choice of number of coefficients and the choice for the cut-off frequencies is slightly more indirect compared to the frequency-sampling method. Details will be explored in the problems at the end of this chapter.

12.2.3 Continuous-frequency representation of a discrete filter

In terms of the frequency response of the filter we can now introduce a different view. The fact that we could use a discrete frequency k and a continuous frequency Ω in the frequency domain will provide us two different views. In case we take the Inverse Discrete Fourier Transform of the filter time-series $h[n]$ using the variable k we get the results we already had and presented in figure 12.2. However if we use the continuous variable Ω in the Inverse Discrete Fourier Transform, we get the result presented by the green line in figure 12.4. The initial filter elements are also indicated (red circles) in the figure. The blue line is the original design.
The number of filter coefficients K is referred to as the number of taps, the kernel length or simply the number of elements.[4]

[4]While we will stick to this terminology, it is good to know that this definition is not completely universal and various authors use slightly different definitions. For example, the term 'kernel length' is sometimes used to refer to $K/2+1$ instead of K. Another term, 'the order of the filter', is so prone to confusion that we will avoid using it entirely.

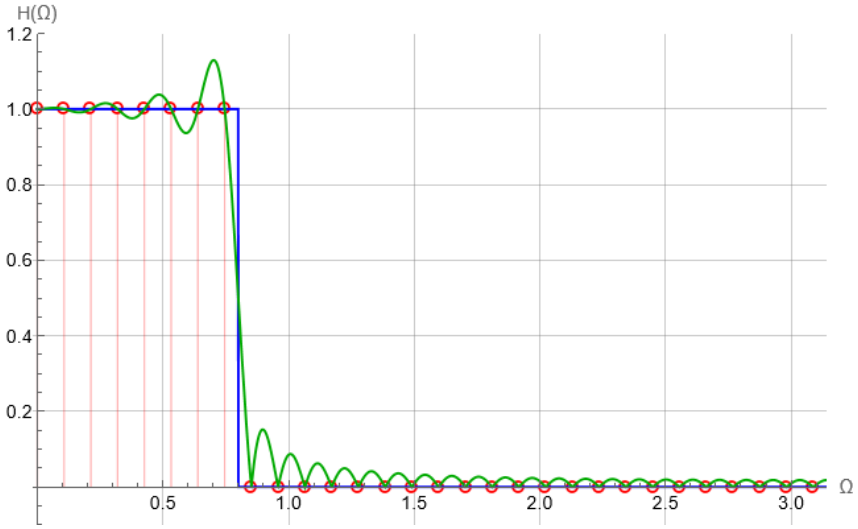


Figure 12.4: Typical response of a FIR filter in the frequency domain after the inverse Discrete Fourier Transform with the continuous frequency variable Ω. The basic shape is preserved, but ripples appear in between the discrete points; at the locations of the frequency samples $H(\Omega)$ takes the right values. Only the positive part is shown here; the graph is mirrored in the $\Omega = 0$ axis to get the full filter.

From the initial design of the filter, a kind of transfer function can be calculated via the Discrete Fourier Transform[5], since we have a finite number of elements and we know the coefficients $h[n]$:

$$\begin{aligned} H(\Omega) &= \sum_{n=-(K-1)/2}^{(K-1)/2} h[n] \mathrm{e}^{-\mathrm{j}\Omega n} \\ &= 1 + \sum_{n=1}^{(K-1)/2} 2h[n] \cos(\Omega n) \end{aligned} \tag{12.7}$$

In the last step the symmetry property of $h[n]$ is used, *i.e.* $h[-n] = h[n]$. Note that the $n = 0$ element is separated from the sum, due to the fact that this element occurs only once in the time-series, whereas the other elements have a counterpart for $n \leftrightarrow -n$.
A small remark has to be made here on the nature of the number of elements in the filter K. The exact implementation differs a bit for an even number of values of K from that of the implementation for an odd number of values of K. In this example, we used an odd value for $K = 59$ in order to make the frequency-series (discrete spectrum) as well as the time-series (impulse response) symmetric, both having an element with index zero and running over negative indices and symmetric positive indices. An even number of elements would either break this symmetry or is missing index zero element ($n = 0$; $k = 0$). Here the homogeneous distribution of K elements over the (cyclic) interval $[-\pi, \pi)$ is the critical condition. See figure 11.2 for an illustration of this distribution of filter coefficients. This leads to the advise:
It is strongly recommended to use an odd number of elements in the design of digital FIR filters.
The finite length and with that the finite number of harmonic functions that build the filter response in the frequency domain cause the ripple in the transmission part as well as in the rejection part of the filter spectrum. Increasing the number of filter coefficients will reduce the ripple in the response, however it is a trade-off between various design criteria. Special techniques are used to reduce the ripple in filters. We will not use those in this course.

[5]Strictly speaking this calculation does not result in the transfer function, since the real transfer function is calculated by means of a z-Transform. This will be treated in chapter13. The use of the Discrete Fourier Transform is a close enough approximation here.

A final remark on the continuous frequency response of the FIR-filters related to the two methods discussed above; the frequency-sampling method and the time-sampling method or windowing method additional to section 12.1.1:

FS: In the frequency-sampling method by definition the recalculated filter coefficients – from the time-domain filter coefficients – in the frequency-domain coincide with the original sampled block function. Additional to this, this these points are the intersection of the block function and the continuous frequency response curve with the ripples.

TS: In the time-sampling method the recalculated filter coefficients – from the time-domain filter coefficients – in the frequency-domain do lie on the continuous frequency response curve, as it should, however do not lie on the original block function. This is illustrated in figure 12.5. Compare this result with the result of the frequency-sampling method in figure 12.4 where the same values for the design parameters are used.

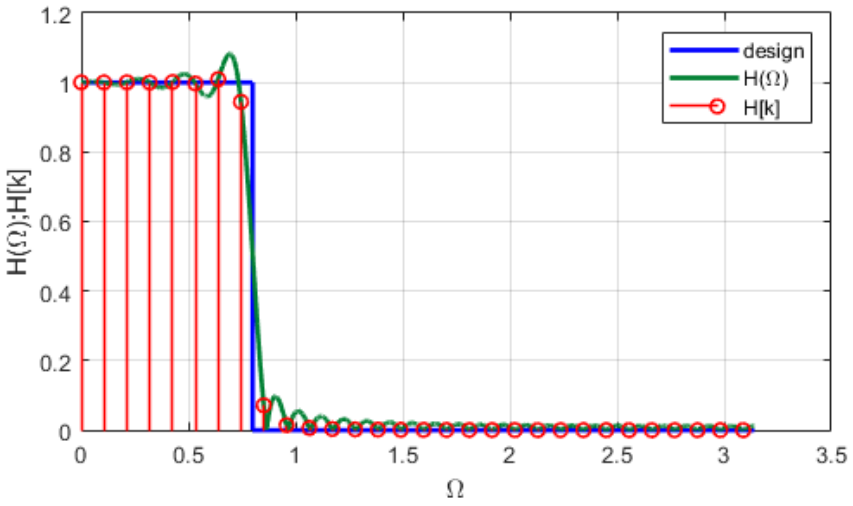


Figure 12.5: Combination of the continuous graph of the filter as a function of Ω and the discrete values with index k of the filter for $K = 59$, $\Omega_c = 0.8$ and $k_c = 7$. This filter has the same design parameters, however is now sampled in the time domain. Compare with figure 12.4 and see text for explanation.

From here onwards we will mainly concentrate on the FIR-filter design with the *frequency-sampling* method. At some points we will make a note about the *time-sampling* method, particular at locations where the time-sampling method deviates in the results from the frequency-sampling method. These will be identified by "**TS**: ≪comments≫" in the text.

12.2.4 Application of the FIR-filter

Having determined the filter coefficients list $h[n]$ either way and calculated its frequency response, it is time to apply the filter on a discrete-time signal.

Application of the filter $h[n]$ on an input signal $x[n]$ is performed by circular convolution between the two time-series. Needless to mention that both time-series have in general a different number of elements. For every element in the input signal, we have to run the sum over K items, the number of elements in the filter. Needless to mention that the length of the filter coefficients list must be smaller than the length of the signal.

$$y[n] = \sum_{m=-(K-1)/2}^{(K-1)/2} h[m]x[n-m]_K \tag{12.8}$$

Although this formula could in principle be applied on an input signal, it will give problems if real-time filtering is required. The fact that the sum starts at negative index, means that time samples of the future of the input are needed to calculate the output. Below we will discus this effect and how to work around this. We will also see that this effect plays an important role in the FIR filter design.

12.2.5 Time delay and run-in effect of the filter

There is still an important problem to solve: The impulse response calculated from the ideal frequency response, represented in figure 12.3 violates causality: The time-series $h[n]$ contains elements with a negative index n. In order to generate the figure, we had to use negative values for the index n (the peak of the filter coefficients list is at $n = 0$). Applying the convolution this results in using elements of the time-series from the future, which is not possible without the violation of causality. The solution to this problem is very simple, but comes with a cost: a time delay of the output of the filter.
By shifting the impulse response of the filter by $(K-1)/2$ steps. In the presentation of the impulse response in figure 12.3 the delay was already present without mentioning. In this figure the index of the time-series started at $n = 0$; just an arbitrary counter of the elements. In practice renumbering of time-series elements is easy. In reality this is a non-problem, since there is no absolute timing between the filter response and the input signal to be filtered.
Application of the filter on a signal (the input), via a convolution with the impulse response of the filter results in an effect that the Kth output element is the first element that is correctly filtered with all filter elements. In many cases this can clearly be observed at the beginning of the output signal. This effect is called the *run-in of the filter*. For a filter of K elements this takes K elements of the output signal. In practice, often the first K elements of an output are dropped after filtering. This effect is important to be included in the design process of FIR-filters. It is a trade-off between resolution and run-in time of the filter.

12.3 Design Criteria for FIR filters

Now we know the principle of FIR filters we can concentrate on the design of FIR filters. In order to design a good filter there are two things you have to realise: you have to know the signal and you have to know what you want to do with the signal. Only with this knowledge, you can design a good filter. In order to know the signal, you need to perform an analysis that is mainly concentrated on the frequency content of the signal. In other words: apply a Discrete Fourier Transform on the time-series. Prior to this step you may need to select a certain part of the time-series to focus on the relevant part if applicable. The second step is not so easy to advise on to you, since the circumstances strongly determine the selections you need to make. So let us assume that a certain frequency range of the signal is of interest and need to be preserved – *pass-band* – whereas the rest of the signal is classified as noise that can be rejected – *stop-band*. The range of frequencies between the pass-band and the stop-band is called the *transition-band*. Depending on the resolution of the filter, also called the steepness of the filter, this band can vary in width. Higher-order filters will have steeper edges and therefore thinner transition bands. The edges of the pass-band are the pass-band frequencies

$\Omega_{p,i}$ and the edges of the stop-band are the stop-band frequencies $\Omega_{s,i}$; the cut-off frequencies $\Omega_{c,i}$ are in the transition-band between the corresponding pass-band frequency and stop-band frequency. In figure 12.6 these definitions are illustrated for a low-pass filter where the low frequencies below a certain frequency are of interest and the rest is noise. Later we will discuss other types of filters, however the principle remains the same. The techniques we need for the design will in principle be transferable to the other types of filters.

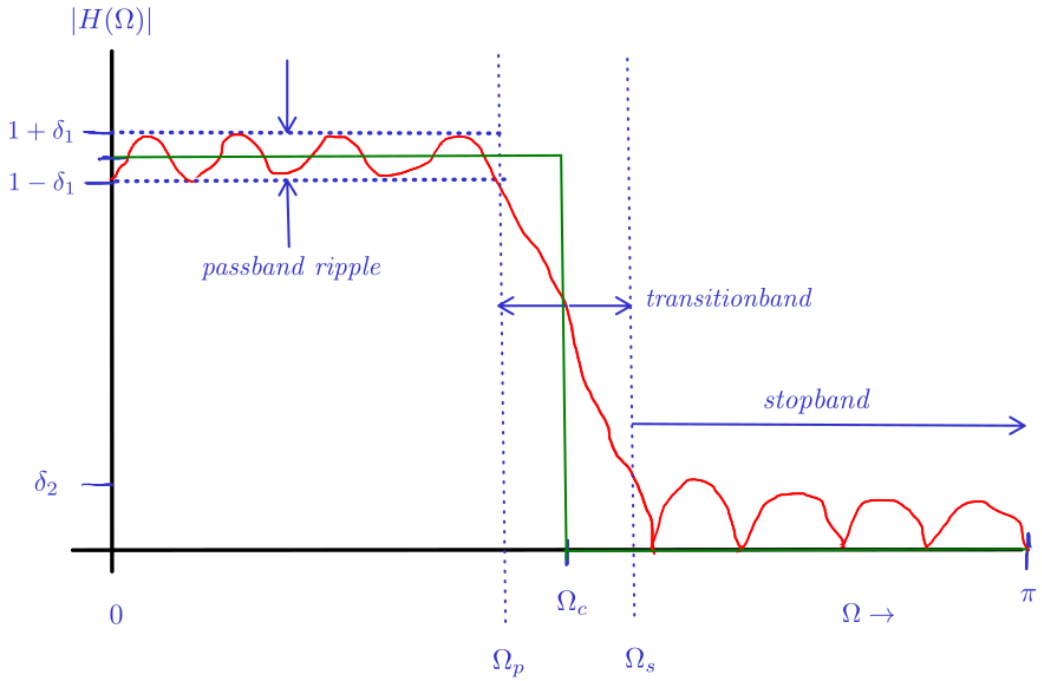


Figure 12.6: Definition of filter parameters for the design process; in this case a low-pass filter. See text for further explanation and generalisation to other filter types.

Since we only consider full transmission or full rejection in the frequency ranges, we only have one degree of freedom to adjust in the design of the filter beside the cut-off frequency: the resolution or the steepness of the filter at the cut-off frequency Ω_c. This resolution is determined by the number of filter elements in the filter K; this also determines the width of the transition-band. Remember that this number of filter elements is equal in both representations: the frequency-series and the time-series (impulse response of the filter).
On the normalised frequency axis $\Omega \in [-\pi, \pi)$, the frequency resolution is equal to:

$$\Delta\Omega = \frac{2\pi}{K}$$

Remember that the normalisation of the frequency axis was done on the sampling frequency of the input signal: $2\pi/f_s$, which gives us the resolution on the real frequency axis:

$$\Delta f = \frac{f_s}{K}$$

This is a very important result, since it states that the resolution of the filter is proportional to the sample frequency and proportional to the inverse of the number of filter elements. This $\Delta\Omega$ or Δf determines the steepness of the filter, since the step from transmission to rejection, or from rejection to transmission of the filter is done in one frequency step. As we have seen above, a continuous representation of the frequency response shows a ripple in both the pass-bans as well as the stop-band, despite the fact that the discrete values of the frequency-series are either one or zero.

Since the sample frequency determines the time step, the number of filter elements determines also the length of the filter in the time-domain and therefore the delay of the filter. Remember that for every sample in the input signal, K elements have to pass for a full filter action. The corresponding time is KT_s (or K time steps) which is the run-in time of the filter (it takes KT_s time to have a fully filtered first element.

Since half of the impulse response have to be shifted to positive times to fulfil causality, half of that time is the delay of the output signal with respect to the input signal. An additional disadvantage is the increasing computational effort for an increasing number of filter elements.

The relation between the various quantities is summarised in table12.1 and figure12.7.

Table 12.1: Filter parameters and time-series parameters.

Quantity	Symb.	D/A	Range	Relation
Sample frequency	f_s	A	-	from input $x[n]$
Time step	T_s	A	-	$T_s = 1/f_s$
Filter length	K	D	-	to be determined
Filter length	t_K	A	-	$t_K = KT_s$
Time index	n	D	$[-(K-1)/2, (K-1)/2]$	independent
Time index (shifted)	n'	D	$[0, K)$	independent
Frequency index	k	D	$[-(K-1)/2, (K-1)/2]$	independent
Frequency index (shifted)	k'	D	$[0, K)$	independent
Frequency	f	A	$[0, f_s/2)$	$f = kf_s/(2K)$
Normalised frequency	Ω	A	$[-\pi, \pi)$	$\Omega = 2\pi k/K$
Frequency step	$\Delta\Omega$	A	-	$\Delta\Omega = 2\pi/K$
Frequency step	Δf	A	-	$f_s/(2K)$

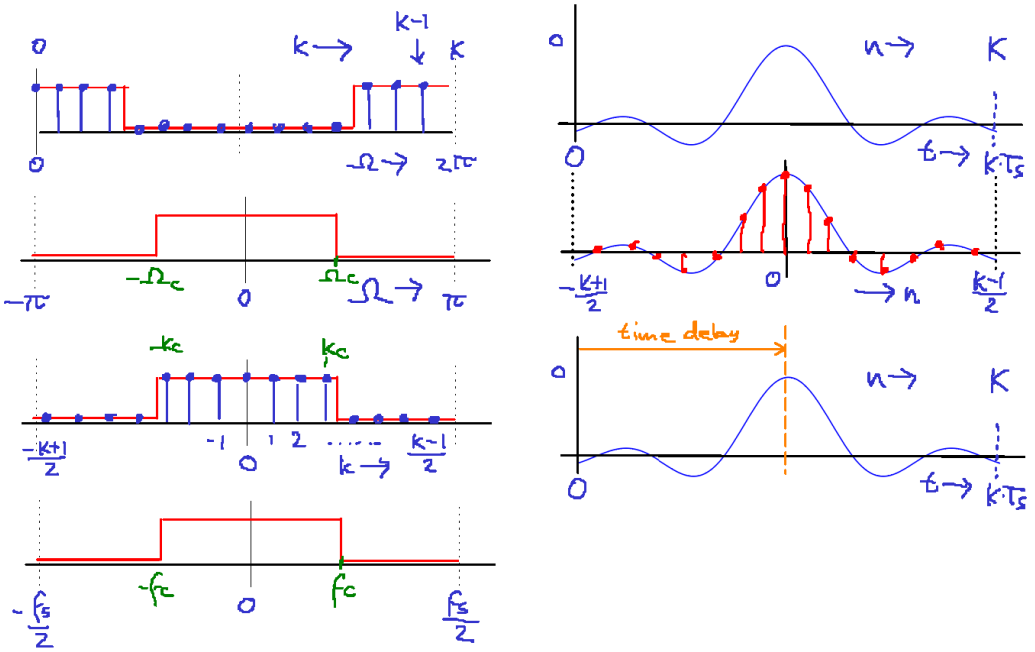


Figure 12.7: Various definitions of the frequency axis and time axis presented in a figure for comparison. See also table 12.1. Note: the frequency axis is modular over the given interval.

In the design process of the filter, we now clearly identified a trade-off: A higher resolution in the frequency domain of the filter, comes with the cost of a larger delay of the output and longer run-in time of the filter. For the design of the FIR filter this is an important optimisation to apply. As a rule of thumb, you can take the minimum number of elements that fulfils the frequency resolution or steepness of the filter to do the job.
The ripples in the continuous frequency response reflect the uncertainty in the filter definition due to the limited number of filter coefficients in combination with the finite sample frequency. The discrete frequency response is still flat. One could also argue that there is limited information in the filter coefficients. More filter coefficients increases the frequency resolution (with a corresponding reduction of the ripples). A larger number of filter coefficients does not lead to a better definition of the ripples but to an increased number of ripples; the continuous filter response is equal to the discrete filter response at the position of the elements. The inherent lack of information in a discrete filter can also be illustrated by the location of the cut-off frequency Ω_c (independently chosen) in relation to the exact positions of the discrete frequency elements Ω_k. With respect to the frequency-series, there is an uncertainty of $\Delta\Omega$. A consequence is that all cut-off frequencies between the same two discrete frequencies result in the same filter.
TS: *In the time-sampling method, Ω_c is exact and will in general be at a position halfway between the pass and stop amplitude of the filter. Here every other cur-off frequency results in a different filter.* For both methods hold that the uncertainty in the filters is about equal, only the appearance is different.
The influence of the filter length K on the sharpness of the response in the frequency domain is illustrated in figure 12.8 where continuous frequency responses of filters that only differ in the number of elements are presented for four different numbers of elements.
The effect of the filter length in the time-domain is presented in figure 12.9. In the upper graph where a low-pass filter with $K = 101$ filter elements is applied on an input signal, a delay of 50 samples is easily observed; in the lower graph, where a high-pass filter with the same number of filter elements is applied on the same input signal, you can not easily identify the delay in the output signal. Large values of K result in long delays in the filter response. So keep this in mind when designing a filter. Especially you should keep the filter length much smaller than the signal to be filtered. For band-pass and high-pass filters analogous derivations can be made.

12.4 Linear-phase response of a FIR filter

An important characteristic of a FIR filter is the linear-phase response in the pass region, the phase delay ϕ is proportional to the frequency. This results in a preservation of the phase relation of the various frequency components even though the these components are attenuated by different amounts. This implicit feature is among other domains especially important for audio applications. Although we only can get the full understanding after we understand the z-Transform, this linear phase transfer characteristic of the FIR filter can be observed in the output signal: see figure 12.9. The top graph of Figure 12.9 shows the result of a filtered multi-sinusoidal signal having the same phase delay, as can be observed by the shifted red curve with respect to the unfiltered blue trace. This corresponds to a linear phase delay – the phase response changes proportional to with the frequency – in order to have a time delay that is equal for all frequencies. The result is shape preservation of the signal. The bottom panel shows a high-pass filter with the same characteristics. The phase shift in the bottom graph is the same, however this is difficult to see because the high-frequency period is shorter than the time delay. Note that this implies that you have to be careful using filtering by only amplitude multiplication in the frequency domain. A similar consequence: you can not retrieve a time signal from its amplitude spectrum $\rightarrow$ no essential phase information.
A phase delay is unavoidable in real-time filtering (whether digital or analogue). However, in cases that do not need real-time processing, a zero phase transfer can be obtained in the output signal by filtering the signal twice with the same filter, however in two different directions. In the second

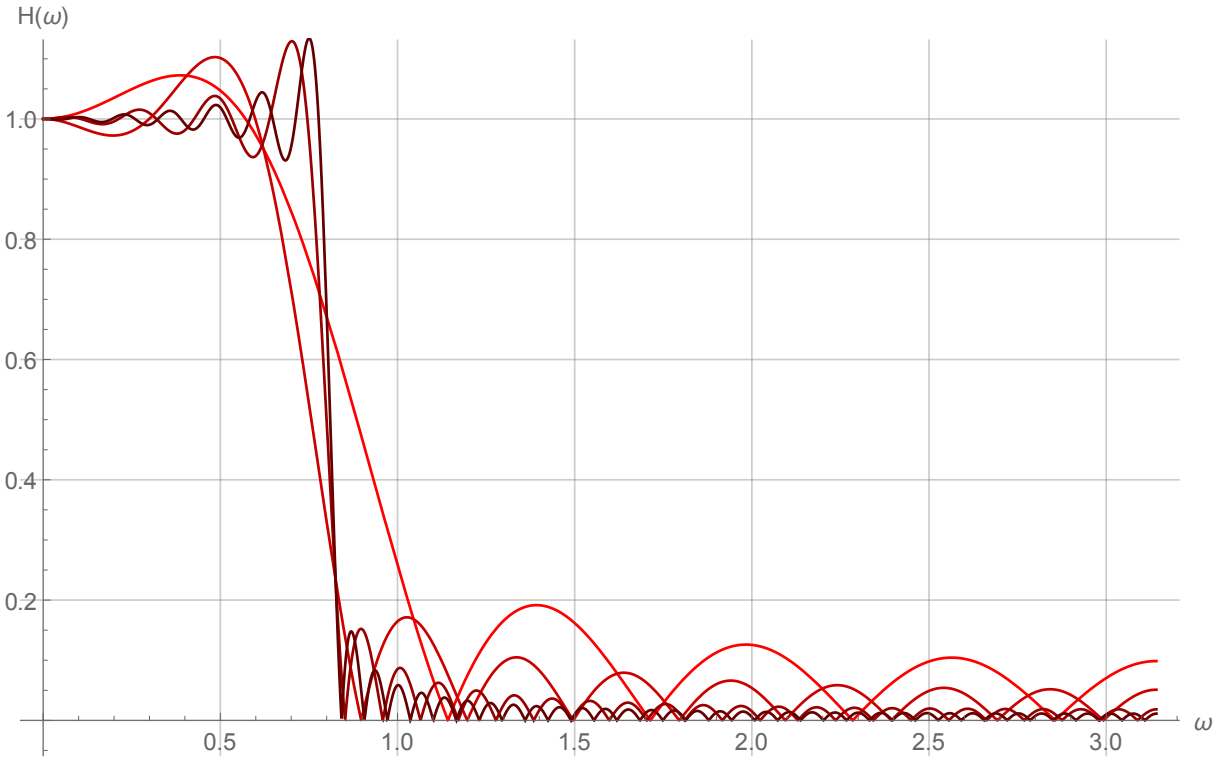


Figure 12.8: Four FIR-filter responses, that differ only in order; 11, 21, 61 and 101 taps filter. The frequency axis is normalized on the interval $0 - \pi$ and the normalized cut-off frequency is $\pi/4$

pass the signal is reversed, giving a negative delay, that compensates the delay of the first pass. The total filter action thus results in a net zero phase change. The method requires a symmetric impulse response of the filter. We will not apply this technique in this course.

Example 12.4.1. *FIR-filter design example to illustrate the effects*

In this section a practical example of the steps in the design of a FIR filter are presented. In parallel a low-pass and a high-pass filter are designed with the same cut-off frequency. For convenience the filters are designed in normalized frequency space $[0, \pi]$. The cut-off frequency is at $\Omega_c = 0.195\pi$; the number of taps is equal to 21.

Implementation of this design results in the block filter given in blue in figure 12.10. The frequency values corresponding to a 21-tap filter are also indicated. The resulting time coefficients are presented in figure 12.11. These coefficients are obtained via the inverse DFT of the desired frequency response.

The frequency spectrum is presented in figure 12.12. From these figures it is clearly seen that the resolution of the cut-off frequency is limited by discretization (spacing of $2\pi/21$ between values of Ω) and has – in this case - a different effect for both filters. The effective cut-off frequency is at about 0.2π for both filters.

Applying the filters on a noisy sinusoidal signal will result in a cleaner sinusoidal signal for the low-pass filter and a noisy signal without the sinusoidal signal for the high-pass filter. These results are presented in figure 12.13. The time delay of 10 time steps is clearly seen from the results. This holds for both filters, since the delay is determined by the number of taps, which is equal for both filters.

This example showed the principle of FIR-filter design via the so-called frequency sampling method.

□

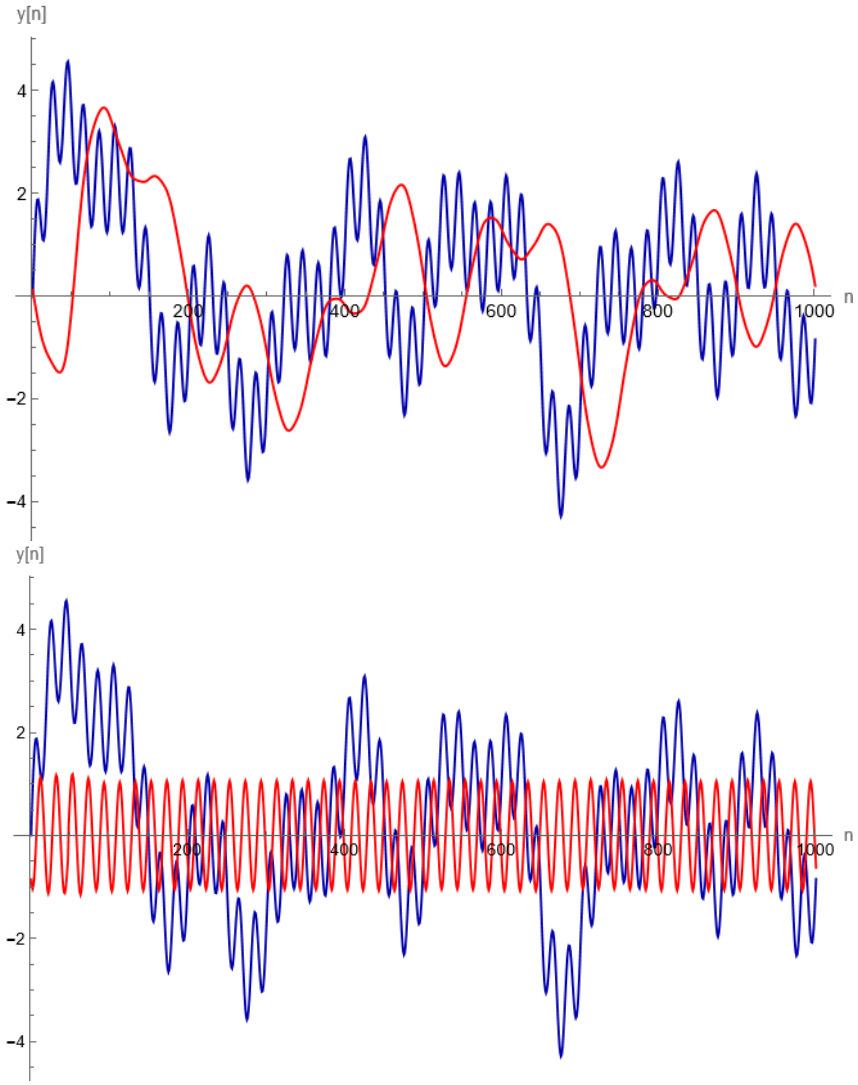


Figure 12.9: Application of two FIR-filters, both with $\Omega_c = 0.06\pi$ and $K = 101$ on a superposition of sine waves to illustrate the linear phase response. Top panel: low-pass filter. Bottom panel: high-pass filter. The shape of the signal remains the same. The time signal has a legacy of 50 samples.

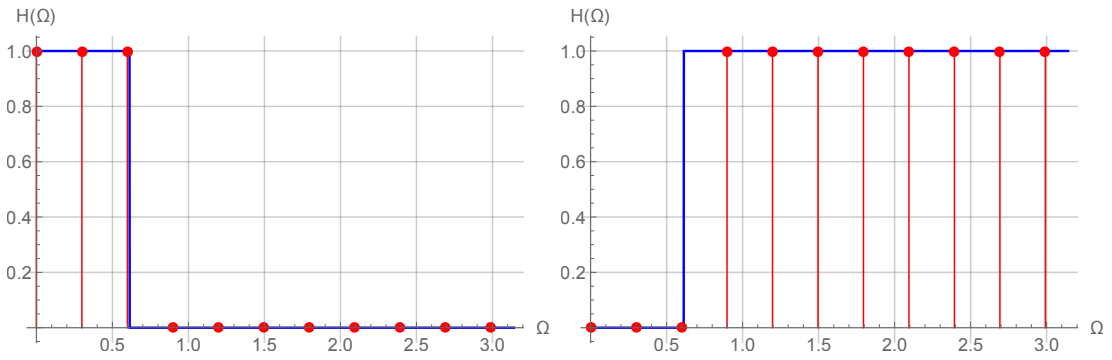


Figure 12.10: Illustration of a FIR-filter Design procedure. Frequency scale is normalized to $0 - \pi$; blue curve is design and stem data are the frequency coefficients. Left: low-pass filter; Right: high-pass filter. Order of the filters is 10, *i.e.* a 21-taps filter.

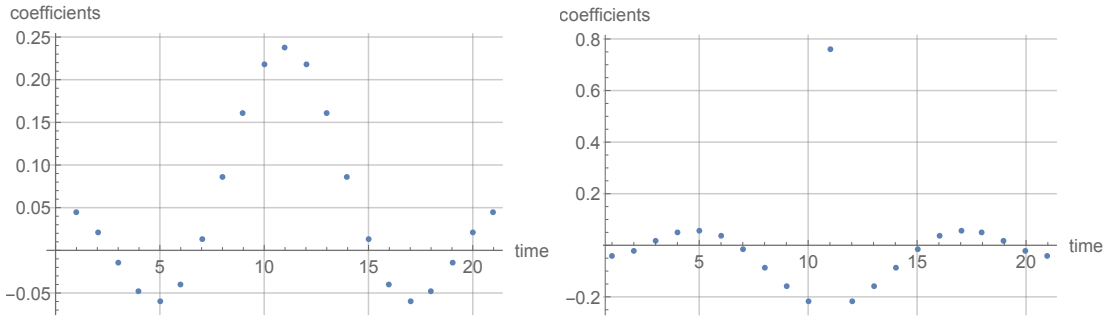


Figure 12.11: Time coefficients of the above designed filters, obtained via a $z-$Transform. feature: 21-taps filter coefficients.

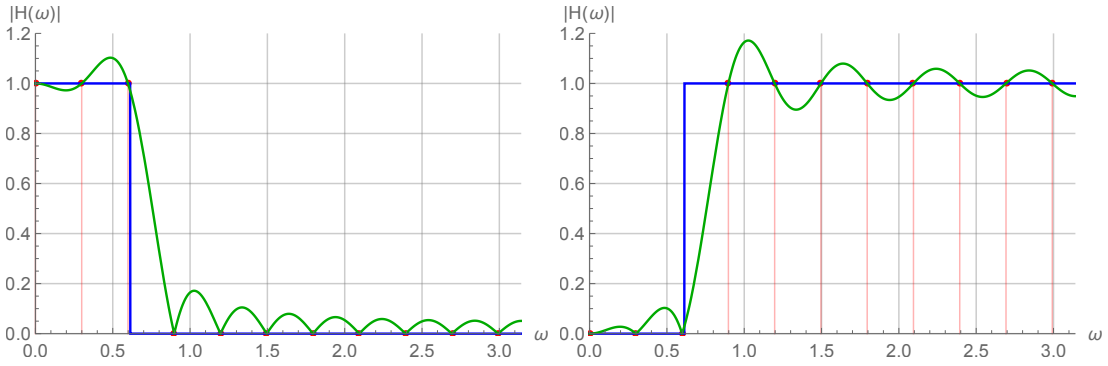


Figure 12.12: Again the freqency spectrum of the designed FIR-filters. Inverse $z-$Transform shows the continuous frequency spectrum.

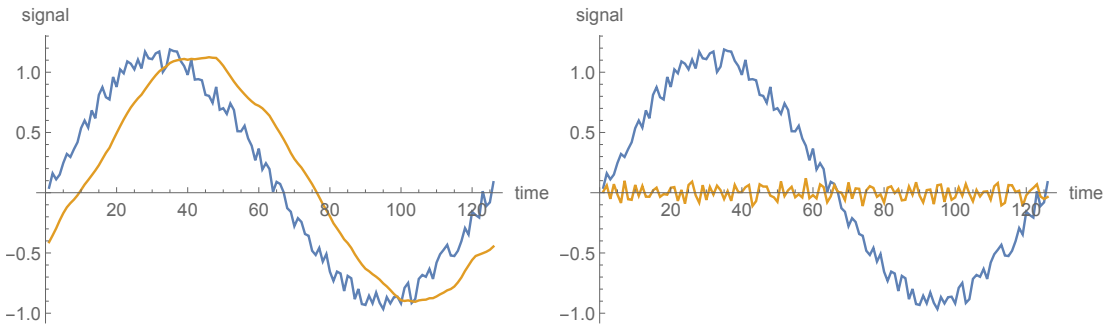


Figure 12.13: Application of the FIR-filters on a one-period sine wave with noise. The low-pass filter takes out the noise, whereas the high-pass filter takes out the sine wave. A clear time shift due to the method of filtering is seen: ± 10 time samples.

12.5 The moving-average filter reconsidered

Let's return to the simple moving-average filter in example 10.7.1, that was given by the following difference equation:

$$y[n] = \frac{1}{M} \sum_{i=0}^{M-1} x[n-i] \tag{12.9}$$

Equation 12.9 can be seen as the convolution of the input signal $x[n]$ with a rectangular block function with area 1. Or, in other words, the rectangular block function $h[k]$ with height $1/M$ and duration M is the impulse response of our filter (why?).
We therefore know that, in the frequency domain, the output for any input $X[k]$ will be given by the product of the corresponding functions in the frequency domain: $Y[k] = H[k]X[k]$, where the frequency-series $H[k]$ (or the equivalent continuous Transfer function $H[\Omega]$) is the Discrete Fourier Transform of the block function. However we have to be careful here, since $H[k]$ only consists of the elements that are ones. The zeros in the series, that are important for the transformation are not present, and need not to be present for the convolution in the time domain. Before we can calculate the transfer function $H[\Omega]$ we have to add leading and trailing zeros, to complete the frequency-series. So the question is how many do we have to add? The DFT of the block in the time domain gives:

$$H[\Omega] = 1 + 2 \sum_{n=1}^{\frac{M-1}{2}} \cos(\Omega n) \tag{12.10}$$

in the frequency domain. This means that M determines the cut-off in normalised frequency range Ω. However if we make Ω discrete, we have to know the length of the filter K and therefore the frequency series is dependent on the number of elements:

$$H[k] = 1 + 2 \sum_{n=1}^{\frac{M-1}{2}} \cos\left(\frac{2\pi k}{K} n\right) \tag{12.11}$$

The value of K, the number of filter elements, determines the resolution of the filter in normalized frequency $2\pi k/K$. In figure 12.14 the normalised continuous frequency response is presented for a number of M values. The effect of the number of elements can be seen from the fact that you have to sample this graph in $(K-1)/2$ elements. Translating this to real frequencies, we see that M as well as f_s determine the cut-off frequency. The combination of K and f_s of the original time-series, determines the actual cut-off frequency. Rewriting this in frequencies:

$$f_c = \frac{1}{M} f_s \tag{12.12}$$

In normalized frequency this becomes:

$$\Omega_c = 2\pi \frac{K}{M}$$

where K is the length of the filter kernel and M is the number over which is averaged in the moving average. In the limit of infinite filter length K and averaging elements M under the restriction to keep Ω_c constant, the frequency response becomes:
With increasing the number of filter elements, in the limit for an infinite number of coefficients the Fourier Transform of the rectangular block time-function being the sinc-function in the frequency domain,

$$H[\Omega] = \mathrm{sinc}\left(\frac{\Omega}{\Omega_c}\right) \tag{12.13}$$

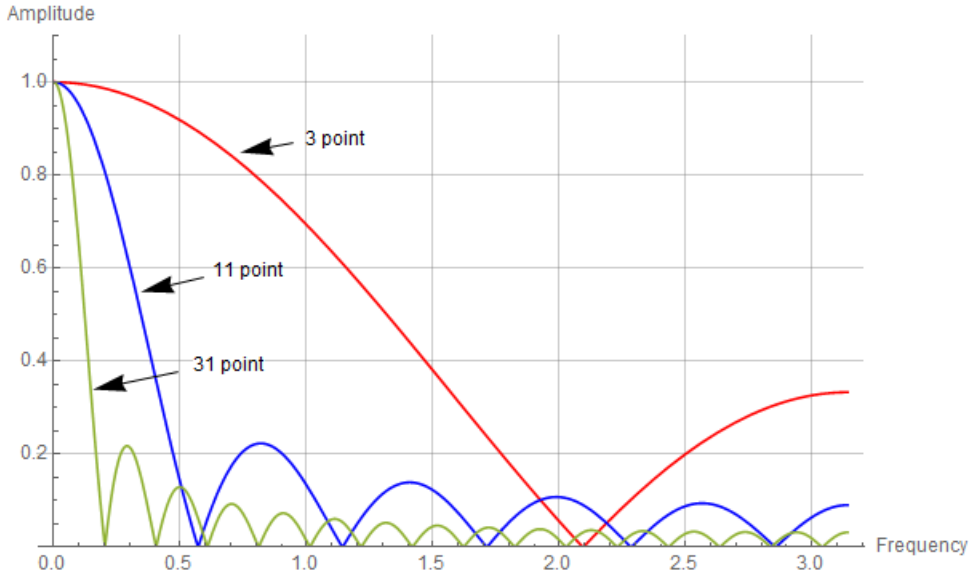


Figure 12.14: Frequency characteristic of a Moving Average filter. The number of averaging points is indicated. With increasing M, the cut-off frequency (the frequency at which signal components start to be suppressed) becomes lower. Oscillations at high frequencies remain, however.

In figure 12.14 the frequency response according to equation 12.13 of the moving average filter is shown for a few different values of number of sample averages M. The frequency axis is normalized so that π is equivalent to the Nyquist frequency $f_s/2$.
Higher number of averaging samples show a steeper fall-off; lower frequencies are transferred, as expected, while higher frequencies are suppressed more and more. The filtering is not very smooth, however: the transfer function exhibits oscillations, with some high frequencies being completely suppressed while nearby frequencies are still transmitted to some degree. Thus, while our filter does the job it was intended for at some basic level, it looks like there is room for improvement.

12.6 Windowing

Working with finite time-series or filter coefficients series always has a drawback, especially if they are cut out of a larger range of data. This is an inherent feature that has to do with the strong relation between the time-domain and the frequency-domain. Any abrupt cut in one of the domains has consequences in the other domain. As an extreme example, we have seen that a moving average filter retains some high frequency components in the output signal.
We have already seen that the impulse response of an ideal low-pass filter results in a discrete sinc-function, that is terminated at the boundaries. This termination is a result of the sample frequency f_s, the cut-off frequency f_c of the filter and the number of elements. This termination in the time domain, has an effect on the ripples in the frequency domain of the filter.
A method to reduce the effect of such edges during averaging is make use of windows that make the edges more gradual. This brief chapter is about the use of windowing on series of data. This is meant as supporting information to the main course material.

12.6.1 Principles of windowing in the time domain

Consider a time-series $a(n)$ with a significant large length say M samples. Taking out a part of length N samples $b(n)$ that is significantly smaller ($N \ll M$) and away from the edges. Mathematically this can be represented by the multiplication of the original time series with a block-function, which

has the value 1 for the data points we have in series $b(n)$ and the value 0 for the rest.

$$f_{sec}[n] = \begin{cases} 1 & n_l \leq n \leq n_r \\ 0 & \textit{otherwise} \end{cases} \tag{12.14}$$

resulting in:

$$b[n] = a[n] \cdot f_{sec}[n] \tag{12.15}$$

This block function transfers to a sinc function in the frequency domain. Using the fact that a product of two functions in the time-domain is equivalent with a convolution of the Fourier transforms in the frequency domain, means that the action of taking out a limited length time series is equivalent with the convolution of the original function's spectrum with a sinc function. A consequence, for example, is that a single frequency (only possible in an infinite time-series) smears out to a sinc-like blob in frequency space. The width of the sinc function in the frequency domain is inversely proportional to the width of the block in the time domain. A relative large sub-series $b[n]$ results in a relative narrow width in the frequency domain. However the sinc function has significant magnitude away from the main peak (all those ripples).
The contribution away from the main sinc peak can be reduced significantly by applying a window over the cut-out section $b[n]$. The essence of a window is reducing the contribution of the edges of the time-series. The basic principle is multiplying the time-series with a function that goes gradually to zero at the edges and has a maximum in the center of the interval. A simple triangle function already does the job and gives a significant improvement in reducing the smearing of frequencies.
Mathematically the original time-series $a[n]$ is not only multiplied by a block function but also with the applied window function $w[n]$:

$$b'[n] = a[n] \cdot f_{sec}[n] \cdot w[n] = b[n] \cdot w[n] \tag{12.16}$$

The dot in the multiplication is used here to indicate that this is an element-wise multiplication. Therefore, the length of the window is equal to the length of the filter. The corresponding spectrum is given by the convolution of the z-transforms of the individual functions:

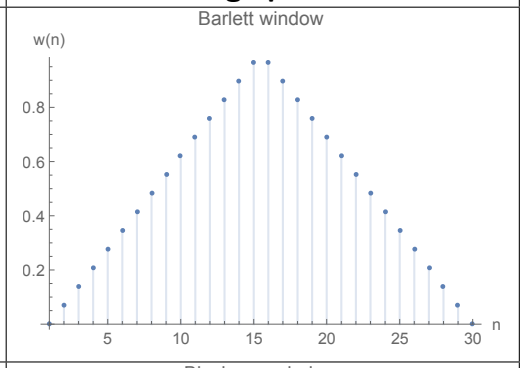
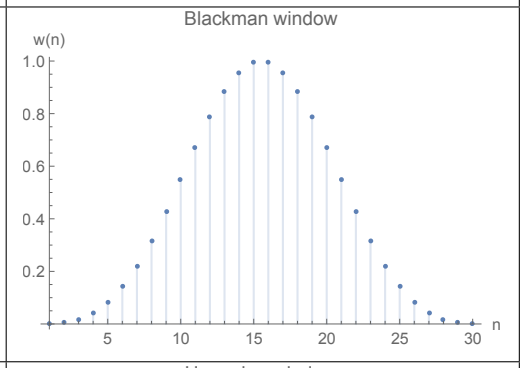
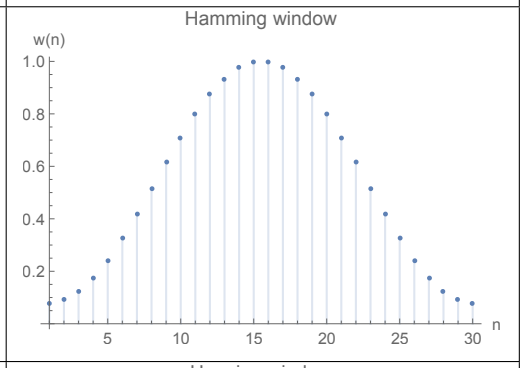
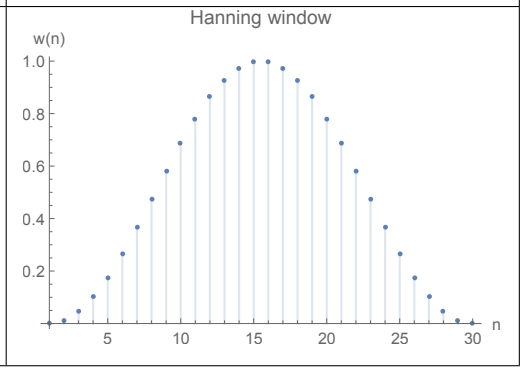
$$B'(z) = A(z) \otimes F_{sec}(z) \otimes W(z) = B(z) \otimes W(z) \tag{12.17}$$

The application of windowing in the time domain is often used in filter design, particular in the FIR-design technique.
Windowing is not only used in the design of filters but also in the determination of spectra. Here again, if you want to calculate the spectrum of a finite time series, the steep edge at the time domain edges will introduce frequency spread in the spectrum. A repetitive extrapolation of the time series is not an option, since it favours frequencies that are harmonics of the fundamental that fits to the time series length. Applying a window is a trade-off between the two extremes.
In table 12.2 four commonly used window functions are presented with there mathematical definition for a length of M samples. In the last column a graphical presentation is given of the windowing functions for a length $M = 30$.

12.6.2 Principles of windowing in the frequency domain

In the frequency domain the same principle applies on a frequency series (that we call a spectrum). The principles are the same: Cutting out part of the spectrum with a block function results in a convolution of the block, *i.e.*, sinc function in the time-domain. So now the time-domain gets smeared out. To reduce this you could apply a window function in the frequency domain on the coefficients.

Table 12.2: Various Window functions

name	**function** $w(n)$, $0 \leq n \leq M-1$	**graph**
Barlett	$w(n) = 1 - \frac{2\left\lvert n - \frac{M-1}{2}\right\rvert}{M-1}$	
Blackman	$w(n) = 0.42 - 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1}$	
Hamming	$w(n) = 0.54 - 0.46\cos\frac{2\pi n}{M-1}$	
Hanning	$w(n) = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{M-1}\right)$	

The mathematical formulation is equivalent. In case we take out a part of the spectrum with block and a window function we get:

$$B'[k] = A[k] \cdot F_{sec}[k] \cdot W[k] = B[k] \cdot W[k] \tag{12.18}$$

The corresponding result in the (discrete) time domain is a convolution of the inverse z-transforms of the frequency functions:

$$b'[n] = a[n] \otimes f_{sec}[n] \otimes w[n] = b[n] \otimes w[n] \tag{12.19}$$

The effects in the time and frequency domain are completely symmetric complementary.
Windowing in the frequency domain is not explicitly used in tis course, however you might face this effect later and you need some means to deal with it.

12.7 Implementation of FIR-filter design in codes

The various mathematical programs each have their own implementation of filter design methods. In the design of the optimal filter a choices for Ω_c and K have to be made based on cut-off frequency, response speed and desired roll-off (steepness of the filter).
☞ *Give some functions from computational programmes or libraries with explanation or an example worked out in the various codes.*

Summaries

Summary Chapter 12a

Key concepts on Finite-Impulse-Response filters:
For FIR-filter design only one method is used: the *Frequency Sampling method*. This method is based on a construction of a discrete frequency spectrum of the filter, an inverse Fourier transform to the time domain. The filter is applied on a discrete signal via convolution. The procedure or the filter design follows the following steps:

- Rescale the cut-off frequencies to the normalized frequency domain: $\Omega \in [-\pi, \pi)$. where

$$\Omega = \frac{2\pi k}{K}$$

- Construct the discrete frequency spectrum in this domain "1"'s for pass and "0"'s for stop regions in total of K (preferably an odd number) elements, the number of *taps* or *filter elements*.
- Use the inverse Discrete Fourier Transform on this list to construct the filter kernel (filter coefficients $h[n]$) in the time domain; this is the impulse response of the filter.
- Include a time shift in the filter to make the filter causal (for real-time operation). This need to be done, since the filter kernel is symmetric around $n = 0$. The filter shift depends on the kernel length and is equal to $(K-1)/2$.
- Apply the filter on the signal using convolution.
- You might reconstruct a continuous frequency spectrum applying a Fourier Transform on the kernel via:

$$H(\Omega) = \sum_{n=-(K-1)/2}^{(K-1)/2} h[n]\mathrm{e}^{-\mathrm{j}\Omega n} = 1 + \sum_{n=1}^{(K-1)/2} 2\,\mathrm{sinc}(\Omega_c n)\cos(\Omega n)$$

- In order to reduce ripple in the pass-band and the stop-band, a window can be applied on the filter kernel – window length is equal to kernel length. See next chapter.

The discrete circular convolution to obtain an output $y[n]$ of a filtered input $x[n]$ with a filter kernel $h[n]$ with length K is defined as:

$$y[n] = (h \otimes x)[n] = \sum_{k=0}^{K-1} h[k]_K x[n-k]$$

Note: The nomenclature of filter kernels is somewhat flexible and different authors use slightly different definitions. Here we only use the total number of filter elements K.

Summary Chapter 12b

Key concepts on Windowing:
Windowing is used on both time signals as on filter kernels, to reduce the edge effect of the abrupt edge effect in the time domains, that cause wobbles in the frequency domain.

- The length of the window is equal to the length of the time-series, or the length of the filter kernel.
- Various windows are known by name, however are defined by a function, that gradually goes to zero at the edges.
- Details on the characteristics of windows are not expected from you.

Problems

Problem 12.1. Frequency spectrum of a filtered random signal
In order to explore the effect of moving-average filtering you have to make a time signal, and filter that with three different filters, that only differ in the length of the filter.

a. Construct a discrete-time signal in MATLAB, MATHEMATICA, or PYTHON consisting of 1000 time-samples with random values between -1 and +1. Make a plot of the resulting time signal.

b. Construct a moving average filter that is described by the difference equation:

$$y[n] = \sum_{m=0}^{8} x[n-m]$$

c. Apply the filter to your original time signal and plot the result.

d. Determine the Fourier transform of the resulting signal and plot it.

e. Construct a new modified moving average filter that is given by:

$$y[n] = \sum_{m=0}^{8} (1 - |\frac{4-m}{5}|)x[n-m]$$

f. Again, apply the filter on the original time signal and plot the result.

g. Apply a Fourier transform on the filtered signal and plot the result.

h. Compare the results of the two filters and give an explanation of the differences, both in the discrete-time domain, as well as in the discrete-frequency domain.

Problem 12.2. The perfect notch filter
A notch filter is a filter designed to allow only a very narrow range of frequencies and reject everything else. At the most extreme it has a transfer function in the continuous domain:

$$H(\omega) = \delta(\omega - \omega_0) + \delta(\omega + \omega_0) \tag{12.20}$$

Can you employ the procedure we developed to build the filter embodied by this equation? Explain your reasoning.

Problem 12.3. Frequency-sampling low-pass filter
We sample a signal $x(t)$ with $f_s = 250$Hz. We want to keep the frequencies at least up to $f = 26$Hz by applying a low pass filter with length $K = 19$.

a. Determine the spectral resolution of the filter $\Delta\Omega$ and give its specification.

b. Determine the filter coefficients of your filter.

c. Sketch the frequency response of this filter. Indicate at which points you specified the filter values.

d. Construct two signals such that $x_1[n] = \cos(\Omega_1 n)$ is expected to be unaltered and $x_2[n] = \cos(\Omega_2 n)$ to be attenuated. Pick $n = 0, \cdots, 127$ and apply the filter to demonstrate the validity of your choice.

Problem 12.4. FIR-filter design I
Design a low-pass FIR-filter with a kernel of 9 elements to filter a digital signal that is sampled at 1 kHz. The cut-off frequency of the filter is at 200 Hz. Give the kernel of the filter, *i.e.* a list of $K = 9$ elements.

Problem 12.5. Frequency-sampling high-pass filter

a. Design a highpass filter of length $K = 19$, having cutoff frequency $\Omega_c = \frac{4\pi}{19}$. Give the filter coefficients.

b. Sketch the frequency response of this filter. Indicate at which points you specified the filter values.

c. Construct two signals such that $x_1[n] = \cos(\Omega_1 n)$ is expected to be attenuated and $x_2[n] = \cos(\Omega_2 n)$ to be unaltered. Pick $n = 0, \cdots, 127$ and apply the filter to demonstrate the validity of your choice.

Problem 12.6. FIR-filter design II
Design a high-pass FIR-filter with a kernel of $K = 21$ taps (elements). The characteristics of the filter are:

- cut-off frequency is $f_c = 1.8$ kHz
- signal sampled at $f_s = 8$ kHz

Use the following steps in the design process:

(i) First relate the cut-off frequency to half the sample frequency and translate it to the discrete frequency domain: k_c or Ω_c.

(ii) Construct the filter coefficient $H[k]$ or $H(\Omega)$ in terms of $k \in [-(K-1)/2, (K-1)/2]$ or $\Omega \in [-\pi, \pi)$: with ones for the transfer section and zero's for the stop section.

(iii) Construct the filter kernel using the inverse DFT or other functions from MATHEMATICA, MATLAB or PYTHON.

(iv) Use convolution to apply the filter on a time-series consisting of 1000 random points with values between -1 and 1.

(v) Check your results by calculating the Fourier spectrum of the signal before application of the filter and after application of the filter.

Problem 12.7. Frequency-sampling band-pass filter

a. Design a bandpass filter of length $K = 19$ specified by

$$|H[3]| = \cdots = |H[7]| = 1,\ |H[12]| = \cdots = |H[16]| = 1,\ 0 \text{ otherwise}$$

b. Sketch the frequency response of this filter. Indicate at which points you specified the filter values.

c. Construct two signals such that both $x_1[n] = \cos(\Omega_1 n)$ and $x_2[n] = \cos(\Omega_2 n)$ are expected to be attenuated. Pick $n = 0, \cdots, 127$ and apply the filter to demonstrate the validity of your choice.

Problem 12.8. FIR-filter design III

Design a band-pass and a band-stop FIR-filter to respectively select and reject the center frequency from a time signal consisting of three harmonic signals. The data signal is given by:

$$x[n] = \sum_{i=1}^{3} \sin\left(\frac{2\pi f_i}{f_s} n\right) \text{ for } f_i = \{800, 1600, 2400\}$$

and is sampled at a frequency of $f_s = 8$ kHz.

a. Use the method of the previous problem, choosing your own cut-off frequencies and a kernel of sufficient length. Keep the filter efficient in terms of length.

b. Prove that you have designed the right filter, by showing the signals in both the time as well as in the frequency domain.

Problem 12.9. Playing with FIR filters

a. Add the filter coefficients of the low-pass and high-pass filter. What do you notice? Explain why this is a general property.

b. Suppose $h[n]$ are the filter coefficients of the lowpass filter. Calculate element wise

$$2\cos\left(\frac{2\pi}{19}\left(n - \frac{19-1}{2}\right)\right) h[n]$$

and verify that they are equal to the filter coefficients of the bandpass filter. Explain in words why this equality should hold.

Problem 12.10. Even and odd filter length

Suppose we sample a signal $x(t)$ which contains a hum around $f_h = 35$Hz with $f_s = 200$Hz. We want to remove the frequency band $[30\text{Hz}, 40\text{Hz}]$ completely, but we want keep the signal up to $f = 25Hz$ and beyond $f = 45Hz$.

a. Describe how an even length filter could be used for this.

b. Suppose you are stubborn and want to use an odd length filter instead. What is smallest odd filter length that can achieve this? Warning: this will not lead to the best filter!

c. Sample the synthetic signal $x(t) = \cos(50\pi t) + \cos(70\pi t) + \cos(100\pi t)$ for 10 seconds at a sampling frequency $f_s = 200$Hz resulting in $x[n]$. Filter this signal with the odd length filter of the previous question, resulting in $y[n]$. Calculate and plot $X[k]$ and $Y[k]$ and plot them. Explain why the hum has not been removed completely.

Problem 12.11. Rectangular versus Bartlett window

What is the advantage of a Bartlett window over a rectangular window?

Problem 12.12. Application of window I
Apply a window on the filter kernel in problem 12.6 and show the effect of the the application of the window on the signal, as well as on the frequency plot of the filter.

Problem 12.13. Application of window II
Apply a window on the filter kernel in problem 12.8 and show the effect of the the application of the window on the signal, as well as on the frequency plot of the filter.

Chapter 13

The z-transform

The z-transform is the digital equivalent of the continuous-time Laplace Transform. It is a powerful formalism for dealing with discrete control problems, and a necessary tool to build IIR filters.
Like the Laplace transform is used to transform continuous-time differential equations to algebraic equations, the z-transform is used to transform discrete-time difference equations into algebraic equations. Therefore we will see many parallels between the z-transform and the Laplace transform. Characteristics will be similar and the concepts are equal. The implementation, however will be different.
For a relation between continuous-time systems and discrete-time systems you are referred to chapter 10. Before we go to the definition of the z-transform it is useful to look to at closely related concept of the Discrete-time Fourier transform of a difference equation.

13.0.1 Difference equations and DFTs

The general form of a difference equation for a single input single output LTI system is:

$$a_0 y[n] + a_1 y[n-1] \dots a_p y[n-p] = b_0 x[n] + b_1 x[n-1] \dots b_q x[n-q] \tag{13.1}$$

Taking the DFT of both sides and applying the time shift property yields:

$$Y[k] \sum_{n=0}^{p} a_n \mathrm{e}^{-\frac{\mathrm{j}2\pi nk}{N}} = X[k] \sum_{n=0}^{q} b_n \mathrm{e}^{-\frac{\mathrm{j}2\pi nk}{N}} \tag{13.2}$$

This equations relates the frequency components from the input to those of the output. Like in the continuous-time domain, we can calculate the ratio to define the transfer function in the discrete-time domain:

$$H[k] \equiv \frac{Y[k]}{X[k]} = \frac{\sum_{n=0}^{q} b_n \mathrm{e}^{-\frac{\mathrm{j}2\pi nk}{N}}}{\sum_{n=0}^{p} a_n \mathrm{e}^{-\frac{\mathrm{j}2\pi nk}{N}}} \tag{13.3}$$

Here the function $H[k]$ in the discrete frequency domain is equivalent to the transfer function $H(s)$ in the continuous frequency domain. The function relates the output to the input signal and is independent of the input signal:

$$Y[k] = H[k]X[k] \tag{13.4}$$

In chapter 12 on FIR-filters similar relations were already derived from the DFT. Equation 13.1 was used where $a_n = 0 \; \forall \; n > 0$ only having $a_0 y[n]$ on the left hand side of the relation. This means that there is no recursion in the equation. The corresponding transfer function in equation 13.3 had only a numerator for FIR-filters.
Similar to the use of transfer functions in the continuous-frequency domain, we also get the output in the discrete-frequency domain by multiplying the input signal with the transfer function.

Note that the transfer function in the continuous-time domain was obtained via the Laplace transformation of the input-output differential equation. So let us first look at the equivalent of the Laplace transform: the z-transform. We will come back to the relation with the discrete Fourier transform later.

13.1 The z-Transform

For the definition of the z-Transform we use the most general, *i.e.* the bi-lateral z-Transform.

Definition 13.1.1. the z-Transform – bi-lateral
For a discrete-time sequence $x[n]$, the bi-lateral (two-sided) z-transform is defined as:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{13.5}$$

Here $z \in \mathbb{C}$ is a continuous complex number, that can also be written as $z = r\mathrm{e}^{-\mathrm{j}\Omega}$, with $\{r, \Omega\} \in \mathbb{R}$.

□

It might be a bit confusing that the frequency domain represented by the variable $z \in \mathbb{C}$ is continuous as a result from a transformation from a discrete time domain represented by the variable $n \in \mathbb{Z}$. Only in the relation to the discrete Fourier components of a signal or the discrete Fourier components of the impulse response of a system, we will relate the continuous z-variable to the discrete frequency components k. We will come back to this soon.
Apart from the bi-lateral z-Transform there is also a uni-lateral z-Transform, that differs in the summation of n over the positive domain only. We will address this here, since in practice we can obtain the same result by multiplying the function with the unit-step function prior to the transformation. Only in solving difference equations with initial conditions you might need to use the uni-lateral z-Transform.
Let us first look at two examples to get an impression of the result of a z-transformation of a discrete signal.

Example 13.1.1. Example 1
Consider the function $x[n] = a^n u[n]$, where $u[n]$ is the unit-step function. Its z-transform is:

$$\sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

Here we made use of the infinite sum

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1 - r} \ (|r| < 1)\,.$$

The z-transform thus only converges when $|a/z| < 1$. Therefore, $|z| > |a|$ is called the Region of Convergence (ROC) of the transform. Similar convergence criteria we also found in the s-domain for the Laplace transform.

□

This concludes this chapter on the z-Transform. In the next chapter we will use the (inverse) z-Transform in the design of IIR-filters.

Example 13.1.2. Example 2
Exploiting the result above, we can now apply the z-transform to a linear combination of two functions.

For a sum of two causal sequences,

$$x[n] = x_1[n] + x_2[n] = 2u[n] - 2\left(\frac{1}{2}\right)^n u[n],$$

we can use the result of example 13.1.1 in combination with the linear property of the z-transform (see later) to get the z-transform of each sequence and combine the z-transforms as

$$\begin{aligned} X(z) &= \mathcal{Z}\{x[n]\} = X_1(z) + X_2(z) \\ &= 2\mathcal{Z}\{u[n]\} - 2\mathcal{Z}\{(1/2)^n u[n]\} \\ &= 2\left.\frac{z}{z-a}\right|_{a=1} - 2\left.\frac{z}{z-a}\right|_{a=\frac{1}{2}} \\ &= \frac{2z}{z-1} - \frac{2z}{z-\frac{1}{2}} \\ &= \frac{2z}{(z-1)(2z-1)} \end{aligned} \tag{13.6}$$

The form of this expression is like what we have seen in the Laplace domain. Using properties of Laplace, we see that the representation of the z-transformed signal is determined by three complex numbers: one zero at $z = 0$ and two poles at $z = 1$ and $z = \frac{1}{2}$ respectively. Using more tools from Laplace, we can make a graphical representation of this function in the form of a pole-zero plot, as is presented in figure 13.1.

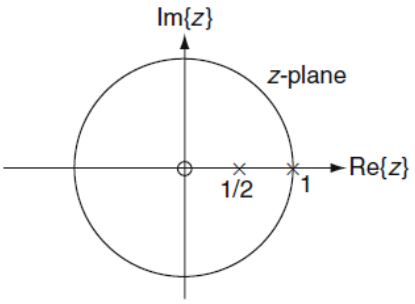


Figure 13.1: Pole-zero diagram of $X(z) = \frac{2z}{(z-1)(2z-1)}$

Note that in this diagram everything outside the unit circle is the ROC because the first term requires $|z| > |1|$. By definition the poles always exist outside the ROC since we know that $X[z]$ diverges at those locations. More generally, while we will not prove it here, the ROC always consists of the region outside the circle comprising the pole furthest from the origin if the sequence is causal ($x[n] = 0 \; \forall \; n < 0$).

□

Example 13.1.3. *Example 3*
Consider now a discrete harmonic signal starting at $n = 0$ given by:

$$x[n] = u[n]\cos\Omega_0 n = \frac{u[n]}{2}\left(\mathrm{e}^{\mathrm{j}\Omega_0 n} + \mathrm{e}^{-\mathrm{j}\Omega_0 n}\right)$$

where Ω_0 is a constant. The signal is graphically represented on the left side of figure 13.2.

Applying the result of example 13.1.1 we have:

$$\begin{aligned} X(z) &= \frac{1}{2}\left(\frac{z}{z-\mathrm{e}^{\mathrm{j}\Omega_0}} + \frac{z}{z-\mathrm{e}^{-\mathrm{j}\Omega_0}}\right) \\ &= \frac{z\,(z-\cos\Omega_0)}{(z-\mathrm{e}^{\mathrm{j}\Omega_0})\,(z-\mathrm{e}^{-\mathrm{j}\Omega_0})} \\ &= \frac{z\,(z-\cos\Omega_0)}{z^2-2z\cos\Omega_0+1}, \end{aligned} \tag{13.7}$$

which is represented graphically on the right side in figure 13.2.

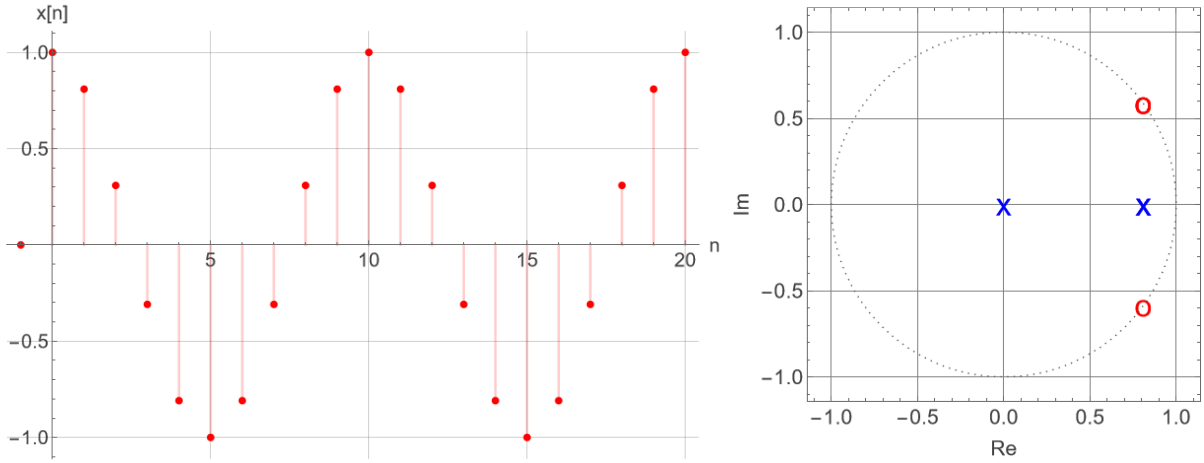


Figure 13.2: Time sequence representation of the signal with $\Omega_0 = 2\pi/10$ and corresponding pole-zero plot of the transfer function representation in the complex domain.

Notes:

- Recall that the constant Ω_0 is in units of the normalized frequency in the range $[-\pi, \pi)$. Looking at the initial function $x[n] = u[n]\cos\Omega_0 n$, we see that $\Omega_0 = (2\pi)/T_0$ is related to the period of the discrete harmonic wave T_0. In the example we have chosen for $T_0 \to 10\,\mathrm{s}$.
- Referring to the function in the z-domain, Ω_0 is the phase part of the poles that appear in complex-conjugated pairs:

  $$z_{1,2} = r\mathrm{e}^{\pm\mathrm{j}\Omega_0}$$

  being the constants of the signal.
- The two poles in the expression for $X[z]$ are complex conjugates of each other. This is referred to as a complex-conjugate pair. This is the result of a real-valued signal and similar as the characteristics we know from the Laplace domain.

. □

With this example we have now enough reference to take a closer look at the properties of z-Transforms.

13.1.1 Properties of the z-Transform

The z-transform has properties that are similar to those of the other transforms. Since we have proven these in detail for other transforms and the logic here is essentially the same, we do not repeat the full arguments here. Summarizing:

Linearity

$$\alpha x[n] + \beta y[n] \overset{\mathcal{Z}}{\leftrightarrow} \alpha X(z) + \beta Y(z) \tag{13.8}$$

Time delay

For a positive delay of the signal by $n_1 > 0$ time samples we have:

$$\begin{aligned} x[n-1] &\overset{\mathcal{Z}}{\longleftrightarrow} x[-1]z^0 + z^{-1}X(z) \\ x[n-2] &\overset{\mathcal{Z}}{\longleftrightarrow} x[-2]z^0 + z[-1]z^{-1} + z^{-2}X(z) \\ &\overset{\mathcal{Z}}{\ldots long \ldots} \\ x[n-n_1] &\overset{\mathcal{Z}}{long} \leftrightarrow z^{-n_1}\left(X(z) + \sum_{m=-n_1}^{-1} x[m]z^{-m}\right) \end{aligned} \tag{13.9}$$

Note that the sum starts at $n = 0$ due to the fact that $x[n]$ is specified for $n \in [infty, \infty]$ terms like $x[-1]$ appear and can be non-zero.

Convolution

The convolution is defined as:

$$y[n] = g[n] \otimes x[n] = \sum_{m=-\infty}^{\infty} g[m]x[n-m]$$

we have:

$$g[n] \otimes x[n] \overset{\mathcal{Z}}{\leftrightarrow} Y(z) = G(z)X(z) \tag{13.10}$$

Initial value theorem

$$x[0] = \lim_{z\to\infty} X(z) \tag{13.11}$$

Final value theorem

$$\lim_{n\to\infty} x[n] = \lim_{z\to 1}(z-1)X(z) = \lim_{z\to 1}(1-z^{-1})X(z) \tag{13.12}$$

13.2 The inverse z-transform

The formal definition for the inverse z-transform is:

Definition 13.2.1. inverse z-Transform
The inverse of the z-transform is defined as:

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} \equiv \frac{1}{2\pi \mathrm{j}} \oint_r X(z)z^{n-1}\mathrm{d}z \tag{13.13}$$

where r is the counter-clockwise contour in the ROC of $X(z)$ encircling the origin of the z-plane. □

Just as for the Laplace transform, applying the inverse z-transform requires mathematics from the field of complex analysis and we will not use this here. Instead we will use the method similar to the one used with the inverse Laplace Transform: if we know the transformation for the time-domain to the complex frequency-domain, we can also find the way back. In order to apply the inverse z-transform we might need to rewrite the function $X(z)$ in a suitable form and applying the table of z-transform pairs and the z-transform properties to get the signal in the discrete-time domain.

When dealing with LTI-systems this again mostly involves doing a partial fraction expansion. The only small new twist is that we get expressions of the form:

$$k_1 \frac{z}{z-a_1} + k_2 \frac{z}{z-a_2} + k_3 \frac{z}{z-a_3} \tag{13.14}$$

whereas before we had a recipe for expressions of the form:

$$k_1 \frac{1}{s-a_1} + k_2 \frac{1}{s-a_2} + k_3 \frac{1}{s-a_3}. \tag{13.15}$$

When looking at the table of z-transform pairs, most of them are rational expressions with a z in the numerator. In order to fulfil this form, it is recommended to determine the residues of $Y(z)/z$ and multiplying the resulting equation by z at the end. We illustrate this in the next example.

Example 13.2.1. inverse z-transform
Typical example of finding the inverse z-transform of a function like:

$$X(z) = \frac{z}{z^3 - 2z^2 + (5/4)z - 1/4} = \frac{z}{(z-1)(z-1/2)^2}$$

with ROC $r = \{z : |z| > 1\}$. We first divide the equation $X(z)$ by z and then expand it into partial fractions using the poles of the function $X(z)$:

$$\frac{X(z)}{z} = \frac{r_1}{z-1} + \frac{r_2}{z-1/2} + \frac{r_3}{(z-1/2)^2} = \frac{4}{z-1} - \frac{4}{z-1/2} - \frac{2}{(z-1/2)^2}$$

where the coefficient of each term can be found from the partial fraction method with a double pole:

$$\begin{aligned} r_1 &= (z-1)\frac{X(z)}{z}|_{z=1} = \frac{1}{(1-1/2)^2} = 4 \\ r_2 &= \frac{\mathrm{d}}{\mathrm{d}z}((z-1/2)^2 \frac{X(z)}{z})|_{z=1/2} = \frac{\mathrm{d}}{\mathrm{d}z}(\frac{1}{z-1})|_{z=1/2} = -\frac{1}{(z-1)^2}|_{z=1/2} = -4 \\ r_3 &= (z-1/2)^2 \frac{X(z)}{z}|_{z=1/2} = \frac{1}{z-1}|_{z=1/2} = -2 \end{aligned} \tag{13.16}$$

Now, multiplying both sides with z yields:

$$X(z) = 4\frac{z}{z-1} - 4\frac{z}{z-1/2} - 4\frac{(1/2)z}{(z-1/2)^2}$$

The back transform now gives:

$$x[n] = \left(4 - 4\left(\frac{1}{2}\right)^n - 4n\left(\frac{1}{2}\right)^n\right) u[n]$$

There is an alternative approach for computing z-transforms called "long division", however in most of the cases we will use tools from MATHEMATICA, PYTHON of MATLAB.

□

13.3 Solving LTI difference equations using the z-transform

Similar to the use of Laplace transforms in solving differential equations, z-transforms are very convenient in solving difference equations with initial value conditions. The strategy is best illustrated via an example.

Example 13.3.1. Solving a difference equation
Find the output $y[n]$ of the causal system whose input-output relationship is described by

$$y[n] - ay[n-1] = x[n-1],\ n \geq 0$$

where $y[-1] = y_0$ and

$$x[n] = b^n u[n] \text{ with } b \neq a.$$

To solve this difference equation for $y[n]$, we apply the time shifting property to write its z-transform as:

$$\begin{aligned} Y(z) - a(z^{-1}Y(z) + y[-1]) &= z^{-1}X(z) + x[-1] \\ (1 - az^{-1})Y(z) &= ay_0 + z^{-1}X(z) \end{aligned} \tag{13.17}$$

since $x_1[-1] = 0$. We can solve this algebraic equation for $Y(z)$ as

$$Y_1(z) = \frac{1}{1 - az^{-1}}\left(ay_0 + z^{-1}X(z)\right) = \frac{z}{z-a}\left(ay_0 + z^{-1}\frac{z}{z-b}\right)$$

To take the inverse z-transform of this expression, we divide both sides by z and perform the partial-fraction decomposition as

$$\begin{aligned} \frac{Y(z)}{z} &= \frac{ay_0}{z-a} + \frac{1}{(z-a)(z-b)} = \frac{ay_0}{z-a} + \frac{1}{a-b}\left(\frac{1}{z-a} - \frac{1}{z-b}\right) \\ Y_1(z) &= \frac{ay_0 z}{z-a} + \frac{1}{a-b}\left(\frac{z}{z-a} - \frac{z}{z-b}\right) \end{aligned} \tag{13.18}$$

Finally we take the inverse z-transform to get the output as

$$\begin{aligned} y[n] &= ay_0 a^n u[n] + \frac{1}{a-b}(a^n - b^n)u[n] \\ &= y_0 a^{n+1}u[n+1] + \frac{a^n - b^n}{a-b}u[n] \end{aligned} \tag{13.19}$$

In the last step we replaced $u[n]$ by $u[n+1]$ to express the existence of the given initial condition $y_{-1} = y_0$.

□

13.4 Relation between the z-Transform and the DFT

Assume $x[n]$ is a sample with length N of a signal. Then the Discrete Fourier Transform (DFT) is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n]\mathrm{e}^{-\mathrm{j}\frac{2\pi k}{N}n}$$

where $2\pi k/N$ are the discrete frequencies.
In order to apply a z-transform, we have to extend the signal to the complete domain of $n \in \mathbb{Z}$. This can be achieved by pre-fixing and suffixing $x[n]$ with zero's, *e.g.* by means of multiplying the signal with the unit-step function: $x[n] \to x[n]u[n]$ and limiting the range of the summation. Applying the z-transform then becomes:

$$X(z) = \sum_{n=\infty}^{\infty} x[n]u[n]z^{-n} = \sum_{n=0}^{N-1} x[n]z^{-n} = X(\mathrm{e}^{\mathrm{j}\frac{2\pi}{N}k} = X[k] \tag{13.20}$$

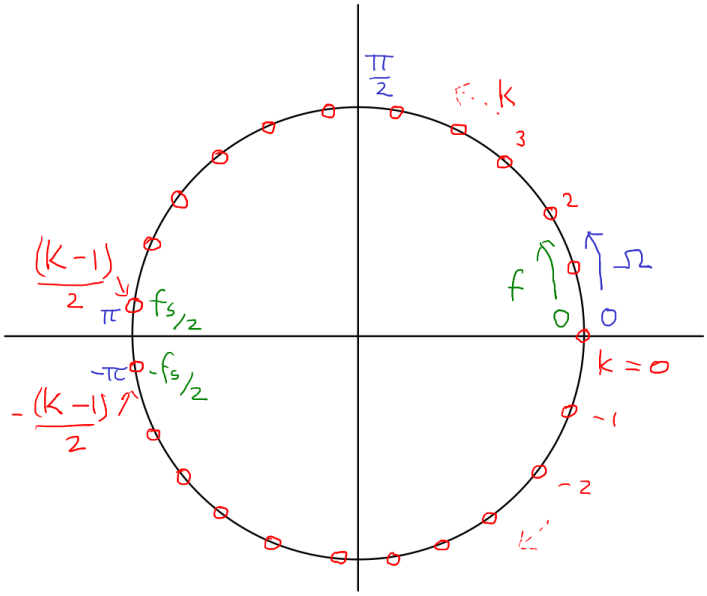


Figure 13.3: The points in z-space where the z-Transform and the Discrete Fourier Transform are equivalent. Here the frequency along the unit circle is $\Omega \in [-\pi, \pi)$ is also presented in discrete points $\Omega_k = (2\pi k)/N$.

Note that here we go from a continuous complex variable z to a integer frequency number k. For clarity:

$$z \overset{polar}{\longleftrightarrow} re^{j\Omega} \overset{r=1}{\longleftrightarrow} e^{j\Omega} \overset{\Omega->\Omega_k}{\longleftrightarrow} e^{j\Omega_k} \overset{\Omega_k=(2\pi k)/N}{\longleftrightarrow} e^{j\frac{2\pi k}{N}} \overset{relevantvariable}{\longleftrightarrow} k$$

The result of this is a finite set of k-values of length $K = N$ on the unit-circle as frequency components Ω_k in the complex plane. This is graphically illustrated in figure 13.3.
This is identical to the definition of the z-transform evaluated at $z = e^{j\Omega_k}$ if we take into consideration that values $n \geq N$ are effectively zero in $x[n]$.
In this sense the DFT can be seen as being embedded in the z-transform. This will prove useful in the design of IIR-filters.

13.5 Solve difference equations with the uni-lateral z-Transform

In this section we shortly address the uni-lateral z-Transform to solve a difference equation with initial conditions.

Definition 13.5.1. the uni-lateral z-Transform
For a discrete-time sequence $x[n]$, the uni-lateral (one-sided) z-transform is defined as:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n] z^{-n} \tag{13.21}$$

Here $z \in \mathbb{C}$ is a continuous complex number, that can also be written as $z = re^{-j\Omega}$, with $\{r, \Omega\} \in \mathbb{R}$.

□

We present the use of the uni-lateral z-Transform in solving difference equations with initial conditions by means of a single example.

Example 13.5.1. Solve a difference equations with initial conditions Consider the difference equations given by:

$$y[n] - \frac{1}{2}y[n-1] - \frac{1}{6}y[n-2] = 0$$

with the initial conditions: $y[-2] = 0$ and $y[-1] = 1$.
Applying the z-Transform to this equation gives:

$$Y(z) - \frac{1}{2}\left(y[-1] + z^{-1}Y(z)\right) - \frac{1}{6}\left(y[-2] + y[-1]z^{-1} + z^{-2}Y(z)\right) = 0$$

Plugging in the initial conditions and collecting terms with $Y(z)$ we obtain:

$$\left(1 - \frac{1}{2}z^{-1} - \frac{1}{6}z^{-2}\right)Y(z) = \frac{1}{2} + \frac{1}{6}z^{-1}$$

from which we get the solution in the z-domain:

$$Y(z) = \frac{3z^2 + z}{6z^2 - 3z - 1}$$

Taking the inverse z-transform gives us the solution to the difference equations with the initial conditions:

$$y[n] = \frac{1}{\sqrt{11}}\left(3^{-\frac{n+2}{2}}4^{-(n+1)}\left((-7+\sqrt{33})(3-\sqrt{33})^n + (7+\sqrt{33})(3+\sqrt{33})^n\right)\right)$$

which is a nasty expression, even if you look at the numerical values, however if we plot the results we get a much better impression of the result as is done in figure 13.4.

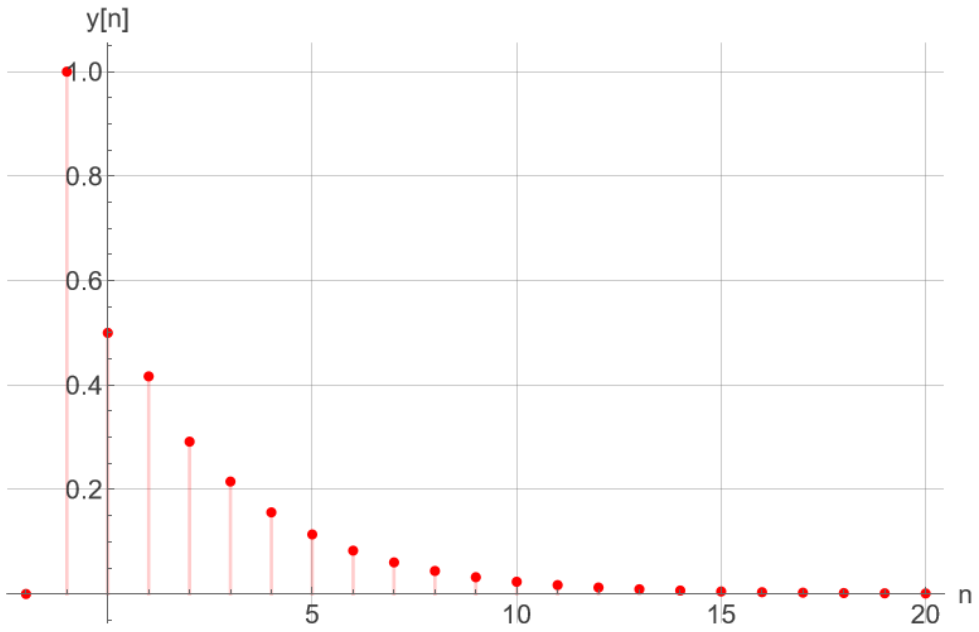


Figure 13.4: Solution of the difference equation with given initial conditions from example 13.5.1

For illustrative purposes, the initial conditions are also plotted for negative values of n.
In this example we did not use the analytical method of long division to write the transfer function in a suitable form and subsequently using the table to calculate the resulting solution. In general this is a tedious job, so instead we used available computational mathematical tools to calculate the result.

□

Summaries

Summary Chapter 13a

Key concepts on z-Transform and its characteristics:

- Definition of the z-transform:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n] z^{-n}$$

- Definition of the inverse z-transform:

$$\begin{aligned}
X(z) &= \frac{Q(z)}{P(z)} = \frac{b_m z^m + \ldots b_1 z + b_0}{a_n z^n + \ldots a_1 z + a_0} \qquad (m < n) \\
&= \left(\sum_{i=1}^{n-l} \frac{r_i}{z - p_i} \right) + \frac{r_{n-l+1}}{z - p_0} + \ldots + \frac{r_n}{(z - p_0)^l} + K \text{ (for } l \text{ times } p_0) \\
r_i &= (z - p_i) \left. \frac{Q(z)}{P(z)} \right|_{z = p_i \neq p_0}, \; i = 1, 2, \ldots, n - l \\
r_{n-k} &= \frac{1}{k!} \frac{\mathrm{d}^k}{\mathrm{d}z^k} \left\{ (z - p_0)^l \frac{Q(z)}{P(z)} \right\} \Bigg|_{z = p_0}, \; k = 0, 1, 2, \ldots, l - 1 \\
x[n] &= \left\{ \sum_{i=1}^{n-l} r_i \mathrm{e}^{p_i n} + r_{n-l+1} \mathrm{e}^{p_0 n} + r_{n-l+2} n \mathrm{e}^{p_0 n} + \ldots + \frac{r_n}{(l-1)!} r^{l-1} \mathrm{e}^{p_0 n} \right\} u_s[n]
\end{aligned}$$

Summary Chapter 13b

z-Transform Properties:

#	Property	$x[n] = \mathcal{Z}^{-1}\{X(z)\}$	$X(z) = \mathcal{Z}\{x[n]\}$
1	Linearity	$\alpha x_1[n] + \beta x_2[n]$	$\alpha X_1(z) + \beta X_2(z)$
2	Differentiation	$x[n] - x[n-1]$	$\left(1 - z^{-1}\right) X(z) = \frac{z-1}{z} X(z)$
3		$-nx[n]$	$z\frac{\mathrm{d}}{\mathrm{d}z} X(z)$
4	Time shift	$x[n - n_0]$	$z^{-n_0} X(z) + z^{-n_0+1} x[-1] + \ldots + x[-n_0]$
5	Frequency shift	$z_1^n x[n]$	$X\left(\frac{z}{z_1}\right)$
6	Initial value T	$x[0] = \lim_{z\to\infty} X(z)$	
7	Final value T	$x[\infty] = \lim_{z\to 1} \frac{z-1}{z} X(z)$	
8	Convolution	$x[n] \otimes y[n]$	$X(z)Y(z)$
9		$x[n]y[n]$	$\frac{1}{2\pi\mathrm{j}} \oint_c X\left(\frac{z}{\nu}\right) Y[\nu]\mathrm{d}\nu$

Summary Chapter 13c

z-Transform pairs:

$x[n] = \mathcal{Z}^{-1}\{X(z)\}$	$X(z) = \mathcal{Z}\{x[n]\}$
$\delta[n]$	1
$\delta[n - n_0]$	z^{-n_0}
$u[n]$	$\frac{z}{z-1}$
$a^n\, u[n]$	$\frac{z}{z-a}$
$n\, a^n\, u[n]$	$\frac{az}{(z-a)^2}$
$\cos[\Omega_0 n]\, u[n]$	$\frac{z(z - \cos(\Omega_0)}{z^2 - 2z\cos(\Omega_0) + 1}$
$\sin[\Omega_0 n]\, u[n]$	$\frac{z\sin(\Omega_0)}{z^2 - 2z\cos(\Omega_0) + 1}$
$\mathrm{e}^{-an}\cos[\Omega_0 n]\, u[n]$	$\frac{z\left(z - \mathrm{e}^{-a}\cos(\Omega_0)\right)}{\left(z - \mathrm{e}^{-a}\cos(\Omega_0)\right)^2 + \mathrm{e}^{-2a}\sin^2(\Omega_0)}$
$\mathrm{e}^{-an}\sin[\Omega_0 n]\, u[n]$	$\frac{z\mathrm{e}^{-a}\sin(\Omega_0)}{\left(z - \mathrm{e}^{-a}\cos(\Omega_0)\right)^2 + \mathrm{e}^{-2a}\sin^2(\Omega_0)}$

Problems

Problem 13.1. Exponential growth
During the lecture it was argued that $f[n] = (1/2)^n$ is an exponential function in just the same way as $f[n] = \mathrm{e}^{kn}$. What is the equivalent value of k in this case?

Problem 13.2. z-transforms
Compute the z-transforms of the following functions:

a. $x[n] = u[n]\sin(2\pi f n)$

b. $x[n] = u[n]na^n$

Problem 13.3. Playing with properties Consider the sequence $x[n] = 1$ for $n = 0, \cdots, 3$, $x[n] = 0$ otherwise.

a. Find $X[z]$.

b. Define $y[n] = x[n-2]$. Find $Y[z]$ from (1) $y[n]$ and from (2) $X(z)$ and verify matching answers.

c. Calculate $z[n] = x[n] \otimes y[n]$.

d. Find $Z[z]$ from (1) $z[n]$ and from (2) $X(z)$ and $Y(z)$ and verify matching answers.

e. Given $X[z]$, how can you retrieve $x[n]$? Hint: use your eyes and do not use your pen.

f. Suppose we want to calculate the DFT $X[k]$ of $x[n]$. Show how you can use $X[z]$.

Problem 13.4. More properties Consider the sequence $x[n] = (-1)^n$ for $n = 0, \cdots, 5$, $x[n] = 0$ otherwise.

a. Write $X[z]$ as a sum using $\sum$.

b. Define $y[n] = (\frac{1}{2})^n x[n]$ for $n = 0, \cdots, 5$, $y[n] = 0$ otherwise. Find $Y[z]$ from (1) $y[n]$ and from (2) $X[z]$ and verify matching answers.

c. Calculate $z[n] = -nx[n]$.

d. Find $Z[z]$ from (1) $z[n]$ and from (2) $X[z]$ and verify matching answers.

Problem 13.5. Inverse z-transforms
Find the inverse z-transform of:

a. $H[z] = \dfrac{z}{(z+1)(z-1)}$

b. $H[z] = \dfrac{z^3 + 2z^2 + 4z}{(z+1)(z-1)^2}$.

Problem 13.6. Inverse z-transform
Compute the inverse z-transforms of the following functions:

a. $X[z] = 1/(z-3)$

b. $X[z] = 3z/(z^2 - z/4 - 1/8)$

Problem 13.7. Inverse z-transform by long division
Consider again the expression

$$X[z] = \frac{z}{z-3}. \tag{13.22}$$

This can be written as

$$\begin{aligned} X(z) &= \frac{z}{z-3} \qquad (13.23)\\ &= \frac{z-3}{z-3} + \frac{3}{z-3} \\ &= 1 + z^{-1}\frac{3z}{z-3} \\ &= 1 + 3z^{-1}\frac{z}{z-3} \\ &= 1 + 3z^{-1}\left(\frac{z-3}{z-3} + \frac{3}{z-3}\right) \\ &= 1 + 3z^{-1} + 3z^{-1}\frac{3}{z-3} \\ &= \ldots \qquad (13.24) \end{aligned}$$

Compute the time-domain values of $x[n]$ for $n = 0, 1, 2, 3$ using this strategy. This approach can be programmed to compute $x[n]$ given a polynomial expression for $X(z)$.

Problem 13.8. A difference equations
Given is the system

$$y[n] - \frac{1}{2}y[n-1] = 3x[n] + x[n-1]$$

Suppose $y[-1] = 2$, $x[n] = 2u[n]$, find $y[n]$ for $n \geq 0$ using the z transform.

Problem 13.9. Another difference equation
Given is the system

$$y[n] + \frac{4}{3}y[n-1] + \frac{1}{3}y[n-2] = x[n] - x[n-1]$$

Suppose $y[-2] = -1,\ y[-1] = 1$, $x[n] = (-1)^n u[n]$, find $y[n]$ for $n \geq 0$ using the z-transform.

Problem 13.10. Solving a difference equation
Consider a causal system whose input-output relationship is described by

$$y[n] - 2y[n-1] = 3x[n] \tag{13.25}$$

with $y[-1] = 1$ and $x[n] = \delta_{n,0}$ (that is, a unit impulse).

a. Show that the z-transform of $x[n]$ is $X[z] = 1$.

b. Solve for $y[n]$.

c. Check your answer by computing the first few time steps by iteration in the difference equation.

Problem 13.11. Creating your very own difference equation
Consider a mass suspended by a Hookian spring. The forces acting on the mass are gravity, and the restoring force of the spring, according to Newton's second law:

$$m\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} = -Kx(t) - mg. \tag{13.26}$$

Here m is the mass of the particle, K is the spring constant of the spring and $g = 9.8$ m/s^2 is the gravitational acceleration at the earth's surface. The mass is initially held in place at position $x = x_0$ and is released at $t = 0$, from then on obeying the equation of motion.

a. Can this be thought of as a LTI system? Explain your reasoning.

b. Write down a difference equation for $x[n]$, a discrete time representation of the same system when the position $x(t)$ is sampled at intervals Δt.

c. Solve for $X[z]$. We could invert this function to get $x[n]$ but it is a bit messy so we will skip this part for now.

Problem 13.12. An obscure table entry
A table entry is

$$\cos(\Omega_0 n)u[n] \leftrightarrow \frac{z(z-\cos(\Omega_0))}{z^2 - 2z\cos(\Omega_0) + 1}$$

Use $a^n u[n] \leftrightarrow \frac{z}{z-a}$ to prove this.

Chapter 14

IIR Filters

Add a special section devoted to bi-quad filters. Those play an important role in digital filter use. Two means reasons:

- frequency band filtering can be designed more or less independently.
- a sequence of bi-quad filters is more stable than a single higher-order filter.

14.1 Introduction

Where FIR filters are described by difference equations that are only dependent on a finite number of input samples, IIR filters are also dependent on previous output values. The latter makes recursive definitions, and therefore infinite responses on finite inputs possible. A IIR filter is described by the full difference equation given by:

$$y[n] + \sum_{j=1}^{M} a_i y\left[n-j\right] \; = \sum_{i=0}^{N} b_i x\left[n-i\right] \tag{14.1}$$

The discrete transfer function of an IIR filter thus has both a numerator and a denominator in contrast to a FIR filter that has only a numerator:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \ldots + b_N z^{-N}}{1 + a_1 z^{-1} + \ldots + a_M z^{-M}} \tag{14.2}$$

This transfer function can be rewritten as a multiplication of the roots of the polynomial equation that constitute the numerator and a denominator.

$$H(z) = \frac{Y(z)}{X(z)} = b_0 \frac{(z-r_1)(z-r_2)etc.}{(z-p_1)(z-p_2)etc} = b_0 \frac{\prod_{i=1}^{N}(z-r_i).}{\prod_{j=1}^{M}(z-p_j).} \tag{14.3}$$

In this the symbol Π is used to indicate a multiplication of terms. The effect of the various zero's r_i and poles p_j on an input signal is very different. This is best understood by replacing z with the discrete frequency it stands for, *i.e.* $z = \exp(\mathrm{j}\Omega)$, with Ω the normalized frequency on the interval $[-\pi, \pi)$, representing the frequency range between $-f_s/2$ and $f_s/2$, where f_s is the sample frequency. In the left part of figure 14.1 two complex zero's (a so called conjugate pair) have been placed on the unit circle at an angle θ (which is in fact the normalised frequency Ω_0 and related to the actual frequency f by $\Omega_0 = 2\pi f_0/f_s$) From here onwards, we will denote the phase of the pole by Ω_0, so $\theta \rightarrow \Omega_0$ to relate the phase directly to the normalised frequency. When the frequency Ω becomes equal to the phase location of the zero, the value of the numerator becomes zero and hence the amplitude of the transfer function goes to zero. The frequency component at this frequency in the signal is thus damped to zero when applying this transfer function.

The opposite occurs with a pole. In the right part of figure 14.1 a conjugate pole pair is again placed at an angle $\theta = \Omega_0$, but now at a radius r that has a value smaller than 1. When the normalized signal frequency Ω approaches Ω_0 (the signal follows the unit circle). The response of the denominator in the transfer function reduces to (1-r). This implies that the signal at frequency Ω_0 is amplified by $1/(1\text{-r})$. This results in an enhancement of the signal at frequencies close to poles.

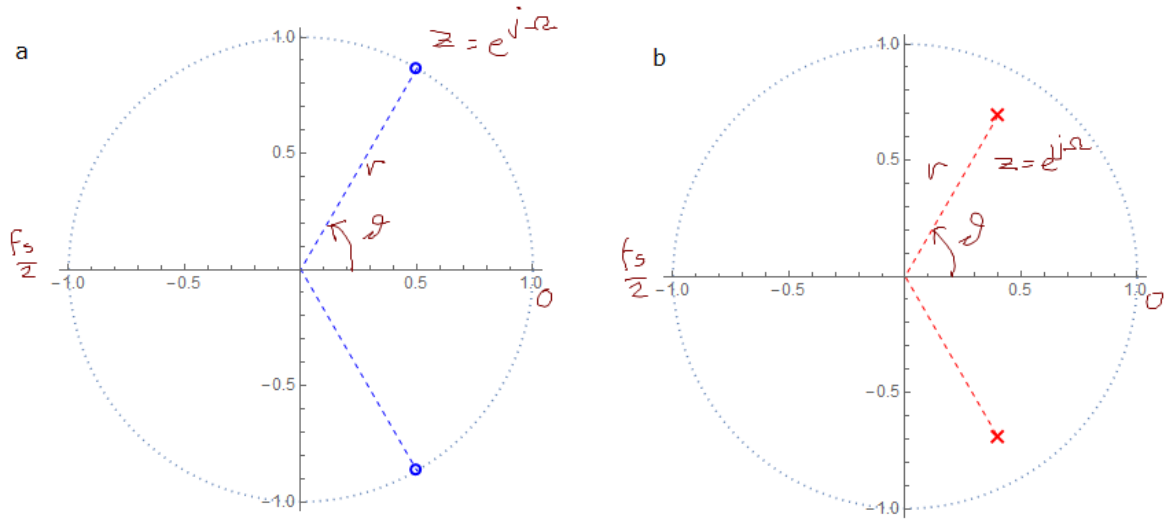


Figure 14.1: The effect of a complex poles and complex zero's.

The action of a pole and zero are thus completely opposite, a pole at an angle Ω_0 enhances the frequency component associated with this angle, whereas a zero at Ω_0 reduces the frequency component to zero. Starting from specifications of a filter, the placement of zero's and poles at specific positions lead by frequency regions that have to be damped or enhanced, a transfer function of an appropriate filter can be constructed. This is a powerful design process that is quite intuitive. Since the zero's and poles are the solutions of polynomials with real coefficients, the zero's and poles appear in conjugate pairs. Only real zero's and poles can appear singly. You should keep that in mind when designing filters, otherwise you will end up with an unphysical filter.

14.2 IIR filter design through pole zero placement

This technique is illustrated with a few basic first and second order filters: low-pass, high-pass, band-pass and band-reject filters. These filters will give you enough basis to feed your intuition to be able to design more complex filters by adding zero's and poles to fulfil more stringent design criteria.

14.2.1 Low-pass filter

In a low pass filter a zero is placed at (-1,0). This attenuates the high frequency side of the spectrum and in the limit of $f_s/2$ (Nyquist limit) the response is zero. With the position of a pole on the positive real axis enhances low frequency part.

$$H(z) = \frac{K(z+1)}{(z-\alpha)}, \; with \; K = \frac{1-\alpha}{2} \tag{14.4}$$

The position of the pole determines the cut-off frequency. In the lectures and exercises, we show that the cutoff (digital) frequency is $\Omega_c \approx d$.

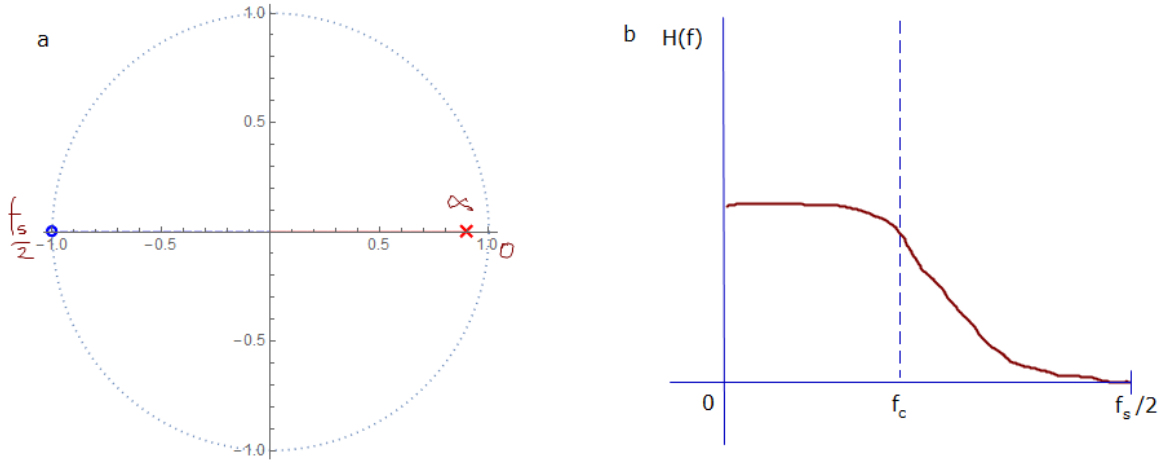


Figure 14.2: A low pass IIR filter.

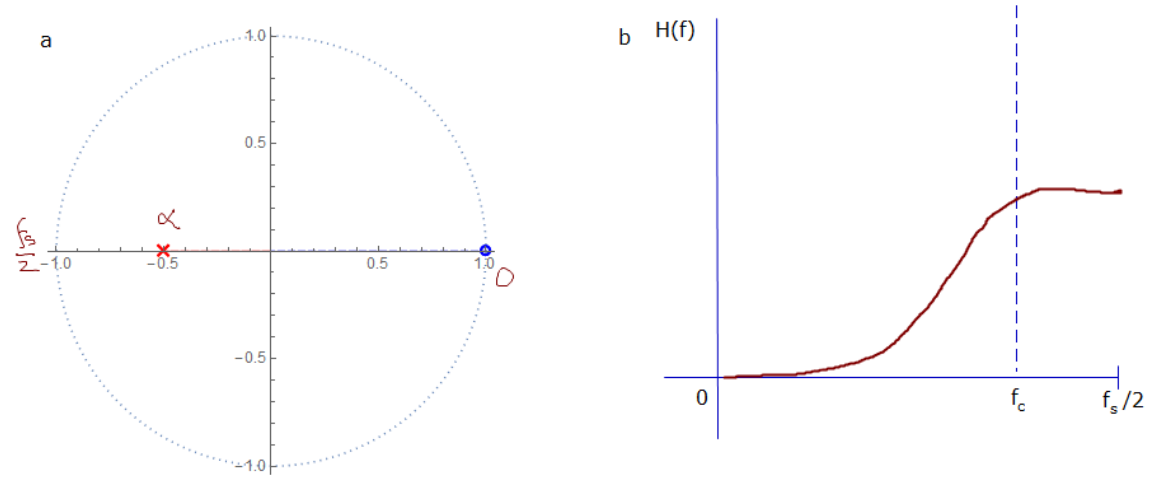


Figure 14.3: A high pass IIR filter.

14.2.2 High-pass filter

In a high pass filter the opposite of the low pass filter is done, *i.e.* a zero at (1,0) and still a pole on the real axis.

$$H(z) = \frac{K(z-1)}{(z-\alpha)},\ with\ K = \frac{1+\alpha}{2} \tag{14.5}$$

14.2.3 Band-pass filter

The limited frequency range required for a bandpass filter is obtained by putting zero's at both the low and high frequency end. An enhanced pass at the desired frequency f_0 is obtained by placing a conjugate pole pair at an angle $\Omega_0 = 2\pi f_0/f_s$. The band width (Γ) of the filter is determined by the radius r of the poles.

$$H(z) = \frac{K(z-1)(z+1)}{(z-re^{j\Omega_0})(z-re^{-j\Omega_0})} = \frac{K(z^2-1)}{z^2-2rzcos(\Omega_0)+r^2},\ with\ K = \frac{(1-r)\sqrt{1-2rcos(2\Omega_0)+r^2}}{2\,|\sin(\Omega_0)|} \tag{14.6}$$

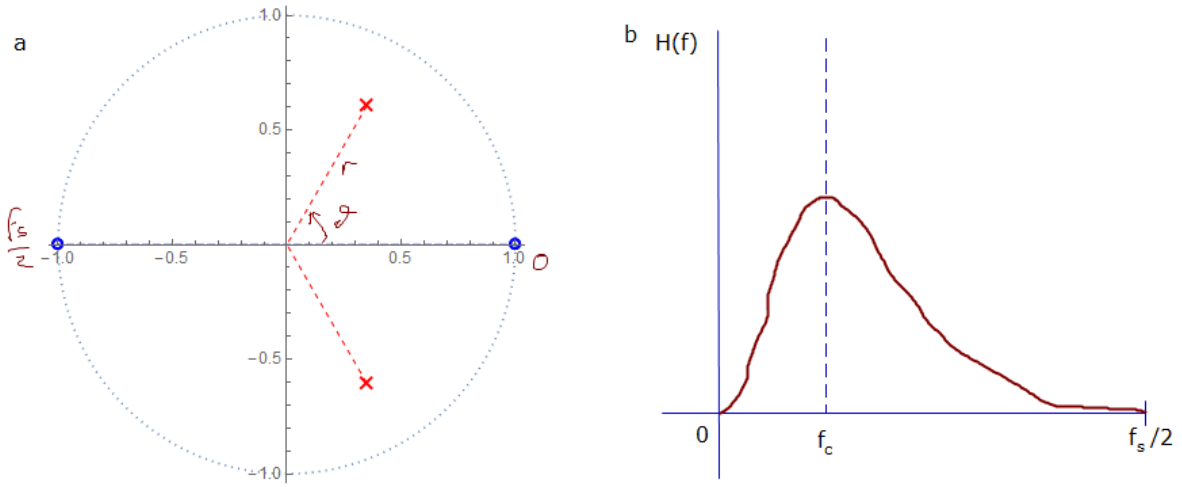


Figure 14.4: A band pass IIR filter.

14.2.4 Band-reject or Notch filter

In a notch filter a certain frequency range is strongly attenuated. This is directly obtained by a conjugated zero pair on the unit circle at the required frequency, $\Omega_0 = 2\pi f_0/f_s$. To make sure that other frequencies are not strongly affected, a conjugated pole pair is placed just in front. The radial distance to the zero's determines the bandwith $Gamma$ of this filter.

$$H(z) = \frac{K(z-e^{j\Omega_0})(z-e^{-j\Omega_0})}{(z-re^{j\Omega_0})(z-re^{-j\Omega_0})} = \frac{K(z^2-2zcos(\Omega_0)+1)}{z^2-2rzcos(\Omega_0)+r^2},\ with\ K = \frac{1-2rcos(2\Omega_0)+r^2}{1-2cos(\Omega_0)} \tag{14.7}$$

14.2.5 Placement of additional poles and zeroes

Knowing the effect of the position of zero's and poles in a basic setting, filter can be made more sophisticated by adding extra poles or zero's. With the addition of additional pole's and zero's the

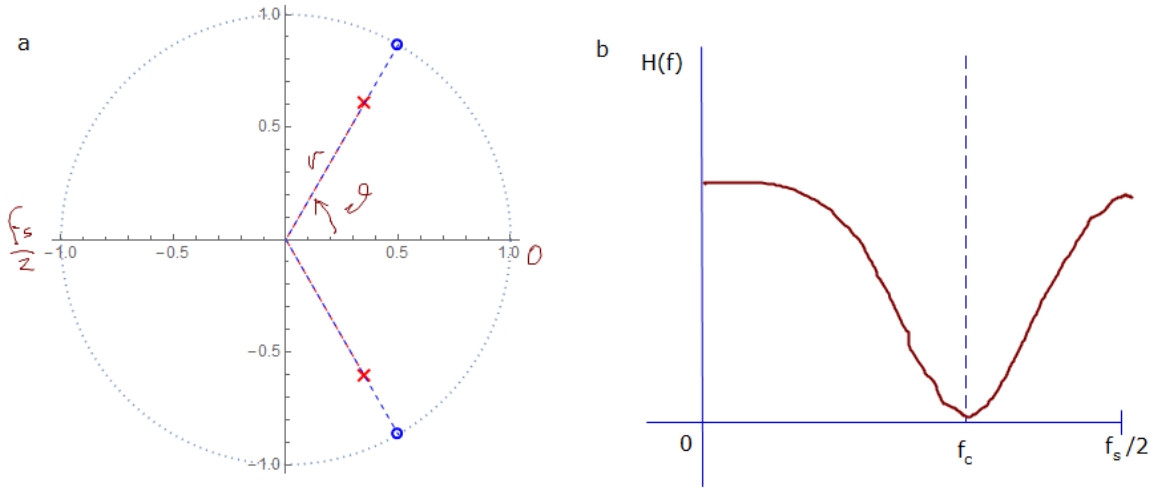


Figure 14.5: A notch IIR filter.

steepness of the filter can be strongly enhanced. However, this comes at a price: the phase behaviour in a region with many poles and/or zero's becomes quite wild: large phase changes and highly non-linear phase response. You should keep this in mind when designing filters. For many applications wild phase behaviour is not an issue, however for some applications phase information is essential (*e.g.* audio applications). Advantages of IIR filters is there compactness: with only a few coefficients you can already create a sophisticated filter. In real-time applications they have an advantage over FIR filters by their much shorter delay times.

The design of filters is always an optimization process. Solely the effect that you have counter directing requirements, means that there is an optimum.

The better you know your signal and the better you know what you want, the better you can make choices in designing your filter.

14.3 Bilinear Transformation

14.3.1 Introduction

Analogue filters such as a Butterworth or a Chebyshev filter have very pleasant frequency characteristics. These characteristics can also be obtained for digital frequencies by transforming the frequency response of these filters from the analogue to the digital domain. In order to do so, first the precise link between analogue and digital frequencies has to be established. This may seem strange at first glance, but it implies that the 0 to ∞ frequency range in the analogue world has to be mapped on the limited digital frequency range of 0 to $f_s/2$, with f_s the sample frequency.

14.3.2 Frequency mapping

The Bilinear transformation is used to map the continuous frequency domain (s) on the discrete frequency domain (z). These frequency domains are derived from the continuous time domain via the Laplace transform and from the discrete time domain via the z-transform. Note that the corresponding basis of the continuous time domain are sets of differential equations, whereas for the discrete time domain sets of difference equations. The relation between the complex frequencies of the continuous and discrete frequency domain can be obtained by working out the form for integration in both domains. In the analogue domain,

$$y(t) = \int_0^t x(t')\mathrm{d}t' \tag{14.8}$$

$$Y(s) = \frac{X(s)}{s} \tag{14.9}$$

$$G(s) = \frac{Y(s)}{X(s)} = \frac{1}{s} \tag{14.10}$$

The corresponding operation in the digital domain with T_s the sampling period that relates the continuous time t to the sampling number n is:

$$t = nT_s \tag{14.11}$$

$$y[n] = y[n-1] + \frac{x[n] + x[n-1]}{2}T_s \tag{14.12}$$

$$Y(z) = z^{-1}Y(z) + \frac{T_s}{2}(X(z) + z^{-1}X(z)) \tag{14.13}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{T_s}{2}\frac{1+z^{-1}}{1-z^{-1}} \tag{14.14}$$

Now the analogue and digital world can be glued together leading to:

$$s = \frac{2}{T_s}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = \frac{2}{T_s}\left(\frac{z-1}{z+1}\right) \tag{14.15}$$

or

$$z = \frac{1+sT_s/2}{1-sT_s/2} = \frac{2f_s+s}{2f_s-s} \tag{14.16}$$

where as usual the sampling period T_s is directly related to the sampling frequency f_s by $f_s = 1/T_s$. Using the definitions for the continuous and the discrete frequency domain respectively:

$$s \equiv \sigma + \mathrm{j}\omega_a \tag{14.17}$$

$$z \equiv r\mathrm{e}^{\mathrm{j}\Omega} \tag{14.18}$$

where σ is the exponential decay factor and ω_a the frequency in the continuous domain on the domain:

$$\omega_a \in [-\infty, \infty] \tag{14.19}$$

and r is the magnitude in the discrete domain, and Ω is the normalized frequency in the discrete domain:

$$\Omega \in [-\pi, \pi) \tag{14.20}$$

The normalized discrete frequency can be converted to the absolute discrete frequency ω_d:

$$\omega_d = 2\Omega\frac{f_s}{2} \tag{14.21}$$

$$= \frac{\Omega}{T_s} \tag{14.22}$$

which is in the domain:

$$\frac{\omega_d}{2\pi} = f_d \in \left[-\frac{f_s}{2}, \frac{f_s}{2}\right) \tag{14.23}$$

where f_s is the sampling frequency.
Substituting the definition of the discrete frequency z in the definition of the bilinear transformation we obtain:

$$s = \frac{2}{T_s}\left(\frac{re^{j\Omega} - 1}{re^{j\Omega} + 1}\right) \tag{14.24}$$

$$= \frac{2}{T_s}\left(\frac{r^2 - 1}{1 + r^2 + 2r\cos\Omega} + \frac{2r\sin\Omega}{1 + r^2 + 2r\cos\Omega}\mathrm{i}\right) \tag{14.25}$$

The first term of this expression must be equal to σ and the second term to ω_a:

$$\sigma = \frac{2}{T_s}\frac{r^2 - 1}{1 + r^2 + 2r\cos\Omega} \tag{14.26}$$

$$\omega_a = \frac{2}{T_s}\frac{2r\sin\Omega}{1 + r^2 + 2r\cos\Omega} \tag{14.27}$$

$$\omega \in [-\infty, \infty] \Leftrightarrow \Omega \in [-\pi, \pi) \tag{14.28}$$

where the frequency Ω is normalized to half the sampling frequency.
From these relations it can be concluded that the left half plane of the complex continuous frequencies is mapped on the inner region of the unit circle in the complex plane of the discrete frequencies, see fig.14.6, since:

$$\forall\, r < 1 \Rightarrow \sigma < 0 \tag{14.29}$$

$$\forall\, r > 1 \Rightarrow \sigma > 0 \tag{14.30}$$

The imaginary axis of the complex plain for the continuous frequencies maps on the unit circle, relating the frequencies in the two domains:

$$r = 1 \Rightarrow \sigma = 0 \tag{14.31}$$

$$\Rightarrow \omega = \frac{2}{T_s}\frac{2\sin\Omega}{2 + 2\cos\Omega} = \frac{2}{T_s}\tan\frac{\Omega}{2} \tag{14.32}$$

$$\Rightarrow \Omega = 2\arctan\frac{\omega_a T_s}{2} \tag{14.33}$$

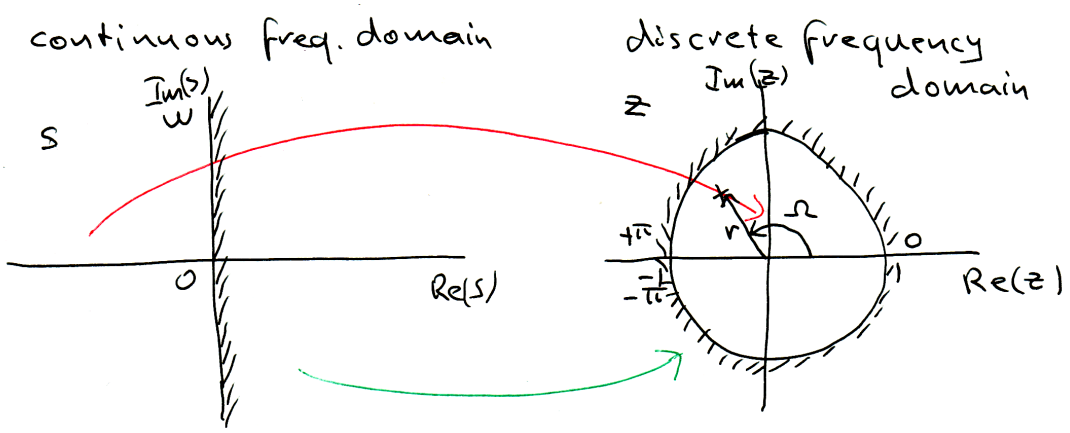


Figure 14.6: Mapping of frequency domain continuous and discrete.

Using the absolute discrete frequency ω_d instead of the normalized frequency Ω these equations become:

$$\omega_a = \frac{2}{T_s} \tan \frac{\omega_d T_s}{2} \tag{14.34}$$

$$\omega_d = \frac{2}{T_s} \arctan \frac{\omega_a T_s}{2} \tag{14.35}$$

This relation is graphically depicted in fig. 14.7 and shows that around a one to one map of the both worlds exist around zero frequency. However high frequencies in the analogue domain are compressed asymptotically to the Nyquist frequency.

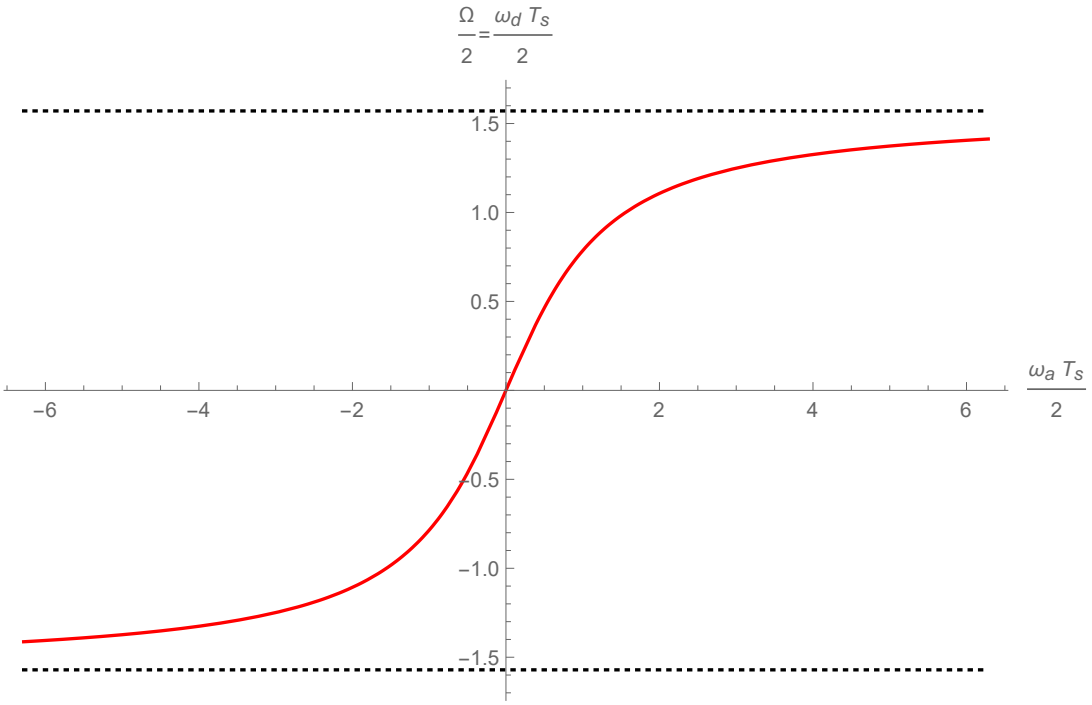


Figure 14.7: Relation between ω_a and Ω or ω_d of the s and z domain respectively.

Summaries

Summary Chapter 14

Key concepts on Infinite-Impulse-Response filters:
The general difference equation for a Linear-Time-Invariant IIR filter is given by:

$$y[n] = \sum_{i=0}^{N} b_i x\left[n-i\right] \; + \; \sum_{j=1}^{M} a_i y\left[n-j\right]$$

with the corresponding discrete transfer function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \ldots + b_M z^{-M}}{1 + a_1 z^{-1} + \ldots + a_N z^{-N}} = \frac{\prod_{i=1}^{N}(z - r_i)}{\prod_{j=1}^{M}(z - p_j)}$$

Two methods of IIR-filter design:

- Pole-Zero placement
- Bilinear Transformation of a known continuous-time filter

Pole-Zero placement
The method of pole-zero placement is based on pacing poles in the complex plane at positions you want pass-band behaviour, and zero's where you want attenuation. The unit circle is used as the normalized frequency axis in the design.

- The angle Ω of the pole/zero determines the frequency
- The magnitude r of the pole/zero determines the strength.

The Bilinear transformation
The bilinear transformation is the mapping of the continuous-time domain to the discrete-time domain, and is mathematically defined as:

$$s = \frac{2}{T_s}\left(\frac{z-1}{z+1}\right) \; \Leftrightarrow \; z = \frac{2f_s + s}{2f_s - s}$$

A transfer function from a known filter in the continuous-time domain can be transformed to a discrete filter in this way. This method is most effective if you already have a good continuous-time filter.

Problems

Problem 14.1. FIR or IIR?
FIR filters are commonly written non recursively

$$y[n] = \sum_{k=0}^{M} b_k x[n-k], \tag{14.36}$$

whereas IIR filters are commonly written recursively

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]. \tag{14.37}$$

Recursion refers to the fact that you need prior values of $y[n]$ to find $y[n]$ in (14.37).

a. Show that the z-transform of (14.36) leads to

$$H[z] = \frac{Y[z]}{X[z]} = \sum_{k=0}^{M} b_k z^{-k}.$$

b. Show that this also can be written as: $H[z] = \sum_{k=0}^{M} h[k] z^{-k}$, where $h[k] = b_k$ is the impulse response.

c. Why can you conclude that (14.36) is always an FIR filter?

d. You can show that

$$y[n] = \frac{3}{4}x[n] + \frac{1}{4}x[n-1] - \frac{1}{2}y[n-1] \tag{14.38}$$

has an infinite impulse response and by definition represents an IIR filter. It is also clear that

$$y[n] = \frac{3}{4}x[n] + \frac{1}{4}x[n-1] \tag{14.39}$$

is a finite impulse response filter (which shouldn't surprise you given **(c)**).
Show that (14.39) can be written in a recursive manner:

$$y[n] - y[n-1] = \frac{3}{4}x[n] - \frac{1}{2}x[n-1] - \frac{1}{4}x[n-2] \tag{14.40}$$

This would mean that apparently the same filter is both FIR (14.39) and IIR (14.40). This is clearly not possible, something cannot be both finite and infinite at the same time. So a recursive formulation does not automatically imply IIR! In the following, we will see a better way to distinguish the two.

e. Calculate the transfer functions $H[z]$ for (14.38), (14.39), and (14.40). Explain how you can distinguish between FIR and IIR filters by inspecting $H[z]$.

Problem 14.2. Transfer functions revisited

a. *This is problem 10.4* To get an impression of the effect of a simple difference equation, sketch the result of the difference equations on the two prototype signals (a DC signal and one fluctuating at the highest frequency possible, $f_s/2$) and describe the filter behaviour of the two digital filters presented in table 14.1.

b. Compute the transfer function $H(z)$ for each filter.

c. Plot the positions of the poles and zeroes on the z-plane. Motivate the link between the qualitative behaviour observed in part a) and the position of the zeroes.

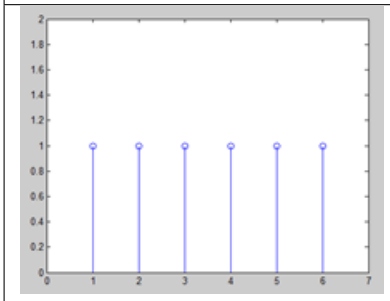
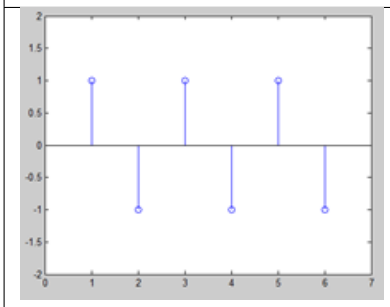
Problem 14.3. Band-pass IIR filter
Consider the simple IIR bandpass filter of figure 14.4. The goal is to allow a narrow band of frequencies around a frequency of interest, Ω_0, while suppressing other frequencies. There are two zeroes at $e^{j\Omega} = \pm 1$ and two complex conjugate poles at angles $\pm\Omega_0 =$ and radius r from the origin.

a. Write down the transfer function $H(z)$ for this system.

b. Suppose that we require $|H(z)| = 1$ at $z = e^{j\Omega_0}$. Normalize $H(z)$ appropriately.

c. Suppose that $r = 1 - d$ with $d \ll 1$ and concentrate on frequencies $\Omega = \Omega_0 \pm \Delta\Omega$, where $\Delta\Omega \ll \Omega_0$. Explain how you can concentrate only on a single pole in this case.

Table 14.1: Two examples of filters, given the difference equations working on two different signals.

Difference equation→ Signal ↓	y[n]=x[n]+x[n-1]	y[n]=x[n]-x[n-1]
	sketch the response	sketch the response
	sketch the response	sketch the response

d. Explain qualitatively how the width of the pass band changes when d is changed.

e. Calculate the width of the pass band, that is, calculate the value of $\Delta\Omega$ for which the transmitted power is halved, for the case $\Omega_0 = \pi/2$. Use your argument from part c) to simplify the calculation.

Problem 14.4. Pole/zero placement low-pass filter
Suppose we have a signal $x(t)$ which contains noise above $f = 40Hz$. We sample at $f_s = 480$Hz.

a. What is Ω_c?

b. Design a high low-pass filter which filters out the noise by placing 2 zeroes and 2 poles. Plot the frequency response. You can use a tool to try various locations of the poles and zeros.

Problem 14.5. Poles and zeroes at the origin

a. Describe the effect that a pole located at the origin ($z = 0$) has on the amplitude and phase of the transfer function $H(\Omega)$.

b. Repeat for a zero at the origin.

Problem 14.6. Filter implementation
Re-consider the filter of problem 14.3 above. Write down the difference equation corresponding to that filter.

Problem 14.7. Bilinear transformation low-pass Butterworth filter by hand
Suppose we have a signal $x(t)$ which contains noise above $f = 40Hz$. We sample at $f_s = 480$Hz.

a. What is Ω_c? How much does it deviate from Ω_c of the previous question?

b. Design a lowpass filter which filters out the noise by warping the analog second order Butterworth low pass filter $H(s) = \frac{\omega_c^2}{s^2+\omega_c\sqrt{2}s+\omega_c^2}$ with cut-off frequency $\omega = \omega_c$. Find the poles and zeros of this transfer function and compare with the previous question.

c. Apply the filter on $x[n] = \cos(\Omega_c n)$ for $n = 0, \cdots, 120$. Comment on your result.

d. Suppose you approximate $\tan(\theta)$ by θ for a small value of θ. Show that $\Omega_c = 2\pi\frac{f}{f_s} = \frac{\pi}{6}$.

e. Select $\Omega_c = \frac{\pi}{6}$. What is the value of ω_c and derive the difference equation using $t = \tan(\frac{\Omega_c}{2}) = \tan(\frac{\pi}{12})$.

Problem 14.8. Bilinear transformation low-pass Butterworth filter using a design tool
Repeat the previous question, but now use a design-tool in MATLAB or PYTHON . Discuss any changes you notice.

Problem 14.9. Bilinear transformation
Consider an analogue filter defined by the Laplace domain transfer function,

$$H(s) = \frac{a}{s+a}. \tag{14.41}$$

a. Compute the impulse response of this system, $h(t)$. To what type of elementary circuit does this correspond?

b. Now use the bilinear transformation to translate this filter into the z domain and obtain the corresponding transfer function, $H[z]$.

c. Compare the form of your result with that of the low-pass filter discussed in the lecture. What is the bandwidth of the discrete filter in terms of a and f_s?

d. Find the inverse z-Transform to determine $h[n]$.

Problem 14.10. Stricter requirements Butterworth filter
Until now we hand picked the order (2) and assume we select the cut-off frequency $\Omega_c = \frac{\pi}{6}$. We did not consider transition band (the frequency band where the pass band transcends into the stopband) nor the behaviour ("ripple/attentuation") of the filter in the pass- and stopbands. It is possible to specify this behaviour as a requirement and to determine the corresponding lowest filter order and cutoff frequency. We want a low pass Butterworth filter with passband frequency $\Omega_p = 0.1\pi$, maximum pass band ripple 3dB, stopband frequency $\Omega_p = 0.2\pi$ and minimum stop band attenuation of 50dB. Plot the frequency response.

Problem 14.11. Strict requirements on any high-pass filter
There exist more analogue filters. Given passband frequency $\Omega_p = 0.1\pi$, maximum pass band ripple 3dB, stop-band frequency $\Omega_s = 0.2\pi$ and minimum stop band attenuation of 50dB, compare Butterworth, Chebyshev Type 1 and elliptic low pass filters. Compare the frequency response and the number of coefficients.

Part III

Addendum

Appendix A

Symbols and Units

Name	Symbol	Unit	Description
General symbols			
Input	u	-	
Output	y	-	
State Variable	x	-	
Time	t	s	
Frequency	ω	rad.s^{-1}	
Damping coefficient	β	-	
Energy	E	J	
Power	P	J.s^{-1}	
Accumulator	C		
Conductor	G	s^{-1}	
Resistor	R	s	
Inductor	L	s^2	
Extensive quantity	Ψ		
Intensive quantity	ψ		
Flow	j	s^{-1}	
Diffusion	D	s^{-1}	

Name	Symbol	Unit	Description
Electrical domain			
Voltage	U	V	
Current	i	A	
Capacitor	C	F	
Inductor	L	H	
Resistor	R	Ω	
Mechanical translation domain			
Force	F	N	D
Velocity	v	$m.s^{-1}$	
Distance	s	m	
Mass	m	kg	D
Spring	k	$N.m^{-1}$	
Damper	b	$N.s.m^{-1}$	
Mechanical rotation domain			
Torque	M	N.m	
Rotation Velocity	Ω	$rad.s^{-1}$	
Angle	θ	rad	
Inertia	I	$kg.m^2$	
Angular Momentum	L	$kg.m^2.rad.s^{-1}$	
Spring (rotation)	K	$N.m.rad^{-1}$	
Damper (rotation)	B	$N.m.s.rad^{-1}$	
Heat flow domain			
Heat Flow	Q_w	$J.s^{-1}$	
Enthalpy	H	J	
Temperature	T	K	
Heat Capacity	c_p	$J.kg^{-1}K^{-1}$	
Heat Transmission coefficient	U	$W.m^{-2}.K^{-1}$	
Heat Transmission coefficient	α	$W.m^{-2}.K^{-1}$	
Heat Conduction coefficient	λ	$W.m^{-1}.K^{-1}$	
Cubic Expansion coefficient	γ	K^{-1}	
Activation Energy	$E_{a\rightarrow b}$	$J.mol^{-1}$	
Concentration component A	c_A	$mol.m^{-3}$	
Reaction Rate	r	$mol.m^{-3}.s^{-1}$	
Chemical Potential	μ_B	$J.mol^{-1}$	
Fluid transport domain			
Pressure	p	Pa	
Volume Flow	Φ_f	$m^3.s^{-1}$	
Mass Density	ρ	$kg.m^{-3}$	
Fluid Capacitor	C_p	$m^5.N^{-1}$	
Fluid Inductor	I_p	$N.s^2.m^{-5}.$	
Fluid Resistor	R_p	$m^5.s^{-1}.N^{-1}$	
Shear Strain	τ	$N.m^{-2}$	
Dynamic Viscosity	μ	$kg.m^{-1}.s^{-1}$	

Appendix B

Constants

Name	Symbol	Value	Unit	Description
Avogadro constant	N_A	$6.022137 \cdot 10^{23}$	mol^{-1}	
Gas Constant	R	8.31451	$J.mol^{-1}K^{-1}$	
Boltzmann constant	k	$1.380658 \cdot 10^{-23}$	$J.K^{-1}$	
Stefan-Boltzmann constant	σ	$5.67051 \cdot 10^{-8}$	$W.m^{-2}.K^{-4}$	
Gravitational Acceleration	g	9.8	$m.s^{-2}$	

Appendix C

A generic approach for modelling dynamical systems

Thermal, electrical and transport systems were introduced in the previous chapters. The dynamic behaviour of these very different systems can all be described mathematically by similar differential equations. This similarity expresses that in all three cases, a spatial difference in a level quantity (temperature, electrical potential, pressure) results in a flow of heat, electrical current or material from one point to another. These level quantities and associated flows can be defined for many domains in physics, chemistry and many other sciences. Several basic elements that control the (differential) relation between level quantity difference and flow were introduced. Below this concept will be generalized to formulate the underlying principles of many dynamic systems. This implies the introduction of the concept of intensive and extensive quantities. The principle of conservation is at the basis for connecting various parts of a system. This conservation principle has as a direct consequence the balance of a quantity.

C.1 Extensive and intensive quantities

The state of a system is described by a number of quantities. This state can change with time, which implies a change in the quantities used to describe a system. First a system and the related quantities will be defined.
System - A system is any region in space or quantity of matter looked upon for analysis. Everything not inside the system is in the surroundings. The system boundary is an infinitesimally thin surface, real or imagined, that separates the system from its surroundings. It has no mass and merely serves as a delineation of the extent of the system.
For modelling purposes, it is useful to classify systems according to the behaviour of their boundaries. Using this approach we define three types of systems: closed, open and isolated systems. An open system is a system whose boundary allows the transfer of one ore more quantities between the system and the surroundings. A system can accumulate quantities like mass, energy and charge. However, some of these quantities can be directly related. The introduction of mass in a system also introduces energy in the form of the associated enthalpy. If energy has to be considered, a system can be defined as closed and only a specific quantity can be exchanged through the boundary. Energy can pass, but mass not. The result is that such a closed system has a fixed and unchanging mass. An isolated system is a system whose boundaries prevent all interactions with the surroundings. Thus, an isolated system exchanges nothing with its surroundings.
Subsystem A system can be divided into appropriate subsystems. These subsystems are by definition open systems. The flow of quantities between the subsystems are known as internal flows of the system.
Quantity - A quantity of a system is any characteristic of a system that has a value at any instance of time. Quantities are classified as either intensive or extensive.

- *Extensive quantities of a system are those quantities that depend on the extent of the system*

This definition holds for both the whole system as well as for any subsystem, regardless of the choice of the subsystem, as long as the subsystem is finite. This implies that extensive quantities have to be strictly additive. The respective extensive quantities of the system S and subsystems 1 and 2 are related by:

$$\Psi_i^S = \Psi_i^{S_1} + \Psi_i^{S_2} \tag{C.1}$$

For example, for a system divided in two subsystems containing energies E_1 and E_2 the energy of the total system is equal to the sum of the energies of the sub-systems: $E_{tot} = E_1 + E_2$.
The value of an extensive quantity for a system only depends upon time. Extensive quantities are for example: mass, charge, energy, volume and entropy.
The flow j is a quantity derived from the extensive quantity and describes how much of it passes through defined area per time unit. Examples of flow are electric current (charge/time) and mechanical force (momentum/time). Sometimes they are simply called fluid flow (mass/time) and heat flow (heat energy/time).

- *Intensive quantities are those quantities that do not depend on the extent of the system*

An intensive quantity has a value at any point at any given time of the system and its value is independent of the extent or size. (To talk about a value at a point, we assume that we are dealing with a continuum where a *point* is physically small enough to have a single value and large enough to contain sufficient particles that the value has statistical significance. This concept is described in most fluid mechanics' textbooks.) Examples of intensive quantities are: temperature, voltage, pressure and velocity. Note that these intensive quantities are always related to some base value. An electrical voltage has only meaning in relation to a zero or ground point. Also pressure, temperature and velocity have a meaning with respect to an agreed zero value.
Intensive quantities tend to equilibrate. A spatial variation of intensive quantities is the driving force for extensive quantities to flow, i.e. a temperature variation induces a heat flow and a pressure variation induces mass transport. If two systems S_1 and S_2 with intensive quantities $\psi_i^{S_1}$ and $\psi_i^{S_2}$ which are not equal and

$$\psi_i^{S_1} < \psi_i^{S_2} \tag{C.2}$$

are brought into contact so that the respective extensive quantities can interchange, then for the intensive quantity of the combined system ψ_i^S one will observe that :

$$\psi_i^{S_1} < \psi_i^S < \psi_i^{S_2} \tag{C.3}$$

A forced flow of an extensive quantity is needed to create an imbalance in an intensive quantity. Work has to be applied to the system for this forced flow to occur (second law of thermodynamics).

State - The state of a system is a complete description of a system in terms of the value of its quantities. Strictly speaking this requires knowledge of all the relevant quantities of a system at a specified time; however, it turns out that we will often only need to know something about a limited number of the quantities. Other quantities can then be deduced from these. The set of quantities that fully describe the state of a system are often combined into a vector which is called the *state-vector* of the system. Typical important state quantities are the amounts of accumulated quantities in all the available accumulators. For instance, the amount of thermal energy stored in a mass. Very often this is simply represented by the temperature of this mass as temperature and thermal energy are directly linked by the heat capacity.

Process - A change in state of a system with time gives the dynamic behaviour of a system and reflects that the system has undergone a process. It is frequently the goal of engineering analysis to predict the path of states that a system passes through when it undergoes a specific process.
A special process is the steady-state process. In this process no accumulation occurs and the value of the state vector does not change. Typically, the analysis of a system undergoing a steady-state process involves the solution of a set of algebraic equations. If the variation of the state vector is much slower than the dynamic quantities of the system, these dynamic quantities can be disregarded. The system makes a transition through quasi steady states that are all such that all accumulated quantities show a negligible variation.

Accounting Principle - Now that we have defined the basic concepts, we can discuss the accounting principle. Experience has taught us that the extensive quantities of a system, *i.e.* the amount of an extensive quantity within a system, may change with time. Based on our observations, we postulate that this change can only occur by two mechanisms:

a. Transport of the extensive quantity across the system boundary

b. In and out flow due to the presence of sources and drains of extensive quantities inside the system.

This provides a simple balance and this is referred to as the accounting principle for an extensive quantity. Although this principle can be applied to a system for any extensive quantity, it will be especially useful for those quantities that are conserved.

C.2 Elemental Lumped Parameter Systems

In the preceding paragraphs, special attention was given to lumping systems with accumulation quantities. These lumped systems have a single intensive variable that describes the amount of extensive quantity accumulated, although the value of this intensive variable is always relative to a known (zero) level. Also resistive elements are identified in this lumping process. These accumulators and resistors are both present in fluid and thermodynamic systems. In many physical domains, like electricity and mechanics, elements with similar behaviour are encountered. In the fluid domain also a very different element, the inductor was described. In the various domains, these lumped systems can be mathematically described in a similar way. This similarity means that the dynamic behaviour in the various domains will be alike and the analysis of the behaviour follows the same lines. It only requires that the model is in the same mathematical language. Below, the general quantities of these different lumped elements are treated. However, a too strict thinking in these elements is a trap.
Strictly speaking we should not speak of elements that can store, dissipate or hold quantities. Instead we should speak about processes of storage, dissipation and holding. An element is too easily associated with a physical object. A physical object can be involved in more than one process. For example a mass can store momentum as well as angular momentum, i.e. it can perform a linear and rotational movement at the same time. These different movements are characterized by two parameters associated with this object: mass m and inertia I. Within the context of this text we talk about two elements and two processes connected to one object. An other example is the boiler. The water in the boiler is both associated with a heat capacity C_h as well as a volume capacity C_f. There is a heat exchange process as well as a volume (or mass) flow associated with the water in the boiler.
Lumped elements should be used used in a way that makes sense. For example it is of no use to connect two capacities without an element that makes the flow from one to the other capacity possible. In such cases the capacities have to be added (directly of reciprocally, depending on the situation: series or parallel). The same arguments hold for inductive elements and resistive elements, although for the latter is might be convenient to determine the separate flows.

It is clear from the discussion above, that a complicated system can be broken up into (small) elements that are described by simple algebraic or differential equations. These equations are connected via the variables that appear in the various equations.

C.2.1 Lumped Element Types

C.2.1.1 Capacitive Elements

Capacitors are elements that are able to store extensive quantities. The amount of accumulated quantity is evaluated from the inflow minus the outflow of this extensive quantity. The amount of accumulated extensive quantity Ψ can be related to it's related intensive quantity ψ by the capacitance C, with respect to the base value ψ_0. At the latter value, no quantity is accumulated. The equation that describes the storage mechanism relates the difference in inflow and outflow of an extensive quantity to the change of the intensive quantity in the capacitive element.

$$C\frac{\mathrm{d}(\psi-\psi_0)}{\mathrm{d}t} = f_C(j_{\mathsf{in}} - j_{\mathsf{out}}) \tag{C.4}$$

The function $f_C(j_{\mathsf{in}} - j_{\mathsf{out}})$ cannot be just any function. On basis of physical interpretation, the function f_C should be a function that depends on the difference between the inflows and the outflows. The function value must be zero when the inflow balances the outflow: $f(0) = 0$, i.e. the amount of accumulated extensive quantity does not change and the derivative is zero. The numerical value of C has to be positive and finite.
The differential relation denotes a dynamical process. It relates the rate of change to other variables in the process. As long as the derivative is not zero the system is not in equilibrium.

C.2.1.2 Resistive Elements

Resistive elements relate the difference in the intensive variables on two sides of the element to the flow through the element. The relation must be an algebraic one, since the relation is direct. There is no time involved in the relationship. A resistive element has no dynamic behaviour on its own. It facilitates or limits flows to or from storage elements. These elements are characterized by a single parameter denoted by the resistance R, or the conductance G with $G = 1/R$.

$$j = Gf_G(\psi_1 - \psi_2) \text{ or } Rj = f_G(\psi_1 - \psi_2) \tag{C.5}$$

The restrictions that apply for function f_C also apply for function f_G, *i.e.* $f_G(0) = 0$ and R is positive and finite. Materials are usually characterized by there specific conductivity $g = G/d$ with d the distance between locations 1 and 2.
An example of a resistance is the heat flow that occurs between two bodies at a different temperatures that are brought in thermal contact with each other. The heat flow between the two bodies is proportional to the difference in temperatures of the two bodies. For resistive elements we have the restriction that only one flow is involved: what flows in, flows out, *e.g.* an electric resistor: the current on both sides of the resistor is equal in value but is opposite in sign. In a fluid system, the amount of fluid that goes in the resistor at all times equals the amount that flows out. The presence of the resistor creates a pressure difference between the entrance and exit.

C.2.1.3 Inductive Elements

Inductive elements also relate the difference in the intensive variables on two sides of the boundary to the flow through the boundary. However, the relation is not an algebraic one as in the case of a resistor, but it describes the influence on the change in flow of the extensive variable. The parameter with which the element is characterized is denoted by L.

$$L\frac{\mathrm{d}j}{\mathrm{d}t} = f_L(\psi_1 - \psi_2) \tag{C.6}$$

This is again a differential form, as a flow has to be built up in an inductor. The restrictions that apply for function f_C and f_G also apply for function f_L, *i.e.* $f_L(0) = 0$ and L is positive and finite. This differential equation describes that an existing flow can not be stopped or started instantaneously. Examples of inductors are: for fluids, the effect that a pressure difference will occur over a long pipe if the fluid flow is changed, for an electric coil, the effect that it shows a voltage over it's terminals if the current is changed and for a mechanical spring a force difference occurs between the two sides when the length of the spring is changed. The differential form is similar to that of the capacitance and suggests that something can be stored in the inductor. However, what is stored is not as obvious as in case of the capacitor. For the coil it is magnetic energy as a result of the magnetic field produced by current through the coil and potential energy for a mechanical spring.

C.2.2 Power and Energy

The exchange of energy is at the hart of system dynamics. Depending on the area of science, the energy involved in the exchange can have a very different name, like heat, kinetic energy, gravitational energy, electrical energy, chemical energy etc. Complex systems usually involve the transfer of energy from one domain to another. Within the system energy can be temporarily stored or released. The temporal accumulation of energy provides the dynamics. Without this, the system will be in a stationary state and all the energy put in this system is, at the same instance, dissipated or coming out of the system. The amount of energy stored in elemental parts of a system is therefore of great interest to understand the relative importance and interconnected behaviour of elemental parts. The amount of energy stored will be calculated from the energy flow, also known as power. The power is directly related to the state variables of a system. The power handled by an element is the product of two complementary (flow of an extensive and intensive variable) state variables [j, ψ], *e.g.* [i, V], [F, v], [T, Ω], [Q_V, p] and [Q_h, T], for electric, translational mechanic, rotational mechanic, fluid systems and thermal powers. In general, these state variables are a function of time, so are the accompanying powers.
Two different types of elements are known: elements that can store energy and elements that dissipate energy. Both the energy dissipated and stored are evaluated by integrating the power flowing through the element over time. The start point is time zero:

$$E_{\text{diss}}(t) = \int_0^t P(t')\mathrm{d}t' = \int_0^t j_i(t')\psi_i(t')\mathrm{d}t' \tag{C.7}$$

Here the index i signifies the fact that the j and ψ should apply to the same element. A similar equations for the stored energy. In general the integration cannot be simplified for dynamic systems. However for elements that are able to store energy that can be retrieved again, the energy is not a function of history and can be calculated directly from the value of the state variables. In other words the stored energy is a function of the state variables of that element *i.e.*

$$\int_0^t j_i(t')\psi_i(t')\mathrm{d}t' = E(j_i(t), \psi_i(t)) \tag{C.8}$$

A capacitor and the inductor element have the following general relation:

$$C\frac{\mathrm{d}\psi_i}{\mathrm{d}t} = f_C(j_i) \tag{C.9}$$

and

$$L\frac{\mathrm{d}j_i}{\mathrm{d}t} = f_L(\psi_i) \tag{C.10}$$

An analytical expression for the energy stored can be derived if a simple linear differential relation between the variables exists:

$$C\frac{\mathrm{d}\psi_i}{\mathrm{d}t} = j_i \tag{C.11}$$

and

$$L\frac{\mathrm{d}j_i}{\mathrm{d}t} = \psi_i \tag{C.12}$$

the corresponding stored energies can be calculated and get a familiar form:

$$E_C = \int_0^t j_i(t')\psi_i(t')\mathrm{d}t' = \int_0^t \psi_i(t')C\frac{\mathrm{d}\psi_i(t')}{\mathrm{d}t'}\mathrm{d}t' = \frac{1}{2}C\psi_i^2(t) \tag{C.13}$$

and

$$E_L = \int_0^t j_i(t')\psi_i(t')\mathrm{d}t' = \int_0^t j_i(t')L\frac{\mathrm{d}j_i(t')}{\mathrm{d}t'}\mathrm{d}t' = \frac{1}{2}Lj_i^2(t) \tag{C.14}$$

respectively for the capacitive and inductive elements.
Equations (C.13) and (C.14) show that both capacitive elements and inductive elements are able to store energy. The origin for this is the differential relation between the two state variables. At any given time, the amount of energy accumulated in a system is directly known from the values of the intensive or extensive flow variables for either capacitive and inductive systems. Note that not for all disciplines inductive elements are relevant, *e.g.* no inductive elements are found in thermal systems. For a resistive-element the power can also be calculated from the product of extensive flow and intensive variable:

$$\int_0^t j_i(t')\psi_i(t')\mathrm{d}t' = \int_0^t \frac{j_i^2(t')}{R}\mathrm{d}t' = \int_0^t R\psi_i^2(t')\mathrm{d}t' \tag{C.15}$$

However, this integral cannot be evaluated further without explicit knowledge of the history of the two variables. This history determines the total energy that is dissipated in the time interval considered and distinguishes the behaviour of these elements from the energy storage elements.

C.2.3 State Variables

The observed dynamics of a systems reflect the energy flows in a system. This crucial role of energy storage and flow implies that to characterize a system knowledge about the location of energy at any given time is needed. This statement implies that the state of capacitive and inductive elements is at the basis of describing system dynamics. The amount of energy available is directly characterized by the respective intensive and extensive flow variables of these basic elements. All other quantities in the system are directly related to the values ψ of the accumulators and j of the inductors. Their values describe at any given time the state of the system and are therefore an excellent choice as state variables. This is not the only choice, but any other choices will be a (linear) combination of these quantities. The element equations of capacitive elements and inductive elements are thus the basis for a set of first order differential equations that link the dynamic behaviour. The number of first order differential equations equals the number of storage and induction processes in the system. This number is called the *order of system* or *degree of freedom*.

C.2.4 Overview of element equations and related energies

For lumped parameter systems the elements from the various disciplines are related through their dynamic character of the associated intensive and extensive flow variables, ψ and j. In Table C.1 an overview of the three type of elements in the various disciplines is presented. Along with the element relations also the associated variables and the energy expression is given.
The heat system forms an exception[1] at several points. There is no equivalent for the inductor in thermal physics. Also, the energy stored in a heat capacitance is not the product of the intensive and extensive variable associated with the element.

[1]This item is a bit tricked to force it in the structure of this systems overview. A more formal treatment would need to include Entropy S into the treatment. Note also that in line of these lecture notes Q_h indicates the *heat flow* and not the heat itself. The heat (energy) is denoted by E_h

System	Capacitor	Inductor	Resistor	Extensive eqn.
General	$\psi,\ q^{\alpha}$	$j,\ q^{\tau}$		$\int j\mathrm{d}t = C\psi$
	$C\frac{\mathrm{d}\psi}{\mathrm{d}t} = j$	$L\frac{\mathrm{d}j}{\mathrm{d}t} = \psi$	$j = \frac{1}{R}\psi$	
	$\frac{1}{2}C\psi^2$	$\frac{1}{2}Lj^2$	$\int j\psi\mathrm{d}t$	
Mechanical (translation)	v	F		$p = mv$
	$m\frac{\mathrm{d}v}{\mathrm{d}t} = F$	$\frac{1}{k}\frac{\mathrm{d}F}{\mathrm{d}t} = v$	$F = bv$	
	$E = \frac{1}{2}mv^2$	$E = \frac{1}{2k}F^2$	$\int Fv\mathrm{d}t$	
Mechanical (rotational)	Ω	T		$L = I\Omega$
	$I\frac{\mathrm{d}\Omega}{\mathrm{d}t} = T$	$\frac{1}{K}\frac{\mathrm{d}T}{\mathrm{d}t} = \Omega$	$T = B\Omega$	
	$E = \frac{1}{2}I\Omega^2$	$E = \frac{1}{2K}T^2$	$\int T\Omega\mathrm{d}t$	
Electrical	U	i		$Q = CU$
	$C\frac{\mathrm{d}U}{\mathrm{d}t} = i$	$L\frac{\mathrm{d}i}{\mathrm{d}t} = U$	$i = \frac{1}{R}U$	
	$E = \frac{1}{2}CU^2$	$E = \frac{1}{2}Li^2$	$\int iU\mathrm{d}t$	
Fluid	p	Φ_p		$V = C_p p$
	$C_p\frac{\mathrm{d}p}{\mathrm{d}t} = \Phi_p$	$I_p\frac{\mathrm{d}\Phi_p}{\mathrm{d}t} = p$	$\Phi_p = \frac{1}{R_p}p$	
	$E = \frac{1}{2}C_p p^2$	$E = \frac{1}{2}I_p\Phi_p^2$	$\int \Phi_p p\mathrm{d}t$	
Heat	T	Q_h		$E_h = C_h T$
	$C_h\frac{\mathrm{d}T}{\mathrm{d}t} = Q_h$	$-$	$Q_h = \frac{1}{R_h}T$	
	$E_h = C_h T$	$-$	$\int Q_h T\mathrm{d}t$	

Table C.1: Overview of the Element Equations with the associated variables and stored energies.

C.3 Laws of Conservation and Balances

The conservation of a quantity is the basic principle that connects different elements in a system. For electrical systems, the obvious conservation of charge results in the two Kirchhoff laws. These two laws are also valid for any other domain in which the behaviour of a system can be described on the basis of accumulators, inductors and resistors. The Kirchhoff flow law has to hold at any junction of flows:

$$\sum_{i}^{\text{junction}} j_i = 0 \tag{C.16}$$

where $[\sum_i^{\text{junction}}]$ denotes the sum over all items that belong to the junction under consideration. The second Kirchhoff law dealing with potential differences can also be generalized to intensive quantities:

$$\sum_{i}^{\text{closed loop}} \psi_{i,i+1} = 0 \tag{C.17}$$

In this case, the summation is over a closed loop of sequential points i and involves the sum over the differences in intensive variable from one point to the next. This law is not based on conservation of energy, but is a simple counting rule. Together, the two generalized Kirchhoff rules link the quantities that describe elemental parts of a system. This enables the formulation of a consistent set of differential equations that describe the dynamics of the system.

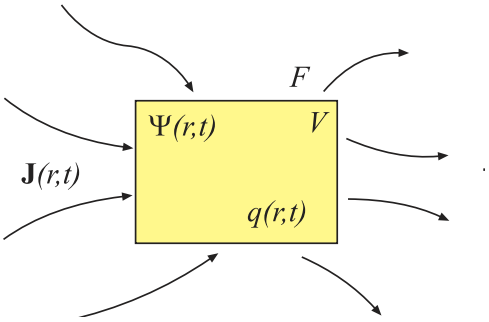


Figure C.1: Model boundaries of a continuous system determined by the volume V and the surface boundary $\mathcal{F}$. Depicted are the position dependent value of the extensive quantity $\Psi(r,t)$, the flow of this quantity across the boundary J(r,t) and sources q(r,t).

The conservation balance can be used to extend the elemental equation for each accumulator:

$$\begin{Bmatrix} net\ change \\ in\ accumulated\ quantity \end{Bmatrix} = \begin{Bmatrix} flow\ in \\ through\ boundary \end{Bmatrix} - \begin{Bmatrix} flow\ out \\ through\ boundary \end{Bmatrix} + \begin{Bmatrix} net \\ generation \end{Bmatrix} - \begin{Bmatrix} net \\ consumption \end{Bmatrix} \tag{C.18}$$

This can be used to evaluate the behaviour of a system with a translation to the mathematical world. The volume-specific form of an extensive quantity $\hat{\Psi}(\vec{r},t)$, its associated flows $J(\vec{r},t)$ and its volume-specific source and drain terms $\hat{q}(\vec{r},t)$ are used for this. All quantities are a function of the three dimensional space coordinates $\vec{r}$ and time t for the volume $\mathcal{V}$ with the boundary surface $\mathcal{F}$. The balance (C.18) translates to:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\iiint_{\mathcal{V}} \hat{\Psi}(r,t)\mathrm{d}V\right\} = -\iint_{\mathcal{F}} \vec{J}(r,t) \cdot \mathrm{d}\vec{f} + \iiint_{\mathcal{V}} \hat{q}(r,t)\mathrm{d}V \tag{C.19}$$

The left-hand side of the equation represents the time variation of the extensive quantity Ψ integrated over the volume $\mathcal{V}$. The integration over space makes this equation a global form of the balance.
The first term on the right-hand side accounts for the flow of the extensive quantity through the boundary of the surface $\mathcal{F}$. The minus sign of this flow term is the consequence of the outward directed definition of the normal of a closed surface surrounding a volume. A positive flow will consequently lower the content of the volume $\mathcal{V}$. The flow $\vec{J}(\vec{r},t)$ is a vector field varying both in magnitude and direction from point to point on the surface $\mathcal{F}$ and is integrated over the whole surface that surrounds the volume V.
The last term in Equation (C.19) represent the sources (generation) and drains (consumption) of extensive quantities within the specified volume $\mathcal{V}$. This term is typically related to energy or species sources and drains associated with reactions in the system.
The conservation balance described in Equation (C.19) is valid for all volumes. With the aid of Gauss's vector theorem the surface integral can be expressed as a volume integral:

$$\iint_{\mathcal{F}} \vec{A} \cdot \mathrm{d}\vec{f} = \iiint_{\mathcal{V}} (\vec{\nabla} \cdot \vec{A})\mathrm{d}V \tag{C.20}$$

For an infinitesimally small volume the balance is transferred to the differential form[2].

$$\frac{\mathrm{d}\hat{\Psi}}{\mathrm{d}t} = -\vec{\nabla} \cdot \vec{J} + \hat{q} \tag{C.21}$$

Equation (C.21) is a local description of the continuity balance (C.19) that holds for all points in the space of the system volume.
The flow $\vec{J}$ of a conserved quantity can be driven by two different sources. First of all the natural tendency to equilibrate differences in intensive quantities within the volume of the system leads to a flow. Such flows are called diffusive flows $\vec{J}_D$. The sensitivity of the diffusion to the spatial difference in intensive quantity is given by the so called diffusion coefficient D. This diffusion coefficient is equal to the earlier introduced specific conductivity g. However, in many situations the term diffusivity is used instead.

$$\vec{J}_D \cong -D\vec{\nabla}\psi(r,t) \tag{C.22}$$

The ∇ operator relates a required spatial difference of an intensive variable to a resulting flow. The negative sign indicates that the flow is from high to low.
Next to this equilibration flow, also a flow is possible that is convective $\vec{J}_C$, *i.e.* that requires a transport mechanism and can for instance reflect that work is done on the system. An example is the central heating system, where a water flow is used to transport heat. The convective flow is directly the result of a movement with a local velocity $\vec{v}(r,t)$.

$$\vec{J}_C = \Psi(r,t)\vec{v}(r,t) \tag{C.23}$$

The final term in the differential conservation equation (C.21) $\hat{q}(r,t)$, is related to sources or drains of mass, energy or other conserved quantities.

[2]Strictly speaking we should have used the partial derivative $\frac{\partial\hat{\Psi}}{\partial t}$ since the equation is also valid in a frame moving with a fluid or any other moving frame.. In that case the divergence of the current $\vec{J}$ applies in the moving frame. This is not a big issue in lumped parameter systems, since all flows are described separately and contribute to the accumulation. When changing from a continuous to a lumped parameter system, all flows will get described explicitly and the rates of change (accumulations) all become full derivatives to time (instead of partial derivatives).

Already with the definition of intensive and extensive quantities, it was mentioned that the amount of accumulated extensive quantity can be expressed with an intensive quantity, i.e. the accumulated thermal energy by temperature, the amount of mass in a tank by the pressure etc. The relation between these two especially holds on a local scale, giving the relation

$$\hat{\Psi}(r,t) = c\psi(r,t) \tag{C.24}$$

Combined with the derived relation of the flows to intensive quantities allows a further rewriting of the balance equation that describes the conservation in local form[3]:

$$c\frac{\mathrm{d}\psi}{\mathrm{d}t} = D\left(\nabla^2\psi(r,t)\right) - \vec{\nabla}\cdot\left(\hat{\psi}(r,t)\vec{v}(r,t)\right) + \hat{q}(r,t) \tag{C.25}$$

This equation is a partial differential equation and describes how the local value of $\psi(r,t)$ depends on spatial fluctuations, forced flows and sources on a local scale. Now it is of course just mathematics to solve this partial differential equation. However, this is not a simple task in general and numerical simulations will have to be done to solve this problem. One of the ways to do this is to subdivide the system in a large number of small lumped systems whose modelling has been studied in the previous chapters.

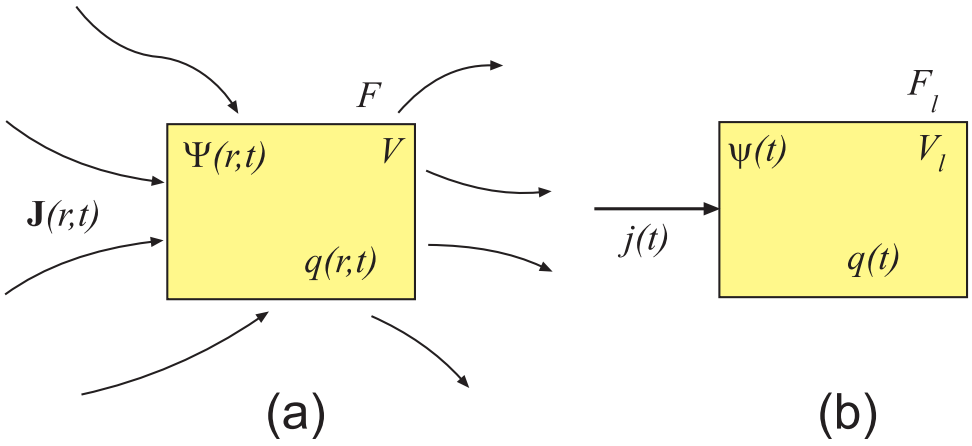


Figure C.2: Model boundaries for (a) a continuous system and (b) a lumped parameter-system consisting of a single compartment.

C.4 Transition from a Continuous to a Lumped System

C.4.1 Transition from a Continuous to a single Lumped System

A semi-general treatment of the transition of a continuous system to a lumped system starts by considering a system as depicted in Fig. C.2. It is characterized by a finite volume $\mathcal{V}$ and is separated from the surrounding by the boundary $\mathcal{F}$. An extensive quantity Ψ can be accumulated in this system. Locally the value of this extensive quantity is given by $\hat{\Psi}(\vec{r},t)$. A flow $\vec{J}(\vec{r},t)$ that passes the system boundary and a source - drain term $\hat{q}(\vec{r},t)$ in the volume complete the general picture. The conservation balance of this system results in an integral equation, as given previously by Eqn. (C.19):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\iiint_{\mathcal{V}_l}\hat{\Psi}(r,t)\mathrm{d}V\right\} = -\iint_{\mathcal{F}}\vec{J}(r,t)\cdot\mathrm{d}\vec{f} + \iiint_{\mathcal{V}_l}\hat{q}(r,t)\mathrm{d}V \tag{C.26}$$

[3]Also here the derivative $\frac{\mathrm{d}\psi}{\mathrm{d}t}$ also applies in partial form $\frac{\partial\psi}{\partial t}$ in case the quantities are described with respect to a moving frame.

The spatial variation of the quantities is eliminated in the transition to a lumped system. Only the averaged values are of interest. These average values are obtained from integration over the volume or surface of the system of the spatially varying quantities. This gives a single value for the lumped system l:

$$\Psi_l(t) = \iiint_{\mathcal{V}} \hat{\Psi}(r,t)\mathrm{d}V \tag{C.27}$$

$$j_l(t) = \iint_{\mathcal{F}} \vec{J}(r,t) \cdot \mathrm{d}\vec{f} \tag{C.28}$$

$$q_l(t) = \iiint_{\mathcal{V}} \hat{q}(r,t)\mathrm{d}V \tag{C.29}$$

These volume and surface averaged quantities only depend on time. The amount of accumulated quantity in the lumped system can be related to an intensive quantity, for instance the amount of enthalpy stored to the temperature, the amount of mass stored in a tank to the pressure etc. The relation between the two is expressed by the capacitance C of the accumulator, i.e. $\psi_l = C\Psi_l$. The intensive quantity, temperature, pressure, voltage etc, is also averaged over the volume. The spatial averaging reduces the integral equation of the conservation balance to a differential equation in which only time determines the value of the relevant variables:

$$c\frac{\mathrm{d}\psi_l(t)}{\mathrm{d}t} = j_l(t) + q_l(t) \tag{C.30}$$

The flows in this equation are either positive or negative; positive for accumulation and negative for depletion. Only the amount of conserved quantity that flows through the boundary and the distinction between *in* and *out* is of interest.
This derivation for an accumulator is a general approach which makes it powerful in its applicability. One does not need to make this derivation separately for every system under consideration. It is even more wise and practical to start on the other end by asking oneself the question: *Can I represent the system by one parameter and describe the dynamics by one dynamic variable*? In that case Equation (C.30) is sufficient, however one should remember its origin.
The accumulation and flow quantities of a single-compartment system can be graphically displayed as shown in Fig. C.3. The depicted cup is the holder of the accumulated quantity (charge, energy, matter) that is filled and emptied through a flow controlled by the resistances R.

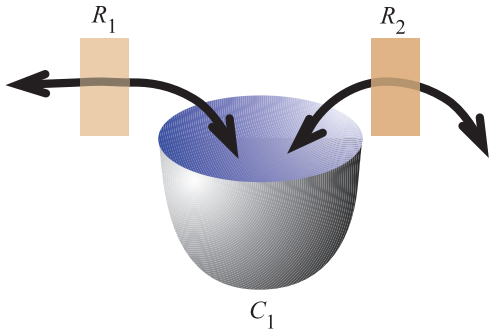


Figure C.3: Model of a lumped system consisting of an accumulator C_i and two flows determined by the resistances R_1 and R_2.

C.4.2 Transition from Continuous to a series of Lumped Systems

An intensive variable like temperature can determine both accumulation and flow through a system part. Because flow is dependent on the gradient of an intensive variable, spatial averaging washes out the driving force for the flow through a system part. A single lumped element can not represent

this behaviour. Breaking up the initial volume in a very large number of much smaller volumes solves this problem. The spatially averaged value is in the very small volume similar to the actual value and small flows connect one volume wit the next. In mathematical sense, a transition from an integral to a summation situation is obtained. However, a too large number would immediately pose a practical numerical problem. It takes too long to solve. Therefore, a limited number of volumes with a finite size will have to be chosen. To make the transition of the continuous description of conserved quantities to a series of lumped parameter systems, the distributed system, the following steps are taken:

- Find regions in the system that have the same character, although the value for intensive and extensive variables might vary.
- Determine whether the variation in variable values in these regions is significantly larger for time variations or for location variations.
  - Time variation larger than location variations: One element can be used to represent the whole region.
  - Time variations smaller than location variations: The region has to be divided up in smaller sections in such a way that for all regions the location variations of the describing variables within all regions is significantly smaller than the time variations of the variables.
- make a chain of alternating storage and flow elements.

Consider the system depicted in Fig. C.4. The spatial variation in the accumulation (extensive quantity) and source and drain terms is such that a division in two subsystems allows an appropriate spatial averaging of these quantities. The volume $\mathcal{V}$ with it's boundary $\mathcal{F}$ is divide in two volumes $\mathcal{V}_1$ and $\mathcal{V}_2$ with their respective boundaries $\mathcal{F}_1$ and $\mathcal{F}_2$, of which the part $\mathcal{F}_{12}$ is the shared boundary. The corresponding quantities are $\hat{\Psi}_1$ and $\hat{\Psi}_2$.

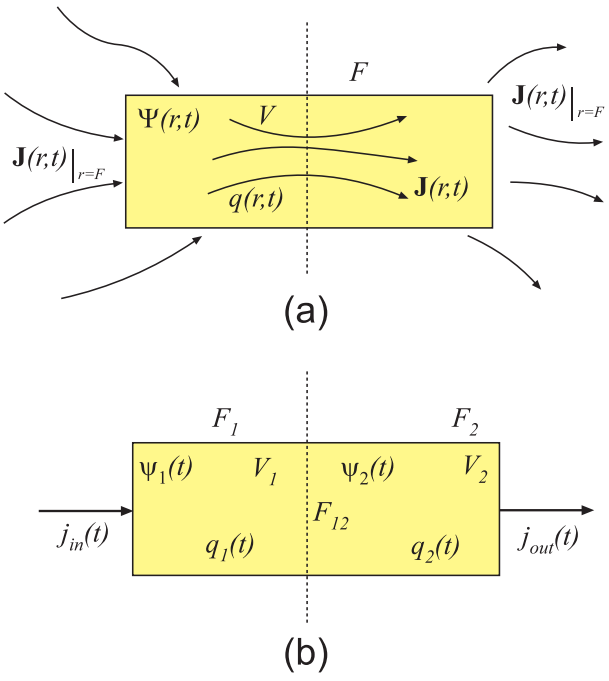


Figure C.4: Model boundaries: case (a) continuous system consisting of one compartment virtually divided into two parts; case (b) two sub-systems with internal boundary.

For the extensive variable the balance has to be made up taking into account flows through the

boundaries and the generation in the two separate volumes:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left\{\iiint_{\mathcal{V}_1}\hat{\Psi}(r,t)\mathrm{d}V\right\} &= -\iint_{\mathcal{F}_1-\mathcal{F}_{12}}\vec{J}(r,t)\cdot\mathrm{d}\vec{f}-\iint_{\mathcal{F}_{12}}\vec{J}(r,t)\cdot\mathrm{d}\vec{f}+\iiint_{\mathcal{V}_1}\hat{q}(r,t)\mathrm{d}V\\ \frac{\mathrm{d}}{\mathrm{d}t}\left\{\iiint_{\mathcal{V}_2}\hat{\Psi}(r,t)\mathrm{d}V\right\} &= -\iint_{\mathcal{F}_2-\mathcal{F}_{12}}\vec{J}(r,t)\cdot\mathrm{d}\vec{f}-\iint_{\mathcal{F}_{12}}\vec{J}(r,t)\cdot\mathrm{d}\vec{f}+\iiint_{\mathcal{V}_2}\hat{q}(r,t)\mathrm{d}V\end{aligned}$$

The integration of the flows over the surface is split in two integrals: one for the flow between the outer world and the volume and one between the two volumes. If the system is divided in more subsystems, than for each interface between two neighboring compartments an integration has to be performed that provides the nett flow between the two compartments. Again the accumulated extensive quantity is related to the accompanying intensive variable, i.e. $\Psi_i = C_i\psi_i$ with C_i the capacitance of volume i:

$$C_1\frac{\mathrm{d}\psi_1(t)}{\mathrm{d}t} = j_{\text{in}}(t) - j_{12}(t) + q_1(t) \tag{C.31}$$

$$C_2\frac{\mathrm{d}\psi_2(t)}{\mathrm{d}t} = -j_{\text{out}}(t) + j_{12}(t) + q_2(t) \tag{C.32}$$

In these equations there is no position dependent variable other than those that are assigned to define the compartments. The flow between the compartments $j_{12}(t)$ appears in both equations with opposite sign in order to fulfill conservation: what flows out of one compartment and remains in the system has to flow into the neighboring compartment.
Figure C.5 depicts the model in the form of two containers that represent the two different accumulation volumes. The flow $j_{12}(t)$ between the two containers is represented by a resistor element. This indicates that this flow is driven by the difference in the intensive variables ψ_1 and ψ_2 averaged over their respective volume:

$$j_{12}(t) = f_R\left(\psi_1 - \psi_2\right)$$

with f_R the function that describes the resistive quantity. Note that the fact that a resistance exists within the volume that controls a flow, is related to the fact that a spatial averaging over the entire system is not sufficient. If no internal resistance would exist, an unlimited flow is possible and the difference in ψ_1 and ψ_2 would immediately be equilibrated.

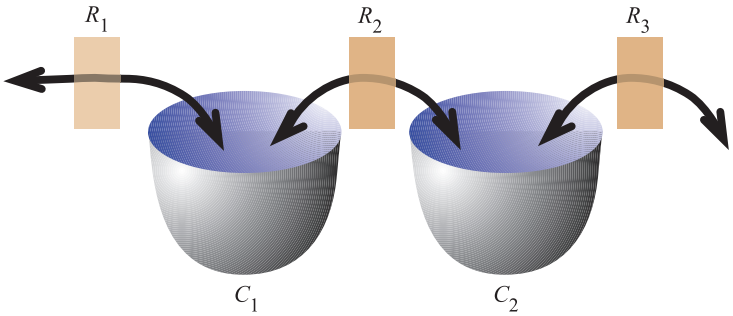


Figure C.5: Model of a 2 container system.

Appendix D

Matlab commands

In this Appendix implementation of continuous and discrete models (filters) in MATLAB is provided. You are expected to know the basics of MATLAB. You are strongly recommended to use the help function and documentation in MATLAB for details on use.

Table D.1: Useful MATLAB commands - continous systems – Modelling

Command	description
`r = roots(a)`	Determines the roots of a polynomial described by the coefficients in vector a. This is a very nice function to solve characteristic equations or to find out the factorization of transfer functions
`[r,p,k] = resdidue(b,a)`	Finds the residues, poles, and direct term of a Partial Fraction Expansion of the ratio of two polynomials with coefficients given by vectors $b = [b_m...b_1b_0]$ and $a = [a_n...a_1a_0]$. This can be used to evaluate an essential and tedious step in the inverse Laplace transform.
`partfrac(expr)`	With Symbolic math the partial fractions of $expr$ can be evaluated. Start symbolic math with the command **`syms s`**.
`laplace(expr)`	Symbolic math Laplace transform of $expr$. Start symbolic math with the command **`syms`** with the appropriate variables.
`ilaplace(Expr)`	Symbolic math inverse Laplace transform of $Expr$. Start symbolic math with the command **`syms`** with the appropriate variables.
`SYS = tf(NUM,DEN)`	This creates a continuous-time transfer function SYS with numerator NUM and denominator DEN. SYS is an object of type tf when NUM,DEN are numeric arrays, of type GENSS when NUM,DEN depend on tunable parameters (see REALP and GENMAT), and of type USS when NUM,DEN are uncertain (requires Robust Control Toolbox).
`SYS = ss(A,B,C,D)`	Creates an object SYS representing the continuous- time state-space model dx/dt = Ax(t) + Bu(t) y(t) = Cx(t) + Du(t)
`SYS = zpk(Z,P,K)`	Creates a continuous-time zero-pole-gain (zpk) model SYS with zeros Z, poles P, and gains K. SYS is an object of class @zpk.
`bode(SYS)`	Draws the Bode plot of the dynamic system SYS. The frequency range and number of points are chosen automatically. bode(SYS1,SYS2) draws in one plot Bode diagrams of systems SYS1 and SYS2.
`nyquist(SYS)`	Draws the Nyquist plot of the dynamic system SYS. The frequency range and number of points are chosen automatically.
`step(SYS)`	Calculates the step response of the dynamic system SYS. The final time can be given. step(SYS1,SYS2) draws in one plot the step response of systems SYS1 and SYS2.
`impulse(SYS)`	Calculates the unit impulse response of a dynamic system SYS.
`[t,y]=lsim(SYS,U,T,X0)`	Calculates the time response of the dynamic system SYS to the input signal described by U and T with initial condition X0. The output is the time array t and matrix (multiple outputs!) y
`minreal(SYS)`	Cancels pole-zero pairs in transfer functions or zero-pole-gain models. After multiplication, division or addition of transfer functions the result might be complex function in which pole and zero values are similar and thus cancel each other. minreal takes care of this elimination.
`[Gm, Pm, Wcg, Wcp] = margin(SYS)`	Computes the gain margin Gm, the phase margin Pm, and the associated frequencies Wcg and Wcp, for the SISO

Table D.2: Useful MATLAB commands – Filtering

Command	**description**
`stem(Y)`	Plots the data sequence Y as stems from the x axis terminated with circles for the data value. Used to show discrete signals.
`[R,P,K] = residue(B,A)`	Finds the residues, poles and direct term of a partial fraction expansion of the ratio of two polynomials B(s)/A(s). Used to solve the signal response to a difference equation.
`zplane(B,A)`	Gives a plot of the poles and zeros of a digital filter function defined by the arrays B (numerator) and A (denominator) of a difference equation.
`Zi=filtic(B,A,Yi,Xi)`	Make initial conditions for 'filter' function defined by the arrays B (numerator) and A (denominator) of a difference equation. Xi and Yi are the initial conditions.
`Y=filter(B,A,x,Zi)`	Filters the data in vector X with the filter described by vectors A and B to create the filtered data Y. Initial conditions in the parameter Zi are optional.
`[H,W] = freqz(B,A,N)`	Gives the N-point complex frequency response of the transfer function (filter) given by the vectors B (numerator) and A (denominator). H is a vector of complex number on the normalized frequency axis W that represent the magnitude and the phase response . The normalized frequency runs from 0 to π. The actual frequency is given by f = W/2π * f_s. with f_s the sampling frequency of the signal.
`B = fir1(N,Wn)`	FIR filter design of an N'th order lowpass FIR digital filter with the window method. It returns the filter coefficients in vector B with length N+1. The cut-off frequency Wn must be between $0 < Wn < 1.0$, with 1.0 corresponding to half the sample rate. See Matlab helpfile for how to design a band, stop or highpass filter.
`[B,A] = butter(n,Wn)`	Returns the transfer function coefficients of an nth-order lowpass digital Butterworth filter with normalized cutoff frequency Wn.
`[B,A] = butter(n,Wn,ftype)`	Designs a lowpass, highpass, bandpass, or bandstop Butterworth filter, depending on the value of ftype and the number of elements of Wn. The resulting bandpass and bandstop designs are of order 2n.
`[B,A] = cheby1(n,Rp,Wp,ftype)`	Designs a lowpass, highpass, bandpass, or bandstop Chebyshev Type I filter, depending on the value of ftype and the number of elements of Wp. The resulting bandpass and bandstop designs are of order 2n.

Table D.3: Useful MATLAB tools/wizards

Tool/Wizard	**description**
`ltiview`	ltiview opens an empty LTI Viewer. The LTI Viewer is an interactive graphical user interface (GUI) for analyzing the time and frequency responses of linear systems and comparing such systems. See LTIMODELS for details on how to model linear systems in the Control System Toolbox.
`rltool`	FVTool is a Graphical User Interface (GUI) that allows you to analyze and tune control systems.
`filterDesigner`	Launches the Filter Design & Analysis Tool. filterDesigner is a Graphical User Interface (GUI) that allows you to design or import, and analyze digital FIR and IIR filters. This can do almost everything in filter design, so the thing to use. However, if you do not know how to handle it, it can easily turn in a GIGO system, garbage in garbage out.

Appendix E

Python commands

Here you find a few tables with PYTHON commands, which might be useful to you in this course. You are strongly recommended to use the help function and documentation in PYTHON for details on use. You are expected to know the basics of PYTHON.

Table E.1: Useful PYTHON commands

Command	**description**

Appendix F

Mathematica Commands

Here you find a few tables with MATHEMATICA commands, which might be useful to you in this course. You are strongly recommended to use the help function and documentation in MATHEMATICA for details on use. You are expected to know the basics of MATHEMATICA.
The tables with commands are for reference, however the documentation of MATHEMATICA has some nice guides and tutorials too on the subjects

Table F.1: Guides

Mathematica guides
`Signal Processing`
`Control Systems`
`Classical Analysis and Design`
`Non-linear Control Systems`
`Fourier Analysis`
`Probability and Statistics`

Table F.2: Tutorials

Mathematica tutorials
`Digital Filter Design`
`Analog Filter Design`
`Audio Processing`
`Integral Transforms`
`Discrete Fourier Transform`

Table F.3: Useful MATHEMATICA commands I

System & Control functions	
Command	**Description**
`StateSpaceModel[{`A`,`B`,`C`,`D`}]`	
	Creates a state-space model object from the matrices A, B, C and D.
`NonlinearStateSpaceModel[]`	
	Creates a non-linear state-space object; non-linear equations should be supplied.
`CreateSystemModel[`sys`]`	
	Creates a model from the system-object sys.
`TransferFunctionModel[`$expr$`,`var`]`	
	Creates a transfer function model object of a function $expr$ of variable var.
`SystemsModelSeriesConnect[`$sys1$`,`$sys2$`]`	
	Creates a system model object from two systems models $sys1$ and $sys2$
`SystemsModelParallelConnect[`$sys1$`,`$sys2$`]`	
	Creates a system model object from two systems models $sys1$ and $sys2$
`SystemsModelFeedbackConnect[`$sys1$`,`$sys2$`]`	
	Creates a feed-back system model object from two systems models $sys1$ and $sys2$
`SystemsModelStateFeedbackConnect[`$sys1$`,`$sys2$`]`	
	Creates a system model object from two state-space models $sys1$ and $sys2$
`SystemsModelMerge[sys1,sys2]`	

Table F.4: Useful MATHEMATICA commands II

System & Control functions – Continued	
Command	**Description**
`NyquistPlot[`sys`]`	
	Generates a Nyquist plot from system sys. Most likely sys is an open-loop. Options like GainPhaseMargin are available for this function.
`BodePlot[`sys`]`	
	Generates Bode plots (Amplitude and phase) of system sys. Options like GainPhaseMargin are available for this function.
`GainPhaseMargins[`sys`]`	
	Calculates the gain margin en phase margin of system sys.
`RootLocusPlot[`sys`]`	
	Generates a root-locus plot of the open-loop system sys. In combination with the function `Manipulate[]` an interactive plot can be constructed.
`TransferFunctionPoles[`sys`]`	
	Returns the poles of a system sys.
`TransferFunctionZeros[`sys`]`	
	Returns the zero's of a system sys.
`OutputResponse[`sys`,`$expr$`,`var`]`	
	Generates an expression that is the output response of system sys with an input $expr$ and independent variable var.
`StateResponse[`sys`,`$expr$`,`var`]`	
	Generates an expression that is the output response of the State-Space system sys with an input $expr$ and independent variable var.

Table F.5: Useful MATHEMATICA commands III

Laplace & Fourier – Continuous-time domain	
Command	**description**
`DSolve[{`$exprs$`},`$var1$`,`$var2$`]`	
	Counter part of `RSolve[]` to solve Differential equations, with initial conditions and input function.
`LaplaceTransform[`$expr$`,`$var1$`,`$var2$`]`	
	Calculates the Laplace transform of $expr$ as a function of $var2$.
`InverseLaplaceTransform[`$expr$`,`$var1$`,`$var2$`]`	
	Calculates the inverse Laplace Transform of $expr$.
`Apart[`$expr$`]`	
	Returns a partial fraction decomposition of $expr$. This function is also of use in other cases, like the z-Transform.
`FourierTransform[`$expr$`,`$var1$`, `$var2$`]`	
	Returns the Fourier Transform of $expr$ as a function of $var2$. The option `FourierParamters` $\rightarrow$ {1,-1} should be used to have the correct normalisation of the Fourier Transforms as is common for Signal analysis.
`InverseFourierTransform[`$expr$`,`$var1$`, `$var2$`]`	
	Returns the inverse Fourier Transform of $expr$ as function of $var2$. The option `FourierParamters` $\rightarrow$ `{1,-1}` should be used to have the correct normalisation of the Fourier Transforms as is common for Signal analysis.
`Convolve[`$expr1$`,`$xpr2$`]`	
	Returns the convolution of two expressions $expr1$ and $expr2$.

Table F.6: Useful MATHEMATICA commands IV

Fourier & z-Transform - Discrete-time domain	
Command	**description**
`RSolve[{`$exprs$`},`$var1$`,`$var2$`]`	
	Counter part of `DSolve[]` to solve difference equations, with initial conditions and input function.
`FourierSequenceTransform[`$expr$`]`	
	Returns the discrete-time Fourier sequence transform of $expr$. The option `FourierParameters` $\rightarrow$ {1,-1} should be used to have the correct normalisation of the Fourier Transforms as is common for Signal analysis.
`InverseFourierSequenceTransform[`$expr$`]`	
	Returns the inverse discrete-time Fourier sequence transform of $expr$.
`DiscreteConvolve`f`,`g`,`n`,`m`]`	
	Returns the convolution of two discrete sequence functions f and g.
`Fourier[`$list$`]`	
	Returns the Discrete Fourier Transform of a list of (complex) numbers $list$.
`InverseFourier[`$list$`]`	
	Returns the inverse Discrete Fourier Transform of a list of (complex) numbers $list$.
`ListConvolve[`ker`,`$list$`]`	
	Returns the filtered signal of a list of numbers $list$ with the filter kernel ker, that is also a list of numbers.
`DiscreteConvolve[`$expr1$`,`$expr2$`,`n`,`m`]`	
	gives the convolution with respect to n of the expressions $expr1$ and $expr2$.
`ZTransform[`$expr$`,`$var1$`,`$var2$`]`	
	Returns the z-Transform of $expr$.
`InverseZTransform[`$expr$`,`$var1$`,`$var2$`]`	
	Returns the Inverse z-Transform of $expr$.

Table F.7: Useful MATHEMATICA commands V

Filter Design: FIR	
Command	**description**
`FrequencySamplingFilterKernel[`$list$`]`	
	Creates a finite impulse response (FIR) filter kernel using a frequency sampling method from amplitude values in $list$
`EquiripplefilterKernel[]`	
	Creates a finite impulse response (FIR) filter kernel of length n with an equiripple amplitude response, given the specified left and right band edge frequencies.
`LeastSquaresFilterKernel[]`	
	Creates a k-band finite impulse response (FIR) filter kernel of length n designed using a least squares method, given the specified frequencies.
Filter Design: IIR	
`ToDiscreteTimeModel[`sys`,`T`,`z`]`	
	Gives the discrete-time approximation, with sampling period T, of the continuous-time systems models sys in terms of variable z.
`RecurrenceFilter[`$\{a_i, b_j\}$`,`x`]`	
	filters x using a linear recurrence equation with coefficients a_i and b_j.
`RecurrenceTable[`$eqns$`,`$expr$`,`$\{n, n_{max}\}$`]`	
	generates a list of values of $expr$ for successive n based on solving the recurrence equations $eqns$.

Table F.8: Useful MATHEMATICA commands VI

Useful functions	
Command	**description**
`DiracDelta[`x`]`	
	Represents the Dirac delta function $\delta(x)$.
`KroneckerDelta[`n`]`	
	Gives the Kronecker delta δ_n, is equal to 1 for $n = 0$, and 0 otherwise.
`UnitStep[`x`]`	
	Represents the unit step function, equal to 0 for $x < 0$ and 1 for $x \geq 0$.
`Interpolation[`$list$`]`	
	Constructs an interpolation of the list values.
Visualisation	
`DiscretePlot[`$expr$`,{`n`,`n_{max}`}]`	
	Generates a plot of the values of $expr$ when n runs from 1 to n_{max}.
`ListPlot[`$list$`]`	
	Plots data from $list$, markers only.
`ListLinePlot[`$list$`]`	
	Plots data from $list$, line.
List functions	
`Table[`$expr$`,`n`]`	
	Generates a list of n copies of $expr$.
`Flatten[`$list$`]`	
	Flattens out nested lists.
`Transpose[`$list$`]`	
	Transposes the first two levels in list.
`Take[`$list$`,`n`]`	
	Gives the first n elements of $list$.
`RotateLeft[`$expr$`,`n`]`	
	Cycles the elements in $expr$ n positions to the left.
`RotateRigh[`$expr$`,`n`]t`	
	cycles the elements in $expr$ n positions to the right.
`TimeSeries[{{`t_1, v_1`},{`t_2, v_2`}...}]`	
	represents a time series specified by time-value pairs $\{t_i, v_i\}$.

Appendix G

Matlab, Python and Mathematica Code Examples

G.1 Introduction

The three systems that are addressed in this reader all have their own implementation of functionalities and integrated in their own environment. Although I have some preferences, all three systems are suitable to do the work. It is hard to give an advice for either one of the three, since that should also depend on your own experiences and preferences. This chapter contains examples of use of the three systems. For most of the examples contain the use of the code for a specific problem, that is worked out in the three different systems, so you are able to make some comparison. Only by working through your own problems with the various systems you can develop your own preferences. A few words on the structure of the systems might help you to get a start on the fly.

G.1.1 The Matlab implementation and environment

G.1.2 The Mathematica implementation and environment

G.1.3 The Python implementation and environment

G.2 Simulation of Pendulum motion

This is a worked out example from chapter 2.
The equation of motion of a pendulum of mass m at the end of a string with length l is expressed in terms of the angle θ with respect to its steady state position. No damping, especially in the pivot point, is assumed. Initially a torque T_c is applied to the system. The non-linear differential equation is given by:

$$\begin{aligned} ml^2\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + mgl\sin\theta &= T_c \\ \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{l}\sin\theta &= \frac{T_c}{ml^2} \end{aligned} \tag{G.1}$$

This is clearly a non-linear equation as a result of the sinusoidal term. For small angles the $\sin\theta$ term can be approximated by linear term θ to obtain a LTI system given by:

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{l}\theta = \frac{T_c}{ml^2} \tag{G.2}$$

From this equation it is obvious that the natural oscillation frequency $\omega_n = \sqrt{g/l}$ and $\zeta = 0$.

The presence of the second derivative requires the introduction of an auxiliary variable, the first derivative. Using a vector notation with $y(1) = \theta$ and $y(2) = \mathrm{d}\theta/\mathrm{d}t$ as the two components of the vector y:

$$\frac{\mathrm{d}y(2)}{\mathrm{d}t} = -\frac{g}{l}\sin y(1) + \frac{T_c}{ml^2} \tag{G.3}$$

$$\frac{\mathrm{d}y(1)}{\mathrm{d}t} = y(2) \tag{G.4}$$

These equations are the basis for the coding in the various programming languages. Codes for the various implementations are given in the sub-sections below.

G.2.1 Numerical simulation of a Pendulum with Matlab

```
% Matlab code
clear all
global g l m Tc % declare global variables
g=9.82;       % m/s^2 gravitational constant
l=1;          % m length of the pendulum
m=0.2;        % kg mass of the pendulum
Tc=1;         % N initial torque on the pendulum
t0=0;         % s start time
tf=10;        % s final time
tspan=[t0 tf]; % vector with time span
y0=[0; 0];    % vector with initial condition, angle and velocity zero


% call integration routine ode23 that evaluates funnction with the aid of
    the defined derivative function
[t_nl,y_nl]=ode23(@pendulumdot_nl,tspan,y0);    %non-linear
[t_lin,y_lin]=ode23(@pendulumdot_lin,tspan,y0);  %linear

% calculate with an smaller initial torque

Tc=0.1;
[t_nl2,y_nl2]=ode23(@pendulumdot_nl,tspan,y0);    %non-linear
[t_lin2,y_lin2]=ode23(@pendulumdot_lin,tspan,y0);  %linear


% plot of the result with conversion from radians to degrees

plot(t_nl,y_nl(:,1)*180/pi,'b-',t_lin,y_lin(:,1)*180/pi,'r-',t_nl2,y_nl2
    (:,1)*180/pi,'k-',t_lin2,y_lin2(:,1)*180/pi,'r--')
xlabel('Time [s]')
ylabel('Angle [deg]')
legend('Non-linear Tc=1N','Linear Tc=1N','Nonlinear Tc=0.1N', 'Linear Tc
    =0.1N')
```

Listing G.1: Matlab code for pendulum I

Define the function that calculates the derivatives for the non-linear pendulum case.

```
function ydot = pendulumdot_nl(t,y)

global g l m Tc     % declare global variables
if t<=1             % determine value input function
    T=Tc;
else T=0;
end
ydot=[0;0];          % assures that ydot is returned as vector
ydot(1)=y(2);
ydot(2)=-(g/l) *sin(y(1)) + T/(m*l^2);
end
```

Listing G.2: Matlab code for pendulum II

Define the function that calculates the derivatives for the linear pendulum case

```
function ydot = pendulumdot_lin(t,y)

global g l m Tc % declare global variables
if t<=1       % determine value input function
    T=Tc;
else T=0;
end
ydot=[0;0];
ydot(1)=y(2);
ydot(2)=- (g/l)* y(1) + T/(m*l^2);
end
```

Listing G.3: Matlab code for pendulum III

The plot with the results is given in figure G.1. For the high torque situation a major difference is observed between the non-linear and linear equation of motion. This is the direct of the consequence that for larger angles the sinus can not be approximated by the angle. For a small initial torque the angles remain small and the linear version is sufficient.

G.2.2 Numerical simulation of a Pendulum with Python

First the various libraries have to be loaded, the parameters set and the functions defined:

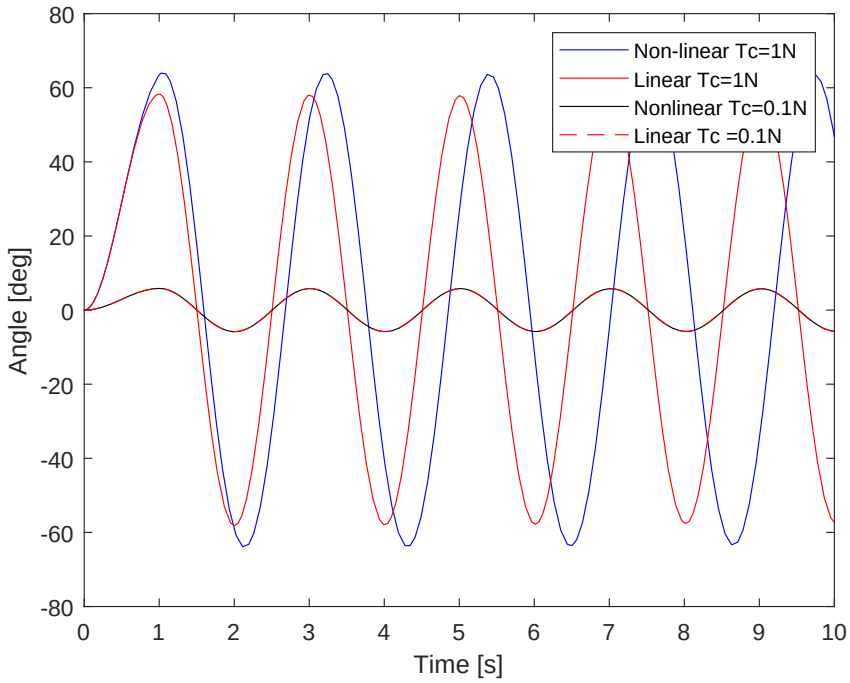


Figure G.1: Response of pendulum after a $1\,\mathrm{s}$ Torque of $1\,\mathrm{N}$ is applied and after $0.1\,\mathrm{N}$ is applied. Simulation is done for both the non-linear equation of motion and the linear version.

```python
# Import the required modules
import numpy as np
import matplotlib.pyplot as plt
# This makes the plots appear inside the notebook
%matplotlib inline
from scipy.integrate import odeint

g=9.82
l=1.0
m=0.2
tc=1.0
t0=0.0
t1=1.0
tf=16.0
y0=np.array([0.0,0.0])

def dTheta_dt(theta, t):
    tci=0.0
    if t<=t1:
        tci=tc
    else:
        tci=0.0
    return [theta[1], -(g/l) * np.sin(theta[0])+ tci/(m * 1**2)]
def dThetal_dt(theta, t):
    tci=0.0
    if t<=t1:
        tci=tc
    else:
        tci=0.0
    return [theta[1], -(g/l) * theta[0] + tci/(m * 1**2)]
```

Listing G.4: Python code for pendulum I

After the initialisation, first both the non-linear model and the linearalized model is run for a large drive, and the results plotted.

```
ts = np.linspace(0.0, tf, 300)
theta0 = [0.0, 0.0]
thetas = odeint(dTheta_dt, theta0, ts)
thetasl = odeint(dThetal_dt, theta0, ts)
thetaOp = thetas[:,0]
thetaOs = thetas[:,1]
thetaOpl = thetasl[:,0]
thetaOsl = thetasl[:,1]

plt.plot(ts, thetaOp, 'r', label="non-linear")
plt.plot(ts, thetaOpl, 'b', label="linear")
plt.xlabel("Time [s]")
plt.ylabel("Angle [rad]")
plt.legend()
plt.grid(True)
plt.title("Response of a Pendulum");
```

Listing G.5: Python code for pendulum II

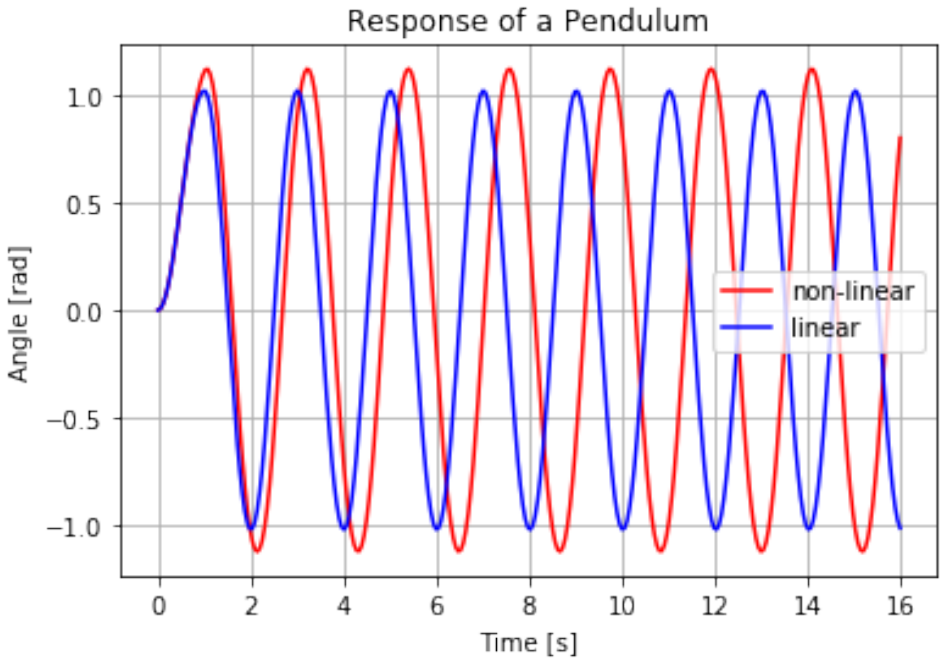


Figure G.2: Mathematica result weak drive

Now both models are run with a small drive. This results in a closer match in the results between both models.

```
tc=0.2
thetas = odeint(dTheta_dt, theta0, ts)
thetasl = odeint(dThetal_dt, theta0, ts)
thetaOp = thetas[:,0]
thetaOs = thetas[:,1]
thetaOpl = thetasl[:,0]
thetaOsl = thetasl[:,1]

plt.figure()
plt.plot(ts, thetaOp, 'r', label="non-linear")
plt.plot(ts, thetaOpl, 'b', label="linear")
plt.xlabel("Time [s]")
plt.ylabel("Angle [rad]")
plt.legend()
plt.grid(True)
plt.title("Response of a Pendulum");
```

Listing G.6: Python code for pendulum III

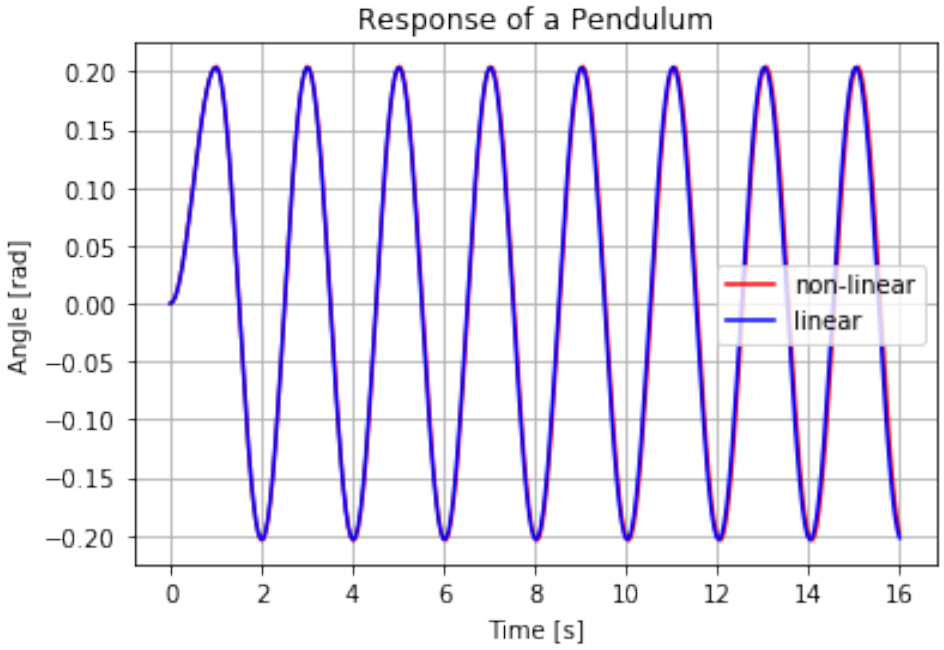


Figure G.3: Mathematica result weak drive

G.2.3 Numerical simulation of a Pendulum with Mathematica

```
In[1]:= Remove["Global`*"]
```

Simulation of Pendulum motion

```
In[2]:= $Assumptions = {m, g, l, tc, tp} ∈ Reals
```

Define the drive (input) function.

```
In[3]:= tdrive[t_, tc_, tp_] = tc (UnitStep[t] - UnitStep[t - tp])
```

Define the non-linear equation:

```
In[4]:= deqn = m l^2 θ''[t] + m g l Sin[θ[t]] == tdrive[t, tc, tp]
```

Define the linearized equation:

```
In[5]:= deqnl = m l^2 θ''[t] + m g l θ[t] == tdrive[t, tc, tp]
```

Set parameter for the strong and weak drive respectively as:

```
In[6]:= param = {g → 9.82, l → 1, m → 0.2, tc → 1, tp → 1}
In[7]:= param2 = {g → 9.82, l → 1, m → 0.2, tc → 0.2, tp → 1}
```

Figure G.4: Mathematica initialisation

Solve both differential equations numerically with a strong drive:

```
In[8]:= dsol = NDSolve[{deqn /. param, θ[0] == 0, θ'[0] == 0}, θ[t], {t, 0, 16}]
In[9]:= dsoll = NDSolve[{deqnl /. param, θ[0] == 0, θ'[0] == 0}, θ[t], {t, 0, 16}]
```

Plot the results:

```
Plot[{θ[t] /. dsol, θ[t] /. dsoll, tdrive[t, 1, 1]}, {t, 0, 16},
 PlotStyle → {Red, Blue, {Black, Thin}},
 Filling → {1 → None, 2 → None, 3 -> Axis},
 FillingStyle → {White, White, LightYellow},
 PlotLegends → {"Exact", "Linearized", "Drive"},
 AxesLabel → {"t [s]", "θ[t] [rad]"},
 Exclusions → None,
 BaseStyle → {FontSize -> 18},
 ImageSize → Large,
 PlotLabel → "Large drive → strong non-linearity"]
```

Figure G.5: Mathematica simulation strong drive

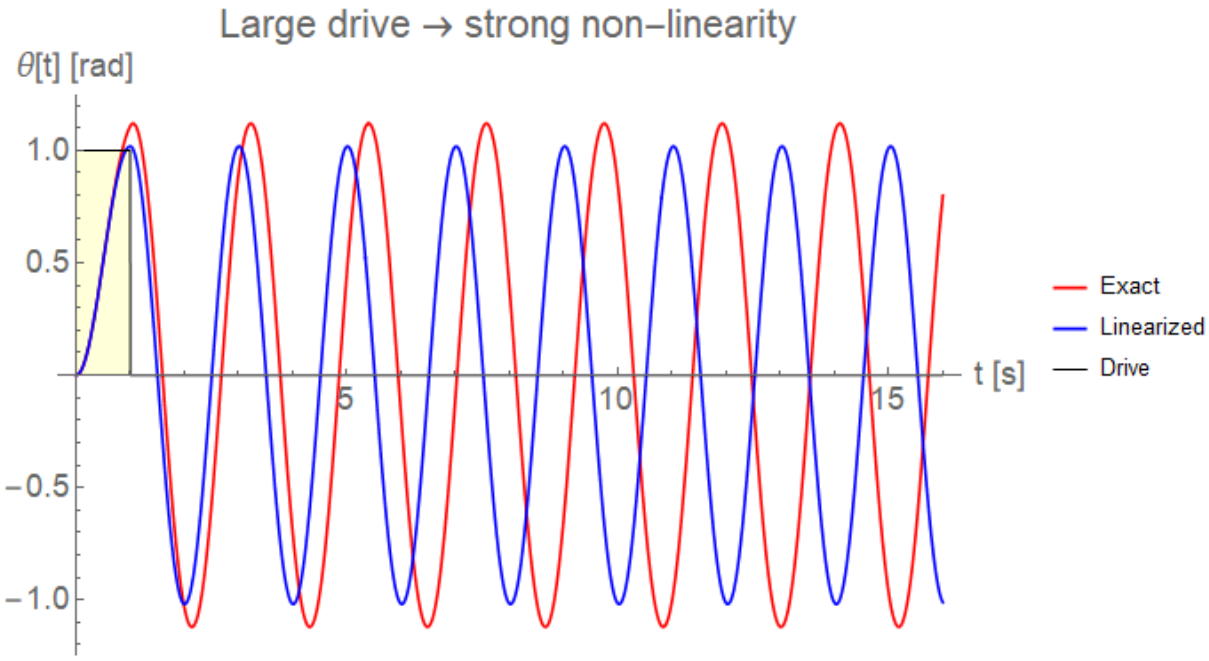


Figure G.6: Mathematica result strong drive

Solve the differential equations again, with a weak drive:

```
dsol2 = NDSolve[{deqn /. param2, θ[0] == 0, θ'[0] == 0}, θ[t], {t, 0, 16}]
```

In[12]:=
```
dsol2l = NDSolve[{deqnl /. param2, θ[0] == 0, θ'[0] == 0}, θ[t], {t, 0, 16}]
```

Plot the results:

```
Plot[{θ[t] /. dsol2, θ[t] /. dsol2l, tdrive[t, 0.2, 1]}, {t, 0, 16},
 PlotStyle → {Red, Blue, {Black, Thin}},
 Filling → {1 → None, 2 → None, 3 -> Axis},
 FillingStyle → {White, White, LightYellow},
 PlotLegends → {"Exact", "Linearized", "Drive"},
 AxesLabel → {"t [s]", "θ[t] [rad]"},
 Exclusions → None,
 BaseStyle → {FontSize -> 18},
 ImageSize → Large,
 PlotLabel → "Small drive → almost linear"]
```

Figure G.7: Mathematica simulation weak drive

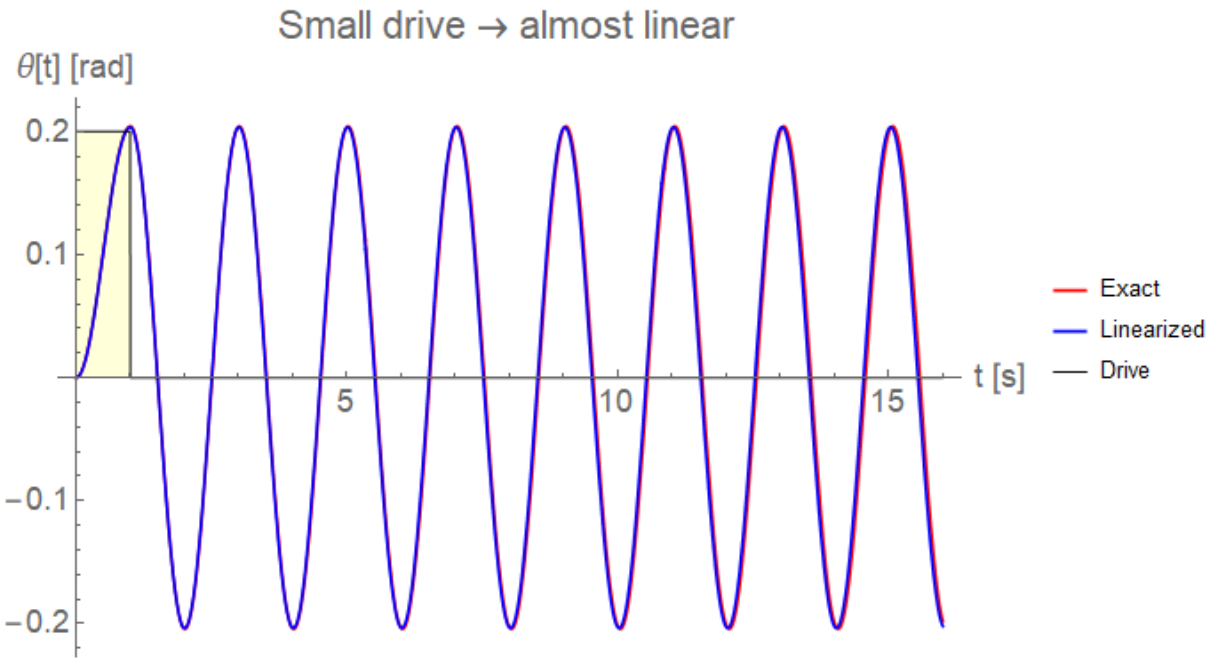


Figure G.8: Mathematica result weak drive

Appendix H

Matlab, Python and Mathematica Function Examples

This Appendix is reserved for examples of the use of dedicated functions related to modelling and Signal processing in the various programs: MATHEMATICA, MATLAB *and* PYTHON. *The information is not yet available in a stable format. A collection will be build up on Canvas and made available later here in this appendix.*

H.1 Introduction

H.2 Mathematica function examples

H.3 Matlab function examples

H.4 Python function examples

Appendix I

Formula Sheet

Formula Sheet

Integrals – indefinite

$$\int \frac{1}{\tau^2 + t^2} \mathrm{d}t = \frac{1}{\tau} \tan^{-1} \frac{t}{\tau}$$

$$\int \frac{t}{\tau^2 + t^2} \mathrm{d}t = \frac{1}{2} \ln(\tau^2 + t^2)$$

$$\int \frac{1}{\tau^2 - t^2} \mathrm{d}t = \frac{1}{2\tau} \ln \frac{\tau + t}{\tau - t}$$

$$\int t \mathrm{e}^{at} \mathrm{d}t = \frac{\mathrm{e}^{at}}{a} \left(t - \frac{1}{a} \right)$$

$$\int t \sin at \mathrm{d}t = \frac{1}{a^2} \sin at - \frac{t}{a} \cos(at)$$

$$\int t \cos at \mathrm{d}t = \frac{1}{a^2} \cos at + \frac{t}{a} \sin(at)$$

$$\int t^2 \sin at \mathrm{d}t = \frac{2t}{a^2} \sin at + \left(\frac{2}{a^3} - \frac{t^2}{a} \right) \cos(at)$$

$$\int t^2 \cos at \mathrm{d}t = \frac{2t}{a^2} \cos at - \left(\frac{2}{a^3} - \frac{t^2}{a} \right) \sin(at)$$

$$\int \cos at \cos bt \mathrm{d}t = \frac{\sin(a-b)t}{2(a-b)} + \frac{\sin(a+b)t}{2(a+b)}$$

$$\int \sin at \cos bt \mathrm{d}t = -\frac{\cos(a-b)t}{2(a-b)} - \frac{\cos(a+b)t}{2(a+b)}$$

Integrals – definite

For the next 4 integrals $n \in \mathbb{N}$ holds.

$$\int_0^\infty t^n \mathrm{e}^{-at} \mathrm{d}t = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty t^{2n} \mathrm{e}^{-a^2 t^2} \mathrm{d}t = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{1}{2a} \right)^{2n+1}$$

$$\int_0^\infty t^{2n+1} \mathrm{e}^{-a^2t^2} \mathrm{d}t = \frac{n!}{2} \frac{1}{a^{2n+2}}$$

$$\int_{-\infty}^\infty \mathrm{e}^{-at^2} \mathrm{d}t = \left(\frac{\pi}{a}\right)^{\frac{1}{2}}$$

$$\int_{-\infty}^\infty \mathrm{e}^{-at^2} \cos bt \mathrm{d}t = \sqrt{\frac{\pi}{a}} \mathrm{e}^{-\frac{b^2}{4a}}$$

Sums

$$\sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{24}$$

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z} \; \forall \; |z| < 1$$

$$\sum_{n=0}^{N-1} z^n = \frac{1-z^N}{1-z} \; \forall \; z \neq 1$$

$$\sum_{n=0}^{\infty} n z^n = \frac{z}{(1-z)^2} \; \forall \; |z| < 1$$

---

end Formula Sheet